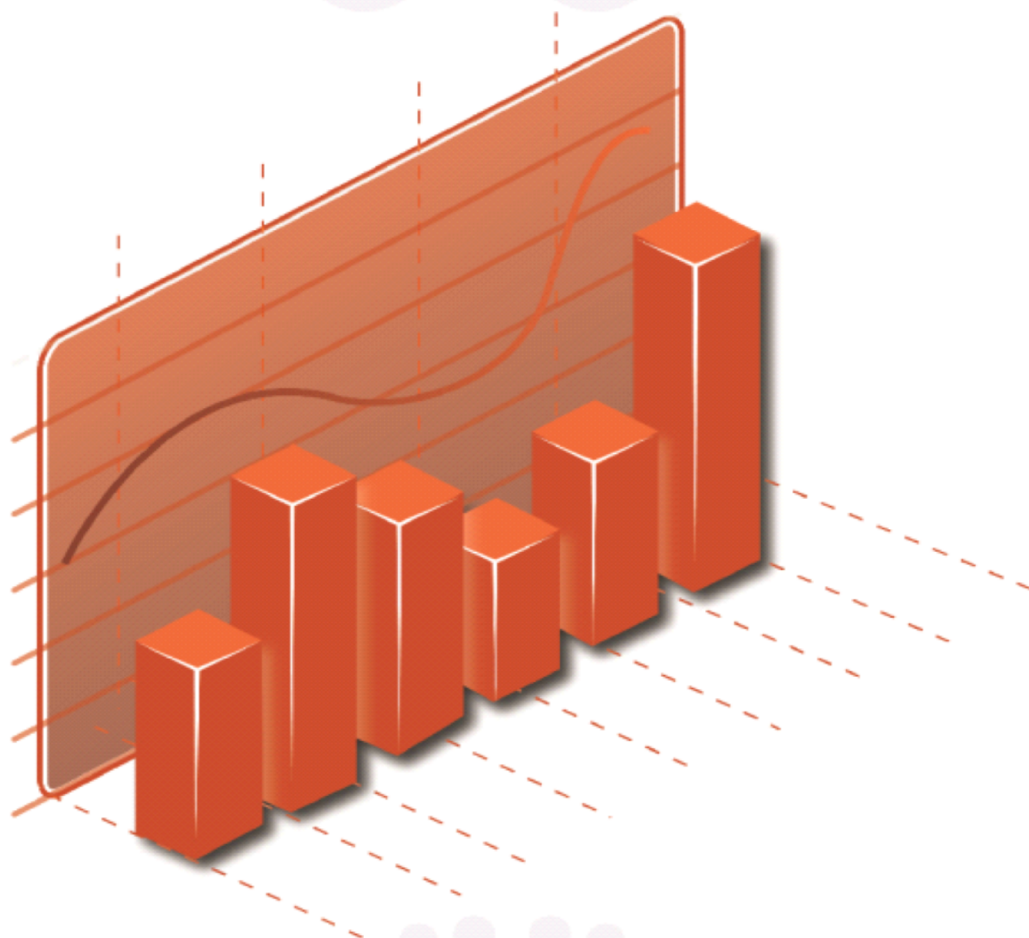


Tatva Practice Book

Integral Calculus

Class 12 **JEE**



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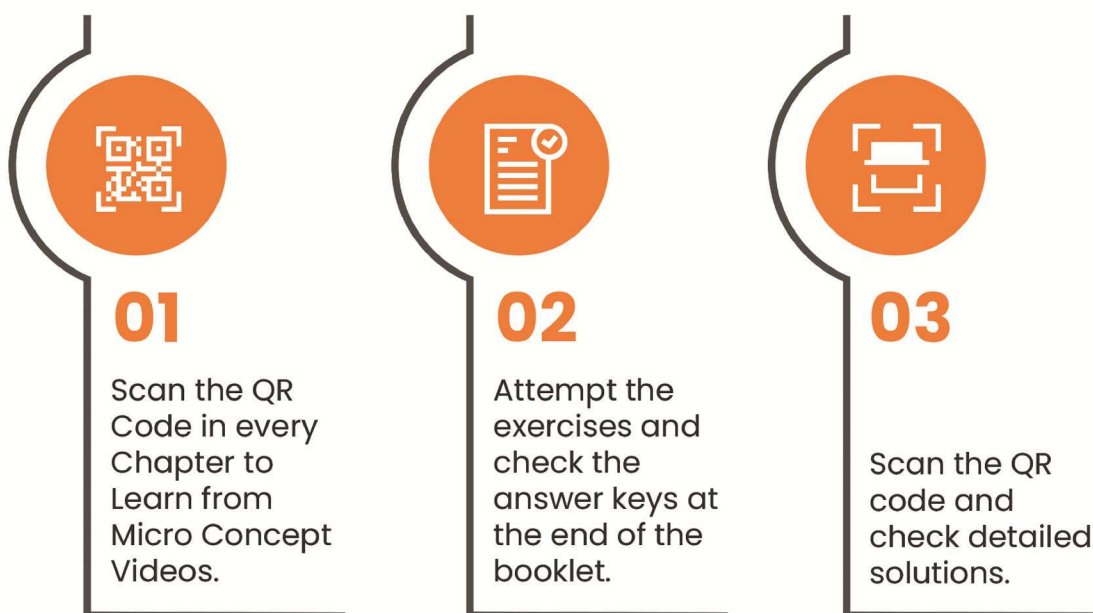


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How to Use ?



Founder's Message



Dear Student,

It gives me immense pleasure to present to you a Ready Reckoner and Practice booklet by Vedantu – 'TATVA'. Tatva in Sanskrit, means a collection of "Core" content/truth and isn't that what Vedantu brings to your table – the Core? Vedantu, at all times, ensures that you have easy accessibility to a collection of essential concepts, theory, derivations, definitions, solved examples, concept videos, and practice questions, important questions from competitive examinations and I am pleased to say that all questions in 'Tatva' come with detailed solutions.

Tatva is the result of the constant endeavour and research done by our highly experienced team of teachers and subject experts who compiled relevant content for you so that you come out with flying colours in IIT JEE/NEET and Olympiads. We strongly believe and vouch for the effectiveness and relevance of this booklet to grab the desired rank in IIT JEE/ NEET.

Key Points on 'How to obtain optimal benefits from 'Tatva' ?

- * We suggest starting with the Practice of 'Tatva questions' at the end of a chapter after revision and Practice of basic questions.
- * For daily practice of questions, we strongly recommend attempting session assignments along with Vedantu live sessions
- * Tatva equips you with Theory, Concept Videos, and Solved examples to help you revise concepts, mark your notes, walk you through the entire summary and eventually clear all your conceptual doubts all by yourself.
- * Attempt Tatva questions post revision of 11th and 12th Grade topics. Take one topic at a time to solve. Attempt any revision test series for NEET/JEE once you have solved Tatva questions. Following this will not only help you during revision but also give a major boost to your confidence.
- * First solve basic level questions and gradually progress to the advanced level.
- * Practice basic level questions regularly all through your journey of preparation.
- * Your success is our dream-come-true

I wish you all the best.

Anand Prakash
Founder, Academic Head
Vedantu



Anand Prakash Sir has been a pioneer in producing Top Ranks in JEE/NEET and Olympiads. He has personally taught and mentored AIR 1,6,7 (JEE ADVANCED) and AIR-1, 7, 9 (AIIMS), and thousands of more students who have successfully cleared these competitive exams in the last few years.



Credits



**“Happiness lies in the joy of achievement
and the thrill of creative effort.”**

—Theodore Roosevelt

Tatva is the brainchild of the creative Vedans who strived tirelessly to weave success stories for you. Our heartfelt thanks to the Super Vedans who give wings to the vision of Vedantu.

Our leaders who have been our guiding light and encouragement in every step:
Vamsi Sir, Anand Prakash Sir and Pulkit Sir

Our gratitude to the insightful leadership and guidance of our leaders who dreamt about ‘Tatva’, steered the project in the right direction and were instrumental in making this dream into a reality:

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Shubam Gupta	Ajay Mittal
Arshad Shahid	Jaideep Sontakke

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Mathematics Team

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INDEFINITE INTEGRATION

INDEFINITE INTEGRATION

Integration is the inverse process of differentiation. That is, the process of finding a function, whose differential coefficient is known, is called integration.

If the differential coefficient of $F(x)$ is $f(x)$,

i.e. $\frac{d}{dx}[F(x)] = f(x)$, then we say that the **antiderivative** or

integral of $f(x)$ is $F(x)$, written as $\int f(x)dx = F(x)$,

Here $\int dx$ is the notation of integration, $f(x)$ is the integrand, x is the variable of integration and dx denotes the integration with respect to x .

1. INDEFINITE INTEGRAL

1.1 Definition

We know that if $\frac{d}{dx}[F(x)] = f(x)$, then $\int f(x)dx = F(x)$.

Also, for any arbitrary constant C ,

$$\frac{d}{dx}[F(x) + C] = \frac{d}{dx}[F(x)] + 0 = f(x).$$

$$\therefore \int f(x)dx = F(x) + C,$$

This shows that $F(x)$ and $F(x) + C$ are both integrals of the same function $f(x)$. Thus, for different values of C , we obtain different integrals of $f(x)$. This implies that the integral of $f(x)$ is not definite. By virtue of this property $F(x)$ is called the indefinite integral of $f(x)$.

1.2 Properties of Indefinite Integration

1. $\frac{d}{dx}\left[\int f(x)dx\right] = f(x)$
2. $\int f'(x)dx = \int \frac{d}{dx}[f(x)]dx = f(x) + c$
3. $\int k f(x)dx = k \int f(x)dx$, where k is any constant

4. If $f_1(x), f_2(x), f_3(x), \dots$ (finite in number) are functions of x , then

$$\begin{aligned} & \int [f_1(x) \pm f_2(x) \pm f_3(x) \dots] dx \\ &= \int f_1(x)dx \pm \int f_2(x)dx \pm \int f_3(x)dx \pm \dots \end{aligned}$$

5. If $\int f(x)dx = F(x) + c$

$$\text{then } \int f(ax \pm b)dx = \frac{1}{a}F(ax \pm b) + c$$

6. Suppose I and J are intervals, $g: J \rightarrow I$ is differentiable and $f: I \rightarrow \mathbb{R}$ has integral with primitive F . Then $(f \circ g).g': J \rightarrow \mathbb{R}$ has an integral and

$$\int ((f \circ g).g')(x) dx = \int f(g(x))g'(x)dx = F(g(x)) + c$$

1.3 Standard Formulae of Integration

The following results are a direct consequence of the definition of an integral.

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1.$
2. $\int \frac{1}{x} dx = \log |x| + C$
3. $\int e^x dx = e^x + C$
4. $\int a^x dx = \frac{a^x}{\log_e a} + C.$
5. $\int \sin x dx = -\cos x + C$
6. $\int \cos x dx = \sin x + C$
7. $\int \sec^2 x dx = \tan x + C$
8. $\int \operatorname{cosec}^2 x dx = -\cot x + C$
9. $\int \sec x \tan x dx = \sec x + C$





10. $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C.$

11. $\int \tan x \, dx = -\log |\cos x| + C = \log |\sec x| + C.$

12. $\int \cot x \, dx = \log |\sin x| + C$

13. $\int \sec x \, dx = \log |\sec x + \tan x| + C$

14. $\int \operatorname{cosec} x \, dx = \log |\operatorname{cosec} x - \cot x| + C$

15. $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C ; |x| < 1$

16. $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

17. $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} |x| + C ; |x| > 1$

$$= \frac{1}{n} f(t) = \frac{1}{n} f(x^n) + c$$

(iii) When the integrand is of the form $[f(x)]^n \cdot f'(x)$, we put $f(x) = t$ and $f'(x) dx = dt$.

Thus, $\int [f(x)]^n f'(x) dx = \int t^n dt = \frac{t^{n+1}}{n+1} = \frac{[f(x)]^{n+1}}{n+1} + c$

(iv) When the integrand is of the form $\frac{f'(x)}{f(x)}$, we put $f(x) = t$ and $f'(x) dx = dt$.

Thus, $\int \frac{f'(x)}{f(x)} dx = \int \frac{dt}{t} = \log |t| = \log |f(x)| + c$

2.2 Some Special Integrals

1. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

2. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$

3. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$

4. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$

5. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$

6. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + C$

7. $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$

8. $\int \sqrt{x^2 + a^2} \, dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C$

9. $\int \sqrt{x^2 - a^2} \, dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$

2. INTEGRATION BY SUBSTITUTION

2.1 Method of Substitution

By suitable substitution, the variable x in $\int f(x) dx$ is changed into another variable t so that the integrand $f(x)$ is changed into $F(t)$ which is some standard integral or algebraic sum of standard integrals.

There is no general rule for finding a proper substitution and the best guide in this matter is experience.

However, the following suggestions will prove useful.

(i) If the integrand is of the form $f'(ax+b)$, then we put $ax+b = t$ and $dx = \frac{1}{a} dt$.

Thus, $\int f'(ax+b) dx = \int f'(t) \frac{dt}{a}$

$$= \frac{1}{a} \int f'(t) dt = \frac{f(t)}{a} = \frac{f(ax+b)}{a} + c$$

(ii) When the integrand is of the form $x^{n-1} f'(x^n)$, we put $x^n = t$ and $nx^{n-1} dx = dt$.

Thus, $\int x^{n-1} f'(x^n) dx = \int f'(t) \frac{dt}{n} = \frac{1}{n} \int f'(t) dt$





2.3 Integrals of the Form

$$(a) \int f(a^2 - x^2) dx,$$

$$(b) \int f(a^2 + x^2) dx,$$

$$(c) \int f(x^2 - a^2) dx,$$

$$(d) \int f\left(\frac{a-x}{a+x}\right) dx,$$

Working Rule

Integral

Substitution

$$\int f(a^2 - x^2) dx, \quad x = a \sin \theta \text{ or } x = a \cos \theta$$

$$\int f(a^2 + x^2) dx, \quad x = a \tan \theta \text{ or } x = a \cot \theta$$

$$\int f(x^2 - a^2) dx, \quad x = a \sec \theta \text{ or } x = a \operatorname{cosec} \theta$$

$$\int f\left(\frac{a-x}{a+x}\right) dx \text{ or } \int f\left(\frac{a+x}{a-x}\right) dx \quad x = a \cos 2\theta$$

2.4 Integrals of the Form

$$(a) \int \frac{dx}{ax^2 + bx + c}, \quad (b) \int \frac{dx}{\sqrt{ax^2 + bx + c}},$$

$$(c) \int \sqrt{ax^2 + bx + c} dx$$

Working Rule

(i) Make the coefficient of x^2 unity by taking the coefficient of x^2 outside the quadratic.

(ii) Complete the square in the terms involving x , i.e. write $ax^2 + bx + c$ in the form

$$a \left[\left(x + \frac{b}{2a} \right)^2 \right] - \frac{(b^2 - 4ac)}{4a}.$$

(iii) The integrand is converted to one of the nine special integrals.

(iv) Integrate the function.

2.5 Integrals of the Form

$$(a) \int \frac{px + q}{ax^2 + bx + c} dx, \quad (b) \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx,$$

$$(c) \int (px + q) \sqrt{ax^2 + bx + c} dx$$

Working Rule

$$(a) \int \frac{px + q}{ax^2 + bx + c} dx$$

Put $px + q = \lambda(2ax + b) + \mu$ or

$px + q = \lambda(\text{derivative of quadratic}) + \mu$.

Comparing the coefficient of x and constant term on both sides, we get

$$p = 2a\lambda \text{ and } q = b\lambda + \mu \Rightarrow \lambda = \frac{p}{2a} \text{ and } \mu = \left(q - \frac{bp}{2a} \right). \text{ Then}$$

integral becomes

$$\begin{aligned} & \int \frac{px + q}{ax^2 + bx + c} dx \\ &= \frac{p}{2a} \int \frac{2ax + b}{ax^2 + bx + c} dx + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{ax^2 + bx + c} \\ &= \frac{p}{2a} \log |ax^2 + bx + c| + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{ax^2 + bx + c} \end{aligned}$$

$$(b) \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx$$

In this case the integral becomes

$$\begin{aligned} & \Rightarrow \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx \\ & \Rightarrow \frac{p}{2a} \int \frac{2ax + b}{\sqrt{ax^2 + bx + c}} dx + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{\sqrt{ax^2 + bx + c}} \\ & \Rightarrow \frac{p}{a} \sqrt{ax^2 + bx + c} + \left(q - \frac{bp}{2a} \right) \int \frac{dx}{\sqrt{ax^2 + bx + c}} \end{aligned}$$

$$(c) \int (px + q) \sqrt{ax^2 + bx + c} dx$$

The integral in this case is converted to





$$\int (px + q)\sqrt{ax^2 + bx + c} \, dx = \frac{p}{2a} \int (2ax + b)\sqrt{ax^2 + bx + c} \, dx + \left(q - \frac{bp}{2a}\right) \int \sqrt{ax^2 + bx + c} \, dx$$

$$= \frac{p}{3a} (ax^2 + bx + c)^{3/2} + \left(q - \frac{bp}{2a}\right) \int \sqrt{ax^2 + bx + c} \, dx$$

2.6 Integrals of the Form

$$\int \frac{P(x)}{\sqrt{ax^2 + bx + c}} \, dx, \text{ where } P(x) \text{ is a polynomial in } x \text{ of degree } n \geq 2.$$

Working Rule:

Write $\int \frac{P(x)}{\sqrt{ax^2 + bx + c}} \, dx =$

$$= (a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1}) \sqrt{ax^2 + bx + c} + k \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

where $k, a_0, a_1, \dots, a_{n-1}$ are constants to be determined by differentiating the above relation and equating the coefficients of various powers of x on both sides.

2.7 Integrals of the Form

$$\int \frac{x^2 + 1}{x^4 + kx^2 + 1} \, dx \quad \text{or} \quad \int \frac{x^2 - 1}{x^4 + kx^2 + 1} \, dx,$$

where k is a constant positive, negative or zero.

Working Rule

- Divide the numerator and denominator by x^2 .
- Put $x - \frac{1}{x} = z$ or $x + \frac{1}{x} = z$, whichever substitution, on differentiation gives, the numerator of the resulting integrand.
- Evaluate the resulting integral in z
- Express the result in terms of x .

2.8 Integrals of the Form

$$\int \frac{dx}{P\sqrt{Q}}, \text{ where } P, Q \text{ are linear or quadratic functions of } x.$$

Integral **Substitution**

$$\int \frac{1}{(ax + b)\sqrt{cx + d}} \, dx \quad cx + d = z^2$$

$$\int \frac{dx}{(ax^2 + bx + c)\sqrt{px + q}} \quad px + q = z^2$$

$$\int \frac{dx}{(px + q)\sqrt{ax^2 + bx + c}} \quad px + q = \frac{1}{z}$$

$$\int \frac{dx}{(ax^2 + b)\sqrt{cx^2 + d}} \quad x = \frac{1}{z}$$

2.9 Integrals of the Form

$$\int R \left(x, x^{\frac{p_1}{q_1}}, x^{\frac{p_2}{q_2}}, \dots, x^{\frac{p_k}{q_k}} \right) dx$$

To evaluate $\int R \left(x, x^{\frac{p_1}{q_1}}, x^{\frac{p_2}{q_2}}, \dots, x^{\frac{p_k}{q_k}} \right) dx$ where R is a rational

function of its variables $x, x^{\frac{p_1}{q_1}}, \dots, x^{\frac{p_k}{q_k}}$, put $x = t^n$ where n is the L.C.M of the denominator of the fractions $p_1/q_1, p_2/q_2, \dots, p_k/q_k$.

3. INTEGRATION BY PARTIAL FRACTIONS

Integrals of the type $\int \frac{p(x)}{g(x)} \, dx$ can be integrated by resolving

the integrand into partial fractions. We proceed as follows:

Check degree of $p(x)$ and $g(x)$.

If degree of $p(x) \geq$ degree of $g(x)$, then divide $p(x)$ by $g(x)$ till its degree is less, i.e. put in the

form $\frac{p(x)}{g(x)} = r(x) + \frac{f(x)}{g(x)}$ where degree of $f(x) <$ degree of $g(x)$.

CASE 1: When the denominator contains non-repeated linear factors. That is





$$g(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n).$$

In such a case write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{(x - \alpha_1)} + \frac{A_2}{(x - \alpha_2)} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 2 : When the denominator contains repeated as well as non-repeated linear factor. That is

$$g(x) = (x - \alpha_1)^2(x - \alpha_3) \dots (x - \alpha_n).$$

In such a case write $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - \alpha_1} + \frac{A_2}{(x - \alpha_1)^2} + \frac{A_3}{x - \alpha_3} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

Note : Corresponding to repeated linear factor $(x - a)^r$ in the denominator, a sum of r partial fractions of the

type $\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \dots + \frac{A_r}{(x - a)^r}$ is taken.

CASE 3 : When the denominator contains a non repeated quadratic factor which cannot be factorised further:

$$g(x) = (ax^2 + bx + c)(x - \alpha_3)(x - \alpha_4) \dots (x - \alpha_n).$$

In such a case express $f(x)$ and $g(x)$ as:

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2 + bx + c} + \frac{A_3}{x - \alpha_3} + \dots + \frac{A_n}{x - \alpha_n}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 4 : When the denominator contains a repeated quadratic factor which cannot be factorised further: That is

$$g(x) = (ax^2 + bx + c)^2(x - \alpha_5)(x - \alpha_6) \dots (x - \alpha_n)$$

In such a case write $f(x)$ and $g(x)$ as

$$\frac{f(x)}{g(x)} = \frac{A_1x + A_2}{ax^2 + bx + c} + \frac{A_3x + A_4}{(ax^2 + bx + c)^2} + \frac{A_5}{x - \alpha_5} + \dots + \frac{A_n}{(x - \alpha_n)}$$

where $A_1, A_2, \dots, A_n$ are constants to be determined by comparing the coefficients of various powers of x on both sides after taking L.C.M.

CASE 5 : If the integrand contains only even powers of x

- (i) Put $x^2 = z$ in the integrand.
- (ii) Resolve the resulting rational expression in z into partial fractions
- (iii) Put $z = x^2$ again in the partial fractions and then integrate both sides.

4. INTEGRATION BY PARTS

The process of integration of the product of two functions is known as integration by parts.

For example, if u and v are two functions of x ,

$$\text{then } \int (uv) dx = u \cdot \int v dx - \int \left(\frac{du}{dx} \cdot \int v dx \right) dx.$$

In words, integral of the product of two functions = first function $\times$ integral of the second – integral of (differential of first $\times$ integral of the second function).

Working Hints

- (i) Choose the first and second function in such a way that derivative of the first function and the integral of the second function can be easily found.
- (ii) In case of integrals of the form $\int f(x) \cdot x^n dx$, take x^n as the first function and $f(x)$ as the second function.
- (iii) In case of integrals of the form $\int (\log x)^n \cdot 1 dx$, take 1 as the second function and $(\log x)^n$ as the first function.
- (iv) Rule of integration by parts may be used repeatedly, if required.
- (v) If the two functions are of different type, we can choose the first function as the one whose initial comes first in the word “ILATE”, where
I — Inverse Trigonometric function
L — Logarithmic function
A — Algebraic function
T — Trigonometric function
E — Exponential function.
- (vi) In case, both the functions are trigonometric, take that function as second function whose integral is simple. If both the functions are algebraic, take that function as first function whose derivative is simpler.





- (vii) If the integral consists of an inverse trigonometric function of an algebraic expression in x , first simplify the integrand by a suitable trigonometric substitution and then integrate the new integrand.

4.1 Integrals of the Form:

Where the initial integrand reappears after integrating by parts.

Working Rule

- Apply the method of integration by parts twice.
- On integrating by parts second time, we will obtain the given integrand again, put it equal to I .
- Transpose and collect terms involving I on one side and evaluate I .

4.2 Integrals of the Form

$$\int e^x [f(x) + f'(x)] dx$$

Working Rule

- Split the integral into two integrals.
- Integrate only the first integral by parts, i.e.

$$\begin{aligned} & \int e^x [f(x) + f'(x)] dx \\ &= \int e^x f(x) dx + \int e^x f'(x) dx \\ &= [f(x) \cdot e^x - \int f'(x) \cdot e^x dx] + \int e^x f'(x) dx \\ &= e^x f(x) + C. \end{aligned}$$

4.3 Integrals of the Form

$$\begin{aligned} & \int (f(x) + xf'(x)) dx \\ &= \int f(x) dx + \int x f'(x) dx \\ &= \int f(x) dx + [x f(x) - \int 1 \cdot f(x) dx] = xf(x) + c \end{aligned}$$

5. INTEGRATION OF VARIOUS TRIGONOMETRIC FUNCTIONS

5.1 Integral of the Form

$$\begin{aligned} \text{(a)} \quad & \int \frac{dx}{a + b \cos x} \quad \text{(b)} \quad \int \frac{dx}{a + b \sin x} \\ \text{(c)} \quad & \int \frac{dx}{a + b \cos x + c \sin x} \end{aligned}$$

Working Rule

(i) Put $\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ and $\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ so that the given

integrand becomes a function of $\tan \frac{x}{2}$.

- Put $\tan \frac{x}{2} = z \Rightarrow \frac{1}{2} \sec^2 \frac{x}{2} dx = dz$
- Integrate the resulting rational algebraic function of z
- In the answer, put $z = \tan \frac{x}{2}$.

5.2 Integrals of the Form

$$\begin{aligned} \text{(a)} \quad & \int \frac{dx}{a + b \cos^2 x} \quad \text{(b)} \quad \int \frac{dx}{a + b \sin^2 x} \\ \text{(c)} \quad & \int \frac{dx}{a \cos^2 x + b \sin x \cos x + c \sin^2 x} \end{aligned}$$

Working Rule

- Divide the numerator and denominator by $\cos^2 x$.
- In the denominator, replace $\sec^2 x$, if any, by $1 + \tan^2 x$.
- Put $\tan x = z \Rightarrow \sec^2 x dx = dz$.
- Integrate the resulting rational algebraic function of z .
- In the answer, put $z = \tan x$.





5.3 Integrals of the Form

$$\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx$$

Working Rule

- (i) Put Numerator = λ (denominator) + μ (derivative of denominator)
 $a \cos x + b \sin x = \lambda (c \cos x + d \sin x) + \mu (-c \sin x + d \cos x)$.
- (ii) Equate coefficients of $\sin x$ and $\cos x$ on both sides and find the values of λ and μ .
- (iii) Split the given integral into two integrals and evaluate each integral separately, i.e.

$$\int \frac{a \cos x + b \sin x}{c \cos x + d \sin x} dx =$$

$$\lambda \int 1 dx + \mu \int \frac{-c \sin x + d \cos x}{c \cos x + d \sin x} dx = \lambda x + \mu \log |c \cos x + d \sin x|.$$

- (iv) Substitute the values of λ and μ found in step 2.

5.4 Integrals of the Form

$$\int \frac{a + b \cos x + c \sin x}{e + f \cos x + g \sin x} dx$$

Working Rule

- (i) Put Numerator = l (denominator) + m (derivative of denominator) + n
 $a + b \cos x + c \sin x = l (e + f \cos x + g \sin x) + m (-f \sin x + g \cos x) + n$
- (ii) Equate coefficients of $\sin x$, $\cos x$ and constant term on both sides and find the values of l , m , n .
- (iii) Split the given integral into three integrals and evaluate each integral separately, i.e.

$$\int \frac{a + b \cos x + c \sin x}{e + f \cos x + g \sin x} dx$$

$$= l \int 1 dx + m \int \frac{-f \sin x + g \cos x}{e + f \cos x + g \sin x} dx + n \int \frac{dx}{e + f \cos x + g \sin x}$$

$$= lx + m \log |e + f \cos x + g \sin x| + n \int \frac{dx}{e + f \cos x + g \sin x}$$

- (iv) Substitute the values of l , m , n found in Step (ii).

5.5 Integrals of the Form

$$\int \sin^m x \cos^n x dx$$

Working Rule

- (i) If the power of $\sin x$ is an odd positive integer, put $\cos x = t$.
- (ii) If the power of $\cos x$ is an odd positive integer, put $\sin x = t$.
- (iii) If the power of $\sin x$ and $\cos x$ are both odd positive integers, put $\sin x = t$ or $\cos x = t$.
- (iv) If the power of $\sin x$ and $\cos x$ are both even positive integers, then express it as sines or cosines of multiple angles. Further integrate term by term.
- (v) If the sum of powers of $\sin x$ and $\cos x$ is an even negative integer, put $\tan x = z$.

5.6 Integrating $\int \tan^m x \sec^n x dx$

1. When m is odd and any n , rewrite the integrand in terms of $\sin x$ and $\cos x$:

$$\tan^m x \sec^n x dx = \left(\frac{\sin x}{\cos x} \right)^m \left(\frac{1}{\cos x} \right)^n dx$$

$$= \frac{\sin^{m-1} x}{\cos^{n+m} x} \sin x dx$$

and then substitute $u = \cos x$, $du = -\sin x dx$

$$\sin^2 x = 1 - \cos^2 x = 1 - u^2.$$

2. Alternatively, if m is odd and $n \geq 1$ move one factor of $\sec x$ $\tan x$ to the side so that you can see $\sec x \tan x dx$ in the integral, and substitute $u = \sec x$. $du = \sec x \tan x dx$ and $\tan^2 x = \sec^2 x - 1 = u^2 - 1$.
3. If n is even with $n \geq 2$, move one factor of $\sec^2 x$ to the side so that you can see $\sec^2 x dx$ in the integral, and substitute $u = \tan x$, $du = \sec^2 x dx$ and $\sec^2 x = 1 + \tan^2 x = 1 + u^2$.
4. When m is even and $n = 0$ – that is the integrand is just an even power of tangent – we can still use the $u = \tan x$ substitution, after using $\tan^2 x = \sec^2 x - 1$ (possibly more than once) to create a $\sec^2 x$.





6. REDUCTION FORMULA

Reduction formulae makes it possible to reduce an integral depending on the index $n > 0$, called the order of the integral, to an integral of the same type with smaller index. (i.e. To reduce the integral into similar integral of order less than or greater than given integral). Application of reduction formula is given with the help of some examples.

6.1 Reduction Formula for $\int \sin^n x \, dx$

$$\text{Let } I_n = \int \sin^n x \, dx = \int \sin^{n-1} x \sin x \, dx$$

I II

$$= -\sin^{n-1} x \cos x + \int (n-1) \sin^{n-2} x \cos^2 x \, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int (\sin^{n-2} x - \sin^n x) \, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore n I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$\Rightarrow I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$$

$$\text{Thus, } \int \sin^n x \, dx = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

6.2 Reduction Formula for $\int \cos^n x \, dx$

$$\text{Let } I_n = \int \cos^n x \, dx = \int \cos^{n-1} x \cos x \, dx$$

$$n I_n = \cos^{n-1} x \sin x + (n-1) I_{n-2}$$

$$\text{or } \int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

6.3 Reduction Formula for $\int \tan^n x \, dx$

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

$$\int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$





SOLVED EXAMPLES

Example – 1

Evaluate : $\int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx$

Sol. $\int \left(x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}} \right) dx$

$$= \int x^3 dx + \int 5x^2 dx - \int 4 dx + \int \frac{7}{x} dx + \int \frac{2}{\sqrt{x}} dx$$

$$= \int x^3 dx + 5 \cdot \int x^2 dx - 4 \cdot \int 1 \cdot dx + 7 \cdot \int \frac{1}{x} dx + 2 \cdot \int x^{-1/2} dx$$

$$= \frac{x^4}{4} + 5 \cdot \frac{x^3}{3} - 4x + 7 \log |x| + 2 \left(\frac{x^{1/2}}{1/2} \right) + C$$

$$= \frac{x^4}{4} + \frac{5}{3} x^3 - 4x + 7 \log |x| + 4\sqrt{x} + C$$

Example – 2

Evaluate : $\int e^{x \log a} + e^{a \log x} + e^{a \log a} dx$

Sol. We have,

$$\int e^{x \log a} + e^{a \log x} + e^{a \log a} dx$$

$$= \int e^{\log a^x} + e^{\log x^a} + e^{\log a^a} dx$$

$$= \int (a^x + x^a + a^a) dx$$

$$= \int a^x dx + \int x^a dx + \int a^a dx$$

$$= \frac{a^x}{\log a} + \frac{x^{a+1}}{a+1} + a^a \cdot x + C.$$

Example – 3

Evaluate : $\int \frac{x^4}{x^2+1} dx$

Sol. $\int \frac{x^4}{x^2+1} dx$

$$= \int \frac{x^4 - 1 + 1}{x^2 + 1} dx = \int \frac{x^4 - 1}{x^2 + 1} + \frac{1}{x^2 + 1} dx$$

$$= \int (x^2 - 1) dx + \int \frac{1}{x^2 + 1} dx = \frac{x^3}{3} - x + \tan^{-1} x + C$$

Example – 4

Evaluate : $\int \frac{2^x + 3^x}{5^x} dx$

Sol. $\int \frac{2^x + 3^x}{5^x} dx$

$$= \int \left(\frac{2^x}{5^x} + \frac{3^x}{5^x} \right) dx$$

$$= \int \left[\left(\frac{2}{5} \right)^x + \left(\frac{3}{5} \right)^x \right] dx = \frac{(2/5)^x}{\log_e 2/5} + \frac{(3/5)^x}{\log_e 3/5} + C$$

Example – 5

Evaluate : $\int x^3 \sin x^4 dx$

Sol. We have

$$I = \int x^3 \sin x^4 dx$$

Let $x^4 = t \Rightarrow d(x^4) = dt$

$$\Rightarrow 4x^3 dx = dt \Rightarrow dx = \frac{1}{4x^3} dt$$

$$I = \frac{1}{4} \int \sin t dt = -\frac{\cos t}{4} + C = -\frac{\cos(x^4)}{4} + C$$

**Example – 6**

Evaluate : $\int \frac{x}{x^4 + x^2 + 1} dx$

Sol. We have,

$$I = \int \frac{x}{x^4 + x^2 + 1} dx = \int \frac{x}{(x^2)^2 + x^2 + 1} dx$$

Let $x^2 = t$, then, $d(x^2) = dt$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow dx = \frac{dt}{2x}$$

$$I = \int \frac{x}{t^2 + t + 1} \cdot \frac{dt}{2x}$$

$$= \frac{1}{2} \int \frac{1}{t^2 + t + 1} dt$$

$$= \frac{1}{2} \int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dt$$

$$= \frac{1}{2} \cdot \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) + C = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x^2+1}{\sqrt{3}} \right) + C.$$

Example – 7

Evaluate : $\int \sqrt{x^2 + 2x + 5} dx$

Sol. We have,

$$\int \sqrt{x^2 + 2x + 5}$$

$$= \int \sqrt{x^2 + 2x + 1 + 4} dx = \int \sqrt{(x+1)^2 + 2^2} dx$$

$$= \frac{1}{2} (x+1) \sqrt{(x+1)^2 + 2^2} + \frac{1}{2} \cdot (2)^2 \log |(x+1)$$

$$+ \sqrt{(x+1)^2 + 2^2}| + C$$

$$= \frac{1}{2} (x+1) \sqrt{x^2 + 2x + 5} + 2 \log |(x+1) + \sqrt{x^2 + 2x + 5}| + C$$

Example – 8

Evaluate : $\int \frac{1}{x^2 - x + 1} dx$

Sol. $\int \frac{1}{x^2 - x + 1} dx$

$$= \int \frac{1}{x^2 - x + \frac{1}{4} - \frac{1}{4} + 1} dx$$

$$= \int \frac{1}{(x - 1/2)^2 + 3/4} dx$$

$$= \int \frac{1}{(x - 1/2)^2 + (\sqrt{3}/2)^2} dx$$

$$= \frac{1}{\sqrt{3}/2} \tan^{-1} \left(\frac{x - 1/2}{\sqrt{3}/2} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{3}} \right) + C.$$

Example – 9

Evaluate : $\int \frac{1}{\sqrt{9 + 8x - x^2}} dx$

Sol. $\int \frac{1}{\sqrt{9 + 8x - x^2}} dx$

$$= \int \frac{1}{\sqrt{-(x^2 - 8x - 9)}} dx$$

$$= \int \frac{1}{\sqrt{-(x^2 - 8x + 16 - 25)}} dx$$

$$= \int \frac{1}{\sqrt{-(x-4)^2 - 5^2}} dx$$

$$= \int \frac{1}{\sqrt{5^2 - (x-4)^2}} dx = \sin^{-1} \left(\frac{x-4}{5} \right) + C$$



Example – 10

Evaluate : $\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$

Sol.
$$\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{(2x+4)-1}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{2x+4}{\sqrt{x^2+4x+1}} dx - \int \frac{1}{\sqrt{x^2+4x+1}} dx$$

$$= \int \frac{dt}{\sqrt{t}} - \int \frac{1}{\sqrt{(x+2)^2 - (\sqrt{3})^2}} dx, \text{ where } t = x^2 + 4x + 1$$

$$= 2\sqrt{t} - \log | (x+2) + \sqrt{x^2+4x+1} | + C$$

$$= 2\sqrt{x^2+4x+1} - \log | x+2 + \sqrt{x^2+4x+1} | + C$$

Example – 11

Evaluate : $\int (x-5)\sqrt{x^2+x} dx$

Sol Let $(x-5) = \lambda \cdot \frac{d}{dx} (x^2+x) + \mu$. Then,
 $x-5 = \lambda(2x+1) + \mu$.
 Comparing coefficients of like powers of x, we get

$$1 = 2\lambda \text{ and } \lambda + \mu = -5 \Rightarrow \lambda = \frac{1}{2} \text{ and } \mu = -\frac{11}{2}$$

$$\int (x-5)\sqrt{x^2+x} dx$$

$$= \int \left(\frac{1}{2}(2x+1) - \frac{11}{2} \right) \sqrt{x^2+x} dx$$

$$= \int \frac{1}{2}(2x+1) \sqrt{x^2+x} dx - \frac{11}{2} \int \sqrt{x^2+x} dx$$

$$= \frac{1}{2} \int (2x+1) \sqrt{x^2+x} dx - \frac{11}{2} \int \sqrt{x^2+x} dx$$

$$= \frac{1}{2} \int \sqrt{t} dt - \frac{11}{2} \int \sqrt{\left(x+\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} dx \text{ where } t = x^2 + x$$

$$= \frac{1}{2} \cdot \frac{t^{3/2}}{3/2} - \frac{11}{2} \left[\frac{1}{2} \left(x + \frac{1}{2}\right) \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \right. \\ \left. - \frac{1}{2} \cdot \left(\frac{1}{2}\right)^2 \log \left[\left(x + \frac{1}{2}\right) + \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \right] \right] + C$$

$$= \frac{1}{3} t^{3/2} - \frac{11}{2} \left[\frac{2x+1}{4} \sqrt{x^2+x} - \frac{1}{8} \ell n \left| \left(x + \frac{1}{2}\right) + \sqrt{x^2+x} \right| \right] + C$$

$$= \frac{1}{3} (x^2+x)^{3/2} - \frac{11}{2} \left[\frac{2x+1}{4} \sqrt{x^2+x} - \frac{1}{8} \ell n \left| \left(x + \frac{1}{2}\right) + \sqrt{x^2+x} \right| \right] + C$$

Example – 12

$$\int \frac{1-x^2}{1+x^2+x^4} dx$$

Sol.
$$\int \frac{-\left(1-\frac{1}{x^2}\right)dx}{x^2+\frac{1}{x^2}+1} \quad (\text{Dividing numerator and denominator by } x^2)$$

Put $x + \frac{1}{x} = t$

$$\Rightarrow -\int \frac{dt}{t^2-1}$$

$$= -\frac{1}{2} \ell n \left| \frac{t-1}{t+1} \right| + C$$

$$= -\frac{1}{2} \ell n \left| \frac{x + \frac{1}{x} - 1}{x + \frac{1}{x} + 1} \right| + C$$

**Example – 13**

Evaluate : $\int \frac{1}{x^4+1} dx$

Sol. We have,

$$I = \int \frac{1}{x^4+1} dx$$

$$\Rightarrow I = \int \frac{\frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{\frac{2}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} - \frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx - \frac{1}{2} \int \frac{1 - \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2} dx$$

$$\Rightarrow I = -\frac{1}{2} \int \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x}\right)^2 - 2} dx$$

Putting $x - \frac{1}{x} = u$ in 1st integral and $x + \frac{1}{x} = v$ in 2nd integral, we get

$$I = \frac{1}{2} \int \frac{du}{u^2 + (\sqrt{2})^2} - \frac{1}{2} \int \frac{dv}{v^2 - (\sqrt{2})^2}$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{u}{\sqrt{2}}\right) - \frac{1}{2} \frac{1}{2\sqrt{2}} \log \left| \frac{v - \sqrt{2}}{v + \sqrt{2}} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x - 1/x}{\sqrt{2}}\right) - \frac{1}{4\sqrt{2}} \log \left| \frac{x + 1/x - \sqrt{2}}{x + 1/x + \sqrt{2}} \right| + C$$

$$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x^2 - 1}{\sqrt{2}x}\right) - \frac{1}{4\sqrt{2}} \log \left| \frac{x^2 - \sqrt{2}x + 1}{x^2 + x\sqrt{2} + 1} \right| + C$$

Example – 14

Evaluate : $\int x \log(1+x) dx$

Sol. $\int x \log(1+x) dx$

$$= \log(x+1) \cdot \frac{x^2}{2} - \int \frac{1}{x+1} \cdot \frac{x^2}{2} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2 - 1 + 1}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \int \frac{x^2 - 1}{x+1} + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \left[\int \left((x-1) + \frac{1}{x+1} \right) dx \right]$$

$$= \frac{x^2}{2} \log(x+1) - \frac{1}{2} \left[\frac{x^2}{2} - x + \log|x+1| \right] + C$$

Example – 15

Evaluate $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$

Sol. $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$

$$= \int \frac{\sin^{-1} \sqrt{x} - (\pi/2 - \sin^{-1} \sqrt{x})}{\pi/2} dx$$

$\{ \because \sin^{-1} \theta + \cos^{-1} \theta = \pi/2 \}$



$$\Rightarrow I = \frac{2}{\pi} \int (2 \sin^{-1} \sqrt{x} - \pi/2) dx$$

$$I = \frac{4}{\pi} \int \sin^{-1} \sqrt{x} dx - \int 1 dx$$

$$I = \frac{4}{\pi} \int \sin^{-1} \sqrt{x} dx - x + c \quad \dots (i)$$

Let $x = \sin^2 \theta$, then $dx = 2 \sin \theta \cos \theta d\theta = \sin 2\theta d\theta$

$$\therefore \int \sin^{-1} \sqrt{x} dx = \int \theta \cdot \sin 2\theta d\theta$$

applying integration by parts

$$\int \sin^{-1} \sqrt{x} dx = -\theta \cdot \frac{\cos 2\theta}{2} + \int \frac{1}{2} \cos 2\theta d\theta$$

$$= \frac{-\theta}{2} \cdot \cos 2\theta + \frac{1}{4} \sin 2\theta$$

$$= \frac{-1 \cdot \theta}{2} \cdot (1 - 2 \sin^2 \theta) + \frac{1}{2} \cdot \sin \theta \cdot \sqrt{1 - \sin^2 \theta}$$

$$= \frac{-1}{2} \sin^{-1} \sqrt{x} (1 - 2x) + \frac{1}{2} \cdot \sqrt{x} \sqrt{1 - x} \quad \dots (ii)$$

from (i) and (ii);

$$I = \frac{4}{\pi} \left\{ \frac{-1}{2} (\sin^{-1} \sqrt{x}) (1 - 2x) + \frac{1}{2} \sqrt{x} \sqrt{1 - x} \right\} - x + c$$

$$= \frac{2}{\pi} \{ \sqrt{x - x^2} - (1 - 2x) \sin^{-1} \sqrt{x} \} - x + c$$

Example – 16

If $\int f(x) dx = \psi(x)$, then $\int x^5 f(x^3) dx$ is equal to

$$(a) \frac{1}{3} [x^3 \psi(x^3) - \int x^2 \psi(x^3) dx] + C$$

$$(b) \frac{1}{3} x^3 \psi(x^3) - 3 \int x^3 \psi(x^3) dx + C$$

$$(c) \frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + C$$

$$(d) \frac{1}{3} [x^3 \psi(x^3) - \int x^3 \psi(x^3) dx] + C$$

Ans. (c)

Sol. Given, $\int f(x) dx = \psi(x)$

$$\text{Let } I = \int x^5 f(x^3) dx$$

$$\text{put } x^3 = t$$

$$\Rightarrow x^2 dx = \frac{dt}{3} \quad \dots (i)$$

$$\therefore I = \frac{1}{3} \int t f(t) dt$$

$$= \frac{1}{3} \left[t \int f(t) dt - \int \left\{ \frac{d}{dt}(t) \int f(t) dt \right\} dt \right]$$

[Integration by parts]

$$= \frac{1}{3} [t \psi(t) - \int \psi(t) dt]$$

$$= \frac{1}{3} [x^3 \psi(x^3) - 3 \int x^2 \psi(x^3) dx] + c \text{ from } \dots (i)$$

$$= \frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + c$$

Example – 17

Evaluate $\int (e^{\log x} + \sin x) \cos x dx$.

Sol. $\int (e^{\log x} + \sin x) \cos x dx$

$$I = \int (x + \sin x) \cos x dx$$

$$I = \int x \cos x dx + \frac{1}{2} \int \sin 2x dx$$

$$= x \sin x - \int \sin x + x - \frac{1}{4} \cos 2x + c$$

$$= x \sin x + \cos x - \frac{1}{4} \cos 2x + c$$

**Example – 18**

Evaluate

$$(i) \int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx \quad (ii) \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx$$

Sol. (i) $I = \int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx$

$$I = \int e^x \left\{ \frac{1}{\cos^2 x} + \frac{\sin x \cos x}{\cos^2 x} \right\} dx$$

$$I = \int e^x \{ \tan x + \sec^2 x \} dx$$

$$I = \int e^x \cdot \tan x dx + \int e^x (\sec^2 x) dx$$

$$I = \tan x \cdot e^x - \int \sec^2 x \cdot e^x dx + \int e^x \cdot \sec^2 x dx + c$$

$$I = e^x \tan x + c.$$

(ii) $I = \int e^{2x} \left\{ \frac{1 + \sin 2x}{1 + \cos 2x} \right\} dx$

$$= \int e^{2x} \left\{ \frac{1 + 2 \sin x \cos x}{2 \cos^2 x} \right\} dx$$

$$= \int e^{2x} \left\{ \frac{1}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right\} dx$$

$$= \int e^{2x} \left\{ \frac{1}{2} \sec^2 x + \tan x \right\} dx$$

$$= \int e^{2x} \cdot \tan x dx + \frac{1}{2} \int e^{2x} \cdot \sec^2 x dx$$

$$= \tan x \cdot \frac{e^{2x}}{2} - \int \sec^2 x \cdot \frac{e^{2x}}{2} dx + \frac{1}{2} \int e^{2x} \cdot \sec^2 x dx$$

$$I = \frac{1}{2} e^{2x} \cdot \tan x + c.$$

Example – 19

Evaluate $\int \frac{(x-1)e^x}{(x+1)^3} dx.$

Ans. $\frac{e^x}{(x+1)^2} + c$

Sol. $I = \int \frac{(x-1)e^x}{(x+1)^3} dx = \int \frac{(x+1-2)e^x}{(x+1)^3} dx$

$$= \int e^x \left[\frac{1}{(x+1)^2} + \frac{-2}{(x+1)^3} \right] dx$$

$$= \frac{e^x}{(x+1)^2} + c$$

Example – 20

Evaluate $\int \frac{3x-5}{(3x-2)(x+1)^2} dx$

Sol. Let $\frac{3x-5}{(3x-2)(x+1)^2} = \frac{A}{3x-2} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

$$\therefore 3x-5 = A(x+1)^2 + B(3x-2)(x+1) + C(3x-2)$$

Putting $x = \frac{2}{3}$, we get $A = -\frac{27}{25}$

Putting $x = -1$, we get $C = \frac{8}{5}$;

Putting $x = 0$, we get $B = \frac{9}{25}$

$$\therefore I = \int \left[-\frac{27}{25} \cdot \frac{1}{3x-2} + \frac{9}{25} \cdot \frac{1}{(x+1)} + \frac{8}{5} \cdot \frac{1}{(x+1)^2} \right] dx$$

$$= -\frac{9}{25} \log |3x-2| + \frac{9}{25} \log |x+1| - \frac{8}{5} \cdot \frac{1}{(x+1)} + c.$$



Example – 21

Evaluate $\int \frac{dx}{(x-1)(x^2+1)}$

Sol. Let $\frac{1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$

$$= \frac{A(x^2+1) + (Bx+C)(x-1)}{(x-1)(x^2+1)}$$

or $1 = A(x^2+1) + (Bx+C)(x-1)$

Putting $x = 1$, we get $A = \frac{1}{2}$;

Putting $x = 0$, we get $A - C = 1 \therefore C = A - 1 = -\frac{1}{2}$

Putting $x = -1$, we get $2A - 2(-B + C) = 1$

or $1 + 2B - 2\left(-\frac{1}{2}\right) = 1 \therefore B = -\frac{1}{2}$

Now $\frac{1}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{1}{2} \cdot \frac{x+1}{x^2+1}$

$$\therefore \int \frac{dx}{(x-1)(x^2+1)} = \int \frac{1}{2(x-1)} dx - \frac{1}{2} \int \frac{x}{x^2+1} dx - \frac{1}{2} \int \frac{dx}{x^2+1}$$

$$= \frac{1}{2} \log |x-1| - \frac{1}{4} \log (x^2+1) - \frac{1}{2} \tan^{-1} x + c$$

Example – 22

Find $\int \frac{x^4}{(x-1)(x^2+1)} dx$

Sol. $x^3 - x^2 + x - 1 \overline{) x^4}$

$$\begin{array}{r} x^4 - x^3 + x^2 - x, \\ + \quad - \quad + \\ \hline x^3 - x^2 + x \\ x^3 - x^2 + x - 1 \\ + \quad - \quad + \\ \hline 1 \end{array}$$

$$\therefore \frac{x^4}{(x-1)(x^2+1)} = x + 1 + \frac{1}{(x-1)(x^2+1)}$$

$$\therefore \int \frac{x^4}{(x-1)(x^2+1)} dx = \int (x+1) dx + \int \frac{dx}{(x-1)(x^2+1)}$$

$$= \frac{x^2}{2} + x + \left(\int \frac{1}{2(x-1)} - \frac{x}{2(x^2+1)} - \frac{1}{2} \frac{1}{x^2+1} \right) dx$$

(By Partial Fraction)

$$= \frac{x^2}{2} + x + \frac{1}{2} \log |x-1| - \frac{1}{4} \log |x^2+1| - \frac{1}{2} \tan^{-1} x + c$$

Example – 23

Evaluate $\int \frac{x^3 + 3x + 2}{(x^2+1)^2(x+1)} dx$

Sol. $I = \int \frac{x(x^2+1) + 2(x+1)}{(x^2+1)^2(x+1)} dx$

$$= \int \frac{x}{(x^2+1)(x+1)} dx + 2 \int \frac{dx}{(1+x^2)^2} \quad \dots (1)$$

Let $\frac{x}{(1+x)(1+x^2)} = \frac{A}{1+x} + \frac{Bx+C}{1+x^2}$

$\therefore x = A(1+x^2) + (Bx+C)(1+x)$

Putting $x = -1$, we get $A = -\frac{1}{2}$

Putting $x = 0$, we get $0 = A + C \Rightarrow C = -A = \frac{1}{2}$

Putting $x = 1$, we get $1 = 2A + 2(B+C)$

$$= 2A + 2B + 2C = -1 + 2B + 1 \therefore B = \frac{1}{2}$$

$$\therefore \int \frac{x}{(1+x)(1+x^2)} dx = \int \left(-\frac{1}{2(1+x)} + \frac{\frac{1}{2}x + \frac{1}{2}}{1+x^2} \right) dx$$

$$= -\frac{1}{2} \log |1+x| + \frac{1}{2} \int \frac{x}{1+x^2} dx + \frac{1}{2} \int \frac{dx}{1+x^2}$$



$$= -\frac{1}{2} \log |1+x| + \frac{1}{4} \log (1+x^2) + \frac{1}{2} \tan^{-1} x \quad \dots (2)$$

To evaluate : $\int \frac{dx}{(1+x^2)^2}$, put $x = \tan \theta$

$$\begin{aligned} \text{Then, } \int \frac{dx}{(1+x^2)^2} &= \int \frac{\sec^2 \theta}{(1+\tan^2 \theta)^2} d\theta \\ &= \int \cos^2 \theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta \\ &= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] = \frac{1}{2} [\theta + \sin \theta \cos \theta] \\ &= \frac{1}{2} \left[\tan^{-1} x + \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} \right] \\ &= \frac{1}{2} \tan^{-1} x + \frac{1}{2} \frac{x}{1+x^2} \end{aligned}$$

Now from (1),

$$I = \frac{3}{2} \tan^{-1} x - \frac{1}{2} \log |1+x| + \frac{1}{4} \log (1+x^2) + \frac{x}{1+x^2} + c$$

Example – 24

Evaluate $\int \frac{dx}{\sin x + \sin 2x}$

Sol. Let the given integral be I, then

$$\begin{aligned} I &= \int \frac{dx}{\sin x + 2 \sin x \cos x} = \int \frac{dx}{\sin x (1 + 2 \cos x)} \\ &= \int \frac{\sin x dx}{\sin^2 x (1 + 2 \cos x)} = \int \frac{\sin x dx}{(1 - \cos^2 x) (1 + 2 \cos x)} \end{aligned}$$

Put $\cos x = y$ so that $-\sin x dx = dy$.

$$\begin{aligned} \therefore I &= \int \frac{-dy}{(1-y^2)(1+2y)} \\ &= - \int \frac{dy}{(1-y)(1+y)(1+2y)} \end{aligned}$$

Now we break $\frac{1}{(1-y)(1+y)(1+2y)}$ into partial fractions

$$\begin{aligned} \text{Let } \frac{1}{(1-y)(1+y)(1+2y)} &= \frac{A}{1-y} + \frac{B}{1+y} + \frac{C}{1+2y} \\ &= \frac{A(1+y)(1+2y) + B(1-y)(1+2y) + C(1-y)(1+y)}{(1-y)(1+y)(1+2y)} \end{aligned}$$

$$\therefore 1 = A(1+y)(1+2y) + B(1-y)(1+2y) + C(1-y)(1+y)$$

Putting $y = 1$, we get $A = \frac{1}{6}$; putting $y = -\frac{1}{2}$, we get $C = \frac{4}{3}$

Putting $y = -1$, we get $B = -\frac{1}{2}$

$$\begin{aligned} \text{Now } I &= - \int \left[\frac{1}{6(1-y)} - \frac{1}{2(1+y)} + \frac{4}{3(1+2y)} \right] dy \\ &= -\frac{1}{6} \log |1-y| + \frac{1}{2} \log |1+y| - \frac{2}{3} \log |1+2y| + c \\ &= -\frac{1}{6} \log (1-\cos x) + \frac{1}{2} \log (1+\cos x) \\ &\quad - \frac{2}{3} \log |1+2\cos x| + c \end{aligned}$$

Example – 25

Evaluate $\int \sin^{-11/3} x \cos^{-1/3} x dx$

Sol. Here, $\int \sin^{-11/3} x, \cos^{-1/3} x dx$

i.e., $\left(-\frac{11}{3} - \frac{1}{3} \right) = -4$

$$\therefore I = \int \frac{\cos^{-1/3}}{\sin^{-11/3} x \cdot \sin^4 x} dx = \int (\cot^{-1/3} x) (\operatorname{cosec}^2 x)^2 dx$$

$$I = \int (\cot^{-1/3} x) (1 + \cot^2 x) \operatorname{cosec}^2 x dx.$$

$$\{\text{let } \cot x = t, -\operatorname{cosec}^2 x dx = dt\}$$

$$= - \int t^{-1/3} (1+t^2) dt = - \int (t^{-1/3} + t^{5/3}) dt$$

$$= - \left\{ \frac{3}{2} t^{2/3} + \frac{3}{8} t^{8/3} \right\} + c$$

$$= - \left\{ \frac{3}{2} (\cot^{2/3} x) + \frac{3}{8} (\cot^{8/3} x) \right\} + c.$$



Example – 26

Evaluate : $\int \frac{1}{1 + \sin x + \cos x} dx$

Sol. $I = \int \frac{1}{1 + \sin x + \cos x} dx$

$$= \int \frac{1}{1 + \frac{2 \tan x/2}{1 + \tan^2 x/2} + \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}} dx$$

$$= \int \frac{1 + \tan^2 x/2}{1 + \tan^2 x/2 + 2 \tan x/2 + 1 - \tan^2 x/2} dx$$

$$= \int \frac{\sec^2 x/2}{2 + 2 \tan x/2} dx$$

Putting $\tan \frac{x}{2} = t$ and $\frac{1}{2} \sec^2 \frac{x}{2} dx = dt$, we get

$$I = \int \frac{1}{t+1} dt = \log |t+1| + C = \log \left| \tan \frac{x}{2} + 1 \right| + C$$

Example – 27

Evaluate : $\int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

Sol. $I = \int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

Let $3 \sin x + 2 \cos x = \lambda \cdot \frac{d}{dx} (3 \cos x + 2 \sin x) +$

$$\mu (3 \cos x + 2 \sin x)$$

$$\Rightarrow 3 \sin x + 2 \cos x = \lambda (-3 \sin x + 2 \cos x) +$$

$$\mu (3 \cos x + 2 \sin x)$$

Comparing the coefficients of $\sin x$ and $\cos x$ on both sides, we get

$$-3\lambda + 2\mu = 3 \text{ and } 2\lambda + 3\mu = 2$$

$$\Rightarrow \mu = \frac{12}{13} \text{ and } \lambda = -\frac{5}{13}$$

$$\therefore I = \int \frac{\lambda(-3 \sin x + 2 \cos x) + \mu(3 \cos x + 2 \sin x)}{3 \cos x + 2 \sin x} dx$$

$$= \mu \int 1 \cdot dx + \lambda \int \frac{-3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$$

$$= \mu x + \lambda \int \frac{dt}{t}, \text{ where } t = 3 \cos x + 2 \sin x$$

$$= \mu x + \lambda \ln |t| + C$$

$$= \frac{12}{13} x + \frac{-5}{13} \ln |3 \cos x + 2 \sin x| + C$$

Example – 28

Evaluate : $\int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$

Sol. We have,

$$I = \int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$$

Let $3 \cos x + 2 = \lambda (\sin x + 2 \cos x + 3) +$

$$\mu (\cos x - 2 \sin x) + v$$

Comparing the coefficients of $\sin x$, $\cos x$ and constant term on both sides, we get

$$\lambda - 2\mu = 0, 2\lambda + \mu = 3, 3\lambda + v = 2$$

$$\Rightarrow \lambda = \frac{6}{5}, \mu = \frac{3}{5} \text{ and } v = -\frac{8}{5}$$

$$\therefore I = \int \frac{\lambda(\sin x + 2 \cos x + 3) + \mu(\cos x - 2 \sin x) + v}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda \int dx + \mu \int \frac{\cos x - 2 \sin x}{\sin x + 2 \cos x + 3} dx +$$

$$v \int \frac{1}{\sin x + 2 \cos x + 3} dx$$

$$\Rightarrow I = \lambda x + \mu \log |\sin x + 2 \cos x + 3| + v I_1, \text{ where}$$

$$I_1 = \int \frac{1}{\sin x + 2 \cos x + 3} dx$$

Putting, $\sin x = \frac{2 \tan x/2}{1 + \tan^2 x/2}$, $\cos x = \frac{1 - \tan^2 x/2}{1 + \tan^2 x/2}$ we get

$$I_1 = \int \frac{1}{\frac{2 \tan x/2}{1 + \tan^2 x/2} + \frac{2(1 - \tan^2 x/2)}{1 + \tan^2 x/2} + 3} dx$$



$$= \int \frac{1 + \tan^2 x / 2}{2 \tan x / 2 + 2 - 2 \tan^2 x / 2 + 3(1 + \tan^2 x / 2)} dx$$

$$= \int \frac{\sec^2 x / 2}{\tan^2 x / 2 + 2 \tan x / 2 + 5} dx$$

Put $\tan x / 2 = t$

$$\sec^2 x / 2 \cdot \frac{1}{2} dx = dt$$

$$\sec^2 x / 2 dx = 2dt = 2 \int \frac{dt}{t^2 + 2t + 5} = 2 \int \frac{dt}{(t+1)^2 + (2)^2}$$

$$= \tan^{-1} \left(\frac{t+1}{2} \right) + C$$

$$= 2 \tan^{-1} \left(\frac{\tan \left(\frac{x}{2} \right) + 1}{2} \right) + C$$

Example – 29

Integrate $\frac{1}{1 - \cot x}$ or $\frac{\sin x}{\sin x - \cos x}$.

Sol. Let $I = \int \frac{\sin x}{\sin x - \cos x} dx$

Again, let $\sin x = A(\cos x + \sin x) + B(\sin x - \cos x)$ then
 $A + B = 1$ and $A - B = 0$

$$\Rightarrow A = \frac{1}{2}, B = \frac{1}{2}$$

$$\therefore I = \int \frac{\frac{1}{2}(\cos x + \sin x) + \frac{1}{2}(\sin x - \cos x)}{(\sin x - \cos x)} dx$$

$$= \frac{1}{2} \int \frac{\cos x + \sin x}{\sin x - \cos x} dx + \frac{1}{2} \int 1 dx + c$$

$$= \frac{1}{2} \log(\sin x - \cos x) + \frac{1}{2} x + c$$

Example – 30

Evaluate $\int \sin^3 x \cdot \cos^5 x dx$

Sol. $I = \int \sin^3 x \cdot \cos^5 x dx$

Let $\cos x = t \Rightarrow -\sin x dx = dt$

$$I = - \int (1 - t^2) \cdot t^5 dt$$

$$I = \int t^7 dt - \int t^5 dt = \frac{t^8}{8} - \frac{t^6}{6} + c$$

$$I = \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + c$$

Aliter : $I = \int R^3(1 - R^2)^2 dR$, if $\sin x = R$, $\cos x dx = dR$.

$$I = \int R^3 dR - \int 2R^5 dR + \int R^7 dR$$

$$I = \frac{\sin^4 x}{4} - \frac{2 \sin^6 x}{6} + \frac{\sin^8 x}{8} + c$$

Example – 31

Evaluate

(i) $\int \frac{1}{\sin(x-a) \cos(x-b)} dx$

(ii) $\int \frac{1}{\cos(x-a) \cos(x-b)} dx$

Sol. (i) $I = \int \frac{1}{\sin(x-a) \cos(x-b)} dx$

$$I = \frac{\cos(a-b)}{\cos(a-b)} \cdot \int \frac{dx}{\sin(x-a) \cos(x-b)}$$

$$= \frac{1}{\cos(a-b)} \cdot \int \frac{\cos\{(x-b) - (x-a)\}}{\sin(x-a) \cos(x-b)} dx$$

$$= \frac{1}{\cos(a-b)} \cdot \int \left\{ \frac{\cos(x-b) \cdot \cos(x-a)}{\sin(x-a) \cos(x-b)} + \frac{\sin(x-b) \cdot \sin(x-a)}{\sin(x-a) \cos(x-b)} \right\} dx$$

$$I = \frac{1}{\cos(a-b)} \int \{\cot(x-a) + \tan(x-b)\} dx$$



$$I = \frac{1}{\cos(a-b)} \{ \log |\sin(x-a)| - \log |\cos(x-b)| \} + c$$

$$I = \frac{1}{\cos(a-b)} \log_e \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + c$$

$$\begin{aligned} \text{(ii)} \quad I &= \int \frac{1}{\cos(x-a) \cos(x-b)} dx \\ &= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b) dx}{\cos(x-a) \cos(x-b)} \\ &= \frac{1}{\sin(a-b)} \int \frac{\sin \{(x-b) - (x-a)\}}{\cos(x-a) \cos(x-b)} dx \\ &= \frac{1}{\sin(a-b)} \int \left\{ \frac{\sin(x-b) \cos(x-a)}{\cos(x-a) \cos(x-b)} - \frac{\cos(x-b) \sin(x-a)}{\cos(x-a) \cos(x-b)} \right\} dx \\ &= \frac{1}{\sin(a-b)} \int \{ \tan(x-b) - \tan(x-a) \} dx \\ &= \frac{1}{\sin(a-b)} [-\log |\cos(x-b)| + \log |\cos(x-a)|] + c \\ &= \frac{1}{\sin(a-b)} \left[\log \left| \frac{\cos(x-a)}{\cos(x-b)} \right| \right] + c \end{aligned}$$

Example – 32

If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then the

value of (A, B) is

- (a) $(-\sin \alpha, \cos \alpha)$ (b) $(\cos \alpha, \sin \alpha)$
(c) $(\sin \alpha, \cos \alpha)$ (d) $(-\cos \alpha, \sin \alpha)$

Ans. (b)

So $\int \frac{\sin x}{\sin(x-\alpha)} dx$

Let $x - \alpha = t \Rightarrow x = t + \alpha \Rightarrow dx = dt$

$$I = \int \frac{\sin(t+\alpha)}{\sin t} dt$$

$$I = \int \frac{\sin t \cos \alpha + \cos t \sin \alpha}{\sin t} dt$$

$$= \int \cos \alpha dt + \int \sin \alpha \cot t dt$$

$$= t \cos \alpha + \sin \alpha \ln |\sin t| + c$$

$$= (x - \alpha) \cos \alpha + \sin \alpha \ln |\sin(x - \alpha)| + c$$

$$\Rightarrow A = \cos \alpha, B = \sin \alpha$$

Example – 33

$$\int \frac{dx}{\cos x - \sin x} \text{ is equal to}$$

(a) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{3\pi}{8} \right) \right| + C$

(b) $\frac{1}{\sqrt{2}} \log \left| \cot \left(\frac{x}{2} \right) \right| + C$

(c) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} - \frac{\pi}{8} \right) \right| + C$

(d) $\frac{1}{\sqrt{2}} \log \left| \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) \right| + C$

Ans. (d)

Sol. $\int \frac{dx}{\cos x - \sin x} = \frac{1}{\sqrt{2}} \int \frac{dx}{\cos \left(x + \frac{\pi}{4} \right)}$

$$\int \frac{dx}{\cos x - \sin x} = \frac{1}{\sqrt{2}} \int \sec \left(x + \frac{\pi}{4} \right) dx = \frac{1}{\sqrt{2}} \log \tan \left(\frac{x}{2} + \frac{3\pi}{8} \right) + c$$

Example – 34

$$\int \frac{\sin^8 x - \cos^8 x}{(1 - 2 \sin^2 x \cos^2 x)} dx \text{ is equal to:}$$

(a) $\frac{1}{2} \sin 2x + c$

(b) $-\frac{1}{2} \sin 2x + c$

(c) $-\frac{1}{2} \sin x + c$

(d) $-\sin^2 x + c$

Ans. (b)

Sol. $= \int \frac{\sin^8 x - \cos^8 x}{1 - 2 \sin^2 x \cos^2 x} dx$



$$\begin{aligned}
 &= \int \frac{(\sin^4 x - \cos^4 x)(\sin^4 x + \cos^4 x)}{1 - 2\sin^2 x \cos^2 x} dx \\
 &= \int \frac{(\sin^2 x - \cos^2 x)(1 - 2\sin^2 x \cos^2 x)}{1 - 2\sin^2 x \cos^2 x} dx \\
 &= \int -\cos 2x dx \\
 &= -\frac{1}{2} \sin 2x + c
 \end{aligned}$$

Example – 35

Evaluate : $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

Sol. Let $I = \int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

Putting $x+1 = t^2$, and $dx = 2t dt$, we get

$$I = \int \frac{(t^2+1) 2t dt}{\{(t^2-1)^2 + 3(t^2-1) + 3\}\sqrt{t^2}}$$

$$\Rightarrow I = 2 \int \frac{(t^2+1)}{t^4+t^2+1} dt = 2 \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}+1} dt$$

$$\Rightarrow I = 2 \int \frac{du}{u^2 + (\sqrt{3})^2} \text{ where } t - \frac{1}{t} = u.$$

$$\Rightarrow I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left\{ \frac{t - \frac{1}{t}}{\sqrt{3}} \right\} + C$$

$$\Rightarrow I = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2-1}{t\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left\{ \frac{x}{\sqrt{3}(x+1)} \right\} + C$$

Example – 36

$$\int \frac{dx}{(x+1)\sqrt{x^2+x+1}}$$

Sol. Putting $x+1 = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$

$$I = \int \frac{-dt}{t^2 \left(\frac{1}{t} \right) \sqrt{\left(\frac{1}{t} - 1 \right)^2 + \frac{1}{t}}} = \int \frac{-dt}{t \sqrt{\frac{1}{t^2} - \frac{1}{t} + 1}}$$

$$= \int \frac{-dt}{\sqrt{t^2 - t + 1}}$$

$$= \int \frac{-dt}{\sqrt{\left(t - \frac{1}{2} \right)^2 + \frac{3}{4}}}$$

$$= -\ln \left(t - \frac{1}{2} + \sqrt{\left(t - \frac{1}{2} \right)^2 + \frac{3}{4}} \right) + C$$

$$= -\ln \left(\frac{1}{x+1} - \frac{1}{2} + \frac{x+1}{\sqrt{x^2+x+1}} \right) + C$$

Example – 37

$$\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$$

Sol. Put $x = \frac{1}{t}$

$$\Rightarrow I = \int \frac{-t dt}{(t^2+1)\sqrt{t^2-1}}$$

put $t^2 - 1 = y^2$

$$\Rightarrow I = -\int \frac{y dy}{(y^2+2)y}$$



$$= -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{y}{\sqrt{2}} \right) + C$$

$$= -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + C$$

Example – 38

If $\int f(x) \cos x \, dx = \frac{1}{2} f^2(x) + c$, then $f(x)$ can be

- (a) x (b) 1
(c) $\cos x$ (d) $\sin x$

Sol. Here, $\int f(x) \cos x \, dx = \frac{1}{2} f^2(x) + c$, differentiating both sides, we get
 $f(x) \cos x = f(x) \cdot f'(x)$

$$\text{i.e., } \cos x = \frac{d}{dx} (f(x))$$

$$\Rightarrow f(x) = \int \cos x \, dx$$

$$\Rightarrow f(x) = \sin x + c.$$

Hence (d) is the correct answer.



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Standard Formulae

1. $\int \sin^2(x/2) dx$ equals

(a) $\frac{1}{2}(x + \sin x) + c$

(b) $\frac{1}{2}(x + \cos x) + c$

(c) $\frac{1}{2}(x - \sin x) + c$

(d) none of these

2. $\int \left(x - \frac{1}{x}\right)^3 dx, (x > 0)$ equals

(a) $\frac{x^3}{3} - \frac{3}{2}x^2 + 3\log x + \frac{1}{2x^2} + C$

(b) $\frac{x^4}{4} - \frac{3}{2}x^2 + 3\log x + \frac{1}{2x^2} + C$

(c) $\frac{x^4}{4} + 3\log x + \frac{1}{2x^2} + C$

(d) none of these

3. The value of $\int \left(\frac{6}{1+x^2} + 10^x\right) dx$ is

(a) $6 \tan^{-1} x + 10^x \log_e 10 + C$

(b) $6 \tan^{-1} x + \frac{10^x}{\log_e 10} + C$

(c) $3 \tan^{-1} x + \frac{10^x}{\log_e 10} + C$

(d) none of these

4. $\int \frac{e^{5\log_e x} - e^{4\log_e x}}{e^{3\log_e x} - e^{2\log_e x}} dx = \frac{x^k}{k} + c$. Then k is equal to

5. $\int (\tan x + \cot x)^2 dx$ is equal to

(a) $\tan x - \cot x + c$

(b) $\tan x + \cot x + c$

(c) $\cot x - \tan x + c$

(d) none of these

6. $\int \frac{x^2 + \cos^2 x}{(1+x^2)\sin^2 x} dx$ is equals to

(a) $\tan^{-1} x + \cot x + c$

(b) $\tan^{-1} x - \cot x + c$

(c) $\cot^{-1} x - \tan x + c$

(d) $-\tan^{-1} x - \cot x + c$

7. $\int \left(\frac{2^x - 5^x}{10^x}\right) dx$ is equal to

(a) $\frac{2^{-x}}{\log_e 2} - \frac{5^{-x}}{\log_e 5} + c$

(b) $\frac{2^x}{\log_e 2} - \frac{5^x}{\log_e 5} + c$

(c) $\frac{2^x}{\log_e 2} + \frac{5^x}{\log_e 5} + c$

(d) $\frac{5^{-x}}{\log_e 5} - \frac{2^{-x}}{\log_e 2} + c$

8. $\int \sec^2(ax+b) dx$ equals

(a) $\tan(ax+b) + C$

(b) $\frac{1}{2} \tan x + C$

(c) $\frac{1}{a} \tan(ax+b) + C$

(d) none of these

Integration by substitution

9. $\int \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$ is equal to

(a) $\frac{(\sin^{-1} x)^2}{2} + c$

(b) $\frac{(\sin^{-1} x)^3}{3} + c$

(c) $\frac{\sin^{-1} x}{x} + c$

(d) $\frac{(\sin^{-1} x)^4}{4} + c$

10. $\int \sec^n x \tan x dx$ is equal to

(a) $\frac{\sec^n x}{n} + c$

(b) $\frac{\sec^2 x}{2} + c$

(c) $\frac{\tan x}{n} + c$

(d) $\frac{(\sec^n x) \tan x}{n} + c$



11. $\int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$ is equal to
 (a) $\log |\cos x| - \sin x + c$ (b) $\log |\sin x| - \sin x + c$
 (c) $\log |\sin x| + \cos x + c$ (d) $\log |\cos x| - \cos x + c$

12. If $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = a \sin \sqrt{x} + c$, then a equals

13. $\int \frac{\log_e x}{x\sqrt{1+\log_e x}} dx =$

- (a) $(1 + \log_e x)^{3/2} + c$
 (b) $\frac{2}{3} (1 + \log_e x) (\log_e x - 2) + c$
 (c) $\frac{2}{3} (1 + \log_e x)^{1/2} (\log_e x - 5) + c$
 (d) $\frac{2}{3} (1 + \log_e x)^{1/2} (\log_e x - 2) + c$

14. $\int \frac{1}{x \log x} dx$ is equal to

- (a) $\log |x \log x| + C$ (b) $\log |\log x + x| + C$
 (c) $\log x + C$ (d) $\log |\log x| + C$

15. $\int \sec^2 x \cos(\tan x) dx$ equals

- (a) $\sin(\cos x) + C$ (b) $\sin(\tan x) + C$
 (c) $\operatorname{cosec}(\tan x) + C$ (d) none of these

16. $\int \tan^n x \sec^2 x dx$ equals

- (a) $\frac{\tan^{n-1} x}{n-1} + C$ (b) $\frac{\tan^{n+1} x}{n+1} + C$
 (c) $\tan^{n+1} x + C$ (d) none of these

17. $\int \frac{\sin 2x}{1 + \cos^4 x} dx$ is equal to

- (a) $\cos^{-1}(\cos^2 x) + c$ (b) $\sin^{-1}(\cos^2 x) + c$
 (c) $\cot^{-1}(\cos^2 x) + c$ (d) none of these

18. $\int \frac{x^2}{1+x^6} dx = \frac{1}{k} \tan^{-1} x^3 + c$. Then k is equal to

19. $\int \frac{dx}{x + \sqrt{x}}$ equals

- (a) $2 \log(\sqrt{x} - 1) + c$ (b) $2 \log(\sqrt{x} + 1) + c$
 (c) $\tan^{-1} x + c$ (d) none of these

20. $\int \frac{x^5}{\sqrt{1+x^3}} dx$ equals

- (a) $\frac{2}{9} (x^3 - 2) \sqrt{1+x^3} + c$ (b) $\frac{2}{9} (x^3 + 2) \sqrt{1+x^3} + c$
 (c) $(x^3 + 2) \sqrt{1+x^3} + c$ (d) none of these

21. $\int \frac{x+1}{\sqrt{x+2}} dx =$

$\frac{a}{b} (x+2)^{\frac{3}{2}} - a \sqrt{x+2} + c$ then ab equals

22. $\int \frac{dx}{3x^2 + 2x + 1}$ equals

(a) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x+1}{\sqrt{2}} \right) + C$

(b) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x-1}{\sqrt{2}} \right) + C$

(c) $-\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3x-1}{\sqrt{2}} \right) + C$

- (d) none of these

23. $\int \sqrt{1+x-2x^2} dx$

$= \frac{1}{2a} (4x-1) \sqrt{1+x-x^2} + \frac{9\sqrt{2}}{b} \sin^{-1} \left(\frac{4x-1}{3} \right) + c$.

Then b/a is equal to



24. $\int \frac{dx}{\sqrt{3-5x-x^2}}$ equals

(a) $\sin^{-1}\left(\frac{2x+5}{\sqrt{37}}\right) + c$ (b) $\cos^{-1}\left(\frac{2x+5}{\sqrt{37}}\right) + c$

(c) $\sin^{-1}(2x+5) + c$ (d) none of these

25. $\int \frac{e^x}{e^{2x}+5e^x+6} dx$ equals

(a) $\log\left(\frac{e^x+3}{e^x+2}\right) + c$ (b) $\log\left(\frac{e^x+2}{e^x+3}\right) + c$

(c) $\frac{1}{2}\log\left(\frac{e^x+2}{e^x+3}\right) + c$ (d) none of these

26. $\int \frac{x^2}{x^2-1} dx$ equals

(a) $x + \log\sqrt{\frac{x-1}{x+1}} + c$ (b) $x + \log\sqrt{\frac{x+1}{x-1}} + c$

(c) $x + \log\left|\frac{x-1}{x+1}\right| + c$ (d) $x + \log\left|\frac{x+1}{x-1}\right| + c$

27. $\int \frac{x+1}{x^2+x+3} dx$

$= \frac{1}{a} \ln|x^2+x+3| + \frac{1}{\sqrt{b}} \tan^{-1}\left(\frac{2x+1}{\sqrt{11}}\right) + c$

then (b - a) equals

28. $\int \frac{(x-x^3)^{1/3}}{x^4} dx = -\frac{3}{k}\left(\frac{1}{x^2}-1\right)^{4/3} + c$. Then k is equal to

29. If m is a non-zero number and

$\int \frac{x^{5m-1} + 2x^{4m-1}}{(x^{2m} + x^m + 1)^3} dx = f(x) + c$, then f(x) is

(a) $\frac{x^{5m}}{2m(x^{2m} + x^m + 1)^2}$ (b) $\frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2}$

(c) $\frac{2m(x^{5m} + x^{4m})}{(x^{2m} + x^m + 1)^2}$ (d) $\frac{(x^{5m} - x^{4m})}{2m(x^{2m} + x^m + 1)^2}$

Integration by Parts

30. $\int \log x dx$ is equal to

(a) $x(\log x) - 1 + c$ (b) $x(\log x) - x + c$
(c) $x(\log x) - 1/x + c$ (d) $x(\log x) + c$

31. $\int \frac{\sin^{-1}\sqrt{x}}{\sqrt{1-x}} dx$ equals

(a) $2[\sqrt{x} - \sqrt{1-x} \sin^{-1}\sqrt{x}] + c$

(b) $2[\sqrt{x} + \sqrt{1-x} \sin^{-1}\sqrt{x}] + c$

(c) $[\sqrt{x} - \sqrt{1-x} \sin^{-1}\sqrt{x}] + c$

(d) none of these

32. $\int x^2 e^x dx$ is equal to

(a) $x^2 e^x - 2[e^{2x} - x e^x] + c$

(b) $x^2 e^x - 2[e^x - x e^x] + c$

(c) $x^2 e^x - 2[x e^{2x} - e^x] + c$

(d) $x^2 e^x - 2[x e^x - e^x] + c$

33. $\int (\log x)^2 dx$ is equal to

(a) $x(\log x)^2 - 2[x \log x - x] + c$

(b) $x(\log x)^2 - 2[\log x - x] + c$

(c) $x(\log x)^2 - 2[\log x^2 - x] + c$

(d) $x(\log x)^2 - 2[\log x - 2x] + c$



34. $\int \frac{\log x}{x^2} dx = -\frac{1}{x}(a + \log bx) + c$. ($a, b \in \text{Integers}$). Then $a + b$ equal to

35. $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$ is equal to

(a) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{2} \log |\sqrt{1-x^2}| + c$

(b) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{2} \log |1-x^2| + c$

(c) $\frac{x}{\sqrt{1-x^2}} \sin^{-1} x + \frac{1}{4} \log |\sqrt{1-x^2}| + c$

(d) none of these

36. $\int x^3 (\log x)^2 dx = \frac{1}{p} x^4 [q (\log x)^2 + r \log x + 1] + c$.

Then $p + q + r$ is equal to

37. The value of $\int x \sec x \tan x dx$ is

(a) $x \sec x + \log |\sec x + \tan x| + c$

(b) $x \sec x - \log |\sec x + \tan x| + c$

(c) $x \sec x + \log |\sec x - \tan x| + c$

(d) none of these

38. $\int \frac{x - \sin x}{1 - \cos x} dx$ is equal to

(a) $-x \cot \frac{x}{2} + c$

(b) $\cot \frac{x}{2} + c$

(c) $-\cot \frac{x}{2} + c$

(d) none of these

39. $\int e^x \frac{(1-x)^2}{(1+x^2)^2} dx$ is equal to

(a) $\frac{1}{x^2+1} + c$

(b) $\frac{e^x}{x^2+1} + c$

(c) $\frac{e^x - 1}{x^2+1} + c$

(d) $\frac{1 - e^x}{x^2+1} + c$

40. $\int \frac{xe^x}{(x+1)^2} dx$ is equal to

(a) $\frac{e^x}{(x+1)^2} + c$

(b) $\frac{e^x}{x+1} + c$

(c) $\frac{e^x}{(x+1)^3} + c$

(d) none of these

41. $\int \frac{x + \sin x}{1 + \cos x} dx = \frac{x}{a} \tan \left(\frac{x}{b} \right) + C$. Then $b - a$ is equal to

42. $\int e^x \frac{x-1}{(x+1)^3} dx$ equals

(a) $-\frac{e^x}{x+1} + C$

(b) $\frac{e^x}{x+1} + C$

(c) $\frac{e^x}{(x+1)^2} + C$

(d) $-\frac{e^x}{(x+1)^2} + C$

43. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to

(a) $\frac{x}{x^2+1} + C$

(b) $\frac{\log x}{(\log x)^2+1} + C$

(c) $\frac{x}{(\log x)^2+1} + C$

(d) $\frac{xe^x}{1+x^2} + C$

44. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ is equal to :

(a) $-xe^{x+\frac{1}{x}} + c$

(b) $(x-1)e^{x+\frac{1}{x}} + c$

(c) $xe^{x+\frac{1}{x}} + c$

(d) $(x+1)e^{x+\frac{1}{x}} + c$



Integration using Partial Fractions

45. $\int \frac{x^3+3}{(x+1)(x^2+1)} dx$ equals.

- (a) $x + \log_e |x+1| - \log_e (x^2+1) + \cot^{-1} x + c$
 (b) $x - \log_e |x+1| + \log_e (x^2+1) + \tan^{-1} x + c$
 (c) $x + \log_e |x+1| - \log_e (x^2+1) + \tan^{-1} x + c$
 (d) $x - \log_e |x+1| - \log_e (x^2+1) - \tan^{-1} x + c$

46. $\int \frac{x^2+1}{(x-1)(x-2)} dx$ equals

- (a) $\log \left| \frac{(x-2)^5}{(x-1)^2} \right| + C$ (b) $x + \log \left| \frac{(x-2)^5}{(x-1)^2} \right| + C$
 (c) $x + \log \left| \frac{(x-1)^5}{(x-2)^5} \right| + C$ (d) none of these

47. The value of $\int \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)}$ is

- (a) $\frac{1}{b^2-a^2} \left[b \tan^{-1} \frac{x}{b} - a \tan^{-1} \frac{x}{a} \right] + C$
 (b) $\frac{1}{b^2-a^2} \left[a \tan^{-1} \frac{x}{b} - b \tan^{-1} \frac{x}{a} \right] + C$
 (c) $\frac{1}{b^2-a^2} \left[b \tan^{-1} \frac{x}{b} + a \tan^{-1} \frac{x}{a} \right] + C$
 (d) none of these

48. $\int \left(\frac{1-x}{1+x} \right)^2 dx = x - a \log |x+1| - \frac{b}{x+1} + c$. Then $a+b$ is equal to

49. If $\int \frac{2x+3}{(x-1)(x^2+1)} dx$

$$= \log \left[|x-1|^{5/2} (x^2+1)^{-\frac{5}{2}} \right] - \frac{1}{2} \tan^{-1} x + C$$

where C is any arbitrary constant, then a is equal to

50. $\int \frac{dx}{1+e^x} =$

- (a) $\log_e \left(\frac{e^x+1}{e^x} \right) + c$ (b) $\log_e \left(\frac{e^x}{e^x+1} \right) + c$
 (c) $x + \log_e (e^x+1) + c$ (d) $e^x + x + c$

Integration of Various Trigonometric Functions

51. $\int \sin^9 x \cdot \cos^3 x dx$ is equal to

- (a) $\frac{\sin^{10} x}{10} - \frac{\sin^{12} x}{12} + c$ (b) $\frac{\cos^6 x}{6} - \frac{\cos^8 x}{8} + c$
 (c) $\frac{\cos^6 x}{6} - \frac{\sin^8 x}{8} + c$ (d) none of these

52. If $\int \frac{\sin^4 x}{\cos x} dx = \frac{1}{2} \log_e \left(\frac{1+\sin x}{1-\sin x} \right) - g(x) + c$ where $g(x)$ equals.

- (a) $\frac{1}{3} \sin^3 x + \sin x$ (b) $\frac{1}{3} \cos^3 x + \cos x$
 (c) $\frac{1}{3} \sin^3 x - \sin x$ (d) $\frac{1}{3} \cos^3 x - \cos x$

53. $\int \frac{dx}{\sin^2 x \cos^2 x}$ is equal to

- (a) $\tan x + \cot x + C$ (b) $(\tan x + \cot x)^2 + C$
 (c) $\tan x - \cot x + C$ (d) $(\tan x - \cot x)^2 + C$

54. $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx$ is equal to

- (a) $2(\sin x + x \cos \theta) + C$
 (b) $2(\sin x - x \cos \theta) + C$
 (c) $2(\sin x + 2x \cos \theta) + C$
 (d) $2(\sin x - 2x \cos \theta) + C$

55. $\int \frac{1-\cos 2x}{1+\cos 2x} dx =$

- (a) $\tan x - x + C$ (b) $x + \tan x + C$
 (c) $x - \tan x + C$ (d) $-x - \cot x + C$



56. $\int \frac{dx}{4\sin^2 x + 4\sin x \cos x + 5\cos^2 x}$ is equal to

- (a) $\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$ (b) $\frac{1}{4}\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$
 (c) $4\tan^{-1}\left(\tan x + \frac{1}{2}\right) + c$ (d) none of these

57. If $\int \frac{dx}{1+\tan x} = px + q \log_e |\cos x + \sin x| + c$
 then $p + q$ equals.

58. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$,
 then a is equal to
 (a) -1 (b) -2
 (c) 1 (d) 2

59. $\int \sqrt[n]{\frac{\sin^n x}{\cos^{n+6} x}} dx$, $n \in \mathbb{N}$ is equal to

- (a) $\frac{3}{n}(\tan x)^{\frac{n}{3}+1} + c$ (b) $\frac{3}{3+n}(\tan x)^{\frac{n}{3}+1} + c$
 (c) $\frac{3}{n}(\cos x)^{n+1} + c$ (d) none of these

60. $\int \frac{\sin^4 x}{\cos^8 x} dx$ is equal to

- (a) $\frac{(1 + \tan^5 x)}{5} - \frac{\tan^7 x}{7} + c$
 (b) $\frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + c$
 (c) $\frac{\tan^5 x}{7} + \frac{\tan^7 x}{5} + c$
 (d) none of these

61. $\int \frac{\sin^{10} x}{\cos^{12} x} dx =$

- (a) $10 \tan^9 x + C$ (b) $\frac{\tan^{11} x}{11} + C$
 (c) $\frac{\tan^{11} x}{11} + C$ (d) none of these

62. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$ is equal to

- (a) $\tan x + \cot x + 3x + c$ (b) $\tan x + \cot x - 3x + c$
 (c) $\tan x - \cot x - 3x + c$ (d) $\tan x - \cot x + 3x + c$

63. If $\int \sin 2x \sin 3x dx$ equals $\frac{1}{k} (5 \sin x - \sin 5x) + c$. Then k is
 equal to

64. $\int \sqrt{\frac{1+\cos x}{1-\cos x}} dx$ equals

- (a) $\log \cos\left(\frac{x}{2}\right) + C$ (b) $2 \log \sin\left(\frac{x}{2}\right) + C$
 (c) $2 \log \sec\left(\frac{x}{2}\right) + C$ (d) none of these

65. $\int \frac{1}{\sin(x-a)\sin(x-b)} dx$ is equal to

- (a) $\sin(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$
 (b) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$
 (c) $\operatorname{cosec}(b-a) \log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| + C$
 (d) $\sin(b-a) \log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| + C$



66. The value of $\sqrt{2} \int \frac{\sin x \, dx}{\sin\left(x - \frac{\pi}{4}\right)}$ is

(a) $x - \log \left| \cos x \left(x - \frac{\pi}{4} \right) \right| + C$

(b) $x + \log \left| \cos \left(x - \frac{\pi}{4} \right) \right| + C$

(c) $x - \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + C$

(d) $x + \log \left| \sin \left(x - \frac{\pi}{4} \right) \right| + C$

67. $\int \frac{dx}{\cos x + \sqrt{3} \sin x}$ equals

(a) $\log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$ (b) $\log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

(c) $\frac{1}{2} \log \tan \left(\frac{x}{2} + \frac{\pi}{12} \right) + C$ (d) $\frac{1}{2} \log \tan \left(\frac{x}{2} - \frac{\pi}{12} \right) + C$

68. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$ equal to:

(a) $\frac{1}{(1 + \cot^3 x)^2} + c$ (b) $-\frac{1}{3(1 + \tan^3 x)} + c$

(c) $\frac{\sin^3 x}{(1 + \cos^3 x)^2} + c$ (d) $-\frac{\cos^3 x}{3(1 + \sin^3 x)} + c$

69. If $\int \tan^4 x \, dx = \lambda \tan^3 x + \mu \tan x + x + C$, then

(a) $\lambda = \frac{1}{3}$ (b) $\mu = 1$

(c) $\lambda = -\frac{1}{3}$ (d) none of these

Integration of irrational Algebraic Functions

70. $\int \sqrt{\frac{1+x}{1-x}} \, dx$ equals

(a) $\sin^{-1} x + \sqrt{1-x^2} + c$ (b) $\sin^{-1} x + \sqrt{x^2-1} + c$

(c) $\sin^{-1} x - \sqrt{1-x^2} + c$ (d) $\sin^{-1} x - \sqrt{x^2-1} + c$



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

1. The integral $\int \frac{dx}{x^2(x^4+1)^{3/4}}$ equal (2015)
- (a) $-(x^4+1)^{1/4}+c$ (b) $-\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$
- (c) $\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$ (d) $(x^4+1)^{1/4}+c$
2. The integral $\int \frac{dx}{(x+1)^{3/4}(x-2)^{5/4}}$ is equal to (2015/Online Set-1)
- (a) $4\left(\frac{x+1}{x-2}\right)^{1/4}+C$ (b) $-\frac{4}{3}\left(\frac{x+1}{x-2}\right)^{1/4}+C$
- (c) $-\frac{4}{3}\left(\frac{x-2}{x+1}\right)^{1/4}+C$ (d) $4\left(\frac{x-2}{x+1}\right)^{1/4}+C$
3. If $\int \frac{\log(t+\sqrt{1+t^2})}{\sqrt{1+t^2}} dt = \frac{1}{2}(g(t))^2 + C$, where C is constant, then g(2) is equal to : (2015/Online Set-2)
- (a) $\frac{1}{\sqrt{5}}\log(2+\sqrt{5})$ (b) $\frac{1}{2}\log(2+\sqrt{5})$
- (c) $2\log(2+\sqrt{5})$ (d) $\log(2+\sqrt{5})$
4. The integral $\int \frac{2x^{12}+5x^9}{(x^5+x^3+1)^3} dx$ is equal to : (2016)
- (a) $\frac{x^{10}}{2(x^5+x^3+1)^2}+C$ (b) $\frac{x^5}{2(x^5+x^3+1)^2}+C$
- (c) $\frac{-x^{10}}{2(x^5+x^3+1)^2}+C$ (d) $\frac{-x^5}{(x^5+x^3+1)^2}+C$
- Where C is an arbitrary constant.
5. If $\int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = (\tan x)^A + C(\tan x)^B + k$, where k is a constant of integration, then A+B+C equals : (2016/Online Set-1)
- (a) $\frac{21}{5}$ (b) $\frac{16}{5}$
- (c) $\frac{10}{7}$ (d) $\frac{27}{10}$
6. The integral $\int \frac{dx}{(1+\sqrt{x})\sqrt{x-x^2}}$ is equal to : (where C is a constant of integration.) (2016/Online Set-2)
- (a) $-2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}+C$ (b) $-2\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}+C$
- (c) $-\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}+C$ (d) $2\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}+C$
7. Let $I_n = \int \tan^n x dx$, ($n > 1$). If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to: (2017)
- (a) $\left(-\frac{1}{5}, 1\right)$ (b) $\left(\frac{1}{5}, 0\right)$
- (c) $\left(\frac{1}{5}, -1\right)$ (d) $\left(-\frac{1}{5}, 0\right)$
8. The integral $\int \sqrt{1+2 \cot x (\operatorname{cosec} x + \cot x)} dx \left(0 < x < \frac{\pi}{2}\right)$ is equal to (where C is a constant of integration) (2017/Online Set-1)
- (a) $4 \log \left(\sin \frac{x}{2}\right) + C$ (b) $2 \log \left(\sin \frac{x}{2}\right) + C$
- (c) $2 \log \left(\cos \frac{x}{2}\right) + C$ (d) $4 \log \left(\cos \frac{x}{2}\right) + C$



9. If $f\left(\frac{3x-4}{3x+4}\right) = x+2$, $x \neq -\frac{4}{3}$, and

$\int f(x) dx = A \log |1-x| + Bx + C$, then the ordered pair

(A, B) is equal to :

(where C is a constant of integration)

(2017/Online Set-2)

(a) $\left(\frac{8}{3}, \frac{2}{3}\right)$ (b) $\left(-\frac{8}{3}, \frac{2}{3}\right)$

(c) $\left(-\frac{8}{3}, -\frac{2}{3}\right)$ (d) $\left(\frac{8}{3}, -\frac{2}{3}\right)$

10. The integral

$\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is

(2018)

(a) $\frac{-1}{1+\cot^3 x} + C$ (b) $\frac{1}{3(1+\tan^3 x)} + C$

(c) $\frac{-1}{3(1+\tan^3 x)} + C$ (d) $\frac{1}{1+\cot^3 x} + C$

(Where C is a constant of integration)

11. If $f\left(\frac{x-4}{x+2}\right) = 2x+1$, $(x \in \mathbb{R} - \{-1, -2\})$, then $\int f(x) dx$ is

equal to

(where C is a constant of integration)

(2018/Online Set-1)

(a) $12 \log_e |1-x| + 3x + C$

(b) $-12 \log_e |1-x| - 3x + C$

(c) $12 \log_e |1-x| - 3x + C$

(d) $-12 \log_e |1-x| + 3x + C$

12. If $\int \frac{2x+5}{\sqrt{7-6x-x^2}} dx = A\sqrt{7-6x-x^2} + B \sin^{-1}\left(\frac{x+3}{4}\right) + C$

(where C is a constant of integration), then the ordered pair (A, B) is equal to :

(2018/Online Set-2)

(a) (2, 1) (b) (-2, -1)

(c) (-2, 1) (d) (2, -1)

13. If $\int \frac{\tan x}{1+\tan x+\tan^2 x} dx = x - \frac{k}{\sqrt{A}} \tan^{-1}\left(\frac{k \tan x + 1}{\sqrt{A}}\right) + C$,

(C is a constant of integration), then the ordered pair (K, A) is equal to :

(2018/Online Set-3)

(a) (2, 1) (b) (-2, 3)

(c) (2, 3) (d) (-2, 1)

14. $\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ is equal to :

(where c is a constant of integration.)

(8-04-2019/Shift-1)

(a) $2x + \sin x + 2 \sin 2x + c$

(b) $x + 2 \sin x + 2 \sin 2x + c$

(c) $x + 2 \sin x + \sin 2x + c$

(d) $2x + \sin x + \sin 2x + c$

15. If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = xf(x)(1+x^6)^{1/3} + C$, where C is a

constant of integration, then the function $f(x)$ is equal to:

(08-04-2019/Shift-2)

(a) $\frac{3}{x^2}$ (b) $-\frac{1}{6x^3}$

(c) $-\frac{1}{2x^2}$ (d) $-\frac{1}{2x^3}$

16. The integral $\int \sec^{\frac{2}{3}} x \operatorname{cosec}^{\frac{4}{3}} x dx$ is equal to:

(9-04-2019/Shift-1)

(a) $-3 \tan^{\frac{-1}{3}} x + C$ (b) $-\frac{3}{4} \tan^{\frac{-4}{3}} x + C$

(c) $-3 \cot^{\frac{-1}{3}} x + C$ (d) $3 \tan^{\frac{-1}{3}} x + C$

(Here C is a constant of integration)



17. If $\int e^{\sec x} (\sec x \tan x f(x) + (\sec x \tan x + \sec^2 x)) dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is:

(9-04-2019/Shift-2)

- (a) $\sec x + \tan x + C$ (b) $\sec x - \tan x - C$
(c) $\sec x + 2 \tan x - C$ (d) $x \sec x + \tan x + C$

18. If $\int x^5 e^{-x^2} dx = g(x) e^{-x^2} + c$, where c is a constant of integration, then $g(-1)$ is equal to:

(10-4-2019/Shift-2)

- (a) -1 (b) 1
(c) $-\frac{5}{2}$ (d) $-\frac{1}{2}$

19. The integral $\int \frac{2x^3 - 1}{x^4 + x} dx$ is equal to: (Here C is a constant of integration)

(12-04-2019/Shift-1)

- (a) $\frac{1}{2} \log_e \left| \frac{x^3 + 1}{x^2} \right| + C$ (b) $\frac{1}{2} \log_e \left| \frac{(x^3 + 1)^2}{x^3} \right| + C$
(c) $\log_e \left| \frac{x^3 + 1}{x} \right| + C$ (d) $\log_e \left| \frac{x^3 + 1}{x^2} \right| + C$

20. Let $\alpha \in \left(0, \frac{\pi}{2}\right)$ be fixed. If the integral

$$\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx = A(x) \cos 2\alpha + B(x) \sin 2\alpha + C,$$

where C is a constant of integration, then the functions $A(x)$ and $B(x)$ are respectively:

(12-04-2019/Shift-2)

- (a) $x + \alpha$ and $\log_e |\sin(x + \alpha)|$
(b) $x - \alpha$ and $\log_e |\sin(x - \alpha)|$
(c) $x - \alpha$ and $\log_e |\cos(x - \alpha)|$
(d) $x + \alpha$ and $\log_e |\sin(x - \alpha)|$

21. For $x^2 \neq n\pi, n \in \mathbb{N}$ (the set of natural numbers), the

$$\text{integral } \int x \sqrt{\frac{2 \sin(x^2 - 1) - \sin 2(x^2 - 1)}{2 \sin(x^2 - 1) + \sin 2(x^2 - 1)}} dx \text{ is equal to:}$$

(where c is a constant of integration)

(9-01-2019/Shift-1)

- (a) $\log_e \left| \frac{1}{2} \sec^2(x^2 - 1) \right| + c$
(b) $\frac{1}{2} \log_e |\sec(x^2 - 1)| + c$
(c) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2 - 1}{2} \right) \right| + c$
(d) $2 \log_e \left| \sec \left(\frac{x^2 - 1}{2} \right) \right| - c$

22. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, (x \geq 0)$, and $f(0) = 0$, then

the value of $f(1)$ is

(9-01-2019/Shift-2)

- (a) $-\frac{1}{2}$ (b) $-\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

23. Let $n \geq 2$ be a natural number and $0 < \theta < \frac{\pi}{2}$ then

$$\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta \text{ is equal to:}$$

(10-1-2019/Shift-1)

- (a) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(b) $\frac{n}{n^2 + 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(c) $\frac{n}{n^2 - 1} \left(1 + \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$
(d) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n+1} \theta} \right)^{\frac{n+1}{n}} + C$

(where C is a constant of integration)



24. If $\int x^5 e^{-4x^3} dx = \frac{1}{48} e^{-4x^3} f(x) + C$, where C is a constant of integration, then $f(x)$ is equal to:
- (10-01-2019/Shift-2)
- (a) $-2x^3 - 1$ (b) $-4x^3 - 1$
(c) $-2x^3 + 1$ (d) $4x^3 + 1$
25. If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x) (\sqrt{1-x^2})^m + C$, for a suitable chosen integer m and a function A(x), where C is a constant of integration, then $(A(x))^m$ equals :
- (11-01-2019/Shift-1)
- (a) $\frac{-1}{27x^9}$ (b) $\frac{-1}{3x^3}$
(c) $\frac{1}{27x^6}$ (d) $\frac{1}{9x^4}$
26. If $\int \frac{x+1}{\sqrt{2x-1}} dx = f(x) \sqrt{2x-1} + C$, where C is a constant of integration, then $f(x)$ is equal to :
- (11-01-2019/Shift-2)
- (a) $\frac{1}{3}(x+1)$ (b) $\frac{2}{3}(x+2)$
(c) $\frac{2}{3}(x-4)$ (d) $\frac{1}{3}(x+4)$
27. The integral $\int \cos(\log_e x) dx$ is equal to (Where C is a constant of integration)
- (12-01-2019/Shift-1)
- (a) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$
(b) $x [\cos(\log_e x) + \sin(\log_e x)] + C$
(c) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$
(d) $x [\cos(\log_e x) - \sin(\log_e x)] + C$
28. The integral $\int \frac{3x^{13} + 2x^{11}}{(2x^4 + 3x^2 + 1)^4} dx$ is equal to (where C is a constant to integration)
- (12-01-2019/Shift-2)
- (a) $\frac{x^4}{6(2x^4 + 3x^2 + 1)^3} + C$ (b) $\frac{x^{12}}{6(2x^4 + 3x^2 + 1)^3} + C$
(c) $\frac{x^4}{(2x^4 + 3x^2 + 1)^3} + C$ (d) $\frac{x^{12}}{(2x^4 + 3x^2 + 1)^3} + C$
29. If $\int \sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) dx = A(x) \tan^{-1}(\sqrt{x}) + B(x) + C$, where C is a constant of integration, then the ordered pair $(A(x), B(x))$ can be :
- (3-09-2020/Shift-2)
- (a) $(x+1, -\sqrt{x})$ (b) $(x-1, -\sqrt{x})$
(c) $(x+1, \sqrt{x})$ (d) $(x-1, \sqrt{x})$
30. The integral $\int \left(\frac{x}{x \sin x + \cos x} \right)^2 dx$ is equal to (where C is a constant of integration) :
- (4-09-2020/Shift-1)
- (a) $\tan x - \frac{x \sec x}{x \sin x + \cos x} + C$
(b) $\sec x - \frac{x \tan x}{x \sin x + \cos x} + C$
(c) $\sec x + \frac{x \tan x}{x \sin x + \cos x} + C$
(d) $\tan x + \frac{x \sec x}{x \sin x + \cos x} + C$
31. If $\int (e^{2x} + 2e^x - e^{-x} - 1) e^{(e^x + e^{-x})} dx$
 $= g(x) e^{(e^x + e^{-x})} + c$ where c is a constant integration, then $g(0)$ is equal to:
- (5-09-2020/Shift-1)
- (a) 2 (b) e
(c) 1 (d) e^2



32. If $\int \frac{\cos \theta}{5 + 7 \sin \theta - 2 \cos^2 \theta} d\theta = A \log_e |B(\theta)| + C$ where c is a constant of integration, then $\frac{B(\theta)}{A}$ can be:

(5-09-2020/Shift-2)

- (a) $\frac{5(2 \sin \theta + 1)}{\sin \theta + 3}$ (b) $\frac{5(\sin \theta + 3)}{2 \sin \theta + 1}$
(c) $\frac{2 \sin \theta + 1}{\sin \theta + 3}$ (d) $\frac{2 \sin \theta + 1}{5(\sin \theta + 3)}$

33. If $\int \frac{\cos x}{\sin^3 x (1 + \sin^6 x)^{\frac{2}{3}}} dx = f(x) (1 + \sin^6 x)^{\frac{1}{2}} + c$,

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to:

(8-01-2020/Shift-1)

- (a) $-\frac{9}{8}$ (b) $\frac{9}{8}$
(c) 2 (d) -2

34. If $f'(x) = \tan^{-1}(\sec x + \tan x)$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $f(0) = 0$, then $f(1)$ is equal to:

(9-01-2020/Shift-1)

- (a) $\frac{\pi+1}{4}$ (b) $\frac{\pi+2}{4}$
(c) $\frac{1}{4}$ (d) $\frac{\pi-1}{4}$

35. The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to: (where C is a constant of integration)

(9-01-2020/Shift-1)

- (a) $-\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (b) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$
(c) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (d) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{13}{7}} + C$

36. If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$

where C is constant if integration, then the ordered pair $(\lambda, f(\theta))$ is equal to:

(9-1-2020/Shift-2)

- (a) $(-1, 1 - \tan \theta)$ (b) $(-1, 1 + \tan \theta)$
(c) $(1, 1 + \tan \theta)$ (d) $(1, 1 - \tan \theta)$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. If $I = \int \frac{dx}{(a^2 - b^2 x^2)^{3/2}}$

(a) $\frac{x}{\sqrt{a^2 - b^2 x^2}} + c$

(b) $\frac{x}{a^2 \sqrt{a^2 - b^2 x^2}} + c$

(c) $\frac{ax}{\sqrt{a^2 - b^2 x^2}} + c$

(d) none of these

2. If $\int \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right)^{1/2} \frac{dx}{x} = 2 \cos^{-1} \sqrt{x} - f(x) + c$, then $f(x)$ equals.

(a) $\log_e \left(\frac{1 + \sqrt{1-x}}{\sqrt{x}} \right)$

(b) $\frac{1}{2} \log_e \left(\frac{1 + \sqrt{1-x}}{\sqrt{x}} \right)$

(c) $2 \log_e \left(\frac{1 - \sqrt{1-x}}{\sqrt{x}} \right)$

(d) $2 \log_e \left(\frac{1 + \sqrt{1-x}}{\sqrt{x}} \right)$

3. If $I = \int \frac{x^3 + x}{x^4 - 9} dx$ then I equals :

(a) $\frac{1}{4} \log |x^4 - 9| + \frac{1}{12} \log \left| \frac{x^2 + 3}{x^2 - 3} \right| + c$

(b) $\frac{1}{4} \log |x^4 - 9| + \frac{1}{12} \log \left| \frac{x^2 - 3}{x^2 + 3} \right| + c$

(c) $\frac{1}{4} \log |x^4 - 9| - \frac{1}{12} \log \left| \frac{x-3}{x+3} \right| + c$

(d) None

4. Evaluate $I = \int \left(\sqrt{\frac{a+x}{a-x}} + \sqrt{\frac{a-x}{a+x}} \right) dx$

(a) $2 \sin^{-1} \left(\frac{x}{a} \right) + c$

(b) $2a \sin^{-1} \left(\frac{x}{a} \right) + c$

(c) $2 \cos^{-1} \left(\frac{x}{a} \right) + c$

(d) $2a \cos^{-1} \left(\frac{x}{a} \right) + c$

5. Evaluate : $\int \frac{\sin x dx}{\sqrt{3 \sin^2 x + 4 \cos^2 x}}$

(a) $-\ln(\cos x + \sqrt{3 + \cos^2 x}) + c$

(b) $\ln(\cos x + \sqrt{3 + \cos^2 x}) + c$

(c) $-\ln(\sin x + \sqrt{3 + \sin^2 x}) + c$

(d) None of these

6. $\int \sqrt{\frac{e^x + a}{e^x - a}} dx$

(a) $\ln(e^x + \sqrt{e^{2x} - a^2}) + \sin^{-1}(ae^{-x}) + c$

(b) $\ln(e^x - \sqrt{e^{2x} - a^2}) + \cos^{-1}(ae^{-x}) + c$

(c) $\ln(e^x + \sqrt{e^{2x} - a^2}) + \cos^{-1}(ae^{-x}) + c$

(d) None

7. $\int \frac{\sqrt{x} - \sqrt{a}}{\sqrt{x+a}} dx$

(a) $\sqrt{ax+x^2} - 2\sqrt{a^2+ax} - a/\ln(\sqrt{x} + \sqrt{a+x}) + c$

(b) $\sqrt{ax+x^2} - 2\sqrt{a+x} - a/\ln(\sqrt{x} + \sqrt{a+x}) + c$

(c) $\sqrt{ax+x^2} - 2\sqrt{a^2+ax} + a/\ln(\sqrt{x} + \sqrt{a-x}) + c$

(d) None of these

8. If $I = \int \frac{x^2 + a^2}{x^4 - a^2 x^2 + a^4} dx$, then $I =$

(a) $\frac{1}{a} \tan^{-1} \left(\frac{ax}{x^2 - a^2} \right) + c$

(b) $\frac{1}{a} \tan^{-1} \left(\frac{x^2 - a^2}{ax} \right) + c$

(c) $\log |x + \sqrt{x^2 - a^2}| + c$

(d) None of these



9. $\int \frac{x^2 - 1}{(x^4 + 3x^2 + 1) \tan^{-1}\left(\frac{x^2 + 1}{x}\right)} dx$ is equal to

(a) $\tan^{-1}\left(x + \frac{1}{x}\right) + c$

(b) $\log_e \left| \tan^{-1}\left(x + \frac{1}{x}\right) \right| + c$

(c) $\log_e \left| \tan\left(\frac{x^2 + 1}{x}\right) \right| + c$

(d) $\left(x + \frac{1}{x}\right) \tan^{-1}\left(x + \frac{1}{x}\right) + c$

10. $\int \sqrt{e^{2x} - 1} dx$ is equal to

(a) $\sqrt{e^{2x} - 1} + \sec^{-1} e^{2x} + c$

(b) $\sqrt{e^{2x} - 1} - \sec^{-1} e^{2x} + c$

(c) $\sqrt{e^{2x} - 1} - \sec^{-1} e^x + c$

(d) none of these

11. Evaluate : $\int \sqrt{x + \sqrt{x^2 + 2}} dx$

(a) $\frac{2}{3} (x + \sqrt{x^2 + 2})^{3/2} - \frac{1}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(b) $\frac{1}{3} (x + \sqrt{x^2 + 2})^{3/2} - \frac{2}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(c) $\frac{1}{3} (x + \sqrt{x^2 + 2})^{1/2} - \frac{2}{\sqrt{x + \sqrt{x^2 + 2}}} + c$

(d) None of these

12. Evaluate : $\int \frac{(1+x)}{x(1+xe^x)^2} dx$

(a) $\ln\left(\frac{xe^x}{1+xe^x}\right) + \frac{1}{1+xe^x} + c$

(b) $\ln\left(\frac{1+xe^x}{xe^x}\right) + \frac{1}{xe^x + 1} + c$

(c) $\ln\left(\frac{xe^x}{xe^x + 1}\right) - \frac{1}{xe^x + 1} + c$

(d) None of these

13. $\int \frac{x-1}{(x+1)\sqrt{x(x^2+x+1)}} dx$ is

(a) $\tan^{-1}\left(\frac{x^2+x+1}{x}\right) + c$ (b) $2\tan^{-1}\left(\frac{x^2+x+1}{x}\right) + c$

(c) $\tan^{-1}\left(\frac{\sqrt{x^2+x+1}}{x}\right) + c$ (d) $2\tan^{-1}\sqrt{x + \frac{1}{x} + 1} + c$

14. $\int \frac{(1+x)\sin x}{(x^2+2x)\cos^2 x - (1+x)\sin 2x} dx = \frac{1}{2} \log_e \left| \frac{t+1}{t-1} \right| + c$

where t is

(a) $(x+1)\cos x - \sin x$ (b) $(x+1)\sin x - \cos x$

(c) $(x+1)\sin x + \cos x$ (d) $(x+1)\cos x + \sin x$

15. $\int \frac{(x^2-1)dx}{2x\sqrt{x^4+4x^3-6x^2+4x+1}}$ is

(a) $\frac{1}{2} \ln \left| x + \frac{1}{x} + 2 + \sqrt{\left(x + \frac{1}{x} + 2\right)^2 - 12} \right| + c$

(b) $\frac{1}{2} \ln \left| x - \frac{1}{x} + 2 + \sqrt{\left(x - \frac{1}{x} + 2\right)^2 - 12} \right| + c$

(c) $\frac{1}{2} \ln \left| x + \frac{1}{x} - 2 + \sqrt{\left(x + \frac{1}{x} - 2\right)^2 - 12} \right| + c$

(d) None of these



16. If $\int \log(x^2 + x) dx = x \log|x| + (x+1) \log|x+1| + k$, then k equals
 (a) $2x + \log|x+1| + c$ (b) $2x - \log|x+1| + c$
 (c) constant (d) None
17. If $I = \int \cos \theta \log \left(\tan \frac{\theta}{2} \right) d\theta$, then I equals
 (a) $\sin \theta \log(\tan \theta/2) + \theta + c$
 (b) $\cos \theta \log(\tan \theta/2) + \theta + c$
 (c) $\sin \theta \log(\tan \theta/2) - \theta + c$
 (d) None of these
18. If $f(x)$ is a polynomial function of the n^{th} degree, then $\int e^x f(x) dx$ is equal to
 (a) $e^x \{f(x) - f'(x) - f''(x) - f'''(x) - \dots - (-1)^n f^{(n)}(x)\}$
 (b) $e^x \{f(x) - f'(x) + f''(x) - f'''(x) + \dots + (-1)^n f^{(n)}(x)\}$
 (c) $e^x \{f(x) - f'(x) + f''(x) - f'''(x) + \dots + (-1)^n f^{(n)}(x)\}$
 (d) none of these
19. If $I = \int e^x (x \cos x + \sin x) dx$ then I equals :
 (a) $\frac{1}{2} e^x (x \sin x - \cos x) + c$
 (b) $\frac{1}{2} e^x (x \sin x + \cos x) + c$
 (c) $\frac{1}{2} e^x (x \cos x - \sin x) + c$
 (d) None
20. $\int \sqrt[3]{x} (\log_e x)^2 dx =$
 (a) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 - \frac{3}{2} \log_e x - \frac{9}{8} \right] + c$
 (b) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 + \frac{3}{2} \log_e x + \frac{9}{8} \right] + c$
 (c) $\frac{3}{4} (x)^{4/3} \left[(\log_e x)^2 - \frac{3}{2} \log_e x + \frac{9}{8} \right] + c$
 (d) $\frac{3}{4} (x)^{1/3} \left[(\log_e x)^2 + \frac{3}{2} \log_e x - \frac{9}{8} \right] + c$
21. If $\int g(x) dx = g(x)$ then $\int g(x) (f(x) + f'(x)) dx$ is equal to
 (a) $g(x) f(x) - g(x) f'(x) + c$ (b) $g(x) f'(x) + c$
 (c) $g(x) f(x) + c$ (d) $g(x) f^2(x) + c$
22. If $\int (x^3 - 2x^2 + 5) e^{3x} dx = e^{3x} (Ax^3 + Bx^2 + Cx + 13/9)$ then which of the following statement is incorrect :
 (a) $3C = 2$ (b) $A + B + \frac{2}{3} = 0$
 (c) $C + 2B = 0$ (d) $A + B + C = 0$
23. Evaluate : $\int \sqrt{\frac{1+x^{2n}}{x^{2n}}} \cdot \frac{\ln(1+x^{2n}) - 2n \ln x}{x^{2n+1}} dx$
 (a) $\frac{2P^3}{9n} (1 - 3 \ln P) + C$ (b) $\frac{P^3}{3n} (3 \ln P - 1) + C$
 (c) $\frac{2P^3}{3n} (3 \ln P - 1) + C$ (d) None of these
 where $P = \left(1 + \frac{1}{x^{2n}} \right)^{1/2}$
24. $\int \frac{e^{2 \tan^{-1} x} (1+x)^2}{(1+x^2)} dx$ is equal to
 (a) $x e^{\tan^{-1} x} + c$ (b) $x e^{2 \tan^{-1} x} + c$
 (c) $2x e^{2 \tan^{-1} x} + c$ (d) none of these
25. Evaluate $I = \int \frac{dx}{\sin^3 x + \cos^3 x}$
 (a) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}+t}{\sqrt{2}-t} \right| + \tan^{-1} t + c$
 (b) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}-t}{\sqrt{2}+t} \right| + \frac{1}{3} \tan^{-1} t + c$
 (c) $\frac{1}{3\sqrt{2}} \log \left| \frac{\sqrt{2}+t}{\sqrt{2}-t} \right| + \frac{2}{3} \tan^{-1} t + c$
 (d) None of these
 where $t = \sin x - \cos x$



26. If $I = \int \sin^7 x \, dx$, then I equals

- (a) $-\cos x + \cos^3 x - \cos^5 x + \frac{1}{7} \cos^7 x + c$
 (b) $-\cos x + \cos^3 x - \frac{3}{5} \cos^5 x + \frac{1}{7} \cos^7 x + c$
 (c) $\tan x - \tan^3 x + \sin x - \frac{1}{3} \cos^3 x + c$
 (d) $-\cos x + \cos^5 x - \frac{1}{3} \cos^3 x + \frac{1}{7} \cos^7 x + c$

27. If $I = \int \frac{\sqrt{\sin^3 2x}}{\sin^5 x} dx$, and $f(x) = (\cot x)^{3/2}$, $g(x) = (\cot x)^{5/2}$, then I equals

- (a) $\frac{2\sqrt{3}}{3} f(x) - \frac{1}{5} g(x) + c$ (b) $-\frac{4\sqrt{2}}{5} g(x) + c$
 (c) $\frac{1}{2\sqrt{3}} f(x) + c$ (d) $\frac{2\sqrt{2}}{3} f(x) + \frac{1}{5} g(x) + c$

28. Evaluate : $\int \frac{\sec x \, dx}{\sqrt{\sin(2x+A) + \sin A}}$

- (a) $2 \cos A \sqrt{2 \cos A + \sin A \tan x} + c$
 (b) $\sqrt{2} \sec A \sqrt{2 \cos A \tan x + 2 \sin A} + c$
 (c) $\sqrt{2} \sec A \sqrt{\cos A \tan x + \sin A} + c$
 (d) None of these

29. $\int \sqrt{\sec x - 1} \, dx$

- (a) $2 \ln(\sqrt{\cos x} + \sqrt{1 + \cos x}) + c$
 (b) $2 \ln(\sqrt{\sin x} + \sqrt{1 + \sin x}) + c$
 (c) $-2 \ln(\sqrt{\cos x} + \sqrt{1 + \cos x}) + c$
 (d) None of these

30. $\int (\sin 4x) e^{\tan^2 x} \, dx =$

- (a) $-2 e^{\tan^2 x} \cos^4 x + c$ (b) $2 e^{\tan^2 x} \sec^4 x + c$
 (c) $-2 e^{\tan^2 x} \sec^2 x + c$ (d) $2 e^{\tan^2 x} \cos^2 x + c$

31. $\int \frac{dx}{\tan x + \cot x + \sec x + \csc x}$ is equal to

- (a) $\frac{1}{2} (\sin x - \cos x + x) + c$
 (b) $\frac{1}{2} (\sin x - \cos x - \tan x + \cot x) + c$
 (c) $\frac{1}{2} (\sin x - \cos x - x) + c$
 (d) $\frac{1}{2} (\sin x + \cos x - \tan x - \cot x + x) + c$

32. $\int \frac{\sin x - \cos x}{(\sin x + \cos x) \sqrt{\sin x \cos x + \sin^2 x \cos^2 x}} dx =$

- (a) $-\sin(\sin 2x + 1) + c$
 (b) $\operatorname{cosec}(\sin 2x + 1)$
 (c) $-\sec^{-1}(\sin 2x + 1) + c$
 (d) $\tan^{-1}(\sin 2x + 1) + c$

33. If $I = \int \frac{(2x+3) dx}{(x^2 + 2x + 3) \sqrt{x^2 + 2x + 4}}$, then I equals

- (a) $\log \left| \frac{\sqrt{x^2 + 2x + 4} - 1}{\sqrt{x^2 + 2x + 4} + 1} \right| + c$
 (b) $\log \left| \frac{\sqrt{x^2 + 2x + 4} - 1}{\sqrt{x^2 + 2x + 4} + 1} \right| + \tan^{-1} \left(\frac{x+2}{3} \right) + c$
 (c) $\log \tan^{-1} \left(\frac{\sqrt{x+3}}{2} \right) + c$
 (d) None

34. If $\int \frac{dx}{(x^2 - 4) \sqrt{x+1}} = \frac{1}{k} \log \left| \frac{\sqrt{x+1} - \sqrt{3}}{\sqrt{x+1} + \sqrt{3}} \right|$

$-\frac{1}{2} \tan^{-1} \sqrt{x+1} + c$ then k equals

- (a) $2\sqrt{3}$ (b) $4\sqrt{3}$
 (c) $\frac{1}{4\sqrt{3}}$ (d) none of these



35. Evaluate $\int \frac{(x+2) dx}{(x^2+3x+3)\sqrt{x+1}}$

(a) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}\sqrt{x+1}}{-x} \right) + c$

(b) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}\sqrt{x+1}}{x} \right) + c$

(c) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}\sqrt{x+1}}{-x} \right) + c$

(d) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}\sqrt{x+1}}{x} \right) + c$

36. Evaluate : $\int \frac{\sqrt{x-1}}{x\sqrt{x+1}} dx$

(a) $\ln(x + \sqrt{x^2+1}) + \sec^{-1} x + c$

(b) $\ln(x - \sqrt{x^2+1}) - \sec^{-1} x + c$

(c) $\ln(x + \sqrt{x^2-1}) - \sec^{-1} x + c$

(d) None of these

37. If $y = \sqrt{x^2 - x + 1}$ and for $n \geq 1$, $I_n = \int x^n / y dx$ and $aI_3 + bI_2 + cI_1 = x^2y$, then (a, b, c) is equal to

(a) $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$ (b) $(1, -1, 1)$

(c) $\left(3, -\frac{5}{2}, 2\right)$ (d) $\left(\frac{1}{2}, -\frac{1}{2}, 1\right)$

38. Let $g(x)$ be an antiderivative of $f(x)$. Then $\ln(1+(g(x))^2)$ is an antiderivative for :

(a) $\frac{2f(x) \cdot g(x)}{1+(f(x))^2}$ (b) $\frac{2f(x) \cdot g(x)}{1+(g(x))^2}$

(c) $\frac{2f(x)}{1+(f(x))^2}$ (d) None

39. If $I_n = \int (\ln x)^n dx$ then $I_n + n I_{n-1} =$

(a) $(\ln x)^n$ (b) $x (\ln x)^n$
(c) $x^n \ln x$ (d) $x (\ln x)^{n-1}$

40. If $I_n = \int \cos^n x dx$ then $I_n - \left(\frac{n-1}{n}\right) I_{n-2} =$

(a) $\cos^{n-1} x \cdot \sin x + c$ (b) $\frac{1}{n} (\cos^{n-1} x \sin x) + c$

(c) $\frac{1}{n} (\cos^n x \sin x) + c$ (d) None of these

41. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$ exists finitely and

$\lim_{x \rightarrow 0} \left(1 + x + \frac{f(x)}{x}\right)^{1/x} = e^3$, where $f(x) = ax^2 + bx + c$

then $\int f(x) \log_e x dx$ is equal to

(a) $\frac{2}{3} x^3 \left(\log_e x - \frac{1}{3}\right) + c$ (b) $\frac{x^3}{3} \left(\log_e x - \frac{1}{3}\right) + c$

(c) $\frac{2}{3} x^3 (\log_e x + 1) + c$ (d) $\frac{2}{3} x^3 (\log_e x - 1) + c$

42. For $0 < x < 1$, let

$f(x) = \lim_{n \rightarrow \infty} (1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n})$

then $\int \frac{f(x)}{1-x} \log_e x dx$ equals

(a) $\log_e \left(\frac{x}{1-x}\right) + c$

(b) $-\log_e \left(\frac{x}{1-x}\right) + \frac{\log_e x}{1-x} + c$

(c) $\frac{\log_e x}{1-x} + \log_e (1-x) + c$

(d) $x \log_e x + \log_e (1-x) + c$



43. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying the following :

(i) $f(-x) = -f(x)$

(ii) $f(x+1) = f(x) + 1$

(iii) $f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2} \quad \forall x \neq 0$

then $\int e^x f(x) dx$ is equal to

(a) $e^x (x-1) + c$

(b) $e^x \log x + c$

(c) $\frac{e^x}{x} + c$

(d) $\frac{e^x}{x+1} + c$

Objective Questions II [One or more than one correct option]

44. $\int \frac{1}{x^2-1} \ln\left(\frac{x-1}{x+1}\right) dx$ equals

(a) $\frac{1}{2} \ln\left(\frac{x-1}{x+1}\right) + c$

(b) $\frac{1}{4} \ln^2\left(\frac{x-1}{x+1}\right) + c$

(c) $\frac{1}{2} \ln^2\left(\frac{x+1}{x-1}\right) + c$

(d) $\frac{1}{4} \ln^2\left(\frac{x+1}{x-1}\right) + c$

45. If primitive of $\sin(\log x)$ is $f(x) (\sin(g(x)) - \cos(h(x)) + c$ then

(a) $\lim_{x \rightarrow 2} f(x) = 1$

(b) $\lim_{x \rightarrow 1} \frac{g(x)}{h(x)} = 1$

(c) $g(e^3) = 3$

(d) $h(e^5) = 5$

46. It $f(x) = \lim_{n \rightarrow \infty} e^{x \tan(1/n) \log(1/n)}$ and $\int \frac{f(x) dx}{\sqrt[3]{\sin^{11} x \cos x}} = g(x) + c$

then

(a) $g\left(\frac{\pi}{4}\right) = \frac{3}{2}$

(b) $g(x)$ is continuous for all x

(c) $g\left(\frac{\pi}{4}\right) = \frac{-15}{8}$

(d) $g(x)$ is non-differentiable at infinitely many points

47. $\int \frac{dx}{5+4\cos x} = \lambda \tan^{-1}\left(m \tan \frac{x}{2}\right) + c$ then

(a) $\lambda = \frac{2}{3}$

(b) $m = \frac{1}{3}$

(c) $\lambda = \frac{1}{3}$

(d) $m = \frac{2}{3}$

Numerical Value Type Questions

48. Let f be a function satisfying $f''(x) = x^{-3/2}$, $f'(4) = 2$ and $f(0) = 0$ then $f(784)$ is equal to

49. If $\int \frac{x^2-1}{(x^2+1)\sqrt{x^4+1}} dx$ is equal to

$\frac{\sqrt{2}}{4} A \tan^{-1}\left(\frac{1}{\sqrt{2}} \sqrt{x^2 + \frac{1}{x^2}}\right) + c$ then A is equal to

50. If the graph of the antiderivative

$F(x)$ of $f(x) = \log(\log x) + (\log x)^{-2}$ passes through $(e, 1998-e)$ then the term independent of x in $F(x)$ is

51. Let $F(x)$ be the antiderivative of

$f(x) = \frac{1}{(3+5\sin x + 3\cos x)}$ whose graph passes through

the point $(0, 0)$ then the value of $F\left(\frac{\pi}{2}\right) - \frac{1}{5} \log \frac{8}{3} + 1982$ is equal to

52. $f(x)$ is the integral of $\frac{2\sin x - \sin 2x}{x^3}$, $x \neq 0$ find

$\lim_{x \rightarrow 0} f'(x)$.



ASSERTION REASON

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
 (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
 (C) If ASSERTION is true, REASON is false.
 (D) If ASSERTION is false, REASON is true.

53. Assertion : If $\int \frac{1}{f(x)} dx = 2 \log |f(x)| + c$, then $f(x) = \frac{x}{2}$

Reason : When $f(x) = \frac{x}{2}$, then

$$\int \frac{1}{f(x)} dx = \int \frac{2}{x} dx = 2 \log |x| + c$$

54. Assertion : $\int \frac{x^2-1}{x^2} e^{\frac{x^2+1}{x}} dx = e^{\frac{x^2+1}{x}} + c$

Reason : $\int f(x) e^{f(x)} dx = e^{f(x)} + c$

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

55. $\int f(x) dx$ when

- | | |
|--------------------------------------|---------------------------------------|
| (a) $f(x) = \frac{1}{e^x + 1}$ | (p) $2 \log (e^{x/2} + e^{-x/2}) + c$ |
| (b) $f(x) = e^{e^x + x}$ | (q) $\log \frac{e^x}{e^x + 1} + c$ |
| (c) $f(x) = \frac{e^x - 1}{e^x + 1}$ | (r) $\frac{1}{4} e^{2x^2} + c$ |
| (d) $f(x) = e^{2x^2 + \log x}$ | (s) $e^{e^x} + c$ |

The correct matching is :

- (a) (a-s, b-q, c-p, d-r)
 (b) (a-q, b-s, c-r, d-p)
 (c) (a-q, b-s, c-p, d-r)
 (d) (a-r, b-s, c-p, d-q)

56. $\int f(x) dx$, when

- | | |
|---|--|
| (a) $f(x) = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}$ | (p) $c - \frac{2}{\sqrt{15}} \log \left \frac{x+6+\sqrt{60x-15x^2}}{2x-3} \right $ |
| (b) $f(x) = \sqrt{\frac{1-x}{1+x}} \cdot \frac{1}{x}$ | (q) $\log \left \frac{\sqrt{x+1}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} \right + 2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} + c$ |
| (c) $f(x) = \frac{1}{\left(x - \frac{3}{2}\right) \sqrt{4x-x^2}}$ | (r) $(\sqrt{x}-2) \sqrt{1-x} - \sin^{-1} \sqrt{x} + c$ |
| (d) $f(x) = \sqrt{\frac{1+\sqrt{x}}{x}}$ | (s) $\frac{4}{3} (1+\sqrt{x})^{3/2} + c$ |

The correct matching is :

- (a) (A-q, B-r, C-p, D-s)
 (b) (A-r, B-q, C-p, D-s)
 (c) (A-r, B-p, C-q, D-s)
 (d) (A-r, B-q, C-s, D-p)

Using the following passage, solve Q.57 and Q.58

Passage – 1

If the integrand is a rational function of x and fractional powers of a linear fractional function of the form $\frac{ax+b}{cx+d}$.

Then rationalization of the integral is affected by the

substitution $\frac{ax+b}{cx+d} = t^m$, where m is the L.C.M. of

fractional powers of $\frac{ax+b}{cx+d}$.

57. If $I = \int \frac{dx}{\sqrt[4]{(x-1)^3 (x+2)^5}} = A \sqrt[4]{\frac{x-1}{x+2}} + c$ then A is equal to

- | | |
|-------------------|-------------------|
| (a) $\frac{1}{3}$ | (b) $\frac{2}{3}$ |
| (c) $\frac{3}{4}$ | (d) $\frac{4}{3}$ |



58. If $I = \int \frac{(2x-3)^{1/2}}{(2x-3)^{1/3} + 1} dx = 3 \left[\frac{1}{7} (2x-3)^{7/6} - \frac{1}{5} (2x-3)^{5/6} + \right.$

$\left. \frac{1}{3} (2x-3)^{1/2} - (2x-3)^{1/6} + g(x) \right] - 1$ then $g(x)$ is equal to

- (a) $\tan^{-1}(2x-3)^{1/6}$ (b) $(2x-3)^{1/2}$
(c) $3 \tan^{-1}(2x-3)^{1/6}$ (d) $4(2x-3)^{1/6}$

Using the following passage, solve Q.59 to Q.62

Passage – 2

We can derive reduction formula for the integration of the form $\int \sin^n x dx$, $\int \cos^n x dx$, $\int \tan^n x dx$ and other integrals of these form using integration by parts. In turn these reduction formulas can be used to compute integrals of higher power of $\sin x$ and $\cos x$.

59. If $\int \sin^5 x dx = -\frac{1}{5} \sin^4 x \cos x + A \sin^2 x \cos x - \frac{8}{15} \cos x + c$

then A is equal to

- (a) $-\frac{2}{15}$ (b) $-\frac{3}{5}$
(c) $-\frac{4}{15}$ (d) $-\frac{1}{15}$

60. If $\int \tan^6 x dx = \frac{1}{5} \tan^5 x + A \tan^3 x + \tan x - x + c$ then A is

equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{2}{3}$ (d) $-\frac{1}{3}$

61. If $I = \int \sec^6 x dx = \frac{1}{5} \tan^5 x + A \tan^3 x + \tan x + c$ then A is equal to

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $-\frac{1}{3}$ (d) $-\frac{2}{3}$

62. If $\int \operatorname{cosec}^n x dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + A \int \operatorname{cosec}^{n-2} x dx$

then A is equal to

- (a) $\frac{1}{n-2}$ (b) $\frac{n}{n-2}$
(c) $\frac{n-1}{n-2}$ (d) $\frac{n-2}{n-1}$

Subjective Type

63. Evaluate $\int \frac{(x+1)}{x(1+xe^x)^2} dx$

64. Integrate $\int \frac{x^3 + 3x + 2}{(x^2 + 1)^2 (x+1)} dx$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

- Let $f(x) = \int e^x (x-1)(x-2) dx$, then f decrease in the interval (2000)
 - $(-\infty, -2)$
 - $(-2, -1)$
 - $(1, 2)$
 - $(2, \infty)$
- $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$ is equal to (2006)
 - $\frac{\sqrt{2x^4-2x^2+1}}{x^2} + c$
 - $\frac{\sqrt{2x^4-2x^2+1}}{x^3} + c$
 - $\frac{\sqrt{2x^4-2x^2+1}}{x} + c$
 - $\frac{\sqrt{2x^4-2x^2+1}}{2x^2} + c$
- Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals (2007)
 - $\frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + c$
 - $\frac{1}{n-1} (1+nx^n)^{1-\frac{1}{n}} + c$
 - $\frac{1}{n(n+1)} (1+nx^n)^{1+\frac{1}{n}} + c$
 - $\frac{1}{n+1} (1+nx^n)^{1+\frac{1}{n}} + c$
- Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant c , the value of $J-I$ equals (2008)
 - $\frac{1}{2} \log \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^x + 1} \right) + c$
 - $\frac{1}{2} \log \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + c$
 - $\frac{1}{2} \log \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$
 - $\frac{1}{2} \log \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + c$
- The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals to (for some arbitrary constant C) (2012)
 - $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$
 - $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$
 - $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$
 - $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + C$



Objective Questions II [One or more than one correct option]

6. Let b be a nonzero real number. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$ for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE? (2020)
- (a) If $b > 0$, then f is an increasing function
 - (b) If $b < 0$, then f is a decreasing function
 - (c) $f(x) \cdot f(-x) = 1$ for all $x \in \mathbb{R}$
 - (d) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

ASSERTION REASON

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
- (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
- (C) If ASSERTION is true, REASON is false.
- (D) If ASSERTION is false, REASON is true.

7. Let $F(x)$ be an indefinite integral of $\sin^2 x$.

Assertion : The function $F(x)$ satisfies $F(x + \pi) = F(x)$ for all real x .

Reason : $\sin^2(x + \pi) = \sin^2 x$ for all real x . (2007)

Subjective Type

8. Evaluate $\int \sin^{-1} \left(\frac{2x+2}{\sqrt{4x^2+8x+13}} \right) dx$ (2000)
9. For any natural number m , evaluate $\int (x^{3m} + x^{2m} + x^m) (2x^{2m} + 3x^m + 6)^{1/m} dx, x > 0$ (2002)



INDEFINITE INTEGRATION



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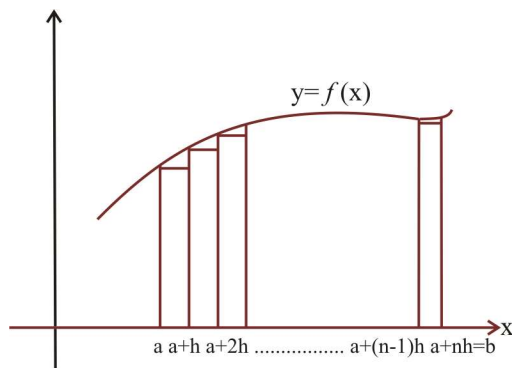
02

DEFINITE INTEGRATION, AREA UNDER CURVES

DEFINITE INTEGRATION, AREA UNDER CURVES

1. DEFINITION

Let $f(x)$ be a continuous real valued function defined on the closed interval $[a, b]$ which is divided into n parts as shown in figure.



The point of division on x-axis are

$a, a+h, a+2h, \dots, a+(n-1)h, a+nh$, where $\frac{b-a}{n} = h$.

Let S_n denotes the area of these n rectangles.

Then, $S_n = hf(a) + hf(a+h) + hf(a+2h) + \dots + hf(a+(n-1)h)$

Clearly, S_n is area very close to the area of the region bounded by curve $y=f(x)$, x-axis and the ordinates $x=a, x=b$.

$$\text{Hence } \int_a^b f(x) dx = \lim_{n \rightarrow \infty} S_n$$

$$\begin{aligned} \int_a^b f(x) dx &= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} hf(a+rh) \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{b-a}{n} \right) f \left(a + \frac{(b-a)r}{n} \right) \end{aligned}$$

NOTES :

1. We can also write

$$S_n = hf(a+h) + hf(a+2h) + \dots + hf(a+nh) \text{ and}$$

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{b-a}{n} \right) f \left(a + \left(\frac{b-a}{n} \right) r \right)$$

$$2. \text{ If } a=0, b=1, \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{n} f \left(\frac{r}{n} \right)$$

2. DEFINITE INTEGRAL AS LIMIT OF SUM/SUM OF INFINITE SERIES USING DEFINITE INTEGRAL

Step 1. Replace $\frac{r}{n}$ by x , $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by $\int$

Step 2. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right)$ by putting least and greatest values of r as lower and upper limits respectively.

$$\text{For example } \lim_{n \rightarrow \infty} \sum_{r=1}^{pn} \frac{1}{n} f \left(\frac{r}{n} \right) = \int_0^p f(x) dx$$

$$\left[\lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \right]_{r=1} = 0, \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) \Big|_{r=np} = p$$

3. SECOND FUNDAMENTAL THEOREM OF CALCULUS

Let $F(x)$ be any antiderivative of $f(x)$, then for any two values of the independent variable x , say a and b , the difference $F(b) - F(a)$ is called the definite integral of $f(x)$ from a to b and is

denoted by $\int_a^b f(x) dx$. Thus $\int_a^b f(x) dx = F(b) - F(a)$,



The numbers a and b are called the limits of integration; a is the lower limit and b is the upper limit. Usually $F(b) - F(a)$ is abbreviated by writing $F(x) \Big|_a^b$.

4. GEOMETRICAL INTERPRETATION OF THE DEFINITE INTEGRAL

4.1

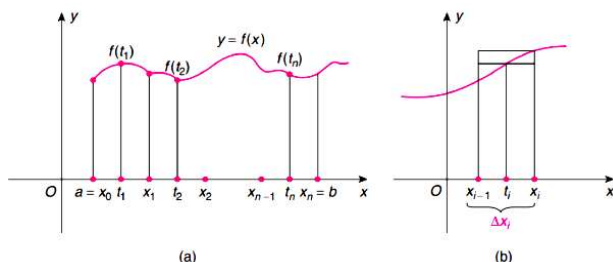


Figure 4.1

Let $f: [a, b] \rightarrow [0, \infty)$ be a function, $P = \{x_0, x_1, x_2, \dots, x_n\}$ be a partition of $[a, b]$ and $t_i \in [x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$. Then (see fig. 4.1) $f(t_i) \Delta x_i = f(t_i)(x_i - x_{i-1}) = \text{Area of the rectangle with width } \Delta x_i \text{ and height } f(t_i)$

Hence

$$S(f, P) = \sum_{i=1}^n f(t_i) \Delta x_i$$

= Sum of the areas of the rectangles with width $\Delta x_i = x_i - x_{i-1}$ and height $f(t_i)$

Thus, the area A enclosed by the x -axis, the lines $x = a$, $x = b$ and the curve $y = f(x)$ is approximately equal to $S(f, P)$. When the width of the rectangles becomes smaller, that is when $\text{Max} \{\Delta x_1, \Delta x_2, \dots, \Delta x_n\}$ is small, the sum of the areas or $S(f, P)$ is very nearly equal to A . If f is integrable, then

$S(f, P)$ converges to $\int_a^b f(x) dx$ and hence

$$A = \int_a^b f(x) dx$$

Thus, definite integral of a non-negative function f , when integrable, may be interpreted over $[a, b]$ as the area enclosed by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x -axis.

4.2 Geometrical Interpretation of the Definite Integral

If $y = f(x)$ is continuous and $\int_a^b f(x) dx = 0$,

then $f(x) = 0$ has at least one real root in (a, b) .

5. PROPERTIES OF DEFINITE INTEGRALS

- $\int_a^b f(x) dx = - \int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^b f(y) dy$
- $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, where c may or may not lie between a and b .
- $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
- $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

NOTES :

$$1. \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$$

$$2. \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}$$

$$6. \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$= \begin{cases} 0 & \text{if } f(2a-x) = -f(x) \\ 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \end{cases}$$





$$7. \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0 & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

8. If $f(x)$ is a periodic function of period 'a', i.e. $f(a+x) = f(x)$, then

$$(a) \int_0^{na} f(x) dx = n \int_0^a f(x) dx$$

$$(b) \int_a^{a+n} f(x) dx = n \int_a^{a+1} f(x) dx$$

$$(c) \int_{na}^{b+na} f(x) dx = \int_0^b f(x) dx, \text{ where } b \in \mathbb{R}$$

$$(d) \int_b^{b+a} f(x) dx \text{ independent of } b.$$

$$(e) \int_b^{b+na} f(x) dx = n \int_0^a f(x) dx, \text{ where } n \in \mathbb{I}$$

9. If $f(x) \geq 0$ on the interval $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

10. If $f(x) \leq g(x)$ on the interval $[a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

$$11. \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

12. If $f(x)$ is continuous on $[a, b]$, m is the least and M is the greatest value of $f(x)$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

13. For any two functions $f(x)$ and $g(x)$, integrable on the interval $[a, b]$, the Schwarz–Bunyakovsky inequality holds

$$\left| \int_a^b f(x) \cdot g(x) dx \right| \leq \sqrt{\int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx}$$

14. If a function $f(x)$ is continuous on the interval $[a, b]$, then there exists a point $c \in (a, b)$ such that

$$\int_a^b f(x) dx = f(c)(b-a), \text{ where } a < c < b.$$

6. DIFFERENTIATION UNDER INTEGRAL SIGN

NEWTON LEIBNITZ'S THEOREM:

If f is continuous on $[a, b]$ and $g(x)$ & $h(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

$$\frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = \frac{d}{dx} [h(x)] \cdot f[h(x)] - \frac{d}{dx} [g(x)] \cdot f[g(x)]$$

7. REDUCTION FORMULAE IN DEFINITE INTEGRALS

$$7.1 \quad \text{If } I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx, \text{ then show that } I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

$$\text{Proof: } I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n = \left[-\sin^{n-1} x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x \cdot \cos^2 x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cdot (1 - \sin^2 x) dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$





NOTES :

$$1. \int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$2. I_n = \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots I_0 \text{ or } I_1$$

according as n is even or odd. $I_0 = \frac{\pi}{2}$, $I_1 = 1$

$$\text{Hence } I_n = \begin{cases} \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{1}{2}\right) \cdot \frac{\pi}{2} & \text{if } n \text{ is even} \\ \left(\frac{n-1}{n}\right) \left(\frac{n-3}{n-2}\right) \left(\frac{n-5}{n-4}\right) \dots \left(\frac{2}{3}\right) \cdot 1 & \text{if } n \text{ is odd} \end{cases}$$

$$7.2 \quad \text{If } I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx, \text{ then show that } I_n + I_{n-2} = \frac{1}{n-1}$$

$$\text{Proof. } I_n = \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \cdot \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} (\sec^2 x - 1) \, dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} (\tan x)^{n-2} \, dx$$

$$= \left[\frac{(\tan x)^{n-1}}{n-1} \right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$7.3 \quad \text{If } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cdot \cos^n x \, dx, \text{ then show that}$$

$$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}$$

$$\text{Proof. } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^{m-1} x (\sin x \cos^n x) \, dx$$

$$= \left[-\frac{\sin^{m-1} x \cdot \cos^{n+1} x}{n+1} \right]_0^{\frac{\pi}{2}} +$$

$$\int_0^{\frac{\pi}{2}} \frac{\cos^{n+1} x}{n+1} (m-1) \sin^{m-2} x \cos x \, dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} \sin^{m-2} x \cdot \cos^n x \cdot \cos^2 x \, dx$$

$$= \left(\frac{m-1}{n+1} \right) \int_0^{\frac{\pi}{2}} (\sin^{m-2} x \cdot \cos^n x - \sin^m x \cdot \cos^n x) \, dx$$

$$= \left(\frac{m-1}{n+1} \right) I_{m-2,n} - \left(\frac{m-1}{n+1} \right) I_{m,n}$$

$$\Rightarrow \left(1 + \frac{m-1}{n+1} \right) I_{m,n} = \left(\frac{m-1}{n+1} \right) I_{m-2,n}$$

$$I_{m,n} = \left(\frac{m-1}{m+n} \right) I_{m-2,n}$$





NOTES :

1. $I_{m,n} = \left(\frac{m-1}{m+n} \right) \left(\frac{m-3}{m+n-2} \right) \left(\frac{m-5}{m+n-4} \right) \dots \dots \dots I_{0,n}$ or $I_{1,n}$
according as m is even or odd.

$$I_{0,n} = \int_0^{\frac{\pi}{2}} \cos^n x \, dx \text{ and } I_{1,n} = \int_0^{\frac{\pi}{2}} \sin x \cdot \cos^n x \, dx = \frac{1}{n+1}$$

2. Walli's Formula

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots \pi}{(m+n)(m+n-2)(m+n-4) \dots \dots \dots 2} & \text{when both } m, n \text{ are even} \\ \frac{(m-1)(m-3)(m-5) \dots (n-1)(n-3)(n-5) \dots \dots \dots}{(m+n)(m+n-2)(m+n-4) \dots \dots \dots} & \text{otherwise} \end{cases}$$

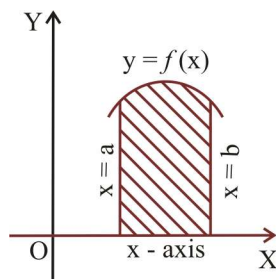
8. AREA OF PLANE REGIONS

1. The area bounded by the curve $y = f(x)$, x -axis and the ordinates $x = a$ and $x = b$ (where $b > a$) is given by

$$A = \int_a^b |y| \, dx = \int_a^b |f(x)| \, dx$$

- (i) If $f(x) > 0 \, \forall x \in [a, b]$

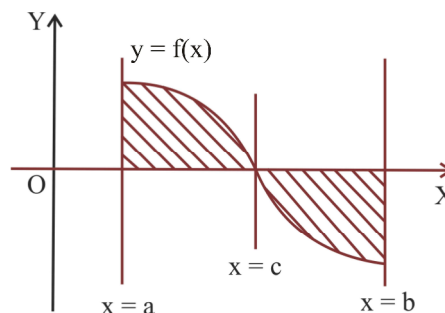
$$\text{Then } A = \int_a^b f(x) \, dx$$



- (ii) If $f(x) > 0 \, \forall x \in [a, c]$ &
 $< 0 \, \forall x \in (c, b]$ Then

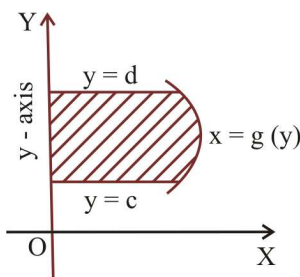
$$A = \left| \int_a^c y \, dx \right| + \left| \int_c^b y \, dx \right| = \int_a^c f(x) \, dx - \int_c^b f(x) \, dx$$

where c is a point in between a and b .



2. The area bounded by the curve $x = g(y)$, y -axis and the abscissae $y = c$ and $y = d$ (where $d > c$) is given by

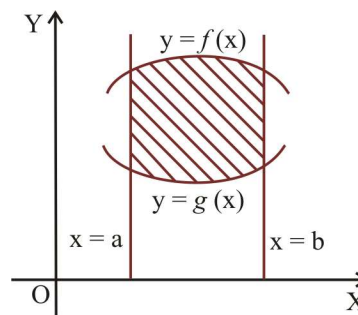
$$A = \int_c^d |x| \, dy = \int_c^d |g(y)| \, dy$$



3. If we have two curve $y = f(x)$ and $y = g(x)$, such that $y = f(x)$ lies above the curve $y = g(x)$ then the area bounded between them and the ordinates $x = a$ and $x = b$ ($b > a$), is given by

$$A = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx$$

i.e. upper curve area – lower curve area.

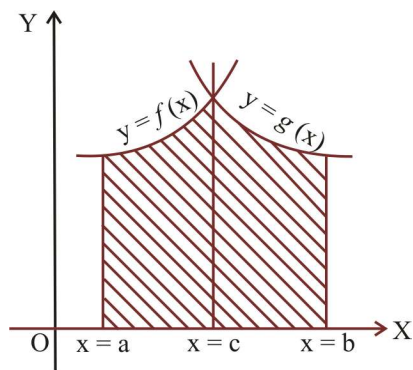




4. The area bounded by the curves $y = f(x)$ and $y = g(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$A = \int_a^c f(x) dx + \int_c^b g(x) dx,$$

where $x = c$ is the point of intersection of the two curves.



9. CURVE TRACING

In order to find the area bounded by several curves, it is important to have rough sketch of the required portion. The following steps are very useful in tracing a cartesian curve, $f(x, y) = 0$.

Step 1 : Symmetry

- The curve is symmetrical about x-axis if all powers of y in the equation of the given curve are even.
- The curve is symmetrical about y-axis if all powers of x in the equation of the given curve are even.
- The curve is symmetrical about the line $y = x$, if the equation of the given curve remains unchanged on interchanging x and y.
- The curve is symmetrical in opposite quadrants, if the equation of the given curve remains unchanged when x and y are replaced by $-x$ and $-y$ respectively.

Step 2 : Origin

If there is no constant term in the equation of the algebraic curve, then the curve passes through the origin.

In that case, the tangents at the origin are given by equating to zero the lowest degree terms in the equation of the given algebraic curve.

For example, the curve $y^3 = x^3 + axy$ passes through the origin and the tangents at the origin are given by $axy = 0$ i.e. $x = 0$ and $y = 0$.

Step 3 : Intersection with the Co-ordinate Axes

- To find the points of intersection of the curve with X-axis, put $y = 0$ in the equation of the given curve and get the corresponding values of x.
- To find the points of intersection of the curve with Y-axis, put $x = 0$ in the equation of the given curve and get the corresponding values of y.

Step 4 : Asymptotes

Find out the asymptotes of the curve.

- The vertical asymptotes or the asymptotes parallel to y-axis of the given algebraic curve are obtained by equating to zero the coefficient of the highest power of y in the equation of the given curve.
- The horizontal asymptotes or the asymptotes parallel to x-axis of the given algebraic curve are obtained by equating to zero the coefficient of the highest power of x in the equation of the given curve.

Step 5 : Region

Find out the regions of the plane in which no part of the curve lies. To determine such regions we solve the given equation for y in terms of x or vice-versa. Suppose that y becomes imaginary for $x > a$, the curve does not lie in the region $x > a$.

Step 6 : Critical Points

Find out the values of x at which $\frac{dy}{dx} = 0$.

At such points y generally changes its character from an increasing function of x to a decreasing function of x or vice-versa.

Step 7: Trace the curve with the help of the above points.





SOLVED EXAMPLES

Example – 1

Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then the value of the integral

$$\int_0^1 f(x)g(x) dx \text{ is}$$

$$(a) e + \frac{e^2}{2} - \frac{2}{2} \quad (b) e - \frac{e^2}{2} - \frac{3}{2}$$

$$(c) e + \frac{e^2}{2} + \frac{5}{2} \quad (d) e - \frac{e^2}{2} - \frac{5}{2}$$

Ans. (b)

Sol. As $f(x) = f'(x)$ and $f(0) = 1$

$$\Rightarrow \frac{f'(x)}{f(x)} = 1$$

$$\Rightarrow \log(f(x)) = x + c \Rightarrow f(x) = e^x + k$$

$$\Rightarrow f(x) = e^x \text{ as } f(0) = 1$$

$$\text{Now } g(x) = x^2 - e^x$$

$$\therefore \int_0^1 f(x)g(x)dx = \int_0^1 e^x (x^2 - e^x) dx$$

$$= \int_0^1 x^2 e^x dx - \int_0^1 e^{2x} dx$$

$$= \left[(x^2 - 2x + 2)e^x \right]_0^1 - \left(\frac{e^{2x}}{2} \right)_0^1$$

$$= (e - 2) - \left(\frac{e^2 - 1}{2} \right) = e - \frac{e^2}{2} - \frac{3}{2}$$

Using $f^n(x)e^x dx = e^x [f^n(x) - f_1^n(x) + f_2^n(x) - \dots + (-1)^n f_n(x)]$

Where $f_1, f_2, \dots, f_n$ are derivatives of first, second $\dots n^{\text{th}}$ order.

Example – 2

Evaluate the following integrals :

$$(i) \int_2^3 x^2 dx \quad (ii) \int_1^3 \frac{x}{(x+1)(x+2)} dx$$

$$\text{Sol. (i) } \int_2^3 x^2 dx = \left[\frac{x^3}{3} \right]_2^3$$

$$= \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$$

$$(ii) \frac{x}{(x+1)(x+2)} = \frac{-1}{x+1} + \frac{2}{x+2}$$

$$\therefore \int_1^3 \frac{x}{(x+1)(x+2)} dx = [-\log|x+1| + 2\log|x+2|]_1^3$$

$$= [-\log|4| + 2\log|5|] - [-\log|2| + 2\log|3|]$$

$$= [-\log 4 + 2\log 5] - [-\log 2 + 2\log 3]$$

$$= -2\log 2 + 2\log 5 + \log 2 - 2\log 3$$

$$= -\log 2 + \log 25 - \log 9 = \log 25 - \log 18$$

$$= \log \frac{25}{18}$$

Example – 3

$$\text{Evaluate : } \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} dx$$

$$\text{Sol. } I = \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} dx$$

$$= \int_0^{\pi/4} \sec x \cdot \sqrt{\frac{1-\sin x}{1+\sin x}} \cdot \sqrt{\frac{1-\sin x}{1-\sin x}} dx$$

$$= \int_0^{\pi/4} \sec x \frac{1-\sin x}{\sqrt{1-\sin^2 x}} dx$$



$$= \int_0^{\pi/4} \sec x \frac{1 - \sin x}{\cos x} dx$$

$$= \int_0^{\pi/4} (\sec^2 x - \sec x \tan x) dx$$

$$= \int_0^{\pi/4} \sec^2 x dx - \int_0^{\pi/4} \sec x \tan x dx$$

$$= [\tan x]_0^{\pi/4} - [\sec x]_0^{\pi/4}$$

$$= \left(\tan \frac{\pi}{4} - \tan 0 \right) - \left(\sec \frac{\pi}{4} - \sec 0 \right)$$

$$= (1 - 0) - (\sqrt{2} - 1) = 2 - \sqrt{2}.$$

Example - 4

Evaluate : $\int_{-1}^1 5x^4 \sqrt{x^5 + 1} dx.$

Sol. Let $I = \int_{-1}^1 5x^4 \sqrt{x^5 + 1} dx$

Put $x^5 = t$ so that $5x^4 dx = dt.$

When $x = -1, t = -1.$ When $x = 1, t = 1.$

$$\therefore I = \int_{-1}^1 \sqrt{t+1} dt$$

$$= \left[\frac{(t+1)^{3/2}}{3/2} \right]_{-1}^1 = \frac{2}{3} [(t+1)^{3/2}]_{-1}^1$$

$$= \frac{2}{3} [2^{3/2} - 0] = \frac{4\sqrt{2}}{3}.$$

Example - 5

Prove that $\int_0^{\pi/2} \sqrt{\sin \phi} \cos^5 \phi d\phi = \frac{64}{231}.$

Sol. $I = \int_0^{\pi/2} \sqrt{\sin \phi} \cos^5 \phi d\phi$

$$= \int_0^{\pi/2} \sqrt{\sin \phi} \cos^4 \phi \cos \phi d\phi$$

$$= \int_0^{\pi/2} \sqrt{\sin \phi} (1 - \sin^2 \phi)^2 \cos \phi d\phi$$

Put $\sin \phi = t$ so that $\cos \phi d\phi = dt.$

When $\phi = 0, \sin 0 = t \Rightarrow t = 0.$

When $\phi = \frac{\pi}{2}, \sin \frac{\pi}{2} = t \Rightarrow t = 1$

$$\therefore I = \int_0^1 \sqrt{t} (1 - t^2)^2 dt = \int_0^1 \sqrt{t} (1 - 2t^2 + t^4) dt$$

$$= \int_0^1 (t^{1/2} - 2t^{5/2} + t^{9/2}) dt$$

$$= \left[\frac{t^{3/2}}{3/2} - 2 \frac{t^{7/2}}{7/2} + \frac{t^{11/2}}{11/2} \right]_0^1$$

$$= \left[\frac{2}{3} t^{3/2} - \frac{4}{7} t^{7/2} + \frac{2}{11} t^{11/2} \right]_0^1$$

$$= \left[\frac{2}{3} (1) - \frac{4}{7} (1) + \frac{2}{11} (1) \right] - [0 - 0 + 0]$$

$$= \frac{2}{3} - \frac{4}{7} + \frac{2}{11}$$

$$= \frac{154 - 132 + 42}{231} = \frac{64}{231}.$$



Example-6

Evaluate : $\int_1^2 \left(\frac{x-1}{x^2} \right) e^x dx$

Or

$$\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Sol. $\int \left(\frac{x-1}{x^2} \right) e^x dx = \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

$$= \int \frac{1}{x} \cdot e^x dx - \int \frac{1}{x^2} \cdot e^x dx$$

$$= \frac{1}{x} \cdot e^x - \int \left(-\frac{1}{x^2} \right) e^x dx - \int \frac{1}{x^2} \cdot e^x dx$$

[Integrating first integral by parts]

$$= \frac{1}{x} \cdot e^x = F(x)$$

$$\therefore \int_1^2 \left(\frac{x-1}{x^2} \right) e^x dx = \left[\frac{e^x}{x} \right]_1^2$$

$$= \frac{1}{2} \cdot e^2 - \frac{1}{1} \cdot e^1 = \frac{1}{2} e^2 - e$$

Example-7

Evaluate : $\int_a^b x^2 dx$ using limit of a sum formula.

Sol. Let $I = \int_a^b x^2 dx = \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} h [(a+h)^2 + (a+2h)^2 + \dots + (a+nh)^2]$

$$= \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} h \left[(a^2 + a^2 + \dots) + (h^2 + 4h^2 + \dots + n^2 h^2) \right. \\ \left. + (2ah + 4ah + \dots + 2anh) \right]$$

$$\Rightarrow I = \lim_{\substack{n \rightarrow \infty \\ h \rightarrow 0}} \left[nha^2 + \frac{2ah^2 n(n+1)}{2} + \frac{h^3 n(n+1)(2n+1)}{6} \right]$$

Using $nh = b - a$, we get :

$$\Rightarrow I = \lim_{n \rightarrow \infty} \left[a^2(b-a) + a(b-a)^2 \left(1 + \frac{1}{n} \right) \right. \\ \left. + (b-a)^3 \frac{1}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right]$$

$$\Rightarrow I = a^2(b-a) + a(b-a)^2 + \frac{(b-a)^3}{6} (2)$$

$$\Rightarrow I = (b-a) \left[a^2 + ab - a^2 + \frac{b^2 + a^2 - 2ab}{3} \right]$$

$$\Rightarrow I = \frac{(b-a)}{3} [a^2 + b^2 + ab] = \frac{b^3 - a^3}{3}$$

Example-8

Evaluate : $\int_0^1 |3x-1| dx$

Sol. We know that $|3x-1| = 3x-1$,

when $x \geq \frac{1}{3}$ i.e. when $3x-1 \geq 0$

and $|3x-1| = -(3x-1)$

when $x < \frac{1}{3}$ i.e. when $3x-1 < 0$.

$$\therefore \int_0^1 |3x-1| dx = \int_0^{1/3} |3x-1| dx + \int_{1/3}^1 |3x-1| dx$$

$$= \int_0^{1/3} -(3x-1) dx + \int_{1/3}^1 (3x-1) dx$$

$$= \left[-\left(\frac{3}{2}x^2 - x \right) \right]_0^{1/3} + \left[\frac{3}{2}x^2 - x \right]_{1/3}^1$$

$$= -\left(\frac{3}{2} \cdot \frac{1}{9} - \frac{1}{3} \right) + \left[\left(\frac{3}{2}(1) - 1 \right) - \left(\frac{3}{2} \cdot \frac{1}{9} - \frac{1}{3} \right) \right]$$

$$= -\left[\frac{1}{6} - \frac{1}{3} \right] + \left[\frac{3}{2} - 1 \right] - \left[\frac{1}{6} - \frac{1}{3} \right]$$

$$= -2 \left(\frac{1}{6} - \frac{1}{3} \right) + \frac{1}{2} = -2 \left(-\frac{1}{6} \right) + \frac{1}{2} = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$



Example – 9

Evaluate : $\int_{0.5}^{3.5} [x] dx$ where $[.]$ is GIF

Sol. Here $f(x) = [x] = \begin{cases} 0, 0.5 \leq x < 1 \\ 1, 1 \leq x < 2 \\ 2, 2 \leq x < 3 \\ 3, 3 \leq x < 3.5 \end{cases}$

$$\begin{aligned} \therefore \int_{0.5}^{3.5} [x] dx &= \int_{0.5}^1 [x] dx + \int_1^2 [x] dx + \int_2^3 [x] dx + \int_3^{3.5} [x] dx \\ &= \int_{0.5}^1 0 dx + \int_1^2 1 dx + \int_2^3 2 dx + \int_3^{3.5} 3 dx \\ &= 0 + [x]_1^2 + 2[x]_2^3 + 3[x]_3^{3.5} \\ &= (2-1) + 2(3-2) + 3(3.5-3) \\ &= 1 + 2 + 1.5 = 4.5. \end{aligned}$$

Example – 10

Evaluate : $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

Sol. Let : $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \dots (i)$

Using property – 4, we have :

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin(\pi/2 - x)}}{\sqrt{\sin(\pi/2 - x)} + \sqrt{\cos(\pi/2 - x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \dots (ii)$$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} dx = \frac{\pi}{2} \quad \Rightarrow I = \frac{\pi}{4}$$

Example – 11

Prove that $\int_0^{\pi/2} \sin 2x \log \tan x dx = 0$.

Sol. Let $I = \int_0^{\pi/2} \sin 2x \log \tan x dx \quad \dots (1)$

$$\therefore I = \int_0^{\pi/2} \sin 2 \left(\frac{\pi}{2} - x \right) \log \tan \left(\frac{\pi}{2} - x \right) dx$$

$$= \int_0^{\pi/2} \sin (\pi - 2x) \log \tan \left(\frac{\pi}{2} - x \right) dx$$

$$= \int_0^{\pi/2} \sin 2x \log \cot x dx \quad \dots (2)$$

Adding (1) and (2) :

$$2I = \int_0^{\pi/2} \sin 2x [\log \tan x + \log \cot x] dx$$

$$= \int_0^{\pi/2} \sin 2x \log (\tan x \cot x) dx$$

$$= \int_0^{\pi/2} \sin 2x \log 1 dx = 0$$

$$[\because \log 1 = 0]$$

Hence $I = 0$.



Example – 12

If $f(a+b-x)=f(x)$, then $\int_a^b x f(x) dx$ is equal to

(a) $\frac{a+b}{2} \int_a^b f(x) dx$

(b) $\frac{b-a}{2} \int_a^b f(x) dx$

(c) $\frac{a+b}{2} \int_a^b f(a+b-x) dx$

(d) $\frac{a+b}{2} \int_a^b f(b-x) dx$

Ans. (a,c)

Sol. Let $I = \int_a^b x f(x) dx$

$$I = \int_a^b (a+b-x) f(a+b-x) dx$$

$$I = \int_a^b (a+b) f(a+b-x) dx - \int_a^b x f(a+b-x) dx$$

$$I = \int_a^b (a+b) f(x) dx - \int_a^b x f(x) dx$$

$$\therefore I = \frac{a+b}{2} \int_a^b f(x) dx = \frac{a+b}{2} \int_a^b f(a+b-x) dx$$

Example – 13

Evaluate : $\int_{-2}^2 f(x) dx$;

$$\text{where } f(x) = \begin{cases} 2x-1, & -2 \leq x < 1 \\ 3x-2, & 1 \leq x < 2 \end{cases}$$

Sol. $\int_{-2}^2 f(x) dx = \int_{-2}^1 f(x) dx + \int_1^2 f(x) dx$

$$= \int_{-2}^1 (2x-1) dx + \int_1^2 (3x-2) dx$$

$$= \left[2 \cdot \frac{x^2}{2} - x \right]_{-2}^1 + \left[\frac{3x^2}{2} - 2x \right]_1^2$$

$$= [x^2 - x]_{-2}^1 + \left[\frac{3}{2}x^2 - 2x \right]_1^2$$

$$= [(1-1) - (4+2)] + \left[\left(\frac{3}{2}(4) - 2(2) \right) - \left(\frac{3}{2}(1) - 2(1) \right) \right]$$

$$= (0-6) + \left(2 + \frac{1}{2} \right) = -6 + \frac{5}{2} = -\frac{7}{2}$$

Example – 14

Evaluate : $\int_0^{\pi/2} \sqrt{1-\sin 2x} dx$.

Sol. Let $I = \int_0^{\pi/2} \sqrt{1-\sin 2x} dx$

$$\Rightarrow I = \int_0^{\pi/2} \sqrt{(\sin x - \cos x)^2} dx$$

$$\Rightarrow I = \int_0^{\pi/2} |\sin x - \cos x| dx$$

$$\Rightarrow I = \int_0^{\pi/4} |\sin x - \cos x| dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$\Rightarrow I = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$\Rightarrow I = [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$\Rightarrow I = \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 \right) + (-1) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$\Rightarrow I = 2\sqrt{2} - 2$$



Example – 15

Evaluate : $\int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$.

Sol. Let $I = \int_0^{\pi} \frac{x}{1 + \cos^2 x} dx$... (i)

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x)}{1 + \cos^2(\pi - x)} dx \quad [\text{using property - 4}]$$

... (ii)

Adding (i) and (ii), we get :

$$\Rightarrow 2I = \int_0^{\pi} \frac{\pi}{1 + \cos^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{dx}{1 + \cos^2 x} = \frac{2\pi}{2} \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x}$$

[using property – 6]

Divide N^r and D^r by $\cos^2 x$ to get :

$$\Rightarrow I = \pi \int_0^{\pi/2} \frac{\sec^2 x}{\sec^2 x + 1} dx$$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$[\sec^2 x = 1 + \tan^2 x]$$

For $x = \pi/2$, $t \rightarrow \infty$ and for $x = 0$, $t = 0$

$$\Rightarrow I = \pi \int_0^{\infty} \frac{dt}{2 + t^2}$$

$$\Rightarrow I = \frac{\pi}{\sqrt{2}} \tan^{-1} \frac{t}{\sqrt{2}} \Big|_0^{\infty} = \frac{\pi}{\sqrt{2}} \times \frac{\pi}{2} = \frac{\pi^2}{2\sqrt{2}}$$

Example – 16

Evaluate : $\int_0^{\pi} \frac{x \sin(2x) \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi}$

Sol. Let $I = \int_0^{\pi} \frac{x \sin(2x) \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi}$... (i)

Apply property – 4 to get

$$\Rightarrow I = \int_0^{\pi} \frac{(\pi - x) \sin(2\pi - 2x) \sin\left(\frac{\pi}{2} \cos(\pi - x)\right) dx}{2(\pi - x) - \pi}$$

$$= \int_0^{\pi} - \frac{(\pi - x) \sin 2x \sin\left(\frac{\pi}{2} \cos x\right) dx}{2x - \pi} \quad \dots (ii)$$

Add (i) and (ii) to get

$$2I = \int_0^{\pi} 2 \sin x \cos x \sin\left[\frac{\pi}{2} \cos x\right] dx$$

$$\text{Let } \frac{\pi}{2} \cos x = t \Rightarrow -\frac{\pi}{2} \sin x dx = dt$$

$$\Rightarrow I = \frac{4}{\pi^2} \int_{-\pi/2}^{\pi/2} t \sin t dt = \frac{8}{\pi^2} \int_0^{\pi/2} t \sin t dt$$

$$\Rightarrow I = \frac{8}{\pi^2} \left[(-t \cos t)_0^{\pi/2} + (\sin t)_0^{\pi/2} \right]$$

$$= \frac{8}{\pi^2} [0 + 1] = \frac{8}{\pi^2}$$



Example – 17

Evaluate : $\int_0^{\pi/2} \log \sin x \, dx$.

Sol. Let $I = \int_0^{\pi/2} \log \sin x \, dx$... (i)

$$\Rightarrow I = \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx \quad [\text{using property-4}]$$

$$\Rightarrow I = \int_0^{\pi/2} \log \cos x \, dx \quad \dots (ii)$$

Adding (i) and (ii) we get :

$$2I = \int_0^{\pi/2} \log (\sin x \cos x) \, dx = \int_0^{\pi/2} \log \left(\frac{\sin 2x}{2} \right) dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log \sin 2x \, dx - \int_0^{\pi/2} \log 2 \, dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log \sin 2x \, dx - \frac{\pi}{2} \log 2 \quad \dots (iii)$$

$$\text{Let } I_1 = \int_0^{\pi/2} \log \sin 2x \, dx$$

$$\text{Put } t = 2x \Rightarrow dt = 2dx$$

$$\text{For } x = \frac{\pi}{2}, t = \pi \text{ and for } x = 0, t = 0$$

$$\Rightarrow I_1 = \frac{1}{2} \int_0^{\pi} \log \sin t \, dt = \frac{2}{2} \int_0^{\pi/2} \log \sin t \, dt$$

$$\Rightarrow I_1 = \int_0^{\pi/2} \log \sin x \, dx \quad [\text{using property-2}]$$

$$\Rightarrow I_1 = I$$

Substituting in (iii) we get :

$$2I = I - \pi/2 \log 2$$

$$\Rightarrow I = -\pi/2 \log 2$$

[learn this result so that you can directly apply it in other difficult problem]

Example – 18

Show that $\int_0^{n\pi+v} |\sin x| \, dx = 2n + 1 - \cos v$, where n is a +ve

integer and $0 \leq v \leq \pi$.

Sol. Let $I = \int_0^{n\pi+v} |\sin x| \, dx = \int_0^{n\pi} |\sin x| \, dx + \int_{n\pi}^{n\pi+v} |\sin x| \, dx$

[using property – 3]

$$\Rightarrow I = I_1 + I_2 \quad \dots (i)$$

Consider I_1 :

$$I_1 = \int_0^{n\pi} |\sin x| \, dx = n \int_0^{\pi} |\sin x| \, dx$$

[using property and period of $|\sin x|$ is π]

$$\Rightarrow I_1 = n \int_0^{\pi} \sin x \, dx$$

[As $\sin x \geq 0$ in $[0, \pi]$, $|\sin x| = \sin x$]

$$\Rightarrow I_1 = -n \cos x \Big|_0^{\pi} = -n [-1 - 1] = 2n$$

$$\text{Consider } I_2 : I_2 = \int_{n\pi}^{n\pi+v} |\sin x| \, dx = \int_0^v |\sin x| \, dx$$

[as period of $|\sin x| = \pi$]

$$\Rightarrow I_2 = \int_0^v \sin x \, dx$$

[as for $0 \leq x \leq \pi$, $\sin x$ is positive]

$$= -\cos x \Big|_0^v = 1 - \cos v$$

On substituting the values of I_1 and I_2 in (i), we get

$$I = 2n + (1 - \cos v) = 2n + 1 - \cos v.$$

Hence proved.



Example – 19

Evaluate $\frac{d}{dx} \left(\int_{1/x}^{\sqrt{x}} \cos t^2 dt \right)$

Sol. Let, $f(x) = \int_{1/x}^{\sqrt{x}} \cos t^2 dt$

$$\therefore \frac{d}{dx} (f(x)) = \cos(\sqrt{x})^2 \cdot \left\{ \frac{d}{dx} (\sqrt{x}) \right\} - \cos\left(\frac{1}{x}\right)^2 \left\{ \frac{d}{dx} \left(\frac{1}{x}\right) \right\}$$

$$= \frac{1}{2\sqrt{x}} \cos x + \frac{1}{x^2} \cdot \cos\left(\frac{1}{x^2}\right) \text{ (Using Leibnitz Rule)}$$

$$\Rightarrow \frac{d}{dx} \left(\int_{1/x}^{\sqrt{x}} \cos t^2 dt \right) = \frac{1}{2\sqrt{x}} \cos x + \frac{1}{x^2} \cos\left(\frac{1}{x^2}\right).$$

Example – 20

Find the points of local minimum and local maximum of the

$$\text{function } \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt.$$

Sol. Let $y = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt = \int_0^{(t-1)(t-4)} \frac{(t-1)(t-4)}{2 + e^t} dt$

For the points of Extremes,

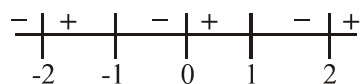
$$\frac{dy}{dx} = 0$$

$$\left[\frac{(x^2-1)(x^2-4)}{2+e^{x^2}} \right] (2x) = 0 \quad [\text{using property under point 3}]$$

$$\Rightarrow x = 0 \text{ or } x^4 - 5x^2 + 4 = 0$$

$$\Rightarrow x = 0 \text{ or } (x-1)(x+1)(x-2)(x+2) = 0$$

$$\Rightarrow x = 0, x = \pm 1 \text{ and } x = \pm 2$$



$f'(x)$ changes sign from $(-)$ to $(+)$ at $x = -2, 0, 2$ where as $f'(x)$ changes sign from $(+)$ to $(-)$ at $x = -1, 1$

$\Rightarrow x = -2, 0, 2$ are points of local minimum and $x = -1, 1$ are points of local maximum.

Example – 21

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function having

$f(2) = 6, f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ equals

- (a) 36 (b) 24
(c) 18 (d) 12

Ans. (c)

Sol. $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$ (0/0) form,

$$= \lim_{x \rightarrow 2} \frac{f'(x) \times 4(f(x))^3}{1}$$

$$= 4f'(2) \times (f(2))^3 = \frac{1}{48} \times 4 \times 6 \times 6 \times 6 = 18$$

Example – 22

If for a continuous function

$\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$, for all $t \geq -\pi$, then $f\left(-\frac{\pi}{3}\right)$ is equal

to:

- (a) π (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

Ans. (a)

Sol. $\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$

Applying Newton-Leibnitz theorem,

$$f(t) + t = -2t$$

$$f\left(-\frac{\pi}{3}\right) - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\Rightarrow f\left(-\frac{\pi}{3}\right) = \pi$$



Example – 23

Find the sum of the series :

$$\lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{6n}.$$

Sol. Let $S = \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+5n}$

Take $1/n$ common from the series i.e.

$$S = \lim_{n \rightarrow \infty} \frac{1}{n} \left[1 + \frac{1}{1+\frac{1}{n}} + \frac{1}{1+\frac{2}{n}} + \dots + \frac{1}{1+\frac{5n}{n}} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{5n} \frac{1}{1+r/n}$$

For the definite integral,

$$\text{Lower limit} = a = \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\text{Upper limit} = b = \lim_{n \rightarrow \infty} \left(\frac{r}{n} \right) = \lim_{n \rightarrow \infty} \frac{5n}{n} = 5$$

Therefore,

$$S = \frac{1}{n} \lim_{n \rightarrow \infty} \sum_{r=0}^{5n} \frac{1}{1+(r/n)}$$

$$= \int_0^5 \frac{dx}{1+x} = [\ln |1+x|]_0^5 = \ln 6 - \ln 1 = \ln 6$$

Example – 24

If $S_n = \frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \frac{1}{3+\sqrt{3n}} + \dots + \frac{1}{n+\sqrt{n^2}}$, then find

$$\lim_{n \rightarrow \infty} S_n.$$

(a) $\log 2$

(b) $2 \log 2$

(c) $3 \log 2$

(d) $4 \log 2$

Ans. (b)

Sol. $S_n = \frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \frac{1}{3+\sqrt{3n}} + \dots + \frac{1}{n+\sqrt{n^2}}$

$$S_n = \sum_{r=1}^n \frac{1}{n} \left(\frac{1}{\frac{r}{n} + \sqrt{\frac{r}{n}}} \right)$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \int_0^1 \frac{1}{x + \sqrt{x}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x}(\sqrt{x}+1)} dx$$

$$= 2 \ln (\sqrt{x}+1) \Big|_0^1$$

$$= 2 \ln 2$$

Example – 25

If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$,

$I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$

then

(a) $I_1 > I_2$

(b) $I_2 > I_1$

(c) $I_3 > I_4$

(d) $I_3 = I_4$

Ans. (a)

Sol. For $0 < x < 1$,

$$x^2 > x^3$$

$$\therefore 2^{x^2} > 2^{x^3}$$

$$\therefore \int_0^1 2^{x^2} dx > \int_0^1 2^{x^3} dx$$

$$\therefore I_1 > I_2$$

and for $1 < x < 2$, $x^3 > x^2 \therefore 2^{x^3} > 2^{x^2}$

$$\text{i.e. } \int 2^{x^2} < \int 2^{x^3} \Rightarrow I_3 < I_4$$



Example – 26

$I_n = \int_0^{\pi/4} \tan^n x \, dx$, then $\lim_{n \rightarrow \infty} n [I_n + I_{n-2}]$ equals

- (a) $1/2$ (b) 1
(c) ∞ (d) 0

Ans. (b)

Sol. $I_n = \int_0^{\pi/4} \tan^n x \, dx$

$$I_{n-2} = \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$\therefore I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x \, dx + \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x - 1) \, dx + \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx = \int_0^1 t^{n-2} \, dt$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$\therefore n(I_n + I_{n-2}) = \frac{1}{1 - \frac{1}{n}}$$

$$\therefore \lim_{n \rightarrow \infty} n(I_n + I_{n-2}) = 1$$

Example – 27

If for $n \geq 1$, $P_n = \int_1^e (\log x)^n \, dx$, then $P_{10} - 90P_8$ is equal to:

- (a) -9 (b) $10e$
(c) $-9e$ (d) 10

Ans. (c)

Sol. $P_n = \int_1^e \underbrace{(\log x)^n}_I \cdot \underbrace{\frac{1}{x}}_{II} \cdot dx$

$$= x(\log x)^n \Big|_1^e - \int_1^e \underbrace{n(\log x)^{n-1}}_I \cdot \underbrace{\frac{1}{x}}_{II} \cdot dx$$

$$= e - n \left[x(\log x)^{n-1} \Big|_1^e - \int_1^e (n-1)(\log x)^{n-2} \, dx \right]$$

$$P_n = e - n[e - (n-1)P_{n-2}]$$

Put $n = 10$,

$$P_{10} = e - 10e + 90P_8$$

$$\Rightarrow P_{10} - 90P_8 = -9e$$

Example – 28

Find the area bounded by the curve

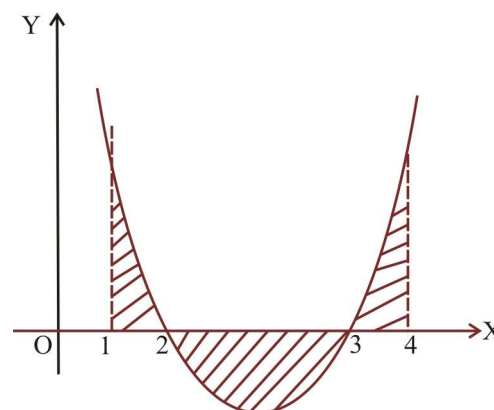
$$y = x^2 - 5x + 6, \text{ X-axis and the lines } x = 1 \text{ and } 4.$$

Sol. For $y = 0$, we get $x^2 - 5x + 6 = 0$

$$\Rightarrow x = 2, 3$$

Hence the curve crosses X-axis at $x = 2, 3$ in the interval $[1, 4]$.

$$\text{Bounded Area} = \left| \int_1^2 y \, dx \right| + \left| \int_2^3 y \, dx \right| + \left| \int_3^4 y \, dx \right|$$



$$\Rightarrow A = \left| \int_1^2 (x^2 - 5x + 6) \, dx \right| + \left| \int_2^3 (x^2 - 5x + 6) \, dx \right|$$

$$+ \left| \int_3^4 (x^2 - 5x + 6) \, dx \right|$$

$$A_1 = \left[\frac{2^3 - 1^3}{3} \right] - 5 \left[\frac{2^2 - 1^2}{2} \right] + [6(2-1)] = \frac{5}{6}$$

$$A_2 = \frac{3^3 - 2^3}{3} - 5 \left[\frac{3^2 - 2^2}{2} \right] + 6(3-2) = -\frac{1}{6}$$

$$A_3 = \frac{4^3 - 3^3}{3} - 5 \left[\frac{4^2 - 3^2}{2} \right] + 6(4-3) = \frac{5}{6}$$

$$\Rightarrow A = \frac{5}{6} + \left| -\frac{1}{6} \right| + \frac{5}{6} = \frac{11}{6} \text{ sq. units.}$$



Example – 29

The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes. If S_1 , S_2 , S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is

- (a) 1 : 2 : 3 (b) 1 : 2 : 1
(c) 1 : 1 : 1 (d) 2 : 1 : 2

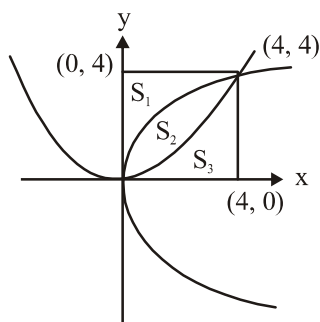
Ans. (c)

Sol. Total area = $4 \times 4 = 16$ sq.units

$$\text{Area of } S_3 = \int_0^4 \frac{x^2}{4} dx = \frac{16}{3} = S_1$$

$$\therefore S_2 = 16 - \frac{16}{3} \times 2 = \frac{16}{3}$$

$$\therefore S_1 : S_2 : S_3 \text{ is } 1 : 1 : 1$$



Example – 30

Find the area bounded by the circle $x^2 + y^2 = a^2$.

Sol. $x^2 + y^2 = a^2 \Rightarrow y = \pm \sqrt{a^2 - x^2}$

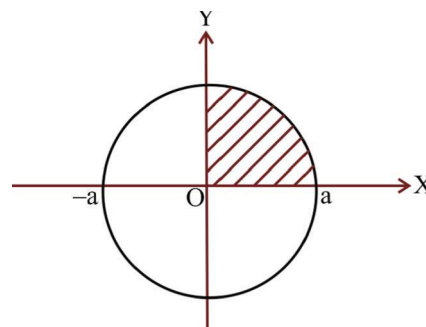
Equation of semicircle above X-axis is $y = + \sqrt{a^2 - x^2}$

Area of circle = 4 (shaded area)

$$= 4 \int_0^a \sqrt{a^2 - x^2} dx$$

$$= 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

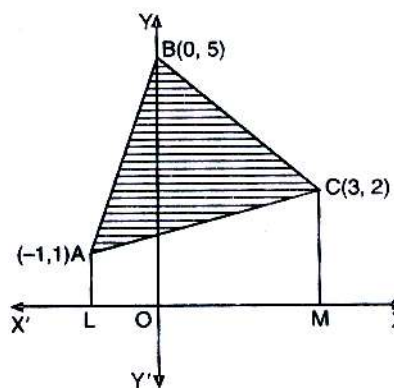
$$= 4 \frac{a^2}{2} \left(\frac{\pi}{2} \right) = \pi a^2$$



Example – 31

Using integration, find the area of the region bounded by: $(-1, 1)$, $(0, 5)$ and $(3, 2)$.

Sol. Let A $(-1, 1)$, B $(0, 5)$ and C $(3, 2)$ be the vertices of the triangle as shown in the following figure :



Equation of AB is :

$$y - 1 = \frac{5 - 1}{0 - (-1)} (x + 1)$$

$$\left[\text{Using } y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \right]$$

$$\Rightarrow y - 1 = 4(x + 1)$$

$$\Rightarrow y = 4x + 5 \quad \dots (1)$$

Equation of BC is :

$$y - 5 = \frac{2 - 5}{3 - 0} (x - 0)$$

$$\Rightarrow 3y - 15 = -3x$$

$$\Rightarrow 3y = 15 - 3x$$

$$\Rightarrow y = 5 - x \quad \dots (2)$$

Equation of AC is :



$$y - 1 = \frac{2-1}{3-(-1)}(x+1)$$

$$\Rightarrow y - 1 = \frac{1}{4}(x+1)$$

$$\Rightarrow 4y - 4 = x + 1$$

$$\Rightarrow 4y = x + 5$$

$$\Rightarrow y = \frac{x}{4} + \frac{5}{4} \quad \dots (3)$$

Now $\text{ar}(\triangle ABC) = \text{ar}(\triangle AOB) + \text{ar}(\triangle OMCB) - \text{ar}(\triangle OMC)$

$$= \int_{-1}^0 (4x+5) dx + \int_0^3 (5-x) dx - \int_{-1}^3 \left(\frac{1}{4}x + \frac{5}{4} \right) dx$$

$$= \left[\frac{4x^2}{2} + 5x \right]_{-1}^0 + \left[5x - \frac{x^2}{2} \right]_0^3 - \left[\frac{x^2}{8} + \frac{5}{4}x \right]_{-1}^3$$

$$= \left[2x^2 + 5x \right]_{-1}^0 + \left[5x - \frac{x^2}{2} \right]_0^3 - \left[\frac{x^2}{8} + \frac{5}{4}x \right]_{-1}^3$$

$$= [(0+0) - (2-5)] + \left[\left(15 - \frac{9}{2} \right) - (0-0) \right]$$

$$- \left[\left(\frac{9}{8} + \frac{15}{4} \right) - \left(\frac{1}{8} - \frac{5}{4} \right) \right]$$

$$= 3 + \frac{21}{2} - \left(\frac{9+30-1+10}{8} \right) = 3 + \frac{21}{2} - \frac{48}{8}$$

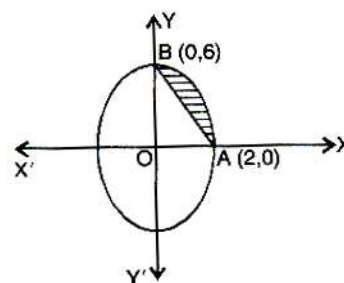
$$= 3 + \frac{21}{2} - 6 = \frac{21}{2} - 3 = \frac{15}{2} = 7.5 \text{ sq. units.}$$

Example – 32

AOBA is the part of the ellipse $9x^2 + y^2 = 36$ in the first quadrant such that $OA = 2$ and $OB = 6$. Find the area between the arc AB and the chord AB.

Sol. The given equation of the ellipse can be written as

$$\frac{x^2}{4} + \frac{y^2}{36} = 1 \text{ i.e. } \frac{x^2}{2^2} + \frac{y^2}{6^2} = 1$$



$\therefore$ A is (2, 0) and B is (0, 6).

$\therefore$ The equation of chord AB is :

$$y - 0 = \frac{6-0}{0-2}(x-2)$$

$$\Rightarrow y = -3x + 6.$$

$\therefore$ Reqd. area (shown shaded)

$$= \int_0^2 \sqrt{4-x^2} dx - \int_0^2 (6-3x) dx$$

$$= 3 \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2 - \left[6x - \frac{3x^2}{2} \right]_0^2$$

$$= 3 \left[\frac{2}{2} (0) + 2 \sin^{-1}(1) \right] - \left[6(2) - \frac{3(4)}{2} \right]$$

$$= 3 \left[2 \times \frac{\pi}{2} \right] - [12 - 6]$$

$$= (3\pi - 6) \text{ sq. units.}$$

Example – 33

Find the area bounded by the curves $y = x^2$ and $x^2 + y^2 = 2$ above X-axis.

Sol. Let us first find the points of intersection of curves.

Solving $y = x^2$ and $x^2 + y^2 = 2$ simultaneously, we get :

$$x^2 + x^4 = 2$$

$$\Rightarrow (x^2 - 1)(x^2 + 2) = 0$$

$$\Rightarrow x^2 = 1 \text{ and } x^2 = -2 \text{ [reject]}$$

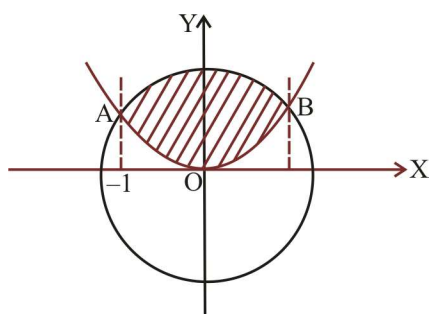
$$\Rightarrow x = \pm 1$$

$$\Rightarrow A = (-1, 1) \text{ and } B = (1, 1)$$

$$\text{Shaded Area} = \int_{-1}^1 \left(\sqrt{2-x^2} - x^2 \right) dx$$



$$\begin{aligned}
 &= \int_{-1}^{+1} \sqrt{2-x^2} dx - \int_{-1}^{+1} x^2 dx \\
 &= 2 \int_0^1 \sqrt{2-x^2} dx - 2 \int_0^1 x^2 dx \\
 &= 2 \left[\frac{x}{2} \sqrt{2-x^2} + \frac{2}{2} \sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 - 2 \left(\frac{1}{3} \right) \\
 &= 2 \left(\frac{1}{2} + \frac{\pi}{4} \right) - \frac{2}{3} = \frac{1}{3} + \frac{\pi}{2} \text{ sq. units.}
 \end{aligned}$$



Example – 34

Find the area of the region included between :

The parabola $y = \frac{3}{4}x^2$ and the line $3x - 2y + 12 = 0$

Sol. The given parabola is $4y = 3x^2$.

i.e. $y = \frac{3}{4}x^2$... (1)

and the given line is $3x - 2y + 12 = 0$... (2)

Putting the value of y from (1) in (2), we get :

$$3x - 2\left(\frac{3}{4}x^2\right) + 12 = 0 \Rightarrow 3x - \frac{3}{2}x^2 + 12 = 0$$

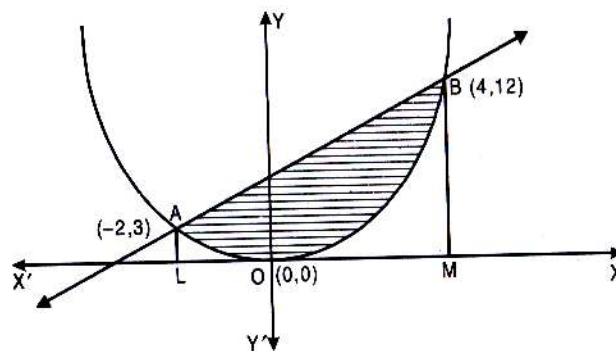
$$\Rightarrow 6x - 3x^2 + 24 = 0$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow (x-4)(x+2) = 0$$

$$\Rightarrow x = 4, -2.$$

Putting in (1), $y = \frac{3}{4}(4)^2 = 12$ and $y = \frac{3}{4}(-2)^2 = 3$.



Hence the line (2) intersects parabola (1) in the points A $(-2, 3)$ and B $(4, 12)$.

$\therefore$ Reqd. area = area ALMB – (area ALO + area OMB)

$$= \int_{-2}^4 \frac{3x+12}{2} dx - \left(\int_{-2}^0 \frac{3}{4}x^2 dx + \int_0^4 \frac{3}{4}x^2 dx \right)$$

$$\left[\because \text{From (2), } y = \frac{3x+12}{2} \right]$$

$$= \frac{1}{2} \left[\frac{3x^2}{2} + 12x \right]_{-2}^4 - \left[\frac{3}{4} \left\{ \frac{x^3}{3} \right\}_{-2}^0 + \frac{3}{4} \left\{ \frac{x^3}{3} \right\}_0^4 \right]$$

$$= \frac{1}{2} [(24+48) - (6-24)] - \left[\frac{3}{4} \left(0 + \frac{8}{3} \right) + \frac{3}{4} \left(\frac{64}{3} - 0 \right) \right]$$

$$= \frac{1}{2} [72+18] - [2+16] = 45-18$$

$$= 27 \text{ sq. units.}$$

Example – 35

Using integration, find the area of the region :

$$\{(x, y) : |x-1| \leq y \leq \sqrt{5-x^2}\}$$

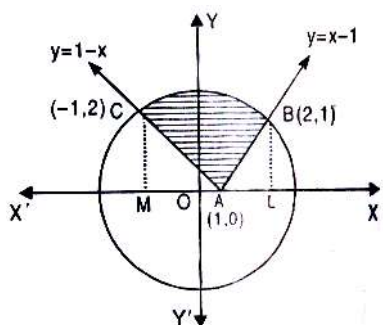
Sol. The given curves are :

$$x^2 + y^2 = 5$$

$$[\because y = \sqrt{5-x^2} \Rightarrow y^2 = 5-x^2 \Rightarrow x^2 + y^2 = 5]$$

and $y = \begin{cases} 1-x, & \text{if } x < 1 \\ x-1, & \text{if } x \geq 1 \end{cases}$

The reqd. region is shown as shaded in the following figure :



$y = x - 1$ meets $x^2 + y^2 = 5$ at B (2, 1)

$y = 1 - x$ meets $x^2 + y^2 = 5$ at C (-1, 2)

$y = x - 1$ and $y = 1 - x$ meet at A (1, 0).

∴ Reqd. area = ar. (MCBLM) – ar (CMAC) – ar (ALBA)

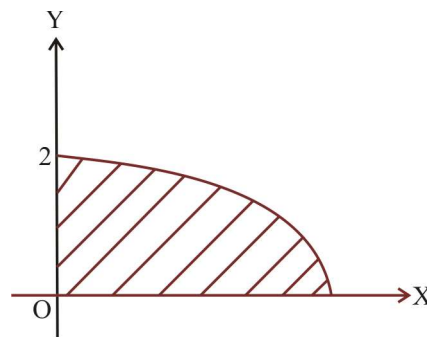
$$\begin{aligned}
 &= \int_{-1}^2 \sqrt{5-x^2} \, dx - \int_{-1}^1 (1-x) \, dx - \int_1^2 (x-1) \, dx \\
 &= \left[\frac{x\sqrt{5-x^2}}{2} + \frac{5}{2} \sin^{-1} \frac{x}{\sqrt{5}} \right]_{-1}^2 - \left[x - \frac{x^2}{2} \right]_{-1}^1 - \left[\frac{x^2}{2} - x \right]_1^2 \\
 &= \left[\left(1 + \frac{5}{2} \sin^{-1} \frac{2}{\sqrt{5}} \right) - \left(-\frac{1}{2} \times 2 + \frac{5}{2} \sin^{-1} \left(-\frac{1}{\sqrt{5}} \right) \right) \right] \\
 &\quad - \left[\left(1 - \frac{1}{2} \right) - \left(-1 - \frac{1}{2} \right) \right] - \left[(2-2) - \left(\frac{1}{2} - 1 \right) \right] \\
 &= 1 + \frac{5}{2} \sin^{-1} \frac{2}{\sqrt{5}} + 1 - \frac{5}{2} \sin^{-1} \left(-\frac{1}{\sqrt{5}} \right) - 2 - \frac{1}{2} \\
 &= -\frac{1}{2} + \frac{5}{2} \left[\sin^{-1} \frac{2}{\sqrt{5}} + \sin^{-1} \left(\frac{1}{\sqrt{5}} \right) \right] \\
 &= -\frac{1}{2} + \frac{5}{2} \times \frac{\pi}{2} = \frac{5\pi}{4} - \frac{1}{2} \text{ sq. units}
 \end{aligned}$$

Example – 36

Find the area bounded by the curve : $y = \sqrt{4-x}$, X-axis and Y-axis.

Sol. Trace the curve $y = \sqrt{4-x}$.

- Put $y = 0$ in the given curve to get $x = 4$ as the point of intersection with X-axis.
Put $x = 0$ in the given curve to get $y = 2$ as the point of intersection with Y-axis.
- For the curve, $y = \sqrt{4-x}$, $4-x \geq 0$
 $\Rightarrow x \leq 4$
 $\Rightarrow$ curve lies only to the left of $x = 4$ line.
- As any y is positive, curve is above X-axis.



Using step 1 to 3, we can draw the rough sketch of $y = \sqrt{4-x}$.

In figure,

$$\begin{aligned}
 \text{Bounded area} &= \int_0^4 \sqrt{4-x} \, dx = \left| \frac{-2}{3} (4-x) \sqrt{4-x} \right|_0^4 \\
 &= \frac{16}{3} \text{ sq. units.}
 \end{aligned}$$



Example – 37

Find the area enclosed by the loop in the curve:

$$4y^2 = 4x^2 - x^3.$$

Sol. The given curve is : $4y^2 = 4x^2 - x^3$

To draw the rough sketch of the given curve, consider the following steps :

(i) On replacing y by $-y$, there is no change in function.

It means the graph is symmetric about Y-axis.

(ii) For $x = 4, y = 0$ and for $x = 0, y = 0$.

(iii) In the given curve, LHS is positive for all values of y .

$$\Rightarrow \text{RHS} \geq 0 \quad \Rightarrow x^2(1-x/4) \geq 0$$

$$\Rightarrow x \leq 4.$$

Hence the curve lies to the left of $x = 4$.

(iv) As $x \rightarrow -\infty, y \rightarrow \pm \infty$

(v) Points of maximum/minimum :

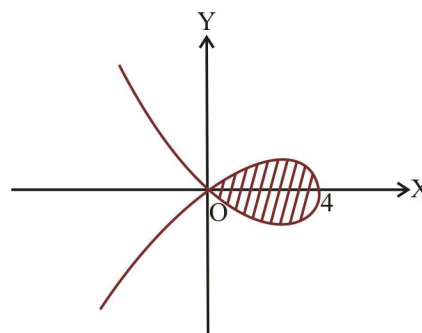
$$8y \frac{dy}{dx} = 8x - 3x^2$$

$$\frac{dy}{dx} = 0 \Rightarrow x = 0, \frac{8}{3}$$

At $x = 0$, derivative is not defined.

By checking for $\frac{d^2y}{dx^2}$, $x = \frac{8}{3}$ is a point of local maximum

(above X-axis).



From graph,

Shaded area (A) = $2 \times$ (area of portion above X-axis)

$$\Rightarrow A = 2 \int_0^4 \frac{x}{2} \sqrt{4-x} \, dx = \int_0^4 x \sqrt{4-x} \, dx$$

$$\Rightarrow A = \int_0^4 (4-x) \sqrt{4-(4-x)} \, dx$$

[using property – 4]

$$\Rightarrow A = \int_0^4 (4-x) \sqrt{x} \, dx$$

$$\Rightarrow A = 4 \left[\frac{2}{3} x \sqrt{x} \right]_0^4 - \left[\frac{2}{5} x^2 \sqrt{x} \right]_0^4$$

$$\Rightarrow A = \frac{128}{15} \text{ sq. units.}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Definite Integration by methods of Indefinite Integration

- The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^2 t \, dt}{x \sin x}$ is
 (a) 2 (b) 1
 (c) 0 (d) 3
- Let $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x} \right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3} dx = F(k) - F(1)$, then one of the possible values of k is
 (a) 16 (b) 63
 (c) 64 (d) 15
- The solution for x of the equation $\int_{\sqrt{2}}^x \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{12}$ is
 (a) $\frac{\sqrt{3}}{2}$ (b) $2\sqrt{2}$
 (c) 2 (d) π
- The integral $\int_0^{\frac{1}{2}} \frac{\ln(1+2x)}{1+4x^2} dx$, equals:
 (a) $\frac{\pi}{4} \ln 2$ (b) $\frac{\pi}{8} \ln 2$
 (c) $\frac{\pi}{16} \ln 2$ (d) $\frac{\pi}{32} \ln 2$
- Let function F be defined as $F(x) = \int_1^x \frac{e^t}{t} dt$, $x > 0$ then the value of the integral $\int_1^x \frac{e^t}{t+a} dt$, where $a > 0$, is:
 (a) $e^a[F(x) - F(1+a)]$ (b) $e^{-a}[F(x+a) - F(a)]$
 (c) $e^a[F(x+a) - F(1+a)]$ (d) $e^{-a}[F(x+a) - F(1+a)]$
- If $\int_0^{\pi/2} \frac{d\theta}{9\sin^2 \theta + 4\cos^2 \theta} = k\pi$, then the value of k is :
 (a) $\frac{1}{16}$ (b) $\frac{1}{12}$
 (c) $\frac{1}{8}$ (d) $\frac{1}{3}$
- The value of $\int_0^\infty \frac{dx}{(1+x)^3}$ is:
 (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
 (c) 0 (d) ∞
- $\int_0^{a/2} \frac{a \, dx}{(x-a)(x-2a)}$ equals to
 (a) $\ln \frac{2}{3}$ (b) $\ln \frac{3}{2}$
 (c) $\ln 6$ (d) none
- $\int_0^\pi \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$ equals $\sqrt{k}$.
 Then the value of k is.
- $\int_0^{\pi/2} \cos^5 \left(\frac{x}{2} \right) \cdot \sin x \, dx$ is equal to
 (a) $\frac{2}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$ (b) $\frac{-4}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$
 (c) $\frac{4}{7} \left(1 - \frac{1}{8\sqrt{2}} \right)$ (d) None of these
- $\int_0^{\pi/4} \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ equals to



12. For $0 < x < \frac{\pi}{2}$, $\int_{1/\sqrt{2}}^{1/2} \cot x \, d(\cos x)$ equals to

- (a) $\frac{\sqrt{3}-\sqrt{2}}{2}$ (b) $\frac{\sqrt{2}-\sqrt{3}}{2}$
(c) $\frac{1-\sqrt{3}}{2}$ (d) none

13. $\int_0^{(\pi/2)^{1/3}} 3x^5 \cdot \sin x^3 \, dx$ equals to

14. $\int_0^{\pi/4} \frac{x \cdot \sin x}{\cos^3 x} \, dx$ equal to

- (a) $\frac{\pi}{4} + \frac{1}{2}$ (b) $\frac{\pi}{4} - \frac{1}{2}$
(c) $\frac{\pi}{4}$ (d) none

15. The value of the integral $I = \int_0^1 x(1-x)^n \, dx$ is:

- (a) $\frac{1}{n+2}$ (b) $\frac{1}{n+1} - \frac{1}{n+2}$
(c) $\frac{1}{n+1} + \frac{1}{n+2}$ (d) $\frac{1}{n+1}$

16. If $\int_0^1 \cot^{-1}(1-x+x^2) \, dx = K \cdot \int_0^1 \tan^{-1} x \, dx$, then K equals to

18. The integral $\int_0^{\pi} \sqrt{1+4\sin^2 \frac{x}{2} - 4\sin \frac{x}{2}} \, dx$ equals :

- (a) $4\sqrt{3} - 4 - \frac{\pi}{3}$ (b) $\pi - 4$
(c) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$ (d) $4\sqrt{3} - 4$

19. If $f(x) = |x| + |x-1| + |x-2|$, $x \in R$ then $\int_0^3 f(x) \, dx$

equals to

- (a) $9/2$ (b) $15/2$
(c) $19/2$ (d) none

20. $\int_0^{1.5} x[x^2] \, dx = 3/k$, where $[]$ denotes greatest integer

function. Then the value of k is.

21. Let $y = \{x\}^{[x]}$ where $\{x\}$ denotes the fractional part of

x & $[x]$ denotes greatest integer $\leq x$ then $\int_0^3 y \, dx =$

- (a) $5/6$ (b) $2/3$
(c) 1 (d) $11/6$

22. $\int_0^{\pi} |1+2\cos x| \, dx$ equals to

- (a) $\frac{2\pi}{3}$ (b) π
(c) 2 (d) $\frac{\pi}{3} + 2\sqrt{3}$

Splitting the integral for Mod, Gint etc

17. $\int_0^{\sqrt{2}} [x^2] \, dx$ where $[]$ is GIF is :

- (a) $2 - \sqrt{2}$ (b) $2 + \sqrt{2}$
(c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$

Use of King's rule: $x \rightarrow (a+b-x)$

23. If the value of $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is $\frac{\pi}{k}$. Then the value of k is



24. $\int_0^{\pi} x f(\sin x) dx$ is equal to :
- (a) $\pi \int_0^{\pi} x f(\cos x) dx$ (b) $\pi \int_0^{\pi} f(\sin x) dx$
- (c) $\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$ (d) $\pi \int_0^{\pi} f(\cos x) dx$
25. The value of the integral, $2 \int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is :
26. If $f(x)$ and $\phi(x)$ are continuous functions on the interval $[0, 4]$ satisfying
- $$f(x) = f(4-x), \phi(x) + \phi(4-x) = 3$$
- and $\int_0^4 f(x) dx = 2$, then $\int_0^4 f(x) \phi(x) dx =$
27. The value of the integral $\int_0^{\pi} x (\sin^4 x \cos^4 x) dx$ is
- (a) $\frac{3\pi^2}{64}$ (b) $\frac{3\pi^2}{128}$
- (c) $\frac{3\pi^2}{256}$ (d) none of these
28. The value of the integral $\int_{\log 1/3}^{\log 3} \log(x + \sqrt{1+x^2}) dx$ is
- (a) $\log 3$ (b) $2 \log 3$
- (c) 0 (d) None
30. If $f(x) = \frac{x^7 - 3x^5 + 7x^3 - x + 1}{\cos^2 x}$ then, $\int_{-\pi/4}^{\pi/4} f(x) dx$ equals to
31. The value of $\int_{-1}^1 \frac{1 + \sin x}{1 + x^2} dx$ is :
- (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$
- (c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

Property on Periodic Functions

32. Let $f(x) = x - [x]$, for every real number x , where $[x]$ is the integral part of x . Then $\int_{-1}^1 f(x) dx$ is:
- (a) 1 (b) 2
- (c) 0 (d) $\frac{1}{2}$
33. $\int_0^{100\pi} \sqrt{1 - \cos 2x} dx$ is equal to
- (a) $100\sqrt{2}$ (b) $50\sqrt{2}$
- (c) $200\sqrt{2}$ (d) $400\sqrt{2}$
34. $\int_{\pi}^{10\pi} |\sin x| dx$ is equal to
- (a) 18 (b) 20
- (c) 40 (d) None
35. $\int_0^{100} (x - [x]) dx$ is equal to
36. $\int_0^{10} e^{x-[x]} dx$ ($[.]$ denotes GIF) is equal to
- (a) $10(e-1)$ (b) $\frac{e-1}{10}$
- (c) $\frac{e^{10}-1}{10}$ (d) $\frac{e^{10}-1}{e-1}$

Property on Even/odd function for limit -a to a

29. $\int_{-\pi/2}^{\pi/2} [(x + \pi)^3 + \cos^2(x + 3\pi)] dx$ is equal to
- (a) $\frac{\pi^4}{32}$ (b) $\frac{\pi^4}{32} + \frac{\pi}{2}$
- (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{2} - 1$



Properties of Definite Integrals

37. $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$ is
 (a) $\pi^2/4$ (b) π^2
 (c) $5/4$ (d) $-3/4$
38. The value of $\int_1^a [x] f'(x) dx$, $a > 1$, where $[x]$ denotes the greatest integer not exceeding x , is
 (a) $a f(a) - \{f(1) + f(2) + \dots + f([a])\}$
 (b) $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$
 (c) $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
 (d) $a f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
39. If $[]$ denotes the greatest integer function, then the integral $\int_0^{\pi} [\cos x] dx$ is equal to:
 (a) $\frac{\pi}{2}$ (b) 0
 (c) -1 (d) $-\frac{\pi}{2}$
40. If $\int_{-1}^{-4} f(x) dx = 4$ and $\int_2^{-4} (3 - f(x)) dx = 7$ then the value of $\int_{-2}^1 f(-x) dx$ is
41. Suppose for every integer n , $\int_n^{n+1} f(x) dx = n^2$. The value of $\int_{-2}^4 f(x) dx$ is
42. $\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$ is equal to
 (a) $\frac{\pi^2}{ab}$ (b) $\frac{\pi^2}{2ab}$
 (c) $\frac{\pi^2}{4ab}$ (d) none of these

43. $\int_{-1}^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx$ equals to
 (a) 2π (b) π
 (c) 4π (d) none

Differentiation of functions by definite integrals: Leibnitz Rule

44. If $f(x) = \int_0^x t \sin t dt$, then $f'(x)$ is
 (a) $\cos x + x \sin x$ (b) $x \cos x + \sin x$
 (c) $x \cos x$ (d) $x \sin x$
45. If $f(x) = \int_1^{x^3} \frac{dt}{1+t^4}$, then $f''(x)$ is equal to
 (a) $\frac{6x(1-5x^{12})}{(1+x^{12})^2}$ (b) $\frac{6x(1+5x^{12})}{(1+x^{12})^2}$
 (c) $-\frac{6x(1-5x^{12})}{(1+x^{12})^2}$ (d) none of these
46. If the variables x and y are connected by the relation $x = \int_1^y \frac{dz}{\sqrt{1+6z^3}}$, then $\frac{d^2y}{dx^2}$ is proportional to
 (a) y (b) y^2
 (c) y^3 (d) none of these
47. $\lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_0^{x^2} \sin \sqrt{t} dt$ is equal to
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $-\frac{1}{3}$ (d) $-\frac{2}{3}$
48. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\tan^{-1} t)^2 dt}{\int_0^{x^4} \sin \sqrt{t} dt}$ is equal to $1/k$. Then the value of k is.



Finding summation of infinite series using definite Integration

Estimation/Approximation of Definite Integrals

49. $\lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sec^2 \frac{1}{n} + \frac{2}{n^2} \sec^2 \frac{2}{n} + \dots + \frac{1}{n} \sec^2 1 \right]$ equals

- (a) $\frac{1}{2} \operatorname{cosec} 1$ (b) $\frac{1}{2} \sec 1$
(c) $\frac{1}{2} \tan 1$ (d) $\tan 1$

50. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right]$ is equal to

- (a) $\frac{3}{8}$ (b) $\frac{1}{4}$
(c) $\frac{1}{8}$ (d) none of these

51. The value of the

$$\lim_{n \rightarrow \infty} \left[\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2-1}} + \frac{1}{\sqrt{n^2-2^2}} + \dots + \frac{1}{\sqrt{n^2-(n-1)^2}} \right] \text{ is}$$

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{2}$ (d) none of these

52. If $S_n = \left[\frac{1}{2n} + \frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \dots + \frac{1}{\sqrt{3n^2+2n-1}} \right]$

then $\lim_{n \rightarrow \infty} S_n$ is equal to

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

53. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} e^{\frac{r}{n}}$ is :

- (a) e (b) $e-1$
(c) $1-e$ (d) $e+1$

54. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true ?

- (a) $I > \frac{2}{3}$ and $J < 2$ (b) $I > \frac{2}{3}$ and $J > 2$
(c) $I < \frac{2}{3}$ and $J < 2$ (d) $I < \frac{2}{3}$ and $J > 2$

55. $I_1 = \int_1^2 e^x dx$ and $I_2 = \int_1^2 \log_e x dx$. Then

- (a) $I_1 > I_2$ (b) $I_1 < I_2$
(c) $I_1 = I_2$ (d) None of these

56. $I = \int_1^2 \frac{e^x}{x} dx$. Then

- (a) $I < e$ (b) $I > \frac{e^2}{2}$
(c) $e < I < \frac{e^2}{2}$ (d) None of these

57. $I_1 = \int_0^{\pi/4} \tan^3 x dx$ and $I_2 = \int_0^{\pi/4} \tan^5 x dx$

$I_3 = \int_0^{\pi/4} \tan^{1/2} x dx$ $I_4 = \int_0^{\pi/4} \tan^{1/3} x dx$ then

- (a) $I_1 < I_2$ (b) $I_1 > I_3$
(c) $I_3 > I_4$ (d) $I_1 > I_2$

58. $\int_0^2 (1+2^{-x^2}) dx =$

- (a) 4 (b) $\frac{17}{2}$
(c) 2 (d) None of these



Reduction Method

59. If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$, where n is a positive integer, then

$n(I_{n-1} + I_{n+1})$ is equal to

- (a) 1 (b) $n-1$
(c) $\frac{1}{n-1}$ (d) none of these

60. If $a_n = \int_{\pi/4}^{\pi/2} \cot^n x dx$, then $a_2 + a_4, a_3 + a_5, a_4 + a_6$ are in

- (a) G.P. (b) A.P.
(c) H.P. (d) None

61. Let $I_n = \int_0^{\pi/4} \tan^n x dx$,

then $\frac{1}{I_2 + I_4}, \frac{1}{I_3 + I_5}, \frac{1}{I_4 + I_6}, \dots$ are in

- (a) A.P. (b) G.P.
(c) H.P. (d) none

62. The value of the integral $\int_0^{\pi} \frac{\sin(2kx)}{\sin x} dx$, when $k \in \mathbb{I}$, is

- (a) $\frac{\pi}{2}$ (b) π
(c) 0 (d) none of these

Bounded between the curve and X-axis or Y-Axis when limits are given

63. Area bounded by the curve $y = x^3$, the x -axis and the ordinates $x = -2$ and $x = 1$ is :

- (a) -9 (b) $-\frac{15}{4}$
(c) $\frac{15}{4}$ (d) $\frac{17}{4}$

64. The area of the region bounded by the curve $y = x - x^2$, x -axis between $x = 0$ and $x = 1$ is :

- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{5}{6}$

65. The area of the region bounded by the curve $y = \sqrt{1 - \cos x}$ between $x = 0$ and $x = \pi$ is :

- (a) $\frac{\sqrt{2}}{2}$ (b) $2\sqrt{2}$
(c) $\sqrt{2}$ (d) 2

66. The area of the region bounded by the curve $y = x \sin x$ between $x = 0$ and $x = 2\pi$ is :

- (a) π (b) 2π
(c) 3π (d) 4π

67. Area of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is :

- (a) 2 (b) $\frac{9}{4}$
(c) $\frac{9}{3}$ (d) $\frac{9}{2}$

68. The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is

- (a) 2 (b) 1
(c) 4 (d) 3

Bounded between two curves when limits are given

69. The area between the curves $y = \tan x$, $y = \cot x$ and x -axis in the interval $[0, \pi/2]$ is $\log k$. Then the value of k is.

70. If $0 \leq x \leq \pi$; then the area bounded by the curve $y = x$ and $y = x + \sin x$ is -



Bounded between two curves when limits are also to be found

71. The area bounded by the curves $y = \ln x$, $y = \ln |x|$, $y = |\ln x|$ and $y = |\ln |x||$ is
 (a) 4 sq. units (b) 6 sq. units
 (c) 10 sq. units (d) none of these
72. The area of the region bounded by the curves $y = |x-1|$ and $y = 3 - |x|$ is
 (a) 3 sq. units (b) 4 sq. units
 (c) 6 sq. units (d) 2 sq. units
73. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is
 (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
 (c) $\frac{2}{3}$ (d)
74. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line $y = 2$ is.
 (a) $20\sqrt{2}$ (b) $\frac{10\sqrt{2}}{3}$
 (c) $\frac{20\sqrt{2}}{3}$ (d) $10\sqrt{2}$
75. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis and lying in the first quadrant is
 (a) 9 (b) 6
 (c) 18 (d) $\frac{27}{4}$
76. The area of the region described by $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is
 (a) $\frac{\pi}{2} + \frac{2}{3}$ (b) $\frac{\pi}{2} + \frac{4}{3}$
 (c) $\frac{\pi}{2} - \frac{4}{3}$ (d) $\frac{\pi}{2} - \frac{2}{3}$
77. Let $A = \{(x, y) : y^2 \leq 4x, y - 2x \geq -4\}$. The area (in square units) of the region A is
 (a) 8 (b) 9
 (c) 10 (d) 11
78. The area of the circle $x^2 + y^2 = 16$ exterior to the parabola $y^2 = 6x$ is :
 (a) $\frac{4}{3}(4\pi - \sqrt{3})$ (b) $\frac{4}{3}(4\pi + \sqrt{3})$
 (c) $\frac{4}{3}(8\pi - \sqrt{3})$ (d) $\frac{4}{3}(8\pi + \sqrt{3})$
79. The area between the curves $y = \cos x$ and the line $y = x + 1$ in the second quadrant is –
 (a) 1 (b) 2
 (c) $3/2$ (d) $1/2$
80. The area bounded by the curves $y = \sin x$, $y = \cos x$ and y-axis in first quadrant is –
 (a) $\sqrt{2} - 1$ (b) $\sqrt{2}$
 (c) $\sqrt{2} + 1$ (d) None of these
81. The area bounded by curve $y = ex \log x$ and $y = \frac{\log x}{ex}$ is –
 (a) $\frac{e^2 - 5}{4e}$ (b) $\frac{e^2 + 5}{4e}$
 (c) $\frac{e^2}{4} - \frac{5}{e}$ (d) None of these
82. The area bounded by $y = x^2 - 4$ and $x + y = 2$ is
 (a) $\frac{75}{6}$ (b) $\frac{100}{6}$
 (c) $\frac{125}{6}$ (d) $\frac{150}{6}$
83. The area bounded by the curves : $y^2 = 4a(x + a)$ and $y^2 = 4b(b - x)$ is $\frac{k}{3}(a + b)\sqrt{4ab}$. Then the value of k is.
84. If the area of the region $\{(x, y) : x^2 \leq y \leq |x|\}$ is $1/k$ sq. units, then the value of k is



Use of standard results

85. Smaller area enclosed by the circle $x^2 + y^2 = 4$ and the line $x + y = 2$ is :

- (a) $2(\pi - 2)$ (b) $\pi - 2$
(c) $2\pi - 1$ (d) $2(\pi + 2)$

86. If the line $y = mx$ bisects the area enclosed by the lines $x = 0$, $y = 0$, $x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, and the value of m is equal to $13/k$. then the value of k is

Determining the unknown parameter for given condition of area

87. If the area enclosed between the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16}{3}$, then value of ab is

88. AOB is the positive quadrant of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; where $OA = a$, $OB = b$. Area between arc AB and chord AB of ellipse is $\frac{(\pi - p)}{q} ab$, then $p + q$ equals

89. If the area bounded by the parabola $x^2 = 4y$, the x -axis and the line $x = 4$ is divided into two equal areas by the line $x = \alpha$, and the value of α is $(32)^{1/k}$, then the value of k is

90. The value of m for which the area included between the curves $y^2 = 4ax$ and $y = mx$ equals $\frac{a^2}{3}$ is



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

1. The area (in sq. units) of the region described by $\{(x, y) : y^2 \leq 2x \text{ and } y \geq 4x - 1\}$ is: (2015)
 - (a) $\frac{15}{64}$
 - (b) $\frac{9}{32}$
 - (c) $\frac{7}{32}$
 - (d) $\frac{5}{64}$
2. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ is equal to: (2015)
 - (a) 1
 - (b) 6
 - (c) 2
 - (d) 4
3. The area (in square units) of the region bounded by the curves $y + 2x^2 = 0$ and $y + 3x^2 = 1$, is equal to : (2015/Online Set-1)
 - (a) $\frac{1}{3}$
 - (b) $\frac{3}{4}$
 - (c) $\frac{3}{5}$
 - (d) $\frac{4}{3}$
4. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1 + a^x} dx$, $a > 0$ is (2015/Online Set-2)
 - (a) π
 - (b) $a\pi$
 - (c) $\pi/2$
 - (d) 2π
5. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to (2015/Online Set-2)
 - (a) $\frac{4}{3}$
 - (b) $\frac{5}{3}$
 - (c) $\frac{1}{3}$
 - (d) $\frac{2}{3}$
6. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is : (2016)
 - (a) $\pi - \frac{8}{3}$
 - (b) $\pi - \frac{4\sqrt{2}}{3}$
 - (c) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$
 - (d) $\pi - \frac{4}{3}$
7. If $2 \int_0^1 \tan^{-1} x dx = \int_0^1 \cot^{-1}(1 - x + x^2) dx$, then $\int_0^1 \tan^{-1}(1 - x + x^2) dx$ is equal to : (2016/Online Set-1)
 - (a) $\log 4$
 - (b) $\frac{\pi}{2} + \log 2$
 - (c) $\log 2$
 - (d) $\frac{\pi}{2} - \log 4$
8. The area (in sq. units) of the region described by $A = \{(x, y) | y \geq x^2 - 5x + 4, x + y \geq 1, y \leq 0\}$ is: (2016/Online Set-1)
 - (a) $\frac{7}{2}$
 - (b) $\frac{19}{6}$
 - (c) $\frac{13}{6}$
 - (d) $\frac{17}{6}$
9. The value of the integral $\int_4^{10} \frac{[x^2] dx}{[x^2 - 28x + 196] + [x^2]}$, where $[x]$ denotes the greatest integer less than or equal to x , is : (2016/Online Set-2)
 - (a) 6
 - (b) 3
 - (c) 7
 - (d) $\frac{1}{3}$



10. For $x \in R$, $x \neq 0$, if $y(x)$ is a differentiable function such that $x \int_1^x y(t) dt = (x+1) \int_1^x t y(t) dt$, then $y(x)$ equals :
(where C is a constant.) (2016/Online Set-2)
- (a) $\frac{C}{x} e^{-\frac{1}{x}}$ (b) $\frac{C}{x^2} e^{-\frac{1}{x}}$
(c) $\frac{C}{x^3} e^{-\frac{1}{x}}$ (d) $Cx^3 e^{\frac{1}{x}}$
11. The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ is equal to: (2017)
- (a) -2 (b) 2
(c) 4 (d) -1
12. The integral $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{8 \cos 2x}{(\tan x + \cot x)^3} dx$ equals :
(2017/Online Set-1)
- (a) $\frac{15}{128}$ (b) $\frac{15}{64}$
(c) $\frac{13}{32}$ (d) $\frac{13}{256}$
13. If $\int_1^2 \frac{dx}{(x^2 - 2x + 4)^{\frac{3}{2}}} = \frac{k}{k+5}$, then k is equal to:
(2017/Online Set-2)
- (a) 1 (b) 2
(c) 3 (d) 4
14. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is:
(2017)
- (a) $\frac{59}{12}$ (b) $\frac{2}{3}$
(c) $\frac{7}{3}$ (d) $\frac{5}{2}$
15. The area (in sq. units) of the smaller portion enclosed between the curves, $x^2 + y^2 = 4$ and $y^2 = 3x$, is :
(2017/Online Set-1)
- (a) $\frac{1}{2\sqrt{3}} + \frac{\pi}{3}$ (b) $\frac{1}{\sqrt{3}} + \frac{2\pi}{3}$
(c) $\frac{1}{2\sqrt{3}} + \frac{2\pi}{3}$ (d) $\frac{1}{\sqrt{3}} + \frac{4\pi}{3}$
16. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ is: (2018)
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{8}$
(c) $\frac{\pi}{2}$ (d) 4π
17. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and $\alpha, \beta (\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve $y = (g \circ f)(x)$ and the lines $x = \alpha, x = \beta$ and $y = 0$, is :
(2018)
- (a) $\frac{1}{2}(\sqrt{2} - 1)$ (b) $\frac{1}{2}(\sqrt{3} - 1)$
(c) $\frac{1}{2}(\sqrt{3} + 1)$ (d) $\frac{1}{2}(\sqrt{3} - \sqrt{2})$
18. The value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x \left(1 + \log \left(\frac{2 + \sin x}{2 - \sin x} \right) \right) dx$ is
(2018/Online Set-1)
- (a) 0 (b) $\frac{3}{4}$
(c) $\frac{3}{8}\pi$ (d) $\frac{3}{16}\pi$



19. The area (in sq. units) of the region $\{x \in R : x \geq 0, y \geq 0, y \geq x-2 \text{ and } y \leq \sqrt{x}\}$ is :
(2018/Online Set-1)
- (a) $\frac{13}{3}$ (b) $\frac{8}{3}$
(c) $\frac{10}{3}$ (d) $\frac{5}{3}$
20. The value of integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1+\sin x} dx$ is :
(2018/Online Set-2)
- (a) $\pi\sqrt{2}$ (b) $\pi(\sqrt{2}-1)$
(c) $\frac{\pi}{2}(\sqrt{2}+1)$ (d) $2\pi(\sqrt{2}-1)$
21. If $I_1 = \int_0^1 e^{-x} \cos^2 x dx$,
 $I_2 = \int_0^1 e^{-x^2} \cos^2 x dx$ and
 $I_3 = \int_0^1 e^{-x^3} dx$; then : (2018/Online Set-2)
- (a) $I_2 > I_3 > I_1$ (b) $I_2 > I_1 > I_3$
(c) $I_3 > I_2 > I_1$ (d) $I_3 > I_1 > I_2$
22. If $f(x) = \int_0^x t(\sin x - \sin t) dt$ then :
(2018/Online Set-3)
- (a) $f'''(x) + f''(x) = \sin x$
(b) $f'''(x) + f''(x) - f'(x) = \cos x$
(c) $f'''(x) + f''(x) = \cos x - 2x \sin x$
(d) $f'''(x) - f''(x) = \cos x - 2x \sin x$
23. If the area of the region bounded by the curves, $y = x^2$, $y = \frac{1}{x}$ and the lines $y = 0$ and $x = t$ ($t > 1$) is 1 sq. unit, then t is equal to : (2018/Online Set-3)
- (a) $e^{\frac{3}{2}}$ (b) $\frac{4}{3}$
(c) $\frac{3}{2}$ (d) $e^{\frac{2}{3}}$
24. The area (in sq. units) of the region $A = \{(x, y) \in R \times R \mid 0 \leq x \leq 3, 0 \leq y \leq 4, y \leq x^2 + 3x\}$ is
(8-04-2019/Shift-1)
- (a) $\frac{53}{6}$ (b) 8
(c) $\frac{59}{6}$ (d) $\frac{26}{3}$
25. If $f(x) = \frac{2-x \cos x}{2+x \cos x}$ and $g(x) = \log_e x$, ($x > 0$) then the value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(f(x)) dx$ is:
(8-04-2019/Shift-1)
- (a) $\log_e 3$ (b) $\log_e e$
(c) $\log_e 2$ (d) $\log_e 1$
26. Let $f(x) = \int_0^x g(t) dt$, where g is a non-zero even function. If $f(x+5) = g(x)$, then $\int_0^x f(t) dt$ equals :
(8-04-2019/Shift-2)
- (a) $\int_{x+5}^5 g(t) dt$ (b) $\int_5^{x+5} g(t) dt$
(c) $2 \int_5^{x+5} g(t) dt$ (d) $5 \int_5^{x+5} g(t) dt$



27. Let $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$. If for a λ , $0 < \lambda < 4$, $A(\lambda) : A(4) = 2:5$, then λ equals : (8-04-2019/Shift-2)
- (a) $2\left(\frac{4}{5}\right)^{\frac{1}{3}}$ (b) $2\left(\frac{2}{5}\right)^{\frac{1}{3}}$
- (c) $4\left(\frac{2}{5}\right)^{\frac{1}{3}}$ (d) $4\left(\frac{4}{25}\right)^{\frac{1}{3}}$
28. The area (in sq. units) of the region $A = \{(x, y) : x^2 \leq y \leq x + 2\}$ is: (09-04-2019/Shift-1)
- (a) $\frac{10}{3}$ (b) $\frac{9}{2}$
- (c) $\frac{31}{6}$ (d) $\frac{13}{6}$
29. The value of the integral $\int_0^1 x \cot^{-1}(1 - x^2 + x^4) dx$ is : (9-04-2019/Shift-2)
- (a) $\frac{\pi}{2} - \frac{1}{2} \log_e 2$ (b) $\frac{\pi}{4} - \log_e 2$
- (c) $\frac{\pi}{2} - \log_e 2$ (d) $\frac{\pi}{4} - \frac{1}{2} \log_e 2$
30. If $f : R \rightarrow R$ is a differentiable function and $f(2) = 6$, then $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{2t dt}{(x-2)}$ is : (9-04-2019/Shift-2)
- (a) $24f'(2)$ (b) $2f'(2)$
- (c) 0 (d) $12f'(2)$
31. The area (in sq. units) of the region $A = \left\{(x, y) : \frac{y^2}{2} \leq x \leq y + 4\right\}$ is: (9-04-2019/Shift-2)
- (a) $\frac{53}{3}$ (b) 30
- (c) 16 (d) 18
32. The value of $\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$, where $[]$ denotes the greatest integer function, is: (10-04-2019/Shift-1)
- (a) π (b) $-\pi$
- (c) -2π (d) 2π
33. $\lim_{n \rightarrow \infty} \left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + \dots + \frac{(2n)^{1/3}}{n^{4/3}} \right)$ is equal to: (10-04-2019/Shift-1)
- (a) $\frac{3}{4}(2)^{4/3} - \frac{3}{4}$ (b) $\frac{4}{3}(2)^{4/3}$
- (c) $\frac{3}{2}(2)^{4/3} - \frac{4}{3}$ (d) $\frac{4}{3}(2)^{3/4}$
34. The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is: (10-4-2019/Shift-2)
- (a) $\ln 2 + \frac{3}{2}$ (b) $\frac{3}{2}$
- (c) $\frac{1}{2}$ (d) $\frac{3}{2} - \frac{1}{\ln 2}$
35. The integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^{\frac{2}{3}} x \operatorname{cosec}^{\frac{4}{3}} x dx$ is equal to: (10-4-2019/Shift-2)
- (a) $3^{\frac{5}{6}} - 3^{\frac{2}{3}}$ (b) $3^{\frac{4}{3}} - 3^{\frac{1}{3}}$
- (c) $3^{\frac{7}{6}} - 3^{\frac{5}{6}}$ (d) $3^{\frac{5}{3}} - 3^{\frac{1}{3}}$
36. If the area (in sq. units) of the region $\{(x, y) : y^2 \leq 4x, x + y \leq 1, x \geq 0, y \geq 0\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to : (12-04-2019/Shift-1)
- (a) $\frac{10}{3}$ (b) 6
- (c) $\frac{8}{3}$ (d) $-\frac{2}{3}$



37. If $\int_0^{\frac{\pi}{2}} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$, then m, n is equal to
(12-04-2019/Shift-1)
- (a) $-\frac{1}{2}$ (b) 1
(c) $\frac{1}{2}$ (d) -1
38. Let $f: R \rightarrow R$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$.
If $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$, then $\lim_{x \rightarrow 2} g(x)$ is equal to
(12-04-2019/Shift-1)
39. A value of α such that
 $\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e \left(\frac{9}{8} \right)$ is _____.
(12-04-2019/Shift-2)
- (a) -2 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) 2
40. If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x, \lambda > 0$ is $\frac{1}{9}$ then λ is equal to _____.
(12-04-2019/Shift-2)
- (a) $2\sqrt{6}$ (b) 48
(c) 24 (d) $4\sqrt{3}$
41. The value of $\int_0^{\pi} |\cos x|^3 dx$ is: (9-01-2019/Shift-1)
- (a) 0 (b) $\frac{4}{3}$
(c) $\frac{2}{3}$ (d) $-\frac{4}{3}$
42. The area of the region
 $A = \{(x, y) : 0 \leq y \leq x | x| + 1 \text{ and } -1 \leq x \leq 1\}$ in sq. units is: (9-01-2019/Shift-2)
- (a) $\frac{2}{3}$ (b) 2
(c) $\frac{4}{3}$ (d) $\frac{1}{3}$
43. If $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta = 1 - \frac{1}{\sqrt{2}}$, ($k > 0$) then the value of k is: (9-01-2019/Shift-2)
- (a) 4 (b) $\frac{1}{2}$
(c) 1 (d) 2
44. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is: (10-1-2019/Shift-1)
- (a) $(0, \sqrt{2})$ (b) $(-\sqrt{2}, 0)$
(c) $(\sqrt{2}, -\sqrt{2})$ (d) $(-\sqrt{2}, \sqrt{2})$
45. If the area enclosed between the curves $y = kx^2$ and $x = ky^2$, ($k > 0$), is 1 square unit. Then k is: (10-1-2019/Shift-1)
- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $\frac{2}{\sqrt{3}}$
46. If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f'(1/2)$ is: (10-01-2019/Shift-2)
- (a) $\frac{24}{25}$ (b) $\frac{18}{25}$
(c) $\frac{4}{5}$ (d) $\frac{6}{25}$



47. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is:
(10-01-2019/Shift-2)
- (a) $\frac{1}{12}(7\pi + 5)$ (b) $\frac{1}{12}(7\pi - 5)$
(c) $\frac{3}{20}(4\pi - 3)$ (d) $\frac{3}{10}(4\pi - 3)$
48. The area (in sq. units) of the region bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$ is
(11-01-2019/Shift-1)
- (a) $\frac{5}{4}$ (b) $\frac{9}{8}$
(c) $\frac{7}{8}$ (d) $\frac{3}{4}$
49. The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ (where $[x]$ denotes the greatest integer less than or equal to x) is :
(11-01-2019/Shift-1)
- (a) 0 (b) $\sin 4$
(c) 4 (d) $4 - \sin 4$
50. The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$ equals :
(11-01-2019/Shift-2)
- (a) $\frac{1}{20} \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right)$ (b) $\frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$
(c) $\frac{\pi}{40}$ (d) $\frac{1}{5} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{3\sqrt{3}} \right) \right)$
51. The area (in sq. units) in the first quadrant bounded by the parabola, $y = x^2 + 1$, the tangent to it at the point $(2, 5)$ and the coordinate axes is :
(11-01-2019/Shift-2)
- (a) $\frac{8}{3}$ (b) $\frac{37}{24}$
(c) $\frac{187}{24}$ (d) $\frac{14}{3}$
52. The maximum area (in sq. units) of a rectangle having its base on the X-axis and its other two vertices on the parabola, $y = 12 - x^2$ such that the rectangle lies inside the parabola, is :
(12-01-2019/Shift-1)
- (a) 36 (b) $20\sqrt{2}$
(c) 32 (d) $18\sqrt{3}$
53. The area (in sq. units) of the region bounded by the parabola, $y = x^2 + 2$ and the lines, $y = x + 1$, $x = 0$ and $x = 3$, is :
(12-01-2019/Shift-1)
- (a) $\frac{15}{4}$ (b) $\frac{21}{2}$
(c) $\frac{17}{4}$ (d) $\frac{15}{2}$
54. Let f and g be continuous functions on $[0, a]$ such that $f(x) = f(a - x)$ and $g(x) + g(a - x) = 4$ then $\int_0^a f(x)g(x)dx$ is equal to
(12-01-2019/Shift-1)
- (a) $4 \int_0^a f(x)dx$ (b) $\int_0^a f(x)dx$
(c) $2 \int_0^a f(x)dx$ (d) $-3 \int_0^a f(x)dx$
55. The integral $\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx$ is equal to
(12-01-2019/Shift-2)
- (a) $\frac{1}{2} - e - \frac{1}{e^2}$ (b) $-\frac{1}{2} + \frac{1}{e} - \frac{1}{2e^2}$
(c) $\frac{3}{2} - \frac{1}{e} - \frac{1}{2e^2}$ (d) $\frac{3}{2} - e - \frac{1}{2e^2}$



56. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{5n} \right)$ is equal to

(12-01-2019/Shift-2)

(a) $\frac{\pi}{4}$

(b) $\tan^{-1}(3)$

(c) $\frac{\pi}{2}$

(d) $\tan^{-1}(2)$

57. Area (in sq. units) of the region outside $\frac{|x|}{2} + \frac{|y|}{3} = 1$

and inside the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is :

(2-9-2020/Shift-1)

(a) $3(\pi - 2)$

(b) $6(\pi - 2)$

(c) $6(4 - \pi)$

(d) $3(4 - \pi)$

58. The integral $\int_0^2 ||x-1| - x| dx$ is equal to :

(2-9-2020/Shift-1)

59. Consider a region $R = \{(x, y) \in R^2 : x^2 \leq y \leq 2x\}$. If a line $y = \alpha$ divides the area of region R into two equal parts, then which of the following is true ?

(2-09-2020/Shift-2)

(a) $\alpha^3 - 6\alpha^2 + 16 = 0$

(b) $3\alpha^2 - 8\alpha^{3/2} + 8 = 0$

(c) $\alpha^3 - 6\alpha^{3/2} - 16$

(d) $3\alpha^2 - 8\alpha + 8 = 0$

60. Let $[t]$ denote the greatest integer less than or equal to t .

Then the value of $\int_1^2 |2x - [3x]| dx$ is

(2-09-2020/Shift-2)

61. $\int_{-\pi}^{\pi} |\pi - |x|| dx$ is equal to :

(03-09-2020/Shift-1)

(a) π^2

(b) $\frac{\pi^2}{2}$

(c) $\sqrt{2}\pi^2$

(d) $2\pi^2$

62. The area (in sq. units) of the region

$\left\{ (x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, \frac{1}{2} \leq x \leq 2 \right\}$ is

(3-09-2020/Shift-1)

(a) $\frac{23}{16}$

(b) $\frac{79}{16}$

(c) $\frac{23}{6}$

(d) $\frac{79}{24}$

63. If the value of the integral $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ is $\frac{k}{6}$, then k is equal to :

(3-09-2020/Shift-2)

(a) $2\sqrt{3} + \pi$

(b) $3\sqrt{2} + \pi$

(c) $3\sqrt{2} - \pi$

(d) $2\sqrt{3} - \pi$

64. Let $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx (x \geq 0)$. Then $f(3) - f(1)$ is equal to :

(4-09-2020/Shift-1)

(a) $-\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

(b) $\frac{\pi}{6} + \frac{1}{2} - \frac{\sqrt{3}}{4}$

(c) $-\frac{\pi}{12} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

(d) $\frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$

65. Let $f(x) = |x - 2|$ and $g(x) = f(f(x))$, $x \in [0, 4]$. Then

$\int_0^3 (g(x) - f(x)) dx$ is equal to :

(4-09-2020/Shift-1)

(a) $\frac{1}{2}$

(b) 0

(c) 1

(d) $\frac{3}{2}$

66. Let f be a twice differentiable function on $(1, 6)$. If $f(2) = 8, f'(2) = 5, f'(x) \geq 1$ and $f''(x) \geq 4$ for all $x \in (1, 6)$ then :

(04-09-2020/Shift-1)

(a) $f(5) + f'(5) \geq 28$

(b) $f'(5) + f''(5) \leq 20$

(c) $f(5) \leq 10$

(d) $f(5) + f'(5) \leq 26$



67. The integral

$$\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$$

is equal to:

(4-9-2020/Shift-2)

(a) $\frac{9}{2}$ (b) $-\frac{1}{18}$

(c) $-\frac{1}{9}$ (d) $\frac{7}{18}$

68. Let $\{x\}$ and $[x]$ denote the fractional part of x and the greatest integer $\leq x$ respectively of real number x . If

$$\int_0^n \{x\} dx, \int_0^n [x] dx \text{ and } 10(n^2 - n), (n \in N, n > 1) \text{ are three}$$

consecutive terms of a G.P., then n is equal to

(4-9-2020/Shift-2)

69. The value of $\int_{-\pi/2}^{\pi/2} \frac{1}{1 + e^{\sin x}} dx$ is:

(5-09-2020/Shift-1)

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$

(c) π (d) $\frac{3\pi}{2}$

70. The area (in sq. units) of the region

$$A = \{(x, y) : (x-1)[x] \leq y \leq 2\sqrt{x}, 0 \leq x \leq 2\} \text{ where } e[t]$$

denotes the greatest integer function, is:

(5-09-2020/Shift-2)

(a) $\frac{4}{3}\sqrt{2} - \frac{1}{2}$ (b) $\frac{8}{3}\sqrt{2} - \frac{1}{2}$

(c) $\frac{8}{3}\sqrt{2} - 1$ (d) $\frac{4}{3}\sqrt{2} + 1$

71. The area (in sq. units) of the region

$$A = \{(x, y) : |x| + |y| \leq 1, 2y^2 \geq |x|\} \text{ (6-09-2020/Shift-1)}$$

(a) $\frac{1}{6}$ (b) $\frac{5}{6}$

(c) $\frac{1}{3}$ (d) $\frac{7}{6}$

72. If $I_1 = \int_0^1 (1-x^{50})^{100} dx$ and $I_2 = \int_0^1 (1-x^{50})^{101} dx$ such that $I_2 = \alpha I_1$ then α equal to :

(6-09-2020/Shift-1)

(a) $\frac{5050}{5049}$ (b) $\frac{5050}{5051}$

(c) $\frac{5051}{5050}$ (d) $\frac{5049}{5050}$

73. The integral $\int_1^2 e^x \cdot x^x (2 + \log_e x) dx$ equal :

(6-09-2020/Shift-2)

(a) $e(4e-1)$ (b) $e(4e+1)$

(c) $4e^2 - 1$ (d) $e(2e-1)$

74. The area (in sq. units) of the region enclosed by the curves

$$y = x^2 - 1 \text{ and } y = 1 - x^2 \text{ is equal to :}$$

(6-09-2020/Shift-2)

(a) $\frac{4}{3}$ (b) $\frac{7}{2}$

(c) $\frac{16}{3}$ (d) $\frac{8}{3}$



75. The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$

(7-01-2020/Shift-1)

- (a) $\frac{1}{3}(12\pi - 1)$ (b) $\frac{1}{3}(6\pi - 1)$
(c) $\frac{1}{6}(12\pi - 1)$ (d) $\frac{1}{6}(24\pi - 1)$

76. If $f(a+b+1-x) = f(x) \forall x$ where a and b are fixed positive real numbers, then

$$\frac{1}{(a+b)} \int_a^b x(f(x) + f(x+1)) dx \text{ is equal to}$$

(7-01-2020/Shift-1)

- (a) $\int_{a-1}^{b-a} f(x) dx$ (b) $\int_{a+1}^{b+1} f(x+1) dx$
(c) $\int_{a-1}^{b-1} f(x+1) dx$ (d) $\int_{a+1}^{b+1} f(x) dx$

77. The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is:

(07-01-2020/Shift-2)

- (a) $\log_e 2$ (b) $\log_e \sqrt{2}$
(c) $\log_e \left(\frac{4}{3}\right)$ (d) $\log_e \left(\frac{3}{2}\right)$

78. If θ_1 and θ_2 be respectively the smallest and the largest values of θ in $[0, 2\pi) - \{\pi\}$ which satisfy the equation,

$$2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0 \text{ then } \int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta \text{ is equal to}$$

(07-01-2020/Shift-2)

- (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{3} + \frac{1}{6}$ (d) $\frac{\pi}{9}$

79. The area (in sq. units) of the region

$$\{(x, y) \in R \mid 4x^2 \leq y \leq 8x + 12\} \text{ is:}$$

(07-01-2020/Shift-2)

- (a) $\frac{125}{3}$ (b) $\frac{128}{3}$
(c) $\frac{124}{3}$ (d) $\frac{127}{3}$

80. For $a > 0$, let the curves $C_1 : y^2 = ax$ and $C_2 : x^2 = ay$

intersect at origin O and a point P. let the line $x = b$ ($0 < b < a$) intersect the chord OP and the x-axis at points Q and R, respectively. If the line $x = b$ bisects the area bounded by the curves, C_1 and C_2 , and the area of

$$\Delta OQR = \frac{1}{2}, \text{ then 'a' satisfies the equation:}$$

(8-01-2020/Shift-1)

- (a) $x^6 - 12x^3 + 4 = 0$ (b) $x^6 - 12x^3 - 4 = 0$
(c) $x^6 + 6x^3 - 4 = 0$ (d) $x^6 - 6x^3 + 4 = 0$

81. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then:

(8-01-2020/Shift-2)

- (a) $\frac{1}{6} < I^2 < \frac{1}{2}$ (b) $\frac{1}{8} < I^2 < \frac{1}{4}$
(c) $\frac{1}{9} < I^2 < \frac{1}{8}$ (d) $\frac{1}{16} < I^2 < \frac{1}{9}$

82. The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 \leq y \leq 3 - 2x\}$, is:

(8-01-2020/Shift-2)

- (a) $\frac{31}{3}$ (b) $\frac{32}{3}$
(c) $\frac{29}{3}$ (d) $\frac{34}{3}$

83. The value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$ is equal to:

(9-01-2020/Shift-1)

- (a) 2π (b) 4π
(c) $2\pi^2$ (d) π^2



84. If for all real triplets (a, b, c) ,

$f(x) = a + bx + cx^2$ then $\int_0^1 f(x) dx$ is equal to:

(9-01-2020/Shift-1)

(a) $2 \left(3f(1) + 2f\left(\frac{1}{2}\right) \right)$

(b) $\frac{1}{3} \left(f(0) + f\left(\frac{1}{2}\right) \right)$

(c) $\frac{1}{2} \left(f(1) + 3f\left(\frac{1}{2}\right) \right)$

(d) $\frac{1}{6} \left(f(0) + f(1) + 4f\left(\frac{1}{2}\right) \right)$

85. Given:

$$f(x) = \begin{cases} x, & 0 \leq x < \frac{1}{2} \\ \frac{1}{2}, & x = \frac{1}{2} \\ 1-x, & \frac{1}{2} < x \leq 1 \end{cases}$$

and $g(x) = \left(x - \frac{1}{2}\right)^2, x \in R$. Then the area (in sq. units) of the region bounded by the curves $y = f(x)$ and $y = g(x)$ between the lines $2x = 1$ to $2x = \sqrt{3}$ is

(9-1-2020/Shift-2)

(a) $\frac{\sqrt{3}}{4} - \frac{1}{3}$

(b) $\frac{1}{3} + \frac{\sqrt{3}}{4}$

(c) $\frac{1}{2} + \frac{\sqrt{3}}{4}$

(d) $\frac{1}{2} - \frac{\sqrt{3}}{4}$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. If $f(x) = A \sin\left(\frac{\pi x}{2}\right) + B$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and

$$\int_0^1 f(x) dx = \frac{2A}{\pi}, \text{ then constants A and B are :}$$

- (a) $\frac{\pi}{2}$ and $\frac{\pi}{2}$ (b) $\frac{2}{\pi}$ and $\frac{3}{\pi}$
(c) 0 and $-\frac{4}{\pi}$ (d) $\frac{4}{\pi}$ and 0

2. The value of $\int_{\pi}^{2\pi} [2 \sin x] dx$ where $[.]$ represents the greatest integral functions, is :

- (a) $-\frac{5\pi}{3}$ (b) $-\pi$
(c) $\frac{5\pi}{3}$ (d) -2π

3. $\int_3^{29} \frac{\sqrt[3]{(x-2)^2}}{3 + \sqrt[3]{(x-2)^2}} dx =$

- (a) $4 + \frac{3\sqrt{3}}{2}\pi$ (b) $2 + \frac{3\sqrt{3}}{2}\pi$
(c) $4 + \frac{\sqrt{3}}{2}\pi$ (d) $8 + \frac{3\sqrt{3}}{2}\pi$

4. $\int_0^{1/\sqrt{3}} \frac{dx}{(2x^2+1)\sqrt{x^2+1}} =$

- (a) $\frac{\pi}{2}$ (b) $\tan^{-1} 2$
(c) $\tan^{-1} 1/2$ (d) π

5. If $a \neq b$ and $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$

for all $x \neq 0$, then

$$\int_1^2 f(x) dx = \frac{1}{a^2 - b^2} \left[a(\log 2 - \alpha) + \beta \left(\frac{b}{2} \right) \right]$$

where $\beta - \alpha$ is equal to

- (a) 12 (b) 5
(c) 7 (d) 2

6. $\int_{-\pi}^{\pi} (\cos px - \sin qx)^2 dx$ is equal to (where $p, q \in \mathbb{Z}$)

- (a) 0 (b) $\frac{\pi}{2}$
(c) π (d) 2π

7. $\int_1^e \frac{1 + \log_{10} x}{x} dx =$

- (a) $\frac{1}{2} \log_{10} e$ (b) $\frac{1 + \log_{10} e}{2}$
(c) $\frac{1}{2} \log_{10} e + 1$ (d) $2 \log_{10} e$

8. $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)} =$

- (a) $2(\tan e - 1)$ (b) $2 \tan e$
(c) 1 (d) $\tan e + \cot e$

9. $\int_0^{\log_5 5} \frac{e^x \sqrt{e^x - 1}}{e^x + 3} dx =$

- (a) $3 + \pi$ (b) $3 - \pi$
(c) $4 + \pi$ (d) $4 - \pi$



10. $\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \pi \frac{\cos \alpha}{1 + \sin^2 \alpha}$

- (a) for no value of α
 (b) for exactly two values of α in $(0, \pi)$
 (c) for at least one α in $\left(\frac{\pi}{2}, \pi\right)$
 (d) for exactly one α in $\left(0, \frac{\pi}{2}\right)$

11. $\int_0^{1/2} e^x \left(\sin^{-1} x - \frac{x}{(1-x^2)^{3/2}} \right) dx =$

- (a) $\sqrt{e} \left(\frac{\pi}{6} - \frac{1}{\sqrt{3}} \right) + 1$ (b) $\sqrt{e} \left(\frac{\pi}{6} + \frac{2}{\sqrt{3}} \right) + 1$
 (c) $\sqrt{e} \left(\frac{\pi}{6} - \frac{2}{\sqrt{3}} \right) + 1$ (d) $\sqrt{e} \left(\frac{\pi}{6} + \frac{1}{\sqrt{3}} \right) + 1$

12. $\int_{-2}^2 |x(x-1)| dx =$

- (a) $\frac{17}{3}$ (b) $\frac{11}{3}$
 (c) $\frac{13}{3}$ (d) $\frac{16}{3}$

13. If $[t]$ stands for the integral part of t , then

$\int_0^{5\pi/12} [\tan x] dx =$

- (a) $\frac{\pi}{2}$ (b) π
 (c) $\frac{\pi}{4}$ (d) 2π

14. If $[t]$ denotes the integral part of t , then

$\int_0^1 \cos(\pi x) \cos([2x]\pi) dx =$

- (a) 1 (b) -1
 (c) $-\frac{2}{\pi}$ (d) $\frac{2}{\pi}$

15. Let $f(x) = \begin{cases} \int_0^x \{5 + |1-y|\} dy & \text{if } x > 2 \\ 5x+1 & \text{if } x \leq 2 \end{cases}$

Then

- (a) $f(x)$ is continuous but not differentiable at $x = 2$
 (b) $f(x)$ is not continuous at $x = 2$
 (c) $f(x)$ is differentiable everywhere
 (d) The right derivative of $f(x)$ at $x = 2$ does not exist

16. Consider the function $f(x) = \int_0^x [t] dt$

where $x > 0$ and $[t]$ is the integral part of t . Then

- (a) $f(x)$ is not defined for $x = 1, 2, 3, \dots$
 (b) $f(x)$ is defined for all $x > 0$ but is not continuous at $x = 1, 2, 3, \dots$
 (c) $f(x)$ is continuous for all $x > 0$
 (d) $f(x)$ is differentiable for all $x > 0$

17. $\int_0^{\pi} \frac{x^2 \sin 2x \sin[(\pi/2)\cos x]}{2x - \pi} dx =$

- (a) $\frac{4}{\pi^2}$ (b) $\frac{\pi^2}{4}$
 (c) $\frac{\pi^2}{8}$ (d) $\frac{8}{\pi}$



18. If $\int_0^{\pi} \frac{x^2}{(1+\sin x)^2} dx = A$ then $\int_0^{\pi} \frac{2x^2 \cos^2(x/2)}{(1+\sin x)^2} dx =$

- (a) $A + \pi - \pi^2$ (b) $A - \pi + \pi^2$
(c) $A - \pi - \pi^2$ (d) $A + 2\pi - \pi^2$

19. Let $f(x) = \frac{e^x}{1+e^x}$

$$I_1 = \int_{f(-a)}^{f(a)} xg(x(1-x)) dx$$

$$I_2 = \int_{f(-a)}^{f(a)} g(x(1-x)) dx$$

then I_2 / I_1 is

- (a) 1 (b) -3
(c) -1 (d) 2

20. $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x + 1} dx =$

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{4\sqrt{3}}$ (d) $\frac{\pi}{2\sqrt{3}}$

21. $\int_0^1 \frac{dx}{(5+2x-2x^2)(1-e^{2-4x})} =$

- (a) $\frac{1}{\sqrt{11}} \log_e \left(\frac{\sqrt{11}+2}{\sqrt{11}} \right)$ (b) $\frac{1}{\sqrt{11}} \log_e \left(\frac{\sqrt{11}+1}{\sqrt{10}} \right)$
(c) $\frac{1}{\sqrt{10}} \log_e \left(\frac{\sqrt{10}+2}{\sqrt{11}} \right)$ (d) $\frac{1}{\sqrt{10}} \log_e \left(\frac{\sqrt{10}+1}{\sqrt{11}} \right)$

22. For any integer n, the integral $\int_0^{\pi} e^{\cos^2 x} \cos^3(2n+1)x dx$ has the value :

- (a) π (b) 1
(c) 0 (d) none of these

23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions. Then the value of the integral

$$\int_{-\pi/2}^{\pi/2} [f(x) + f(-x)][g(x) - g(-x)] dx \text{ is :}$$

- (a) π (b) 1
(c) -1 (d) 0

24. $\int_0^{\sin^2 x} (\sin^{-1} \sqrt{t}) dt + \int_0^{\cos^2 x} (\cos^{-1} \sqrt{t}) dt =$

- (a) π (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2\sqrt{2}}$

25. For each positive integer n, define

$$f_n(x) = \min \left(\frac{x^n}{n!}, \frac{(1-x)^n}{n!} \right) \text{ for } 0 \leq x \leq 1. \text{ Let}$$

$$I_n = \int_0^1 f_n(x) dx \text{ for } n \geq 1. \text{ Then the value of } \sum_{n=1}^{\infty} I_n \text{ is}$$

- (a) $2\sqrt{e} - 3$ (b) $2\sqrt{e} - 2$
(c) $2\sqrt{e} - 1$ (d) $2\sqrt{e}$

26. $\int_0^{2\pi} \frac{e^{|\sin x|} \cos x}{1 + e^{\tan x}} dx =$

- (a) e^{π} (b) 1
(c) $e^{\pi} - 1$ (d) 0

27. If $m = \int_{-2}^0 \frac{|\sin x|}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$ and $n = \int_0^2 \frac{|\sin x|}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$, where $[.]$

represents greatest integer function, then

- (a) $m = n$ (b) $m = -n$
(c) $m = 2n$ (d) $m = -2n$

28. $\int_0^{\pi/2} f(\sin 2x) \sin x dx = k \int_0^{\pi/4} f(\cos 2x) \cos x dx$ where k

equals

- (a) 2 (b) 4
(c) $\sqrt{2}$ (d) $2\sqrt{2}$



29. If $I_1 = \int_{-4}^{-5} e^{(x+5)^2} dx$ and $I_2 = 3 \int_{1/3}^{2/3} e^{9[x-(2/3)]^2} dx$ then the value of $I_1 + I_2$ is
- (a) 0 (b) 1
(c) e^{-1} (d) e
30. If $I = \int_0^1 \frac{e^t}{t+1} dt$, then $\int_{a-1}^a \frac{e^{-t}}{t-a-1} dt =$
- (a) Ie^a (b) $(-I)e^a$
(c) $(-I)e^{-a}$ (d) Ie^{-a}
31. The maximum value of $\int_{a-1}^{a+1} e^{-(x-1)^2} dx$ is attained (a is real) at
- (a) $a = 2$ (b) $a = 1$
(c) $a = -1$ (d) $a = 0$
32. If $f(x)$ is differentiable & defined on R^+ such that $\int_0^{t^2} xf(x) dx = \frac{2}{3} t^5$ then $f(4/25) =$
- (a) $\frac{2}{3}$ (b) $-\frac{3}{2}$
(c) 1 (d) $\frac{3}{2}$
33. The function $F(x) = \int_{\pi/6}^x (4 \sin t + 3 \cos t) dt$ attains least value on $[\pi/4, 3\pi/4]$ at x equals.
- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{3}$
(c) $\frac{3\pi}{4}$ (d) $\frac{\pi}{4}$
34. If $f(x) = \int_{\pi^2/16}^{x^2} \frac{\cos x \cos \sqrt{t}}{1 + \sin^2 \sqrt{t}} dt$ then $f'(\pi)$ is equal to
- (a) 0 (b) π
(c) 2π (d) $\frac{\pi}{2}$
35. If $f(t) = \begin{cases} at-1 & t < 1 \\ t^2+b & t \geq 1 \end{cases}$ then possible set of value of (a, b) so that $\int_0^x f(x) dx$ is differentiable for all $x \geq 0$ is
- (a) (5, 1) (b) (1, 3)
(c) (4, 2) (d) none of these
36. If $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1+t^4} dt$ then $f'(2)$ has the value equal to
- (a) $2/17$ (b) 0
(c) 1 (d) cannot be determined
37. The value of the function $f(x) = 1 + x + \int_1^x (ln^2 t + 2 lnt) dt$ where $f'(x)$ vanishes is
- (a) e^{-1} (b) 0
(c) $2e^{-1}$ (d) $1 + 2e^{-1}$
38. $\lim_{h \rightarrow 0} \frac{\int_a^{x+h} ln^2 t dt - \int_a^x ln^2 t dt}{h}$ equals to
- (a) 0 (b) $ln^2 x$
(c) $\frac{2 lnx}{x}$ (d) does not exist
39. Let $\lambda = \int_0^1 \frac{dx}{1+x^3}$, $p = \lim_{n \rightarrow \infty} \left[\frac{\prod_{r=1}^n (n^3 + r^3)}{n^{3n}} \right]^{1/n}$, then $\ln p$ is equal to
- (a) $\ln 2 - 1 + \lambda$ (b) $\ln 2 - 3 + 3\lambda$
(c) $2 \ln 2 - \lambda$ (d) $\ln 4 - 3 + 3\lambda$
40. The value of the definite integral $\int_0^1 (1 + e^{-x^2}) dx$ is :
- (a) -1 (b) 2
(c) $1 + e^{-1}$ (d) none of these



41. Consider the integrals

$$I_1 = \int_0^1 e^{-x} \cos^2 x \, dx, I_2 = \int_0^1 e^{-x^2} \cos^2 x \, dx,$$

$$I_3 = \int_0^1 e^{-\frac{x^2}{2}} \cos^2 x \, dx, I_4 = \int_0^1 e^{-\frac{x^2}{2}} \, dx \text{ Then}$$

$$(a) I_2 > I_4 > I_1 > I_3 \quad (b) I_2 < I_4 < I_1 < I_3$$

$$(c) I_1 < I_2 < I_3 < I_4 \quad (d) I_1 > I_2 > I_3 > I_4$$

42. If $I_n = \int_1^e (\log_e x)^n \, dx$ (n is a positive integer), then

$$I_{2012} + (2012)I_{2011} =$$

$$(a) I_{2011} + (2010)I_{2010} \quad (b) I_{2013} + (2013)I_{2012}$$

$$(c) I_{2011} + (2010)I_{2009} \quad (d) I_{2012} - (2012)I_{2011}$$

43. If $\int_0^x e^{zx} \cdot e^{-z^2} \, dz = f(x) \int_0^x e^{-z^2/4} \, dz$

$$\text{then } \int e^x \left(\log_e (f(x)) + \frac{x}{2} \right) dx =$$

$$(a) \frac{xe^x}{2} + c \quad (b) \frac{x^2 e^x}{4} + c$$

$$(c) \frac{x^2 e^x}{2} + c \quad (d) \frac{xe^x}{4} + c$$

44. If $\int_0^x f(t) \, dt = x + \int_x^1 t f(t) \, dt$

then $f(1)$ is

$$(a) \frac{1}{2} \quad (b) 0$$

$$(c) 1 \quad (d) \frac{-1}{2}$$

45. A function $f(x)$ which satisfies the relation

$$f(x) = e^x + \int_0^1 e^x f(t) \, dt, \text{ then } f(x) \text{ is}$$

$$(a) \frac{e^x}{2-e} \quad (b) (e-2)e^x$$

$$(c) 2e^x \quad (d) \frac{e^x}{2}$$

46. The area bounded by the curve $y = 3 + 2x - x^2$, $y = 0$ & the ordinate at $x = 1$ & $x = 4$ is

$$(a) 25/3 \quad (b) 23/3$$

$$(c) 19/3 \quad (d) \text{none}$$

47. The area bounded by the curve $y = x(1 - \ln x)$ and positive x -axis between $x = e^{-1}$ and $x = e$ is

$$(a) \left(\frac{e^2 - 4e^{-2}}{5} \right) \quad (b) \left(\frac{e^2 - 5e^{-2}}{4} \right)$$

$$(c) \left(\frac{4e^2 - e^{-2}}{5} \right) \quad (d) \left(\frac{5e^2 - e^{-2}}{4} \right)$$

48. The area enclosed by $y = x^3$, its normal at $(1,1)$ and x -axis is equal to

$$(a) \frac{7}{4} \quad (b) \frac{9}{4}$$

$$(c) \frac{5}{4} \quad (d) \frac{8}{4}$$

49. Area of the region bounded by $x = 0$, $y = 0$, $x = 2$, $y = 2$, $y \leq e^x$ and $y \geq \ln x$, is

$$(a) 6 - 4 \ln 2 \quad (b) 4 \ln 2 - 2$$

$$(c) 2 \ln 2 - 4 \quad (d) 6 - 2 \ln 2$$

50. The area bounded by the curve $y = e^x$ and the lines $y = |x-1|$, $x = 2$ is given by

$$(a) e^2 + 1 \quad (b) e^2 - 1$$

$$(c) e^2 - 2 \quad (d) \text{none}$$



51. The area enclosed by the curves

$$y = \sqrt{4-x^2}, y \geq \sqrt{2} \sin\left(\frac{x\pi}{2\sqrt{2}}\right) \text{ and } x\text{-axis is divided}$$

by y -axis in the ratio

- (a) $\frac{\pi^2-8}{\pi^2+8}$ (b) $\frac{\pi^2-4}{\pi^2+4}$
(c) $\frac{\pi-4}{\pi+4}$ (d) $\frac{2\pi^2}{2\pi+\pi^2-8}$

52. The area of the closed figure bounded by the curves

$$y = \sqrt{x}, y = \sqrt{4-3x} \text{ \& } y = 0 \text{ is}$$

- (a) $4/9$ (b) $8/9$
(c) $16/9$ (d) none

53. The area of the closed figure bounded by the curves

$$y = \cos x, y = 1 + \frac{2}{\pi}x \text{ \& } x = \frac{\pi}{2} \text{ is}$$

- (a) $\frac{\pi+4}{4}$ (b) $\frac{3\pi}{4}$
(c) $\frac{3\pi+4}{4}$ (d) $\frac{3\pi-4}{4}$

54. The ratio in which the curve $y = x^2$ divides the region

bounded by the curve; $y = \sin\left(\frac{\pi x}{2}\right)$ \& the x -axis as

x varies from 0 to 1, is

- (a) $2 : \pi$ (b) $1 : 3$
(c) $3 : \pi$ (d) $(6-\pi) : \pi$

55. The area bounded by $y = 2 - |2-x|$ and $y = \frac{3}{|x|}$ is

- (a) $\frac{4+3\ln 3}{2}$ (b) $\frac{4-3\ln 3}{2}$
(c) $\frac{3}{2} + \ln 3$ (d) $\frac{1}{2} + \ln 3$

56. The area of the region enclosed between the curves $7x^2 + 9y + 9 = 0$ and $5x^2 + 9y + 27 = 0$ is

- (a) 2 (b) 4
(c) 8 (d) 16

57. The area of the plane figure bounded in first quadrant by $y = x^{1/3}$; $y = -x^2 + 2x + 3$; $y = 2x - 1$ and the axis of ordinates is

- (a) $12/55$ (b) $55/12$
(c) $32/55$ (d) none

58. If $f(x) = \sin x \forall x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) =$

$$2 \forall x \in \left(\frac{\pi}{2}, \pi\right] \text{ and } f(x) = f(2\pi - x) \forall x \in (\pi, 2\pi],$$

then the area enclosed by $y = f(x)$ and x -axis is

- (a) π (b) 2π
(c) 2 (d) 4

59. Value of the parameter a such that the area bounded by $y = a^2x^2 + ax + 1$, co-ordinate axes and the line $x = 1$, attains its least value, is equal to

- (a) $-\frac{1}{4}$ (b) $-\frac{1}{2}$
(c) $-\frac{3}{4}$ (d) -1

Objective Questions II [One or more than one correct option]

60. $\int_0^\infty \frac{x}{(1+x)(1+x^2)} dx$

- (a) $\pi/4$
(b) $\pi/2$
(c) is same as $\int_0^\infty \frac{dx}{(1+x)(1+x^2)}$
(d) cannot be evaluated



61. If $f(2-x) = f(2+x)$ and $f(4-x) = f(4+x)$ and

$f(x)$ is a function for which $\int_0^2 f(x) dx = 5$, then

$\int_0^{50} f(x) dx$ is equal to

(a) 125 (b) $\int_{-4}^{46} f(x) dx$

(c) $\int_1^{51} f(x) dx$ (d) $\int_2^{52} f(x) dx$

62. If $F(x) = \frac{1}{x^2} \int_4^x (4t^2 - 2F'(t)) dt$, then $F'(4)$ equals –

(a) $\frac{32}{9}$ (b) $\frac{64}{9}$

(c) $\frac{(2F(4))^5}{9}$ (d) $\frac{11F(8)}{28}$

63. If $f(x)$ is integrable over $[1, 2]$, then $\int_1^2 f(x) dx$ is equal

to

(a) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$ (b) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=n+1}^{2n} f\left(\frac{r}{n}\right)$

(c) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r+n}{n}\right)$ (d) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} f\left(\frac{r}{n}\right)$

64. If $I = \int_0^{\pi/2} e^{-\alpha \sin x} dx$, where $\alpha \in (0, \infty)$, then

(a) $I < \frac{\pi}{2}$ (b) $I > \frac{\pi}{2} (e^{-\alpha} + 1)$

(c) $I > \frac{\pi}{2} e^{-\alpha}$ (d) $I > 0$

65. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, $n \in N$, then which of the following statements hold good ?

(a) $2nI_{n+1} = 2^{-n} + (2n-1)I_n$

(b) $I_2 = \frac{\pi}{8} + \frac{1}{4}$

(c) $I_2 = \frac{\pi}{8} - \frac{1}{4}$

(d) $I_3 = \frac{\pi}{16} - \frac{5}{48}$

Numerical Value Type Questions

66. Determine a positive integer $n \leq 5$, such that

$$\int_0^1 e^x (x-1)^n dx = 16 - 6e$$

ASSERTION REASON

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
 (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
 (C) If ASSERTION is true, REASON is false.
 (D) If ASSERTION is false, REASON is true.

67. Assertion :

$$\int e^{\sin^{-1} x} \left(1 - \frac{x}{\sqrt{1-x^2}} \right) dx = e^{\sin^{-1} x} \cdot \sqrt{1-x^2} + c$$

Reason : $\int e^{g(x)} (g'(x)f(x) + f'(x)) dx = e^{g(x)} f(x) + c$

- (a) A (b) B (c) C (d) D

68. Assertion : $\int_0^{\pi} x \sin x \cos^2 x dx = \frac{\pi}{2} \int_0^{\pi} \sin x \cos^2 x dx$

Reason : $\int_a^b x f(x) dx = \frac{a+b}{2} \int_a^b f(x) dx$

- (a) A (b) B (c) C (d) D



69. Assertion : $1 \leq \int_0^{\pi/2} \frac{\sin x}{x} dx \leq \frac{\pi}{2}$

Reason : If $f(x)$ is continuous in $[a, b]$ and m and l are greatest and least value of $f(x)$ in $[a, b]$, then

$$l(b-a) \leq \int_a^b f(x) dx \leq m(b-a)$$

(a) A (b) B (c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

70. Column - I Column - II

(A) If $\int_0^1 \frac{\cos x}{1+x} dx = k$ and (P) 4

$$\int_{6\pi-3}^{6\pi} \frac{\cos(x/3)}{6\pi+3-x} dx = mk, \text{ then } m \text{ is}$$

(B) $\int_{-1}^1 \left(\sin^{-1} \left[x + \frac{3}{4} \right] \right) dx = \frac{\pi}{k}$, then (Q) 1

k is {where $[.]$ denotes GIF}

(C) If $f(x) = \max(x - |x|, x + [x])$ and (R) 3

$$\int_{-3}^3 f(x) dx = -\frac{3}{k}, \text{ then } k \text{ is}$$

{where $[.]$ denotes GIF}

(D) If $\int_0^{20} \sqrt{1 - \cos \pi x} dx = \frac{10k\sqrt{2}}{\pi}$, (S) 2

then k is

The correct matching is :

- (a) A-Q; B-P; C-S; D-P
- (b) A-Q; B-P; C-S; D-R
- (c) A-Q; B-R; C-Q; D-P
- (d) A-S; B-P; C-S; D-P

71. Column - I Column - II

(A) The area bounded by the curve (P) 2

$y = x + \sin x$ and its inverse function between the ordinates $x = 0$ to $x = 2\pi$ is $4s$. Then the value of s is

(B) The area bounded by $y = xe^{|x|}$ (Q) 1

and lines $|x| = 1, y = 0$ is

(C) The area bounded by the curves (R) $\frac{32}{5}$

$y^2 = x^3$ and $|y| = 2x$ is

(D) The smaller area included (S) $\frac{1}{3}$

between the curves $\sqrt{x} + \sqrt{y} = 1$

and $|x| + |y| = 1$ is

The correct matching is :

- (a) A-P; B-Q; C-R; D-S
- (b) A-Q; B-P; C-R; D-S
- (c) A-P; B-R; C-S; D-P
- (d) A-P; B-P; C-R; D-S

Using the following passage, solve Q.72 to Q.74

Passage - 1

Using integral $\int_0^{\pi/2} \ln(\sin x) dx$

$$= - \int_0^{\pi/2} \ln(\sec x) dx = -\frac{\pi}{2} \ln 2,$$

$$\int_0^{\pi/2} \ln(\tan x) dx = 0 \text{ and } \int_0^{\pi/4} \ln(1 + \tan x) dx = \frac{\pi}{8} \ln 2.$$

72. Evaluate $\int_0^{\pi/4} \ln(\sin 2x) dx =$

- (a) $\frac{-\pi \ln 2}{2}$ (b) $\pi \ln 2$
- (c) $\frac{\pi \ln 2}{4}$ (d) none of these



73. Evaluate $\int_{-\pi/4}^{\pi/4} \ln \left(\frac{\sin x + \cos x}{\cos x - \sin x} \right) dx =$

- (a) $\pi \ln 2$ (b) $\frac{\pi \ln 2}{2}$
(c) 0 (d) $-\pi \ln 2$

74. Evaluate $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx =$

- (a) $\frac{\pi \ln 2}{2}$ (b) $\frac{-\pi \ln 2}{4}$
(c) $\pi \ln 2$ (d) 0

Using the following passage, solve Q.75 to Q.77

Passage – 1

Consider the curve defined implicitly by the equation

$$y^2 - 2ye^{\sin^{-1} x} + x^2 - 1 + [x] + e^{2\sin^{-1} x} = 0, \text{ Where } [x]$$

denotes the greatest integer function

75. The area of the region bounded by the curve between the lines $x = -1$ and $x = 0$ is

- (a) $\frac{\pi}{2} + 1$ (b) $\pi - 1$
(c) $\pi + 1$ (d) $\frac{\pi}{2} - 1$

76. The area of the region bounded by the curve between the lines $x = 0$ & $x = 1$ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{2} - 1$
(c) $\frac{\pi}{2} + 1$ (d) $\pi + 1$

77. The Area of the region bounded by the curve between the lines $x = 0$ & $x = \frac{1}{2}$ is

- (a) $\frac{\sqrt{3}}{4} + \frac{\pi}{6}$ (b) $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$
(c) $\frac{\sqrt{3}}{4} - \frac{\pi}{6}$ (d) $\frac{\sqrt{3}}{2} - \frac{\pi}{6}$

Subjective Type

78. Evaluate $\int_2^3 \frac{2x^5 + x^4 - 2x^3 + 2x^2 + 1}{(x^2 + 1)(x^4 - 1)} dx.$

79. Determine the value of $\int_{-\pi}^{\pi} \frac{2x(1 + \sin x)}{1 + \cos^2 x} dx$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. Let $g(x) = \int_0^x f(t) dt$, where f is such that $\frac{1}{2} \leq f(t) \leq 1$ for $t \in [0, 1]$ and $0 \leq f(t) \leq \frac{1}{2}$ for $t \in [1, 2]$. Then $g(2)$ satisfies the inequality. (2000)

(a) $-\frac{3}{2} \leq g(2) < \frac{1}{2}$ (b) $0 \leq g(2) < 2$

(c) $\frac{3}{2} < g(2) \leq 5/2$ (d) $2 < g(2) < 4$

2. The value of the integral $\int_{e^{-1}}^{e^2} \left| \frac{\log_e x}{x} \right| dx$ is : (2000)

(a) $3/2$ (b) $5/2$
(c) 3 (d) 5

3. If $f(x) = \begin{cases} e^{\cos x} \sin x, & \text{for } |x| \leq 2, \\ = 2, & \text{otherwise} \end{cases}$

then $\int_{-2}^3 f(x) dx$, is equal to (2000)

(a) 0 (b) 1
(c) 2 (d) 3

4. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$ is : (2001)

(a) π (b) $a\pi$
(c) $\pi/2$ (d) 2π

5. Let $f: (0, \infty) \rightarrow \mathbb{R}$ and $F(x) = \int_0^x f(t) dt$. If

$F(x^2) = x^2(1+x)$, then $f(4)$ equals : (2001)

(a) $5/4$ (b) 7
(c) 4 (d) 2

6. Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Then the real roots of the equation

$x^2 - f'(x) = 0$ are : (2002)

(a) ± 1 (b) $\pm \frac{1}{\sqrt{2}}$

(c) $\pm \frac{1}{2}$ (d) 0 and 1

7. Let $T > 0$ be a fixed real number, Suppose f is a continuous function such that for all $x \in \mathbb{R}$, $f(x+T) = f(x)$. If

$I = \int_0^T f(x) dx$, then the value of $\int_3^{3+3T} f(2x) dx$ is :

(2002)

(a) $3/2 I$ (b) I

(c) $3 I$ (d) $6 I$

8. The integral $\int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ equals (2002)

(a) $-\frac{1}{2}$ (b) 0

(c) 1 (d) $2 \ln \left(\frac{1}{2} \right)$

9. If $I(m, n) = \int_0^1 t^m (1+t)^n dt$, then the expression for $I(m, n)$

in terms of $I(m+1, n-1)$ is : (2003)

(a) $\frac{2^n}{m+1} - \frac{n}{m+1} I(m+1, n-1)$

(b) $\frac{n}{m+1} I(m+1, n-1)$

(c) $\frac{2^n}{m+1} + \frac{n}{m+1} I(m+1, n-1)$

(d) $\frac{m}{m+1} I(m+1, n-1)$



10. If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$, then $f(x)$ increases in : (2003)

- (a) (2, 2) (b) no value of x
(c) (0, ∞) (d) $(-\infty, 0)$

11. The value of the integral $\int_0^1 \sqrt{\frac{1-x}{1+x}} dx$ is (2004)

- (a) $\frac{\pi}{2} + 1$ (b) $\frac{\pi}{2} - 1$
(c) -1 (d) 1

12. If $f(x)$ is differentiable and $\int_0^{t^2} x f(x) dx = \frac{2}{5} t^5$, then $f(4/25)$ equals : (2004)

- (a) $2/5$ (b) $-5/2$
(c) 1 (d) $5/2$

13. The value of $\int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1) \cos(x+1)] dx$ is: (2005)

- (a) 0 (b) 3
(c) 4 (d) 1

14. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x \forall x \in (0, \pi/2)$ then $f\left(\frac{1}{\sqrt{3}}\right)$ is : (2005)

- (a) 3 (b) $\sqrt{3}$
(c) $1/3$ (d) none of these

15. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$ and $f(0) = 0$, then (2009)

- (a) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
(b) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$
(c) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$
(d) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

16. The value of the integral

$$\int_{-\pi/2}^{\pi/2} \left(x^2 + \log \frac{\pi-x}{\pi+x} \right) \cos x dx \text{ is } (2012)$$

- (a) 0 (b) $\frac{\pi^2}{2} - 4$
(c) $\frac{\pi^2}{2} + 4$ (d) $\frac{\pi^2}{2}$

17. Let $f: \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right) = 1$. Then, the value of $\int_{1/2}^1 f(x) dx$ lies in the interval (2013)

- (a) $(2e-1, 2e)$ (b) $(e-1, 2e-1)$
(c) $\left(\frac{e-1}{2}, e-1\right)$ (d) $\left(0, \frac{e-1}{2}\right)$

18. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is (2013)

- (a) $4(\sqrt{2}-1)$ (b) $2\sqrt{2}(\sqrt{2}-1)$
(c) $4(\sqrt{2}+1)$ (d) $2\sqrt{2}(\sqrt{2}+1)$

19. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to (2014)

- (a) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$
(b) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$
(c) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$
(d) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$



20. Let $f'(x) = \frac{192x^3}{2+\sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If

$m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and

M are

(2015)

(a) $m = 13, M = 24$

(b) $m = \frac{1}{4}, M = \frac{1}{2}$

(c) $m = -11, M = 0$

(d) $m = 1, M = 12$

21. The value of $\int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx$ is equal to

(2016)

(a) $\frac{\pi^2}{4} - 2$

(b) $\frac{\pi^2}{4} + 2$

(c) $\pi^2 - e^{\pi/2}$

(d) $\pi^2 + e^{\pi/2}$

22. Area of the region

$\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\}$ is equal to

(2016)

(a) $\frac{1}{6}$

(b) $\frac{4}{3}$

(c) $\frac{3}{2}$

(d) $\frac{5}{3}$

23. The area of the region $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is

(2019)

(a) $16 \log_2 2 - 6$

(b) $8 \log_e 2 - \frac{7}{3}$

(c) $16 \log_e 2 - \frac{14}{3}$

(d) $8 \log_e 2 - \frac{14}{3}$

24. Let the functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = e^{x-1} - e^{-|x-1|} \text{ and } g(x) = \frac{1}{2}(e^{x-1} + e^{1-x}).$$

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is (2020)

(a) $(2 - \sqrt{3}) + \frac{1}{2}(e - e^{-1})$

(b) $(2 + \sqrt{3}) + \frac{1}{2}(e - e^{-1})$

(c) $(2 - \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

(d) $(2 + \sqrt{3}) + \frac{1}{2}(e + e^{-1})$

Objective Questions II [One or more than one correct option]

25. Let $S_n = \sum_{k=1}^n \frac{n}{n^2 + kn + k^2}$ and $T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2}$, for

$n = 1, 2, 3, \dots$, then

(2008)

(a) $S_n < \frac{\pi}{3\sqrt{3}}$

(b) $S_n > \frac{\pi}{3\sqrt{3}}$

(c) $T_n < \frac{\pi}{3\sqrt{3}}$

(d) $T_n > \frac{\pi}{3\sqrt{3}}$

26. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$, $n = 0, 1, 2, \dots$, then (2009)

(a) $I_n = I_{n+2}$

(b) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$

(c) $\sum_{m=1}^{10} I_{2m} = 0$

(d) $I_n = I_{n+1}$

27. If S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$ and $x = 1$. Then, (2012)

(a) $S \geq \frac{1}{e}$

(b) $S \geq 1 - \frac{1}{e}$

(c) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$

(d) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$



28. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$,

$$\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$$

Then, a is equal to

(2013)

- (a) 5 (b) 7
(c) $\frac{-15}{2}$ (d) $\frac{-17}{2}$

29. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$.

Then

(2014)

- (a) $f(x)$ is monotonically increasing on $[1, \infty)$
(b) $f(x)$ is monotonically decreasing on $(0, 1)$

(c) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$

(d) $f(2^x)$ is an odd function of x on $\mathbb{R}$

30. The option(s) with the value of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\frac{\pi}{2}} e^t (\sin^6 at + \cos^4 at) dt} = L ?$$

(2015)

- (a) $a = 2, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (b) $a = 2, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$
(c) $a = 4, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (d) $a = 4, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$

31. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n) \left(x+\frac{n}{2}\right) \dots \left(x+\frac{n}{n}\right)}{n! \left(x^2+n^2\right) \left(x^2+\frac{n^2}{4}\right) \dots \left(x^2+\frac{n^2}{n^2}\right)} \right)^{\frac{x}{n}}$,

for all $x > 0$. Then

(2016)

- (a) $f\left(\frac{1}{2}\right) \geq f(1)$ (b) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$
(c) $f'(2) \leq 0$ (d) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

32. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then (2017)

- (a) $I > \frac{49}{50}$ (b) $I < \frac{49}{50}$
(c) $I < \log_e 99$ (d) $I > \log_e 99$

33. If the line $x = \alpha$ divided the area of region

$$R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$$

into two equal parts, then

(2017)

- (a) $0 < \alpha \leq \frac{1}{2}$ (b) $2\alpha^4 - 4\alpha^2 + 1 = 0$
(c) $\alpha^4 + 4\alpha^2 - 1 = 0$ (d) $\frac{1}{2} < \alpha < 1$

34. For $\lim_{n \rightarrow \infty} \frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(na+1)^2} + \frac{1}{(na+2)^2} + \dots + \frac{1}{(na+n)^2} \right)} = 54$

$a \in \mathbb{R}$, $|a| > 1$, Then possible value(s) of a is/are –

(2019)

- (a) 8 (b) -9
(c) 7 (d) -6

35. Which of the following inequalities is/are TRUE ?

(2020)

- (a) $\int_0^1 x \cos x dx \geq \frac{3}{8}$ (b) $\int_0^1 x \sin x dx \geq \frac{3}{10}$
(c) $\int_0^1 x^2 \cos x dx \geq \frac{1}{2}$ (d) $\int_0^1 x^2 \sin x dx \geq \frac{2}{9}$

Numerical Value Type Questions

36. The value of $\frac{(5050) \int_0^1 (1-x^{50})^{100} dx}{\int_0^1 (1-x^{50})^{101} dx}$ is (2006)

37. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is (2014)



38. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$

where $[x]$ is the greatest integer less than or equal to x , if

$$I = \int_{-1}^2 \frac{xf(x^2)}{2+f(x+1)} dx, \text{ then the value of } (4I - 1) \text{ is } \quad (2015)$$

39. Let $F(x) = \int_x^{x^2 + \frac{\pi}{6}} 2\cos^2 t (dt)$ for all $x \in \mathbb{R}$ and

$$f: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty) \text{ be a continuous function.}$$

For $a \in \left[0, \frac{1}{2}\right]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$, then $f(0)$ is

(2015)

40. If $\alpha = \int_0^1 \left(e^{9x+3\tan^{-1}x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$ where $\tan^{-1} x$ takes only principal values, then the value of

$$\left(\log_e |1 + \alpha| - \frac{3\pi}{4} \right) \text{ is } \quad (2015)$$

41. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$$F(x) = \int_{-1}^x f(t) dt \text{ for all } x \in [-1, 2] \text{ and}$$

$$G(x) = \int_{-1}^x t |f(f(t))| dt \text{ for all } x \in [-1, 2]. \text{ If } \lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14},$$

then the value of $f\left(\frac{1}{2}\right)$ is (2015)

42. The total number of distinct $x \in [0, 1]$ for which

$$\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1 \text{ is } \quad (2016)$$

43. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR , then the value of n is (2018)

44. The value of the integral $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{\frac{1}{4}}} dx$ is (2018)

45. $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)}$ then find $27I^2$ equals

(2019)

46. The value of the integral

$$\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta \text{ equals } \quad (2019)$$

47. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$.

If $F: [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if

$$\int_0^{\pi} (f'(x) + F(x)) \cos x dx = 2 \text{ then the value of } f(0) \text{ is}$$

(2020)



Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

48. Match the conditions/expressions in **Column I** with statement in **Column II**. (2007)

Column-I

Column-II

(A) $\int_{-1}^1 \frac{dx}{1+x^2}$

(P) $\frac{1}{2} \log\left(\frac{2}{3}\right)$

(B) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

(Q) $2 \log\left(\frac{2}{3}\right)$

(C) $\int_2^3 \frac{dx}{1-x^2}$

(R) $\frac{\pi}{3}$

(D) $\int_1^2 \frac{dx}{x\sqrt{x^2-1}}$

(S) $\frac{\pi}{2}$

The correct matching is:

- (a) (A-S; B-S; C-P; D-R)
 (b) (A-Q; B-S; C-P; D-R)
 (c) (A-S; B-Q; C-P; D-R)
 (d) (A-S; B-R; C-P; D-S)

49. List I List II

- P. The number of polynomials $f(x)$ 1. 8
 with non-negative integer
 coefficients of degree ≤ 2 ,
 satisfying $f(0) = 0$ and

$\int_0^1 f(x) dx = 1$, is

- Q. The number of points in the 2. 2

interval $[-\sqrt{13}, \sqrt{13}]$ at which

$f(x) = \sin(x^2) + \cos(x^2)$ attains
 its maximum value is

- R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals 3. 4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals 4. 0 (2014)

	P	Q	R	S
(a)	3	2	4	1
(b)	2	3	4	1
(c)	3	2	1	4
(d)	2	3	1	4

Using the following passage, solve Q.50 and 51

Passage – 1

Suppose we define definite integral using formula

$$\int_a^b f(x) dx = \left(\frac{b-a}{2}\right) \{f(a) + f(b)\},$$

for more accurate results for $c \in (a, b)$,

$$F(c) = \frac{c-a}{2} [f(a) - f(c)] + \frac{b-c}{2} [f(b) - f(c)]$$

When $c = \frac{a+b}{2}$

$$\int_a^b f(x) dx = \frac{b-a}{4} \{f(a) + f(b) + 2f(c)\} \quad (2006)$$

50. Good approximation of $\int_0^{\pi/2} \sin x dx$, is

- (a) $\pi/4$ (b) $\pi(\sqrt{2}+1)/4$
 (c) $\pi(\sqrt{2}+1)/8$ (d) $\pi/8$

51. If $\lim_{t \rightarrow a} \frac{\int_a^t f(x) dx - \frac{(t-a)}{2} \{f(t) + f(a)\}}{(t-a)^3} = 0$,

then degree of polynomial function $f(x)$ at-most is

- (a) 0 (b) 1
 (c) 3 (d) 2



Using the following passage, solve Q.52 and 53

Passage – 2

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1} dt$

exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$. (2014)

52. The value of $g\left(\frac{1}{2}\right)$ is

- (a) π (b) 2π
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

53. The value of $g'\left(\frac{1}{2}\right)$ is

- (a) $\frac{\pi}{2}$ (b) π
(c) $-\frac{\pi}{2}$ (d) 0

Using the following passage, solve Q.54 and 55

Passage – 3

Let $F: R \rightarrow R$ be a thrice differentiable function. Suppose that $F(1) = 0$, $F(3) = -4$ and $F'(x) < 0$ for all

$x \in \left(\frac{1}{2}, 3\right)$. Let $f(x) = xF(x)$ for all $x \in R$. (2015)

54. The correct statement(s) is (are).

- (a) $f'(1) < 0$
(b) $f(2) < 0$
(c) $f'(x) \neq 0$ for any $x \in (1, 3)$
(d) $f'(x) = 0$ for some $x \in (1, 3)$

55. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression(s) is (are)

- (a) $9f'(3) + f'(1) - 32 = 0$
(b) $\int_1^3 f(x) dx = 12$
(c) $9f'(3) - f'(1) + 32 = 0$
(d) $\int_1^3 f(x) dx = -12$

Subjective Questions

56. For $x > 0$, let $f(x) = \int_1^x \frac{\ln t}{1+t} dt$. Find the function $f(x) + f(1/x)$ and show that $f(e) + f(1/e) = 1/2$. Here, $\ln t = \log_e t$ (2000)

57. Evaluate $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$. (2004)

58. Evaluate

$$\int_0^\pi e^{|\cos x|} \left(2 \sin\left(\frac{1}{2} \cos x\right) + 3 \cos\left(\frac{1}{2} \cos x\right) \right) \sin x dx$$

(2005)



Note:



**DEFINITE INTEGRATION,
AREA UNDER CURVES**



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Teacher's Note:



Go down deep into anything and you will find mathematics, if you integrate anything you will find mathematics. If you differentiate anything you will find mathematics.

*"Samajh ayega to mazaa ayega,
Mazaa ayega to samajh ayega".*



Pulkit Sir loves teaching Mathematics, he has been a pioneer in producing Top Ranks in JEE and Olympiads. He has personally taught and mentored AIR 1,6,7 (JEE ADVANCED), and thousands of more students who have successfully cleared these competitive exams in the last few years.





DIFFERENTIAL EQUATIONS

DIFFERENTIAL EQUATIONS



1. INTRODUCTION

Differential equation constitute a very important part of mathematics as it has many applications in real life. Various laws of physics are often in the form of equations involving rate of change of one quantity with respect to another. As the mathematical equivalent of a rate is a derivative, differential equation arise very naturally in real life and methods for solving them acquire paramount importance.

1.1 Definition

An equation involving the dependent variable and independent variable and also the derivatives of the dependable variable is known as differential equation.

For example:

$$\begin{aligned} \text{(i)} \quad \frac{dy}{dx} &= \frac{x}{y^{1/3}(1+x^{1/3})} & \text{(ii)} \quad \frac{d^2y}{dx^2} &= -p^2y \\ \text{(iii)} \quad \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} &= 3 \frac{d^2y}{dx^2} & \text{(iv)} \quad x^2 \left(\frac{dy}{dx}\right)^2 &= y^2 + 1 \end{aligned}$$

Differential equations which involve only one independent variable are called ordinary differential equation.

2. ORDER AND DEGREE OF DIFFERENTIAL EQUATIONS

2.1 Order

The order of a differential equation is the order of the highest derivative involved in the differential equation

For example:

$$\text{(i)} \quad \left(\frac{dy}{dx}\right)^3 + \left(\frac{dy}{dx}\right)^2 + 4x = 0 \text{ is the differential equation of the first order because maximum derivative of } y \text{ with respect to } x \text{ is } \frac{dy}{dx}$$

$$\text{(ii)} \quad \frac{d^2y}{dx^2} = -p^2y \text{ is the differential equation of the second order because maximum derivative of } y \text{ w.r.t } x \text{ is } \frac{d^2y}{dx^2}$$

$$\text{(iii)} \quad \left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^3 + 2 = 0 \text{ is the differential equation of the third order because maximum derivative of } y \text{ w.r.t } x \text{ is } \frac{d^3y}{dx^3}$$

2.2 Degree

The degree of a differential equation is the degree of the highest differential coefficient when the equation has been made rational and integral as far as the differential coefficients are concerned.

For example:

$$\text{(i)} \quad \frac{dy}{dx} = \frac{x}{y^{1/3}(1+x^{1/3})} \text{ is the differential equation of first degree, because power of the highest order derivative } \frac{dy}{dx} \text{ is 1.}$$

$$\text{(ii)} \quad \left(\frac{d^3y}{dx^3}\right)^2 - 3\left(\frac{dy}{dx}\right)^3 + 2 = 0 \text{ is the differential equation of second degree, because power of highest order derivative } \frac{d^3y}{dx^3} \text{ is 2.}$$

$$\text{(iii)} \quad \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{2/3} = 3 \frac{d^2y}{dx^2} \text{ is the differential equation of third degree, because power of highest}$$





order derivative $\frac{d^2y}{dx^2}$ is 3 (after cubing)

Illustration 1:

Find the order and degree of the following differential equations.

(i) $\sqrt{\frac{d^2y}{dx^2}} = \sqrt[3]{\frac{dy}{dx} + 3}$

(ii) $\frac{d^2y}{dx^2} = \left\{ 1 + \left(\frac{dy}{dx} \right)^4 \right\}^{5/3}$

(iii) $y = px + \sqrt{a^2p^2 + b^2}$ where $p = \frac{dy}{dx}$

Sol. (i) The given differential equation can be written as

$$\left(\frac{d^2y}{dx^2} \right)^3 = \left(\frac{dy}{dx} + 3 \right)^2$$

Hence order = 2, degree = 3

(ii) The given differential equation can be written as

$$\left(\frac{d^2y}{dx^2} \right)^3 = \left[1 + \left(\frac{dy}{dx} \right)^4 \right]^{5/3}$$

Hence order = 2, degree = 3

(iii) The given differential equation can be written as

$$\left(y - x \frac{dy}{dx} \right)^2 = a^2 \left(\frac{dy}{dx} \right)^2 + b^2$$

Hence order = 1, degree = 2

3. FORMATION OF ORDINARY DIFFERENTIAL EQUATION

An ordinary differential equation is formed in an attempt to eliminate certain arbitrary constants from a relation in the variables and constants. Consider an equation containing n arbitrary constants. Differentiating this equation n times we get n additional equations containing n arbitrary constants and derivatives. Eliminating n arbitrary constants from the above $(n + 1)$ equations, we obtain differential equation involving n^{th} derivative.

Thus if an equation contains n arbitrary constants, the resulting differential equation obtained by eliminating these constants will be a differential equation of n^{th} order.

i.e., an equation of the form

$$\phi \left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n} \right) = 0$$

Illustration 2 :

Find the differential equation of the family of all circles which pass through the origin and whose centre lie on y -axis

Sol. Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

If it passes through $(0, 0)$, then $c = 0$

$\therefore$ The equation of circle is $x^2 + y^2 + 2gx + 2fy = 0$

Since the centre of the circle lies on y -axis then $g = 0$

$\therefore$ The equation of the circle is

$$x^2 + y^2 + 2fy = 0 \quad \dots(i)$$

This represents family of circles.

Differentiating, we get

$$2x + 2y \frac{dy}{dx} + 2f \frac{dy}{dx} = 0 \quad \dots(ii)$$

From (i) and (ii), we get

or, $(x^2 - y^2) \frac{dy}{dx} - 2xy = 0$ Which is the required differential equation.

4. SOLUTION OF A DIFFERENTIAL EQUATION

The solution of the differential equation is a relation between the independent and dependent variable free from derivatives satisfying the given differential equation.

Thus the solution of $dy/dx = m$ could be obtained by simply integrating both sides i.e., $y = mx + c$, where c is arbitrary constant.

(a) General solution (or complete primitive)

The general solution of a differential equation is the relation between the variables (not involving the derivatives) which contain the same number of the arbitrary constants as the order of the differential equation.





Thus the general solution of the differential equation

$\frac{d^2 y}{dx^2} = 4y$ is $y = A \sin 2x + B \cos 2x$, where A and B are the constants.

(b) Particular solution or Integral

A solution which is obtained by giving particular values to the arbitrary constants in the general solution is called a particular solution.

Illustration 3:

Show that $v = \frac{A}{r} + B$ is the general solution of the second

order differential equation $\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0$, where A and

B are arbitrary constant.

Sol. Given $v = \frac{A}{r} + B$

Differentiating twice $\frac{d^2 v}{dr^2} = \frac{2A}{r^3}$... (i)

From (i) $\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = \frac{2A}{r^3} + \frac{2}{r} \left(-\frac{A}{r^2} \right) = \frac{2A}{r^3} - \frac{2A}{r^3} = 0$

Putting $A = 4$, $B = 5$ in $v = \frac{A}{r} + B$ we get a particular solution of the differential equation

$\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0$ is $v = \frac{4}{r} + 5$.

Illustration 4:

Show that $y = ae^x + be^{2x} + ce^{-3x}$ is a solution of the

equation $\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$

Sol. We have

$y = ae^x + be^{2x} + ce^{-3x}$... (i)

Differentiating, we get

$y_1 = ae^x + 2be^{2x} - 3ce^{-3x}$... (ii)

$\therefore y_1 - y = be^{2x} - 4ce^{-3x}$... (iii)

Differentiating (ii), we get

$y_2 = ae^x + 4be^{2x} + 9ce^{-3x}$

$\therefore y_2 - y_1 = 2be^{2x} + 12ce^{-3x}$... (iv)

Now, (iv) - 2 (iii) $\Rightarrow y_2 - y_1 - 2(y_1 - y) = 20ce^{-3x}$

or, $y_2 - 3y_1 + 2y = 20ce^{-3x}$... (v)

Differentiating, $y_2 = ae^x + 4be^{2x} + 9ce^{-3x}$, we get

$y_3 = ae^x + 8be^{2x} - 27ce^{-3x}$

Now $y_3 - 3y_2 + 2y_1 = -60ce^{-3x}$... (vi)

And (vi) + 3(v) $\Rightarrow y_3 - 3y_2 + 2y_1 + 3(y_2 - 3y_1 + 2y) = 0$

or, $y_3 - 7y_1 + 6y = 0$

i.e., $\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$, which is the required differential equation

5. METHOD OF SOLVING AN EQUATION OF THE FIRST ORDER AND FIRST DEGREE

A differential equation of the first order and first degree can be written in the form

$$\frac{dy}{dx} = f(x, y)$$

or, $M dx + N dy = 0$, where M and N are functions of x and y

5.1 Variable Separation:

The general form of such an equation is

$$f(x)dx + f(y)dy = 0 \quad \dots (i)$$

Integrating, we get

$\int f(x)dx + \int f(y)dy = c$ which is the solution of (i)

5.2 Solution of differential equation of the type

$$\frac{dy}{dx} = f(ax + by + c) :$$

Consider the differential equation $\frac{dy}{dx} = f(ax + by + c)$

... (i)

Where $f(ax + by + c)$ is some function of $ax + by + c$.

Let $z = ax + by + c$





$$\therefore \frac{dz}{dx} = a + b \frac{dy}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{\frac{dz}{dx} - a}{b}$$

$$\text{From (i) } \frac{\frac{dz}{dx} - a}{b} = f(z) \quad \text{or, } \frac{dz}{dx} = b f(z) + a$$

$$\text{or, } \frac{dz}{b f(z) + a} = dx \quad \dots(\text{ii})$$

In the differential equation (ii), the variables x and z are separated.

Integrating, we get

$$\int \frac{dx}{b f(z) + a} = \int dx + c$$

$$\text{or, } \int \frac{dx}{b f(z) + a} = x + c, \text{ where } z = ax + by + c$$

This represents the general solution of the differential equation (i)

Illustration 5:

$$\text{Solve } (x - y)^2 \frac{dy}{dx} = a^2$$

Sol. Putting $x - y = v$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{dv}{dx} \Rightarrow dx = \frac{v^2}{v^2 - a^2} dv, \text{ variable have been separated}$$

$$\text{Integrating, we get } \int dx = \int \frac{v^2}{v^2 - a^2} dv$$

$$\text{or, } 2y + k = a \log \frac{x - y - a}{x - y + a}$$

Illustration 6:

$$\text{Solve, } \frac{dy}{dx} = \sin(x + y) + \cos(x + y)$$

Sol. Let $z = x + y$

$$\therefore \frac{dz}{dx} = 1 + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$$

$$\frac{dz}{dx} - 1 = \sin z + \cos z$$

$$\text{or, } dx = \frac{dz}{\sin z + \cos z + 1}$$

Integrating, we get

$$\int dx = \int \frac{dz}{\sin z + \cos z + 1} = \int \frac{dt}{t + 1}, \text{ putting } t = \tan \frac{z}{2}$$

i.e., $x + c = \log |t + 1|$ This is the required general solution.

5.3 Solution of differential equation of the type

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}, \text{ where } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\text{Here } \frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2} \text{ where } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \quad \dots(\text{i})$$

$$\text{Let } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \lambda \text{ (say)}$$

$$\therefore a_1 = \lambda a_2, b_1 = \lambda b_2$$

$$\begin{aligned} \text{From (i), } \frac{dy}{dx} &= \frac{\lambda a_2 x + \lambda b_2 y + c_1}{a_2 x + b_2 y + c_2} \\ &= \frac{\lambda(a_2 x + b_2 y) + c_1}{a_2 x + b_2 y + c_2} \quad \dots(\text{ii}) \end{aligned}$$

$$\text{Let } z = a_2 x + b_2 y$$

$$\therefore \frac{dz}{dx} = a_2 + b_2 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\frac{dz}{dx} - a_2}{b_2} \quad \dots(\text{iii})$$





From (ii) and (iii), we get $\frac{dz}{dx} - a_2 = \frac{\lambda z + c_1}{z + c_2}$

or, $\frac{dz}{dx} = \frac{b_2(\lambda z + c_1)}{z + c_2} + a_2 = \frac{\lambda b_2 z + b_2 c_1 + a_2 z + a_2 c_2}{z + c_2}$

or, $dx = \frac{z + c_2}{(\lambda b_2 + a_2)z + b_2 c_1 + a_2 c_2} dz$, where x and z are

separated

Integrating, we get

$$x + c = \int \frac{z + c_2}{(\lambda b_2 + a_2)z + b_2 c_1 + a_2 c_2} dz \text{ where } z = a_2 x + b_2 y$$

6. HOMOGENEOUS DIFFERENTIAL EQUATION

A function $f(x, y)$ is called homogeneous function of degree n if

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

For example:

- (a) $f(x, y) = x^2 y^2 - xy^3$ is a homogeneous function of degree four, since

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda^2 x^2)(\lambda^2 y^2) - (\lambda x)(\lambda^3 y^3) \\ &= \lambda^4 (x^2 y^2 - xy^3) \\ &= \lambda^4 f(x, y) \end{aligned}$$

- (b) $f(x, y) = x^2 e^{x/y} + \frac{x^3}{y} + y^2 \log\left(\frac{y}{x}\right)$ is a homogeneous function of degree two, since

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda^2 x^2) e^{\lambda x / \lambda y} + \frac{\lambda^3 x^3}{\lambda y} + (\lambda^2 y^2) \log\left(\frac{\lambda y}{\lambda x}\right) \\ &= \lambda^2 \left[x^2 e^{x/y} + \frac{x^3}{y} + y^2 \log\left(\frac{y}{x}\right) \right] \\ &= \lambda^2 f(x, y) \end{aligned}$$

A differential equation of the form $\frac{dy}{dx} = f(x, y)$, where

$f(x, y)$ is a homogeneous polynomial of degree zero is

called a homogeneous differential equation. Such

equations are solved by substituting $v = \frac{y}{x}$ or $\frac{x}{y}$ and

then separating the variables.

Illustration 7 :

Solve $\frac{dy}{dx} = \frac{y(2y - x)}{x(2y + x)}$... (i)

Sol. Since each of the functions $y(2y - x)$ and $x(2y + x)$ is a homogeneous function of degree 2, so the given equation is a homogeneous differential equation.

$\therefore$ Putting $y = vx$

Differentiating w.r.t x , we get $\frac{dy}{dx} = v + x \frac{dv}{dx}$

From (i),

$$v + x \frac{dv}{dx} = \frac{vx(2vx - x)}{x(2vx + x)} = \frac{v(2v - 1)}{2v + 1}$$

$$\Rightarrow 2dv + \frac{dv}{v} + 2 \frac{dx}{x} = 0$$

Integrating, we get

$$2v + \log v + \log x^2 = \log k \quad \text{or,} \quad xy = ke^{-2y/x}$$

6.1 Differential equation reducible to homogeneous forms:

Equation of the form $\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$ where $\frac{a}{a'} \neq \frac{b}{b'}$

can be reduced to homogeneous form by changing the variables x, y to x', y' by equations $x = x' + h$ and $y = y' + k$ where h and k are constants to be chosen so as to make the given equation homogeneous, we have

$$dx = dx' \text{ and } dy = dy'$$

$\therefore$ The given equation becomes

$$\frac{dy'}{dx'} = \frac{a(x' + h) + b(y' + k) + c}{a'(x' + h) + b'(y' + k) + c'}$$





$$= \frac{ax' + by' + (ah + bk + c)}{a'x' + b'y' + (a'h + b'k + c')}$$

Now, we choose h and k so that

$$ah + bk + c = 0$$

$$\text{and } a'h + b'k + c' = 0$$

From these equation we get the values of h and k in terms of the coefficients.

Then the given equation reduces to

$$\frac{dy'}{dx'} = \frac{ax' + by'}{a'x' + b'y'}$$

Which is the homogeneous form.

7. LINEAR DIFFERENTIAL EQUATION

A differential equation is said to be linear if the dependent variable y and its derivative occur in the first degree.

$$\text{An equation of the form } \frac{dy}{dx} + Py = Q \quad \dots(i)$$

where P and Q are functions of x only or constant is called a linear equation of the first order

$$\text{Similarly } \frac{dx}{dy} + Px = Q \text{ is a linear differential equation}$$

where P and Q are functions of y only. To get the general solution of the above equations we shall determine a function R of x called Integrating function (I.F). We shall multiply both sides of the given equation by R

$$\therefore \text{ where, } R = e^{\int P dx} = \text{I.F} \quad \dots(ii)$$

From (i) and (ii), we get

$$e^{\int P dx} \cdot \frac{dy}{dx} + P y e^{\int P dx} = Q \cdot e^{\int P dx} \quad \text{or,}$$

$$\frac{d}{dx} \left(y e^{\int P dx} \right) = Q \cdot e^{\int P dx}$$

Integrating, we get

$$y e^{\int P dx} = \int Q \cdot e^{\int P dx} dx + c \text{ is the required solution.}$$

NOTES :

We remember the solution of the above equation as

$$y(\text{I.F}) = \int Q(\text{I.F}) dx + c$$

Illustration 8 :

$$\text{Solve } 2x \frac{dy}{dx} = y + 6x^{5/2} - 2\sqrt{x}$$

Sol. The given equation can be written as

$$\frac{dy}{dx} + \left(\frac{-1}{2x} \right) y = 3x^{3/2} - \frac{1}{\sqrt{x}} \quad \dots(i)$$

$$\text{This is the form of } \frac{dy}{dx} + Py = Q$$

$$\text{Hence I.F} = e^{\int \frac{-1}{2x} dx} = e^{\frac{-1}{2} \log x} = \frac{1}{\sqrt{x}} \quad \dots(ii)$$

$$\text{From (i) and (ii), we get } y = \frac{3}{2} x^{5/2} - \sqrt{x} \log x + c\sqrt{x}$$

7.1 Differential equation reducible to the linear form:

Sometimes equations which are not linear can be reduced to the linear form by suitable transformation.

$$\text{Here, } f'(y) \frac{dy}{dx} + f(y) P(x) = Q(x) \quad \dots(i)$$

$$\text{Let, } f(y) = u \Rightarrow f'(y) dy = du$$

Then (i) reduces to

$$\frac{du}{dx} + u P(x) = Q(x) \text{ Which is of the linear differential equation form.}$$

Illustration 9:

$$\text{Solve } \sec^2 \theta d\theta + \tan \theta (1 - r \tan \theta) dr = 0$$

Sol. The given equation can be written as

$$\frac{d\theta}{dr} + \frac{\tan \theta}{\sec^2 \theta} = \frac{r \tan^2 \theta}{\sec^2 \theta}$$





$$\text{or, } \left(\frac{\sec^2 \theta}{\tan^2 \theta} \right) \frac{d\theta}{dr} + \frac{1}{\tan \theta} = r$$

$$\text{or, } \cos \theta \frac{d\theta}{dr} + \cot \theta = r \quad \dots(i)$$

$$\text{Let } \cot \theta = u$$

$$\Rightarrow -\cos^2 \theta \, d\theta = du$$

Then (i) reduces to

$$-\frac{du}{dr} + u = r \quad \text{or, } \frac{du}{dr} - u = -r \quad \dots(ii)$$

Which is a linear differential equation.

$$\text{So, } I.F = e^{\int -1 \, dr} = e^{-r} \quad \dots(iii)$$

Form (ii) and (iii), we get

$$u e^{-r} = -\int r e^{-r} \, dr = r e^{-r} + \int e^{-r} \, dr, \text{ by parts}$$

$$= r e^{-r} - e^{-r} + c \text{ or } u = r - 1 + c e^r$$

$$\cot \theta = r - 1 + c e^r$$

7.2 Extended form of linear equations :

Bernoulli's equation:

An equation of the form $\frac{dy}{dx} + P y = Q y^n$, where P and

Q are function of x alone or constants and n is constant, other than 0 and 1, is called a Bernoulli's equation.

$$\text{Here } \frac{dy}{dx} + P y = Q y^n$$

Dividing by y^n , we get

$$\frac{1}{y^n} \frac{dy}{dx} + P \cdot \frac{1}{y^{n-1}} = Q$$

Putting $\frac{1}{y^{n-1}} = v$ and differentiating w.r.t x,

$$\text{we get } -\frac{(n-1)}{y^n} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{1}{y^n} \frac{dy}{dx} = \frac{-1}{n-1} \frac{dv}{dx}$$

the equation becomes

$$\frac{dv}{dx} + (1-n) P v = Q(1-n)$$

Which is a linear equation with v as independent variable.

8. EXACT DIFFERENTIAL EQUATION

A differential equation is said to be exact if it can be derived from its solution (primitive) directly by differentiation, without any elimination, multiplication etc.

For example, the differential equation $x \, dy + y \, dx = 0$ is an exact differential equation as it is derived by direct differentiation for its solution, the function $xy = c$

Illustration 10 :

$$\text{Solve } (1 + xy) y \, dx + (1 - xy) x \, dy = 0$$

Sol. The given equation can be written as

$$y \, dx + xy^2 \, dx + x \, dy - x^2 y \, dy = 0$$

$$\text{or, } (y \, dx + x \, dy) + xy(y \, dx - x \, dy) = 0$$

$$\text{or, } d(xy) + xy(y \, dx - x \, dy) = 0$$

Dividing by $x^2 y^2$, we get

$$\frac{d(xy)}{x^2 y^2} + \frac{y \, dx - x \, dy}{xy} = 0 \quad \text{or,}$$

$$\frac{d(xy)}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0$$

Integrating, we get

$$-\frac{1}{xy} + \log x - \log y = c$$

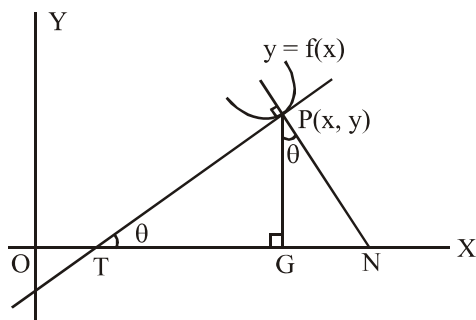
Which is the required solution.





9. APPLICATION OF DIFFERENTIAL EQUATIONS

In solving some geometrical problems, the following results are very helpful.



Let PT and PN be the tangent and the normal at $P(x, y)$.
Let the tangent at P makes an angle θ with the x-axis.

Then the slope of the tangent at $P = \tan \theta = \left(\frac{dy}{dx} \right)_P$ and

the slope of the normal at $P = -\frac{1}{\left(\frac{dy}{dx} \right)_P}$.

Equation of the tangent at $P(x, y)$ is

$$Y - y = \left(\frac{dy}{dx} \right)_P (X - x)$$

Equation of the normal at $P(x, y)$ is

$$Y - y = -\frac{1}{\left(\frac{dy}{dx} \right)_P} (X - x)$$

From ΔPGT $\sin \theta = \frac{PG}{PT} = \frac{y}{PT}$

$\therefore$ $PT = y \operatorname{cosec} \theta$ (length of the tangent)

$$= y \frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta} = y \frac{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}}{\frac{dy}{dx}}$$

And, $\tan \theta = \frac{PG}{TG} = \frac{y}{TG}$

$$\Rightarrow TG = y \cot \theta \text{ (length of the sub tangent)} = \frac{y}{\frac{dy}{dx}}$$

From ΔPGN $\cos \theta = \frac{PG}{PN} = \frac{y}{PN}$

$$\Rightarrow PN = y \sec \theta \text{ (length of the normal)}$$

$$= y \sqrt{1 + \tan^2 \theta} = y \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

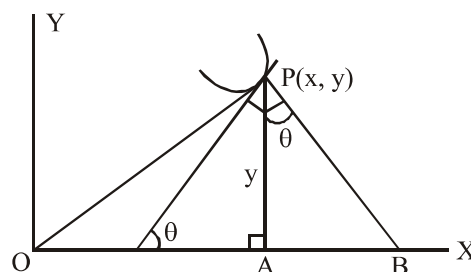
$$\tan \theta = \frac{GN}{y}$$

$$\Rightarrow GN = y \tan \theta = y \frac{dy}{dx} \text{ (length of the sub normal)}$$

Illustration 11 :

If the length of the sub-normal at any point P on the curve is directly proportional to OP^2 , where O is the origin, then form the differential equation of the family of curves and hence find the family of curves.

Sol. Here $AB = y \tan \theta = y \frac{dy}{dx}$



Also $OP^2 = x^2 + y^2$

Given, length of the subnormal = $k \cdot OP^2$

or, $y \frac{dy}{dx} = k(x^2 + y^2)$

or, $2y \frac{dy}{dx} - 2ky^2 = 2kx^2 \dots (i)$

Let $y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx} \dots (ii)$

From (i) and (ii), we get $\frac{dt}{dx} - 2kt = 2kx^2$

Which is a linear differential equation.





$$\therefore I.F = e^{\int -2k \, dx} = e^{-2kx}$$

$\therefore$ The solution is

$$t \cdot e^{-2kx} = \int 2kx^2 e^{-2kx} \, dx + c$$

$$= 2k \left[x^2 \frac{e^{-2kx}}{-2k} + \frac{2}{2k} \int x e^{-2kx} \, dx \right]$$

$$= 2k \left[x^2 \frac{e^{-2kx}}{-2k} + \frac{1}{k} \left\{ x \frac{e^{-2kx}}{-2k} + \frac{1}{2k} \int e^{-2kx} \, dx \right\} \right]$$

$$= -x^2 e^{-2kx} - \frac{x e^{-2kx}}{k} - \frac{1}{k} \frac{e^{-2kx}}{2k} + c \text{ or,}$$

$$y^2 = -x^2 - \frac{x}{k} - \frac{1}{2k^2} + c e^{2kx}$$

10. MISCELLANEOUS

10.1 A special type of second order differential equation

$$\frac{d^2 y}{dx^2} = f(x) \quad \dots (1)$$

Equation (1) may be re-written as

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = f(x) \Rightarrow d \left(\frac{dy}{dx} \right) = f(x) \, dx$$

$$\text{Integrating, } \frac{dy}{dx} = \int f(x) \, dx + c_1 \text{ i.e. } \frac{dy}{dx} = F(x) + c_1 \dots (ii)$$

$$\text{Where } F(x) = \int f(x) \, dx$$

$$\text{From (ii), } dy = \int F(x) \, dx + c_1 \, dx$$

$$\text{Integrating, } y = \int F(x) \, dx + c_1 x + c_2$$

$$\therefore y = H(x) + c_1 x + c_2$$

where $H(x) = \int F(x) \, dx$ c_1 and c_2 are arbitrary constants.

10.2 Particular solution type problems

To solve such a problem, we proceed according to the type of the problem (i.e. variable - separable, linear, exact, homogeneous etc.) and then we apply the given conditions to find the particular values of the arbitrary constants.





SOLVED EXAMPLES

Example – 1

Find the order and degree of the differential equation

$$\left[4 + \left(\frac{dy}{dx} \right)^2 \right]^{2/3} = \frac{d^2y}{dx^2}$$

Sol. Here power on the differential coefficient is fractional, therefore change it into positive integer, so

$$\left[4 + \left(\frac{dy}{dx} \right)^2 \right]^{2/3} = \frac{d^2y}{dx^2}$$

$$\Rightarrow \left[4 + \left(\frac{dy}{dx} \right)^2 \right]^2 = \left[\frac{d^2y}{dx^2} \right]^3$$

Hence order is 2 and degree is 3.

Example – 2

Find the degree of the differential equation

$$\frac{d^2y}{dx^2} - \sqrt{\frac{dy}{dx}} - 3 = x$$

Sol. $\frac{d^2y}{dx^2} - \sqrt{\frac{dy}{dx}} - 3 = x$

$$\Rightarrow \frac{d^2y}{dx^2} - x = \sqrt{\frac{dy}{dx}} - 3$$

Squaring both sides, we get

$$\left(\frac{d^2y}{dx^2} - x \right)^2 = \left(\frac{dy}{dx} - 3 \right)$$

$$\Rightarrow \left(\frac{d^2y}{dx^2} \right)^2 + x^2 - 2x \frac{d^2y}{dx^2} = \frac{dy}{dx} - 3$$

Clearly, degree = 2.

Example – 3

The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant, is

(a) $2(x^2 - y^2)y' = xy$ (b) $2(x^2 + y^2)y' = xy$

(c) $(x^2 - y^2)y' = 2xy$ (d) $(x^2 + y^2)y' = 2xy$

Ans: (c)

Sol: Given family of curve is $x^2 + y^2 - 2ay = 0$... (i)

$$\Rightarrow 2a = \frac{x^2 + y^2}{y}$$

Also from (i), $2x + 2yy' - 2ay' = 0$

$$\Rightarrow 2x + 2yy' - \left(\frac{x^2 + y^2}{y} \right) y' = 0$$

$$\Rightarrow 2x + y' \left(\frac{2y^2 - x^2 - y^2}{y} \right) = 0 \Rightarrow y'(x^2 - y^2) = 2xy$$

Example – 4

If the differential equation representing the family of all circles touching x -axis at the origin is

$(x^2 - y^2) \frac{dy}{dx} = g(x)y$, then $g(x)$ equals:

(a) $\frac{1}{2}x$

(b) $2x^2$

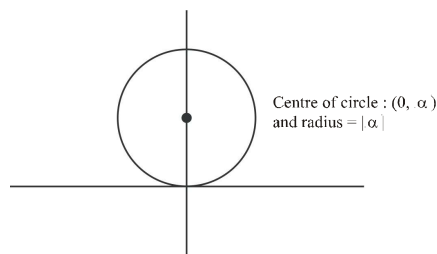
(c) $2x$

(d) $\frac{1}{2}x^2$

Ans: (c)



Sol:



Center of circle = $(0, \alpha)$

and radius = $|\alpha|$

Equation of circle :

$$x^2 + (y - \alpha)^2 = \alpha^2$$

$$\Rightarrow x^2 + y^2 - 2\alpha y = 0 \quad \dots(1)$$

Differentiate w.r.t.x

$$2x + 2y \frac{dy}{dx} - 2\alpha \frac{dy}{dx} = 0$$

$$\Rightarrow \alpha = \frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}}$$

put in (1)

$$x^2 + y^2 - 2y \left(\frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}} \right) = 0$$

$$\Rightarrow (x^2 + y^2) \frac{dy}{dx} - 2xy - 2y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (x^2 - y^2) \frac{dy}{dx} = 2xy$$

$$\Rightarrow g(x) = 2x$$

Example – 5

Find the differential equation for all the straight lines which are at a unit distance from the origin

Sol. Since the equation of lines whose distance from origin is unit, is given by $x \cos \alpha + y \sin \alpha = 1 \dots (i)$

Differentiate w.r.t. x, we get $\cos \alpha + \frac{dy}{dx} \sin \alpha = 0 \dots (ii)$

On eliminating the ' α ' with the help of (i) and (ii) i.e.,

(i) – (x × (ii))

$$\Rightarrow \sin \alpha \left(y - x \frac{dy}{dx} \right) = 1 \Rightarrow \left(y - x \frac{dy}{dx} \right) = \operatorname{cosec} \alpha \dots (iii)$$

$$\text{Also (ii)} \Rightarrow \frac{dy}{dx} = -\cot \alpha \Rightarrow \left(\frac{dy}{dx} \right)^2 = \cot^2 \alpha \dots (iv)$$

$$\text{Therefore by (iii) and (iv), } 1 + \left(\frac{dy}{dx} \right)^2 = \left(y - x \frac{dy}{dx} \right)^2.$$

Example – 6

Form the differential equation represented by family of curves $y = e^x (A \cos x + B \sin x)$.

Sol. Given $y = e^x A \cos x + e^x B \sin x$

$$\frac{dy}{dx} = Ae^x \cos x - Ae^x \sin x + Be^x \sin x + Be^x \cos x$$

$$\frac{dy}{dx} = (A + B) e^x \cos x + (B - A) e^x \sin x$$

$$\frac{d^2y}{dx^2} = (A + B) e^x \cos x - e^x \sin x (A + B) +$$

$$(B - A) e^x \sin x + (B - A) e^x \cos x$$

$$\frac{d^2y}{dx^2} = 2Be^x \cos x - 2Ae^x \sin x.$$

$$\text{Hence } \frac{d^2y}{dx^2} = 2 \frac{dy}{dx} - 2y.$$



Example – 7

Find the solution of the differential equation
 $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$

Sol. $\frac{dy}{dx} = -\frac{\cos x - \sin x}{\sin x + \cos x} \Rightarrow dy = -\left(\frac{\cos x - \sin x}{\sin x + \cos x}\right) dx$

On integrating both sides, we get

$$\Rightarrow y = -\log(\sin x + \cos x) + \log c$$

$$\Rightarrow y = \log\left(\frac{c}{\sin x + \cos x}\right) \Rightarrow e^y(\sin x + \cos x) = c.$$

Example – 8

Find the solution of differential equation $x \frac{dy}{dx} + y = y^2$

Sol. $x \frac{dy}{dx} + y = y^2 \Rightarrow x \frac{dy}{dx} = y^2 - y$

$$\Rightarrow \frac{dy}{y^2 - y} = \frac{dx}{x} \Rightarrow \left[\frac{1}{y-1} - \frac{1}{y} \right] dy = \frac{dx}{x}$$

On integrating, we get

$$\log(y-1) - \log y = \log x + \log c$$

$$\Rightarrow \frac{y-1}{y} = xc \Rightarrow y = 1 + cxy.$$

Example – 9

Find the solution of the equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

Sol. $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0 \Rightarrow \int \frac{dy}{\sqrt{1-y^2}} = -\int \frac{dx}{\sqrt{1-x^2}}$

$$\Rightarrow \sin^{-1} y = -\sin^{-1} x + \sin^{-1} c$$

$$\Rightarrow \sin^{-1} \left[x\sqrt{1-y^2} + y\sqrt{1-x^2} \right] = \sin^{-1} c$$

$$\Rightarrow x\sqrt{1-y^2} + y\sqrt{1-x^2} = c$$

Example – 10

Find the solution of the differential equation

$$(1+x^2)(1+y) dy + (1+x)(1+y^2) dx = 0$$

Sol. Given equation $(1+x^2)(1+y)dy + (1+x)(1+y^2)dx = 0$

$$\Rightarrow \frac{(1+y)}{(1+y^2)} dy = -\frac{(1+x)}{(1+x^2)} dx$$

$$\Rightarrow \int \left[\frac{1}{1+y^2} + \frac{y}{1+y^2} \right] dy + \int \left[\frac{1}{1+x^2} + \frac{x}{1+x^2} \right] dx + c = 0$$

$$\Rightarrow \tan^{-1} y + \frac{1}{2} \log(1+y^2) + \tan^{-1} x + \frac{1}{2} \log(1+x^2) = c.$$

Example – 11

Find the solution of the differential equation

$$\sqrt{a+x} \frac{dy}{dx} + xy = 0 \text{ is}$$

Sol. Given $\frac{dy}{dx} + \frac{xy}{\sqrt{a+x}} = 0 \Rightarrow \frac{dy}{y} = \frac{-x dx}{\sqrt{a+x}}$

Integrating both sides, $\int \frac{dy}{y} = \int \frac{-x dx}{\sqrt{a+x}}$

$$\log y = -\int \frac{x+a-a}{\sqrt{x+a}} dx = -\int \sqrt{x+a} dx + \int \frac{a}{\sqrt{x+a}} dx$$

$$\Rightarrow \log y = -\frac{2}{3} (x+a)^{\frac{3}{2}} + 2a\sqrt{x+a} + \log A$$

$$= 2\sqrt{x+a} \left(\frac{-(x+a)}{3} + a \right) + \log A$$

$$= 2\sqrt{x+a} \left(\frac{2a-x}{3} \right) + \log A$$

$$\Rightarrow y = Ae^{\left[\frac{2}{3} \sqrt{x+a} (2a-x) \right]}.$$



Example – 12

Find the solution of $(x + y - 1)dx + (2x + 2y - 3)dy = 0$

Sol. Given equation is $\frac{dy}{dx} = -\left(\frac{x + y - 1}{2x + 2y - 3}\right)$

Put $x + y = t \Rightarrow \frac{dy}{dx} = \frac{dt}{dx} - 1$

$$\therefore \frac{dt}{dx} - 1 = \frac{1 - t}{2t - 3} \Rightarrow \frac{dt}{dx} = \frac{t - 2}{2t - 3}$$

$$\Rightarrow \frac{2t - 3}{t - 2} dt = dx.$$

Integrating both sides, we get

$$\int \frac{2t - 4}{t - 2} dt - \int \frac{3 - 4}{t - 2} dt = \int 1 dx$$

$$\Rightarrow 2t + \log(t - 2) = x + c$$

Substituting $t = x + y$, we get,

$$\Rightarrow 2(x + y) + \log(x + y - 2) = x + c$$

$$\Rightarrow 2y + x + \log(x + y - 2) = c.$$

Example – 13

Find the solution of the differential equation

$$(x + y)^2 \frac{dy}{dx} = a^2$$

Sol. Put $x + y = v \Rightarrow 1 + \frac{dy}{dx} = \frac{dv}{dx} \Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 1$

$$\therefore v^2 \left(\frac{dv}{dx} - 1 \right) = a^2$$

$$\Rightarrow \frac{dv}{dx} = \frac{a^2}{v^2} + 1 = \frac{a^2 + v^2}{v^2} \Rightarrow \frac{v^2}{a^2 + v^2} dv = dx$$

$$\Rightarrow \left(1 - \frac{a^2}{a^2 + v^2} \right) dv = dx \Rightarrow v - a \tan^{-1} \frac{v}{a} = x + c$$

Substituting $v = x + y$, we get,

$$\Rightarrow y = a \tan^{-1} \left(\frac{x + y}{a} \right) + c$$

Example – 14

Find the solution of the equation $\frac{dy}{dx} = \frac{x + y}{x - y}$

Sol. Given equation, $\frac{dy}{dx} = \frac{x + y}{x - y}$

It is a homogeneous equation as $f(\lambda x, \lambda y) = f(x, y)$ so putting $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}, \text{ we get}$$

$$v + x \frac{dv}{dx} = \frac{x + vx}{x - vx} = \frac{1 + v}{1 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{1 - v}$$

$$\Rightarrow \frac{1}{x} dx = \left(\frac{1}{1 + v^2} - \frac{v}{1 + v^2} \right) dv$$

$$\Rightarrow \log_e x = \tan^{-1} v - \frac{1}{2} \log(1 + v^2) - \log_e c$$

Substituting $v = \frac{y}{x}$, we get

$$\log_e x = \tan^{-1} \frac{y}{x} - \frac{1}{2} \log \left[1 + \left(\frac{y}{x} \right)^2 \right] - \log_e c$$

$$\Rightarrow c(x^2 + y^2)^{1/2} = e^{\tan^{-1}(y/x)}.$$

Example – 15

Find the solution of the differential equation

$$x + y \frac{dy}{dx} = 2y \text{ is}$$

Sol. Given $x + y \frac{dy}{dx} = 2y \Rightarrow \frac{x}{y} + \frac{dy}{dx} = 2$

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore \frac{1}{v} + v + x \frac{dv}{dx} = 2 \Rightarrow v + x \frac{dv}{dx} = \frac{2v - 1}{v}$$



$$\Rightarrow \frac{v}{(v-1)^2} dv = -\frac{dx}{x} \Rightarrow \frac{v-1+1}{(v-1)^2} dv = -\frac{dx}{x}$$

$$\left[\frac{1}{(v-1)} + \frac{1}{(v-1)^2} \right] dv = -\frac{dx}{x}$$

Integrating both sides,

$$\Rightarrow \log(v-1) - \frac{1}{v-1} = -\log x + c$$

$$\text{Substituting } v = \frac{y}{x} \Rightarrow \log(y-x) = \frac{x}{y-x} + c$$

Example - 16

Find the general solution of the differential equation

$$(2x - y + 1)dx + (2y - x + 1)dy = 0$$

Sol. $(2x - y + 1)dx + (2y - x + 1)dy = 0$

$$\frac{dy}{dx} = \frac{2x - y + 1}{x - 2y - 1}, \text{ put } x = X + h, y = Y + k$$

$$\frac{dY}{dX} = \frac{2X - Y + 2h - k + 1}{X - 2Y + h - 2k - 1}$$

$$\text{Put } 2h - k + 1 = 0 \text{ and } h - 2k - 1 = 0$$

$$\text{On solving } h = -1, k = -1;$$

$$\text{So, } X = x + 1, Y = y + 1$$

$$\therefore \frac{dY}{dX} = \frac{2X - Y}{X - 2Y}$$

$$\text{Put } Y = vX; \Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

$$v + X \frac{dv}{dX} = \frac{2X - vX}{X - 2vX} = \frac{2 - v}{1 - 2v}$$

$$X \frac{dv}{dX} = \frac{2 - 2v + 2v^2}{1 - 2v} = \frac{2(v^2 - v + 1)}{1 - 2v}$$

$$\therefore \frac{dX}{X} = \frac{(1 - 2v)}{2(v^2 - v + 1)} dv$$

$$\text{Put } v^2 - v + 1 = t \Rightarrow (2v - 1) dv = dt$$

$$\therefore \frac{dX}{X} = -\frac{dt}{2t}$$

Integrating both sides,

$$\log X = \log t^{-1/2} + \log c$$

$$\therefore X = t^{-1/2} c \Rightarrow X = (v^2 - v + 1)^{-1/2} \cdot c$$

$$X^2(v^2 - v + 1) = \text{constant}$$

$$\text{Plugging } X = x + 1, v = \frac{Y}{X} = \frac{y + 1}{x + 1}, \text{ we get}$$

$$(x + 1)^2 \left(\frac{(y + 1)^2}{(x + 1)^2} - \frac{(y + 1)}{x + 1} + 1 \right) = \text{constant}$$

$$(y + 1)^2 - (y + 1)(x + 1) + (x + 1)^2 = \text{constant}$$

$$y^2 + x^2 - xy + x + y = c.$$

Example - 17

$$\text{Solve the differential equation } (1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}.$$

Sol. Given differential equation is $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$

Dividing throughout by $1 + x^2$, we get

$$\frac{dy}{dx} + \frac{1}{1 + x^2} y = \frac{1}{1 + x^2} e^{\tan^{-1} x},$$

$$\text{which is of the form } \frac{dy}{dx} + Py = Q.$$

$$\text{Here, } P = \frac{1}{1 + x^2} \text{ and } Q = \frac{1}{1 + x^2} e^{\tan^{-1} x}.$$

So, the given differential equation is linear in y .

$$I.F. = e^{\int P dx} = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1} x}$$

$$\therefore \text{Solution is given by } ye^{\tan^{-1} x} = \int e^{\tan^{-1} x} \cdot \frac{e^{\tan^{-1} x}}{1 + x^2} dx + C$$

$$\text{On R.H.S., substitute } \tan^{-1} x = t \Rightarrow \frac{1}{1 + x^2} dx = dt$$

$$\text{So, we obtain } ye^{\tan^{-1} x} = \int e^t e^t dt + C$$

$$\Rightarrow ye^{\tan^{-1} x} = \int e^{2t} dt + C$$

$$\Rightarrow ye^{\tan^{-1} x} = \frac{e^{2t}}{2} + C$$



$$\Rightarrow ye^{\tan^{-1}x} = \frac{1}{2}e^{2\tan^{-1}x} + C$$

$$\text{or } y = \frac{1}{2}e^{\tan^{-1}x} + Ce^{-\tan^{-1}x}, \text{ C being an arbitrary constant.}$$

Example – 18

$$\text{Solve } \frac{dy}{dx} - 2y \cos x = -2 \sin 2x.$$

$$\text{Sol. Given } \frac{dy}{dx} - 2 \cos x \cdot y = -2 \sin 2x. \quad \dots (1)$$

It is a linear in y with 'P' = $-2 \cos x$ and 'Q' = $-2 \sin 2x$.

$$\text{I.F.} = e^{\int -2 \cos x \, dx} = e^{-2 \sin x}, \quad \left(\because \text{I.F.} = e^{\int P \, dx} \right)$$

$\therefore$ Solution of (1) is

$$\begin{aligned} y \cdot e^{-2 \sin x} &= \int -2 \sin 2x \cdot e^{-2 \sin x} \, dx + C \\ &= -4 \int e^{-2 \sin x} \sin x \cos x \, dx + C, \end{aligned}$$

$$\text{On R.H.S. put } -2 \sin x = t \Rightarrow \cos x \, dx = -\frac{1}{2} dt$$

$$\begin{aligned} \therefore ye^{-2 \sin x} &= -4 \int e^t \left(-\frac{t}{2} \right) \left(-\frac{1}{2} dt \right) + C \\ &= -\int t e^t \, dt + C = -\left(te^t - \int 1 \cdot e^t \, dt \right) + C \\ &= -(te^t - e^t) + C = -(-2 \sin x - 1) e^{-2 \sin x} + C \\ \Rightarrow y &= 2 \sin x + 1 + C \cdot e^{2 \sin x}. \end{aligned}$$

Example – 19

$$\text{Solve the equation } \frac{dy}{dx} + \frac{1}{x}y = x^2y^6, x > 0$$

$$\text{Sol. Given equation is } \frac{dy}{dx} + \frac{1}{x}y = x^2y^6,$$

dividing throughout by y^6 ,

$$\text{we get } y^{-6} \frac{dy}{dx} + \frac{1}{x}y^{-5} = x^2 \quad \dots (1)$$

Substitute $y^{-5} = z$

$$\Rightarrow -5y^{-6} \frac{dy}{dx} = \frac{dz}{dx} \Rightarrow y^{-6} \frac{dy}{dx} = -\frac{1}{5} \frac{dz}{dx}$$

The given equation becomes

$$-\frac{1}{5} \left(\frac{dz}{dx} \right) + \frac{1}{x}z = x^2 \text{ or } \frac{dz}{dx} - \frac{5}{x}z = -5x^2,$$

which is linear in z with $P = -\frac{5}{x}$ and $Q = -5x^2$.

$$\text{I.F.} = e^{\int P \, dx} = e^{\int -\frac{5}{x} \, dx} = e^{-5 \log x} = (e^{\log x})^{-5} = x^{-5}$$

Hence, the solution of the transformed equation is given by

$$zx^{-5} = \int x^{-5}(-5x^2) \, dx + C \quad \text{or}$$

$$y^{-5}x^{-5} = -5 \int x^{-3} \, dx + C \quad \left(\because z = y^{-5} \right)$$

$$\text{or } \frac{1}{(xy)^5} = -5 \left(\frac{x^{-2}}{-2} \right) + C$$

$$\text{or } \frac{1}{y^5} = \frac{5}{2}x^3 + Cx^5,$$

which is the required solution of the given differential equation.

Example – 20

Solve the differential equation

$$\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

$$\text{Sol. Given } \sec^2 y \cdot \frac{dy}{dx} + 2x \tan y = x^3 \quad \dots (1)$$

$$\text{Put } \tan y = z \Rightarrow \sec^2 y \frac{dy}{dx} = \frac{dz}{dx},$$

$$\therefore (1) \text{ becomes } \frac{dz}{dx} + 2x \cdot z = x^3 \quad \dots (2)$$

which is linear in z with 'P' = $2x$.

$$\text{I.F.} = e^{\int 2x \, dx} = e^{x^2},$$

$\therefore$ solution of (2) is given by

$$ze^{x^2} = \int x^3 \cdot e^{x^2} \, dx + C,$$

$$\left(\text{put } x^2 = t \Rightarrow x \, dx = \frac{1}{2} dt \right)$$



$$= \int t \cdot e^t \cdot \frac{1}{2} dt + C = \frac{1}{2} (te^t - \int 1 \cdot e^t dt) + C$$

$$= \frac{1}{2} (te^t - e^t) + C = \frac{1}{2} (x^2 - 1) e^{x^2} + C$$

$$\therefore z = \frac{1}{2} (x^2 - 1) + C \cdot e^{-x^2}$$

$$\therefore \text{Solution of (1) is } \tan y = \frac{1}{2} (x^2 - 1) + C e^{-x^2}.$$

Example – 21

Find the solution of $(1 + xy)y dx + (1 - xy)x dy = 0$ is

Sol. $ydx + xdy + xy^2 dx - x^2 y dy = 0$

$$\frac{ydx + xdy}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0. \text{ On integrating, we get}$$

$$\Rightarrow \frac{d(xy)}{x^2 y^2} + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\Rightarrow -\frac{1}{xy} + \log x - \log y = k \Rightarrow \log \frac{x}{y} = \frac{1}{xy} + k.$$

Example – 22

Find the solution of $ye^{-x/y} dx - (xe^{-x/y} + y^3) dy = 0$

Sol. $ye^{-x/y} dx - (xe^{-x/y} + y^3) dy = 0$

$$e^{-x/y} (ydx - xdy) = y^3 dy \Rightarrow e^{-x/y} \frac{(ydx - xdy)}{y^2} = y dy$$

$$\Rightarrow e^{-x/y} d\left(\frac{x}{y}\right) = y dy.$$

Integrating both sides, we get

$$k - e^{-x/y} = \frac{y^2}{2}$$

$$\Rightarrow \frac{y^2}{2} + e^{-x/y} = k$$

Example – 23

Find the equation of a curve passing through the point $(0, -2)$ given that at any point (x, y) on the curve the product of the slope of its tangent and y coordinate of the point is equal to the x coordinate of the point.

Sol. We are given that $y \frac{dy}{dx} = x$... (1)

$\Rightarrow y dy = x dx$, integrating, we obtain

$$\frac{y^2}{2} = \frac{x^2}{2} + C \quad \dots (2)$$

Since $(0, -2)$ lies on this curve, therefore,

$$\frac{(-2)^2}{2} = 0 + C \Rightarrow C = \frac{4}{2} = 2$$

Substituting this value of C in (2), we obtain the required equation of the curve as

$$\frac{y^2}{2} = \frac{x^2}{2} + 2 \quad \text{or} \quad x^2 - y^2 + 4 = 0.$$

Example – 24

If the slope of the tangent at (x, y) to a curve passing through $\left(1, \frac{\pi}{4}\right)$ is given by $\frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$, then find the equation of the curve is

Sol. We have $\frac{dy}{dx} = \frac{y}{x} - \cos^2\left(\frac{y}{x}\right)$

Putting $y = vx$ so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$, we get

$$v + x \frac{dv}{dx} = v - \cos^2 v$$

$$\Rightarrow \frac{dv}{\cos^2 v} = -\frac{dx}{x}$$

On integrating, we get

$$\tan v = -\log x + \log c$$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log c$$



This passes through $\left(1, \frac{\pi}{4}\right)$, therefore $1 = \log c$

or $\log c = \log e$

$$\Rightarrow \tan\left(\frac{y}{x}\right) = -\log x + \log e \Rightarrow y = x \tan^{-1}\left[\log\left(\frac{e}{x}\right)\right].$$

Example – 25

The population $p(t)$ at time t of a certain mouse species

satisfies the differential equation $\frac{dp(t)}{dt} = 0.5 p(t) - 450$. If $\Rightarrow$

$p(0) = 850$, then the time at which the population becomes zero is

(a) $2 \log 18$ (b) $\log 9$

(c) $\frac{1}{2} \log 18$ (d) $\log 18$

Ans: (a)

Sol: $\frac{dp(t)}{dt} = \frac{1}{2} p(t) - 450$

$$\frac{dp(t)}{dt} = \frac{P(t) - 900}{2}$$

$$\therefore 2 \int \frac{d(p(t))}{p(t) - 900} = \int dt$$

$$\therefore 2 \ln |p(t) - 900| = t + c$$

putting $t = 0$

$$\Rightarrow 2 \ln 50 = c$$

$$\therefore 2 \ln |p(t) - 900| = t + 2 \ln 50$$

$$p(t) = 0$$

$$\Rightarrow 2 \ln 900 = t + 2 \ln 50$$

$$t = 2(\ln 900 - \ln 50)$$

$$\Rightarrow t = 2 \ln \left(\frac{900}{50}\right)$$

$$\Rightarrow t \Rightarrow 2 \ln 18$$

Example – 26

In a bank principal increases at the rate of 5% per year.

An amount of Rs 1000 is deposited with this bank, how much will it be worth after 10 years ($e^{0.5} = 1.648$).

Sol. We are given that $\frac{dP}{dt} = \frac{5}{100} P$

where Rs P is the amount at the end of t years.

$$\frac{dP}{P} = \frac{1}{20} dt$$

$$\Rightarrow \int \frac{dP}{P} = \int \frac{1}{20} dt + C$$

$$\Rightarrow \log P = \frac{1}{20} t + C \quad \dots (1)$$

When $t = 0$, $P = 1000$

$$\Rightarrow \log 1000 = C \quad \dots (2)$$

From (1) and (2), we obtain

$$\log P = \frac{1}{20} t + \log 1000$$

$$\Rightarrow \log \left(\frac{P}{1000}\right) = \frac{1}{20} t$$

When $t = 10$, then

$$\log \frac{P}{1000} = \frac{1}{20} \times 10$$

$$\Rightarrow \frac{P}{1000} = e^{1/2} = 1.648$$

$$\Rightarrow P = 1000 \times 1.648$$

$$\Rightarrow P = 1648.$$



Example – 27

If the general solution of the differential equation

$$y' = \frac{y}{x} + \Phi\left(\frac{x}{y}\right), \text{ for some function } \Phi, \text{ is given by}$$

$y \ln |cx| = x$, where c is an arbitrary constant, then $\Phi(2)$ is equal to

- (a) 4 (b) $\frac{1}{4}$
(c) -4 (d) $-\frac{1}{4}$

Ans: (d)

Sol: Given $y \ln |cx| = x$ (1)

differentiate w.r.t.x

$$\frac{y}{cx} \cdot c + \ln |cx| \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{y}{x} + \frac{x}{y} \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{x}{y} \frac{dy}{dx} = 1 - \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \left(\frac{y}{x}\right)^2 \quad \dots (i)$$

Comparing (i) with

$$\frac{dy}{dx} = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$$

$$\Rightarrow \phi\left(\frac{x}{y}\right) = -\frac{1}{\left(\frac{x}{y}\right)^2}$$

$$\phi(2) = -\frac{1}{4}$$

Example – 28

Solve the differential equation $\frac{d^2y}{dx^2} = x + \sin x$ given

that $y = 0, \frac{dy}{dx} = -1$ when $x = 0$.

Sol. Given equation is $\frac{d^2y}{dx^2} = x + \sin x$... (1)

Integrating both sides w.r.t.x, we get

$$\frac{dy}{dx} = \frac{x^2}{2} - \cos x + C_1 \quad \dots (2)$$

when $x = 0, \frac{dy}{dx} = -1$, therefore

$$-1 = 0 - \cos 0 + C_1 \Rightarrow C_1 = 0$$

Substituting this value of C_1 in (2), we get

$$\frac{dy}{dx} = \frac{x^2}{2} - \cos x \quad \dots (3)$$

Again integrating w.r.t.x, we have

$$y = \frac{x^3}{6} - \sin x + C_2 \quad \dots (4)$$

when $x = 0, y = 0$, therefore $0 = 0 - 0 + C_2 \Rightarrow C_2 = 0$

Substituting this value of C_2 in (4), we get

$$y = \frac{x^3}{6} - \sin x, \text{ which is the required solution.}$$

Teacher's Note:



Mathematics is the most beautiful and most powerful creation of human being, you love mathematics and it will open the doors to success and happiness in your life.

*"Samajh ayega to mazaa ayega,
Mazaa ayega to samajh ayega".*



Pulkit Sir loves teaching Mathematics, he has been a pioneer in producing Top Ranks in JEE and Olympiads. He has personally taught and mentored AIR 1,6,7 (JEE ADVANCED), and thousands of more students who have successfully cleared these competitive exams in the last few years.





EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Order and Degree

1. The order and degree of the following differential equation

$$\left(\frac{d^2s}{dt^2}\right)^2 + 3\left(\frac{ds}{dt}\right)^3 + 4 = 0 \text{ are}$$

- (a) 2, 2 (b) 2, 3
(c) 3, 2 (d) none of these

2. The degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{d^2y}{dx^2}\right) \text{ is}$$

- (a) 1 (b) 2
(c) 3 (d) none of these

3. The degree of the differential equation

$$\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} = 0 \text{ is}$$

4. Which of the following differential equations has the same order and degree-

(a) $\frac{d^4y}{dx^4} + 8\left(\frac{dy}{dx}\right)^6 + 5y = e^x$

(b) $5\left(\frac{d^3y}{dx^3}\right)^4 + 8\left(1 + \frac{dy}{dx}\right)^2 + 5y = x^8$

(c) $\left[1 + \left(\frac{dy}{dx}\right)^3\right]^{2/3} = 4\frac{d^3y}{dx^3}$

(d) $y = x^2 \frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Formation of Differential Equations

5. The degree of the differential equation satisfying the

relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is

6. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$, where c_1 and c_2 are arbitrary constants is

- (a) $y' = y^2$ (b) $y'' = y'y$
(c) $yy'' = y'$ (d) $yy'' = (y')^2$

7. The differential equation of all circles passing through the origin and having their centres on the x-axis is

- (a) $x^2 = y^2 + xy \frac{dy}{dx}$ (b) $x^2 = y^2 + 3xy \frac{dy}{dx}$
(c) $y^2 = x^2 + 2xy \frac{dy}{dx}$ (d) $y^2 = x^2 - 2xy \frac{dy}{dx}$

8. The sum of order and degree of the differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$, is a parameter, is

9. The sum of the degree and order of the differential equation of the family of all parabolas whose axis is x-axis, is.

10. The differential equation of all non-vertical lines in a plane is

- (a) $\frac{d^2y}{dx^2} = 0$ (b) $\frac{d^2x}{dy^2} = 0$
(c) $\frac{dy}{dx} = 0$ (d) $\frac{dx}{dy} = 0$

11. The degree of the differential equation corresponding to the family of curves $y = a(x+a)^2$, where a is an arbitrary constant, is

- (a) 1 (b) 2
(c) 3 (d) none of these

12. Form the differential equation representing the family of curves $y = A \cos 2x + B \sin 2x$, where A and B are constants.

- (a) $\frac{d^2y}{dx^2} + 4y = 0$ (b) $\frac{d^2y}{dx^2} - 4y = 0$
(c) $y \cdot \frac{dy}{dx} + 4x = 0$ (d) $\frac{dy}{dx} + 4x = 0$



13. The differential equation from $y = ke^{\sin^{-1} x} + 3$ is

- (a) $\sqrt{1-x^2} \frac{dy}{dx} = y+3$ (b) $\sqrt{1-x^2} \frac{dy}{dx} = y-3$
 (c) $\sqrt{1+x^2} \frac{dy}{dx} = y+3$ (d) none of these

Variable Separable Method

14. The general solution of the differential equation $(1+y^2)dx + (1+x^2)dy = 0$ is

- (a) $x-y = C(1-xy)$ (b) $x-y = C(1+xy)$
 (c) $x+y = C(1-xy)$ (d) $x+y = C(1+xy)$

15. The solution of the differential equation $\frac{dy}{dx} + \frac{1+x^2}{x} = 0$ is

- (a) $y = -\frac{1}{2}\tan^{-1} x + c$ (b) $y + \log x + \frac{x^2}{2} + c = 0$
 (c) $y = \frac{1}{2}\tan^{-1} x + c$ (d) $y - \log x - \frac{x^2}{2} = c$

16. The general solution of the differential equation $(e^x + e^{-x}) \frac{dy}{dx} = (e^x - e^{-x})$ is

- (a) $y = \log |e^x + e^{-x}| + c$ (b) $y = \log |e^x - e^{-x}| - c$
 (c) $y = -\log |e^x - e^{-x}| + c$ (d) none of these

17. The solution of $\frac{dy}{dx} = 1+x+y+xy$ is

- (a) $\log(1-y) = x + \frac{x^3}{2} + C$
 (b) $\log(1+y) = x + \frac{x^2}{2} + C$
 (c) $\log(1+y) = x - \frac{x^2}{2} - C$
 (d) none of these

18. The solution of the equation $\frac{dy}{dx} = \frac{(1+x)y}{(y-1)x}$ is :

- (a) $\log xy + x + y = c$ (d) $\log\left(\frac{x}{y}\right) + x - y = c$
 (c) $\log xy + x - y = c$ (d) none of these

19. If the solution of the differential equation $\frac{dy}{dx} = \frac{ax+3}{2y+f}$ represents a circle, then the value of 'a' is

20. The solution of the differential equation

- $\cos y \log(\sec x + \tan x) dx = \cos x \log(\sec y + \tan y) dy$ is
 (a) $\sec^2 x + \sec^2 y = c$ (b) $\sec x + \sec y = c$
 (c) $\sec x - \sec y = c$ (d) None of these

21. The solution of $\frac{dy}{dx} = \frac{e^x(\sin^2 x + \sin 2x)}{y(2 \log y + 1)}$ is-

- (a) $y^2(\log y) - e^x \sin^2 x + c = 0$
 (b) $y^2(\log y) - e^x \cos^2 x + c = 0$
 (c) $y^2(\log y) + e^x \cos^2 x + c = 0$
 (d) None of these

22. The solution of $(\operatorname{cosec} x \log y)dy + (x^2 y)dx = 0$ is -

- (a) $\frac{\log y}{2} + (2-x^2) \cos x + 2 \sin x = c$
 (b) $\left(\frac{\log y}{2}\right)^2 + (2-x^2) \cos x + 2x \sin x = c$
 (c) $\frac{(\log y)^2}{2} + (2-x^2) \cos x + 2x \sin x = c$
 (d) None of these

23. The solution of the differential equation

- $xy \frac{dy}{dx} = \frac{(1+y^2)(1+x+x^2)}{(1+x^2)}$ is -
 (a) $\frac{1}{2} \log(1+y^2) = \log x - \tan^{-1} x + c$
 (b) $\frac{1}{2} \log(1+y^2) = \log x + \tan^{-1} x + c$
 (c) $\log(1+y^2) = \log x - \tan^{-1} x + c$
 (d) $\log(1+y^2) = \log x + \tan^{-1} x + c$



24. The solution of $(x\sqrt{1+y^2})dx + (y\sqrt{1+x^2})dy = 0$ is -
- (a) $\sqrt{1+x^2} + \sqrt{1+y^2} = c$
 (b) $\sqrt{1+x^2} - \sqrt{1+y^2} = c$
 (c) $(1+x^2)^{3/2} + (1+y^2)^{3/2} = c$
 (d) None of these
25. The Solution of the equation $(1-x^2)dy + xydx = xy^2dx$ is-
- (a) $(y-1)^2(1-x^2) = 0$
 (b) $(y-1)^2(1-x^2) = c^2y^2$
 (c) $(y-1)^2(1+x^2) = c^2y^2$
 (d) None of these
26. The solution of differential equation $\frac{dy}{dx} = \sec(x+y)$ is
- (a) $y - \tan \frac{x+y}{2} = c$ (b) $y + \tan \frac{x+y}{2} = c$
 (c) $y + 2 \tan \frac{x+y}{2} = c$ (d) none of these
29. The solution of differential equation $x^2ydx - (x^3 + y^3)dy = 0$ is
- (a) $\frac{1}{3} \frac{x^3}{y^3} + \log y = C$ (b) $\frac{1}{3} \frac{x^3}{y^3} - \log y = C$
 (c) $\frac{x^3}{y^3} + \log y = -C$ (d) none of these
30. The solution of Differential equation $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ is
- (a) $\frac{(x-1)\left(1+\frac{y-1}{x-1}\right)^{1/2}}{\left(1-\frac{y-1}{x-1}\right)^{3/2}} = C$ (b) $\frac{\left(1+\frac{y-1}{x-1}\right)^{1/2}}{\left(1-\frac{y-1}{x-1}\right)^{3/2}} = C$
 (c) $\frac{\left(1+\frac{y-1}{x-1}\right)^{1/2}}{(x-1)\left(1-\frac{y-1}{x-1}\right)^{3/2}} = C$ (d) none of these

Homogeneous Differential Equation

27. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is
- (a) $y = x \log x + x$ (b) $y = \log x + x$
 (c) $y = x \log x + x^2$ (d) $y = xe^{(x-1)}$
28. If $x \frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is
- (a) $\log\left(\frac{x}{y}\right) = cy$ (b) $\log\left(\frac{y}{x}\right) = cx$
 (c) $x \log\left(\frac{y}{x}\right) = cy$ (d) $y \log\left(\frac{x}{y}\right) = cx$

Linear Differential Equation

31. The solution of differential equation $(x^2-1) \frac{dy}{dx} + 2xy = \frac{1}{x^2-1}$ is
- (a) $y(x^2-1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + C$
 (b) $y(x^2+1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| - C$
 (c) $y(x^2-1) = \frac{5}{2} \log \left| \frac{x-1}{x+1} \right| + C$
 (d) none of these
32. If $\frac{dy}{dx} + y \tan x = \sin 2x$ and $y(0) = 1$ and $y(\pi)$ is equal to k . Then the value of $-k$ is



33. The general solution of the differential equation,

$$\sin 2x \left(\frac{dy}{dx} - \sqrt{\tan x} \right) - y = 0, \text{ is:}$$

- (a) $y\sqrt{\tan x} = x + c$ (b) $y\sqrt{\cot x} = \tan x + c$
(c) $y\sqrt{\tan x} = \cot x + c$ (d) $y\sqrt{\cot x} = x + c$

34. If $\phi(x)$ is a differentiable function, then the solution of the differential equation

$$dy + \{y\phi'(x) - \phi(x)\phi'(x)\} dx = 0 \text{ is}$$

- (a) $y = \{\phi(x) - 1\} + ce^{-\phi(x)}$ (b) $y\phi(x) = \{\phi(x)\}^2 + c$
(c) $ye^{\phi(x)} = \phi(x)e^{\phi(x)} + c$ (d) none of these

35. Consider the differential equation $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$.

If $y(1) = 1$, then x is given by

- (a) $1 - \frac{1}{y} + \frac{e^{1/y}}{e}$ (b) $4 - \frac{2}{y} - \frac{e^{1/y}}{e}$
(c) $3 - \frac{1}{y} + \frac{e^{1/y}}{e}$ (d) $1 + \frac{1}{y} - \frac{e^{1/y}}{e}$

36. The solution of the differential equation

$$(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0, \text{ is}$$

- (a) $(x - 2) = ce^{-2\tan^{-1} y}$ (b) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + c$
(c) $xe^{\tan^{-1} y} = \tan^{-1} y + c$ (d) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + c$

37. The solution of differential equation

$$(2x - 10y^3) \frac{dy}{dx} + y = 0 \text{ is:}$$

- (a) $xy^2 = y^5 + c$ (b) $xy^2 + 2y^5 = c$
(c) $xy^2 = 2y^5 + c$ (d) none of these

38. The integrating factor of the differentiable equation

$$(xy - 1) \frac{dy}{dx} + y^2 = 0 \text{ is}$$

- (a) y (b) $\frac{1}{y}$
(c) $\frac{1}{xy}$ (d) xy

39. If $\sin x$ is an integrating factor of the differential equation

$$\frac{dy}{dx} + Py = Q, \text{ then } P \text{ can be}$$

- (a) $\log \sin x$ (b) $\cot x$
(c) $\sin x$ (d) $\log \cos x$

40. The solution of differential equation

$$\frac{dy}{dx} - y \tan x = -y^2 \sec x \text{ is:}$$

- (a) $y^{-1} \sec x = \cot x + c$ (b) $y^{-1} \cos x = \tan x + c$
(c) $y^{-1} \sec x = \tan x + c$ (d) none of these

41. The solution of the equation $x^3 \frac{dy}{dx} + 4x^2 \tan y = e^x \sec y$

satisfying $y(1) = 0$, is

- (a) $\tan y = (x - 2)e^x \log 3$ (b) $\sin y = e^x (x - 1)x^{-4}$
(c) $\tan y = (x - 1)e^x x^{-3}$ (d) $\sin y = e^x (x - 1)x^{-3}$

42. The general solution of the differential equation

$$\left[2\sqrt{xy} - x \right] dy + y dx = 0 \text{ is}$$

- (a) $\log x + \sqrt{\frac{y}{x}} = c$ (b) $\log y - \sqrt{\frac{x}{y}} = c$
(c) $\log y + \sqrt{\frac{x}{y}} = c$ (d) none of these

43. The general solution of the differential equation

$$y(x^2 y + e^x) dx - e^x dy = 0 \text{ is}$$

- (a) $x^3 y - 3e^x = cy$ (b) $x^3 y + 3e^x = 3cy$
(c) $y^3 x - 3e^y = cx$ (d) $y^3 x + 3e^y = cx$



Exact Differentials

44. Solution of the differential equation

$$\cos x \, dy = y (\sin x - y) \, dx, 0 < x < \frac{\pi}{2}, \text{ is}$$

(a) $\sec x = (\tan x + c) y$ (b) $y \sec x = \tan x + c$

(c) $y \tan x = \sec x + c$ (d) $\tan x = (\sec x + c) y$

45. The solution of the differential equation $y \, dx + (x + x^2 y) \, dy = 0$ is

(a) $-\frac{1}{xy} = c$ (b) $-\frac{1}{xy} + \log y = c$

(c) $\frac{1}{xy} + \log y = c$ (d) $\log y = cx$

46. The solution of the differential equation

$$\frac{x}{x^2 + y^2} \, dy = \left(\frac{y}{x^2 + y^2} - 1 \right) \, dx \text{ is}$$

(a) $y = x \cot (c - x)$ (b) $\cos^{-1} \frac{y}{x} = (-x + c)$

(c) $y = x \tan (c - x)$ (d) $\frac{y^2}{x^2} = x \tan (c - x)$

Applications of Differential Equation

47. Equation of curve through point (1, 0) which satisfies the differential equation

$$(1 + y^2) \, dx - xy \, dy = 0, \text{ is}$$

(a) $x^2 + y^2 = 1$ (b) $x^2 - y^2 = 1$

(c) $2x^2 + y^2 = 2$ (d) none of these

48. The equation of a curve passing through $\left(2, \frac{7}{2}\right)$ and

$$\text{having gradient } 1 - \frac{1}{x^2} \text{ at } (x, y) \text{ is}$$

(a) $y = x^2 + x + 1$ (b) $xy = x^2 + x + 1$

(c) $xy = x + 1$ (d) none of these

49. The curve passing through the point (0, 1) and satisfying

$$\text{the equation } \sin \left(\frac{dy}{dx} \right) = a \text{ is :}$$

(a) $\cos \left(\frac{y+1}{x} \right) = a$ (b) $\cos \left(\frac{x}{y+1} \right) = a$

(c) $\sin \left(\frac{y-1}{x} \right) = a$ (d) $\sin \left(\frac{x}{y-1} \right) = a$

50. A curve passes through the point (0, 1) and the gradient at (x, y) on it is $y(xy - 1)$. The equation of the curve is

(a) $y(x - 1) = 1$ (b) $y(x + 1) = 1$

(c) $x(y + 1) = 1$ (d) $x(y - 1) = 1$

51. A tangent to the curve $y = f(x)$ cuts the line $y = x$ at a point which is at a distance of 1 unit from y-axis. The equation of the curve is

(a) $\frac{x-1}{y-1} = c$ (b) $\frac{x}{y} = c$

(c) $xy = c$ (d) none of these

52. The equation of the curve for which the tangent at P (x, y) cuts the y-axis at (0, y^3) is

(a) $xy^2 = x + y$ (b) $x^2(y^2 - 1) = cy^2$

(c) $y(x^2 - 1) = cx^2$ (d) $yx + x^2 = c$

53. The slope of a curve at any point is the reciprocal of twice the ordinate at the point and it passes through the point (4, 3). The equation of the curve is-

(a) $x^2 = y + 5$ (b) $y^2 = x - 5$

(c) $y^2 = x + 5$ (d) $x^2 = y - 5$

54. The curve in which the portion of the tangent included between the coordinate axes is bisected at the point of contact, is

(a) a parabola (b) an ellipse

(c) a circle (d) a hyperbola

55. The equation of the curve in which subnormal varies as the square of the ordinate is (λ is constant of proportionality)

(a) $y = ce^{2\lambda x}$ (b) $y = ce^{\lambda x}$

(c) $\frac{y^2}{2} + \lambda x = c$ (d) $y^2 + \lambda x^2 = c$



56. The slope of the tangent at (x, y) to a curve passing through $(2, 1)$ is $\frac{x^2 + y^2}{2xy}$, then the equation of the curve is
- (a) $2(x^2 - y^2) = 3x$ (b) $2(x^2 - y^2) = 6y$
 (c) $x(x^2 - y^2) = 6$ (d) $x(x^2 + y^2) = 10$
57. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P with respect to additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is 100 k. Then the value of k is
58. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then $p(t)$ equals :
- (a) $400 - 300e^{-t/2}$ (b) $400 - 300e^{t/2}$
 (c) $300 - 200e^{-t/2}$ (d) $600 - 500e^{t/2}$
59. In a bank, principal increases at the rate of 5% per year. If Rs. 1000 doubles itself in $k \log_e 2$ years. Then the value of k is
60. Water is dropped at the rate of $2 \text{ m}^3/\text{sec}$ into a cone of semi vertical angle of 45° . The rate at which periphery of water surface changes when height of water in the cone is 2 meters is
- (a) 2 m/sec (b) 1 m/sec
 (c) 3 m/sec (d) 4 m/sec
- Miscellaneous**
61. The solution of the equation $\frac{d^2y}{dx^2} = e^{-2x}$ is
- (a) $\frac{e^{-2x}}{4}$ (b) $\frac{e^{-2x}}{4} + cx + d$
 (c) $\frac{1}{4}e^{-2x} + cx^2 + d$ (d) $\frac{1}{4}e^{-2x} + c + d$
62. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is $(x + a)^b$. The value of $a + b$ is
63. The solution of the differential equation $x \frac{d^2y}{dx^2} = 1$, given that $y = 1, \frac{dy}{dx} = 0$, when $x = 1$, is
- (a) $y = x \log x + x + 2$ (b) $y = x \log x - x + 2$
 (c) $y = x \log x + x$ (d) $y = x \log x - x$



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

- Let $y(x)$ be the solution of the differential equation $(x \log x) \frac{dy}{dx} + y = 2x \log x$, $(x \geq 1)$. Then $y(e)$ is equal to: (2015)
 - 2
 - $2e$
 - e
 - 0
- If $y(x)$ is the solution of the differential equation $(x+2) \frac{dy}{dx} = x^2 + 4x - 9$, $x \neq -2$ and $y(0) = 0$, then $y(-4)$ is equal to: (2015/Online Set-1)
 - 2
 - 0
 - 1
 - 1
- The solution of the differential equation $ydx - (x + 2y^2)dy = 0$ is $x = f(y)$. If $f(-1) = 1$, then $f(1)$ is equal to: (2015/Online Set-2)
 - 4
 - 3
 - 1
 - 2
- If a curve $y = f(x)$ passes through the point $(1, -1)$ and satisfies the differential equation, $y(1 + xy) dx = x dy$, then $f\left(-\frac{1}{2}\right)$ is equal to: (2016)
 - $-\frac{4}{5}$
 - $\frac{2}{5}$
 - $\frac{4}{5}$
 - $-\frac{2}{5}$
- The solution of the differential equation $\frac{dy}{dx} + \frac{y}{2} \sec x = \frac{\tan x}{2y}$, where $0 \leq x < \frac{\pi}{2}$, and $y(0) = 1$, is given by: (2016/Online Set-2)
 - $y = 1 - \frac{x}{\sec x + \tan x}$
 - $y^2 = 1 + \frac{x}{\sec x + \tan x}$
 - $y^2 = 1 - \frac{x}{\sec x + \tan x}$
 - $y = 1 + \frac{x}{\sec x + \tan x}$
- If $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ and $y(0) = 1$, then $y\left(\frac{\pi}{2}\right)$ is equal to: (2017)
 - $\frac{1}{3}$
 - $-\frac{2}{3}$
 - $-\frac{1}{3}$
 - $\frac{4}{3}$
- The curve satisfying the differential equation, $ydx - (x + 3y^2) dy = 0$ and passing through the point $(1, 1)$, also passes through the point. (2017/Online Set-1)
 - $\left(\frac{1}{4}, -\frac{1}{2}\right)$
 - $\left(-\frac{1}{3}, \frac{1}{3}\right)$
 - $\left(\frac{1}{3}, -\frac{1}{3}\right)$
 - $\left(\frac{1}{4}, \frac{1}{2}\right)$
- Let $y = y(x)$ be the solution of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 4x$, $x \in (0, \pi)$. if $y\left(\frac{\pi}{2}\right) = 0$ then $y\left(\frac{\pi}{6}\right)$ is equal to: (2018)
 - $-\frac{4}{9}\pi^2$
 - $\frac{4}{9\sqrt{3}}\pi^2$
 - $\frac{-8}{9\sqrt{3}}\pi^2$
 - $-\frac{8}{9}\pi^2$
- Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 2y = f(x)$, where $f(x) = \begin{cases} 1, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$. If $y(0) = 0$, then $y\left(\frac{3}{2}\right)$ is: (2018/Online Set-1)
 - $\frac{e^2 + 1}{2e^4}$
 - $\frac{1}{2e}$
 - $\frac{e^2 - 1}{e^3}$
 - $\frac{e^2 - 1}{2e^3}$



10. The differential equation representing the family of ellipses having foci either on the x-axis or on the y-axis, centre at the origin and passing through the point (0, 3) is: **(2018/Online Set-3)**

(a) $xy y'' + x(y')^2 - y y' = 0$

(b) $x + y y'' = 0$

(c) $xy y' + y^2 - 9 = 0$

(d) $xy y' - y^2 + 9 = 0$

11. Let $y = y(x)$ be the solution of the differential equation, $(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1$ such that $y(0) = 0$. If $\sqrt{a}y(1) = \frac{\pi}{32}$, then the value of 'a' is:

(8-04-2019/Shift-1)

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) 1

(d) $\frac{1}{16}$

12. Given that the slope of the tangent to a curve $y = y(x)$ at any point (x, y) is $\frac{2y}{x^2}$. If the curve passes through the centre of the circle $x^2 + y^2 - 2x - 2y = 0$, then its equation is:

(8-04-2019/Shift-2)

(a) $x \log_e |y| = 2(x-1)$

(b) $x \log_e |y| = -4(x-1)$

(c) $x \log_e |y| = -2(x-1)$

(d) $x \log_e |y| = x-1$

13. The solution of the differential equation $x \frac{dy}{dx} + 2y = x^2 (x \neq 0)$ with $y(1) = 1$, is:

(9-04-2019/Shift-1)

(a) $y = \frac{4}{5}x^3 + \frac{1}{5x^2}$

(b) $y = \frac{x^3}{5} + \frac{1}{5x^2}$

(c) $y = \frac{x^2}{4} + \frac{3}{4x^2}$

(d) $y = \frac{3}{4}x^2 + \frac{1}{4x^2}$

14. If $\cos x \frac{dy}{dx} - y \sin x = 6x, \left(0 < x < \frac{\pi}{2}\right)$

and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

(9-04-2019/Shift-2)

(a) $\frac{\pi^2}{2\sqrt{3}}$

(b) $-\frac{\pi^2}{2}$

(c) $-\frac{\pi^2}{2\sqrt{3}}$

(d) $-\frac{\pi^2}{4\sqrt{3}}$

15. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} = (\tan x - y) \sec^2 x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that

$y(0) = 0$, then $y\left(-\frac{\pi}{4}\right)$ is equal to:

(10-04-2019/Shift-1)

(a) $e - 2$

(b) $\frac{1}{2} - e$

(c) $2 + \frac{1}{e}$

(d) $\frac{1}{e} - 2$

16. Let $y = y(x)$ be the solution of the differential equation, $\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y(0) = 1$, then:

(10-04-2019/Shift-2)

(a) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$

(b) $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = -\sqrt{2}$

(c) $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$

(d) $y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$



17. Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$.
If value of y is 1 when $x = 1$, then the value of x for which $y = 2$, is **(12-04-2019/Shift-1)**
- (a) $\frac{5}{2} + \frac{1}{\sqrt{e}}$ (b) $\frac{3}{2} - \frac{1}{\sqrt{e}}$
(c) $\frac{1}{2} + \frac{1}{\sqrt{e}}$ (d) $\frac{3}{2} - \sqrt{e}$
18. The general solution of the differential equations $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) is _____. (Where c is a constant of integration) **(12-04-2019/Shift-2)**
- (a) $y^2 - 2x^2 + cx^3 = 0$ (b) $y^2 + 2x^3 + cx^2 = 0$
(c) $y^2 + 2x^2 + cx^3 = 0$ (d) $y^2 - 2x^3 + cx^2 = 0$
19. If $y = y(x)$ is the solution of the differential equation, $x \frac{dy}{dx} + 2y = x^2$ satisfying $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to: **(9-01-2019/Shift-1)**
- (a) $\frac{7}{64}$ (b) $\frac{1}{4}$
(c) $\frac{49}{16}$ (d) $\frac{13}{16}$
20. Let $f: [0, 1] \rightarrow \mathbb{R}$ be such that $f(xy) = f(x) \cdot f(y)$, for all $x, y \in [0, 1]$ and $f(0) \neq 0$. If $y = y(x)$ satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with $y(0) = 1$,
They $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to: **(9-01-2019/Shift-2)**
- (a) 3 (b) 4
(c) 2 (d) 5
21. If $\frac{dy}{dx} + \frac{3}{\cos^2 x} y = \frac{1}{\cos^2 x}$, $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$, and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals: **(10-1-2019/Shift-1)**
- (a) $\frac{1}{3} + e^6$ (b) $\frac{1}{3}$
(c) $-\frac{4}{3}$ (d) $\frac{1}{3} + e^3$
22. The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2) dx + 2xy dy = 0$ which passes through $(1, 1)$, is: **(10-01-2019/Shift-2)**
- (a) a circle with centre on the x-axis.
(b) an ellipse with major axis along the y-axis.
(c) a circle with centre on the y-axis.
(d) a hyperbola with transverse axis along the x-axis.
23. Let f be a differentiable such that $f'(x) = 7 - \frac{3f(x)}{4x}$, ($x > 0$) and $f(1) \neq 4$. Then $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right)$ **(10-01-2019/Shift-2)**
- (a) exists and equals $\frac{4}{7}$ (b) exists and equals 4
(c) does not exist (d) exists and equals 0
24. If $y(x)$ is the solution of the differential equation $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$, $x > 0$, where $y(1) = \frac{1}{2}e^{-2}$, then : **(11-01-2019/Shift-1)**
- (a) $y(\log_e 2) = \log_e 4$
(b) $y(\log_e 2) = \frac{\log_e 2}{4}$
(c) $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$
(d) $y(x)$ is decreasing in $(0, 1)$



25. The solution of the differential equation,

$$\frac{dy}{dx} = (x-y)^2 \text{ when } y(1) = 1, \text{ is } \quad (11-01-2019/\text{Shift-2})$$

(a) $\log_e \left| \frac{2-y}{2-x} \right| = x-y$

(b) $-\log_e \left| \frac{1-x+y}{1+x-y} \right| = 2(x-1)$

(c) $-\log_e \left| \frac{1+x-y}{1-x+y} \right| = x+y-2$

(d) $\log_e \left| \frac{2-y}{2-x} \right| = 2(y-1)$

26. Let $y = y(x)$ be the solution of the differential equation $x \frac{dy}{dx} + y = x \log_e x, (x > 1)$. If $2y(2) = \log_e 4 - 1$, then $y(e)$ is equal to _____. (12-01-2019/Shift-1)

(a) $-\frac{e}{2}$

(b) $-\frac{e^2}{2}$

(c) $\frac{e}{4}$

(d) $\frac{e^2}{4}$

27. Let f be a differentiable function such that $f(1) = 2$ and $f'(x) = f(x)$ for all $x \in \mathbb{R}$

If $h(x) = f(f(x))$ then $h'(1)$ is equal

(12-01-2019/Shift-2)

(a) $2e^2$

(b) $4e$

(c) $2e$

(d) $4e^2$

28. If a curve passes through the point $(1, -2)$ and has slope of the tangent at any point (x, y) on it as $\frac{x^2 - 2y}{x}$ then the curve also passes through the point :

(12-01-2019/Shift-2)

(a) $(3, 0)$

(b) $(\sqrt{3}, 0)$

(c) $(-1, 2)$

(d) $(-\sqrt{2}, 1)$

29. Let $y = y(x)$ be the solution of the differential equation,

$$\frac{2 + \sin x}{y+1} \cdot \frac{dy}{dx} = -\cos x, y > 0, y(0) = 1.$$

If $y(\pi) = a$, and $\frac{dy}{dx}$ at $x = \pi$ is b , then the ordered pair

(a, b) is equal to : (2-9-2020/Shift-1)

(a) $\left(2, \frac{3}{2}\right)$

(b) $(1, 1)$

(c) $(2, 1)$

(d) $(1, -1)$

30. If a curve $y = f(x)$, passing through the point $(1, 2)$ is the solution of the differential equation,

$$2x^2 dy = (2xy + y^2) dx, \text{ then } f\left(\frac{1}{2}\right) \text{ is equal to :}$$

(2-09-2020/Shift-2)

(a) $\frac{-1}{1 + \log_e 2}$

(b) $1 + \log_e 2$

(c) $\frac{1}{1 + \log_e 2}$

(d) $\frac{1}{1 - \log_e 2}$

31. The solution curve of the differential equation,

$$(1 + e^{-x})(1 + y^2) \frac{dy}{dx} = y^2, \text{ which passes through the point}$$

$(0, 1)$, is : (3-09-2020/Shift-1)

(a) $y^2 = 1 + y \log_e \left(\frac{1 + e^{-x}}{2} \right)$

(b) $y^2 + 1 = y \left(\log_e \left(\frac{1 + e^{-x}}{2} \right) + 2 \right)$

(c) $y^2 + 1 = y \left(\log_e \left(\frac{1 + e^x}{2} \right) + 2 \right)$

(d) $y^2 = 1 + y \left(\log_e \left(\frac{1 + e^x}{2} \right) \right)$



32. If $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to : **(3-09-2020/Shift-2)**
- (a) $\frac{\sqrt{e}}{2}$ (b) $\frac{3}{2}\sqrt{e}$
 (c) $\frac{1}{2} + \sqrt{e}$ (d) $\frac{3}{2} + \sqrt{e}$
33. Let $y = y(x)$ be the solution of the differential equation, $xy' - y = x^2(x \cos x + \sin x)$, $x > 0$. If $y(\pi) = \pi$, then $y''\left(\frac{\pi}{2}\right) + y\left(\frac{\pi}{2}\right)$ is equal to **(4-09-2020/Shift-1)**
- (a) $2 + \frac{\pi}{2} + \frac{\pi^2}{4}$ (b) $2 + \frac{\pi}{2}$
 (c) $1 + \frac{\pi}{2}$ (d) $1 + \frac{\pi}{2} + \frac{\pi^2}{4}$
34. The solution of the differential equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is :
 (where c is a constant of integration) **(4-9-2020/Shift-2)**
- (a) $x - \log_e(y+3x) = C$
 (b) $x - \frac{1}{2}(\log_e(y+3x))^2 = C$
 (c) $x - 2\log_e(y+3x) = C$
 (d) $y+3x - \frac{1}{2}(\log_e x)^2 = C$
35. If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y} \cdot \frac{dy}{dx} + e^x = 0$ satisfying $y(0) = 1$, then a value of $y(\log_e 13)$ is: **(5-09-2020/Shift-1)**
- (a) 1 (b) 0
 (c) 2 (d) -1
36. Let $y = y(x)$ be the solution of the differential equation $\cos x \frac{dy}{dx} + 2y \sin x = \sin 2x$, $x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to: **(5-09-2020/Shift-2)**
- (a) $2 + \sqrt{2}$ (b) $\sqrt{2} - 2$
 (c) $\frac{1}{\sqrt{2}} - 1$ (d) $2 - \sqrt{2}$
37. The general solution of the differential equation $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$
 (where C is a constant of integration) is: **(6-09-2020/Shift-1)**
- (a) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$
 (b) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$
 (c) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$
 (d) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$
38. If $y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$ is the solution of the differential equation, $\frac{dy}{dx} + p(x)y = \frac{2}{\pi} \operatorname{cosec} x$, $0 < x < \frac{\pi}{2}$, then the function $p(x)$ is equal to : **(6-09-2020/Shift-2)**
- (a) $\operatorname{cosec} x$ (b) $\cot x$
 (c) $\tan x$ (d) $\sec x$



39. If $y = y(x)$ is the solution of the differential equation, $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ such that $y(0) = 0$, then $y(1)$ is equal to (7-01-2020/Shift-1)
- (a) $\log_e 2$ (b) 2_e
(c) $2 + \log_e 2$ (d) $1 + \log_e 2$
40. Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x) \frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x -axis at a point whose abscissa is : (7-01-2020/Shift-2)
- (a) $2 + e$ (b) 2
(c) $2 - e$ (d) $-e$
41. Let $y = y(x)$ be a solution of the differential equation, $\sqrt{1-x^2} \frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1$.
If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to: (8-01-2020/Shift-1)
- (a) $-\frac{1}{\sqrt{2}}$ (b) $-\frac{\sqrt{3}}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$
42. The differential equation of the family of curves, $x^2 = 4b(y+b), b \in R$ is (8-01-2020/Shift-2)
- (a) $xy'' = y'$ (b) $x(y')^2 = x + 2yy'$
(c) $x(y')^2 = x - 2yy'$ (d) $x(y')^2 = 2yy' - x$
43. If for $x \geq 0, y = y(x)$ is the solution of the differential equation $(1+x)dy = [(1+x)^2 + y - 3]dx, y(2) = 0$ then $y(3)$ is equal to _____. (9-01-2020/Shift-1)
44. If $\frac{dy}{dx} = \frac{xy}{x^2 + y^2} : y(1) = 1$; then a value of x satisfying $y(x) = e$ is (9-1-2020/Shift-2)
- (a) $\sqrt{3}e$ (b) $\frac{1}{2}\sqrt{3}e$
(c) $\sqrt{2}e$ (d) $\frac{e}{\sqrt{2}}$



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. The differential equation of all the ellipses centred at the origin and having their axes as co-ordinate axes is :

(a) $y^2 + xy_1^2 - yy_1 = 0$ (b) $xyy_2 + xy_1^2 - yy_1 = 0$

(c) $yy_2 + xy_1^2 - xy_1 = 0$ (d) None of these

2. The differential equation whose general solution is given

by, $y = (c_1 \cos(x + c_2)) - (c_3 e^{(-x+c_4)}) + (c_5 \sin x)$, where

c_1, c_2, c_3, c_4, c_5 are arbitrary constants, is

(a) $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} + y = 0$

(b) $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0$

(c) $\frac{d^5 y}{dx^5} + y = 0$

(d) $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$

3. If the function $y = e^{4x} + 2e^{-x}$ is a solution of the differential

equation $\frac{\frac{d^3 y}{dx^3} - 13 \frac{dy}{dx}}{y} = K$ then the value of K is

(a) 4

(b) 6

(c) 9

(d) 12

4. A solution of the differential equation

$\left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + y = 0$ is:

(a) $y = 2$

(b) $y = 2x$

(c) $y = 2x - 4$

(d) $y = 2x^2 - 4$

5. A curve $y = f(x)$ passing through the point $\left(1, \frac{1}{\sqrt{e}}\right)$ satisfies

the differential equation $\frac{dy}{dx} + xe^{-\frac{x^2}{2}} = 0$. Then which of the

following does not hold good?

(a) $f(x)$ is differentiable at $x = 0$.

(b) $f(x)$ is symmetric w.r.t. the origin.

(c) $f(x)$ is increasing for $x < 0$ and decreasing for $x > 0$.

(d) $f(x)$ has two inflection points.

6. Solution of the differential equation

$\frac{dy}{dx} + \sin\left(\frac{x+y}{2}\right) = \sin \frac{x-y}{2}$ is

(a) $\ln \left| \tan \frac{y}{4} \right| = C - 2 \sin \frac{x}{2}$

(b) $\ln \left| \tan \frac{x}{4} \right| = C - 2 \sin \frac{y}{2}$

(c) $\ln \left| \tan \frac{y}{2} \right| = C - 2 \sin \frac{x}{4}$

(d) $\ln |\tan y| = C + 2 \sin \frac{y}{2}$

7. Solution of the differential equation

$\left(\frac{x+y-1}{x+y-2}\right) \frac{dy}{dx} = \left(\frac{x+y+1}{x+y+2}\right)$, given that $y = 1$ when $x = 1$, is

(a) $\log \left| \frac{(x-y)^2 - 2}{2} \right| = 2(x+y)$

(b) $\log \left| \frac{(x-y)^2 + 2}{2} \right| = 2(x-y)$

(c) $\log \left| \frac{(x+y)^2 + 2}{2} \right| = 2(x-y)$

(d) none of these



8. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi(y/x)}{\phi'(y/x)}$ is

(a) $x\phi\left(\frac{y}{x}\right) = k$ (b) $\phi\left(\frac{y}{x}\right) = kx$
 (c) $y\phi\left(\frac{y}{x}\right) = k$ (d) $\phi\left(\frac{y}{x}\right) = ky$

9. The solution of the differential equation $(x^3 - 3xy^2) dx = (y^3 - 3x^2y) dy$ is

(a) $x^2 - y^2 = c(x^2 + y^2)^2$ (b) $y^2 - \frac{x^2}{2} = c(x^2 + y^2)^2$
 (c) $(x^2 - y^2)^2 = c(x^2 + y^2)$ (d) none of these

10. Solution of differential equation

$$x^3 \frac{dy}{dx} = y^3 + y^2 \sqrt{y^2 - x^2} \text{ is}$$

(a) $xy = c(y + \sqrt{y^2 - x^2})$
 (b) $xy = c(y + \sqrt{x^2 - y^2})$
 (c) $xy = c(x + \sqrt{x^2 + y^2})$
 (d) None

11. Solution of the differential equation

$$y^2(ydx + 2xdy) - x^2(2ydx + xdy) = 0 \text{ is}$$

(a) $x^2y^2(x^2 + y^2) = c$ (b) $x^2y^2 = (x^2 - y^2) + c$
 (c) $x^2y^2 = c$ (d) None

12. The substitution $y = z^\alpha$ transforms the differential equation $(x^2y^2 - 1)dy + 2xy^3dx = 0$ into a homogeneous differential equation for

(a) $\alpha = -1$ (b) 0
 (c) $\alpha = 1$ (d) no value of α .

13. The solution of the differential equation

$$\frac{dy}{dx} + \frac{\cos x (3 \cos y - 7 \sin x - 3)}{\sin y (3 \sin x - 7 \cos y + 7)} = 0 \text{ is}$$

(a) $(\cos y - \sin x - 1)(\cos y + \sin x - 1)^5 = c$
 (b) $(\cos x - \sin y + 1)(\cos y + \sin x - 1)^5 = c$
 (c) $(\cos y - \sin x - 1)^5(\cos y + \sin x - 1)^2 = c$
 (d) none of these

14. Solution of the differential equation $\frac{\sqrt{x} dx + \sqrt{y} dy}{\sqrt{x} dx - \sqrt{y} dy} = \sqrt{\frac{y^3}{x^3}}$

is given by

(a) $\frac{3}{2} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1}\left(\frac{y}{x}\right)^{3/2} + c = 0$

(b) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left|\frac{x^{3/2} + y^{3/2}}{x^{3/2}}\right| + \tan^{-1} \frac{y}{x} + c = 0$

(c) $\frac{2}{3} \log\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right) + c = 0$

(d) none of the above

15. A function $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx} - y = \cos x - \sin x \text{ with initial condition that } y \text{ is bounded when } x \rightarrow \infty. \text{ The area enclosed by } y = f(x), y = \cos x \text{ and the } y\text{-axis in first quadrant is}$$

(a) $\sqrt{2} - 1$ (b) $\sqrt{2}$
 (c) 1 (d) $\frac{1}{\sqrt{2}}$

16. A function $y = f(x)$ satisfies

$$(x+1) \cdot f'(x) - 2(x^2 + x)f(x) = \frac{e^{x^2}}{(x+1)}, \forall x > -1$$

If $f(0) = 5$, then $f(x)$ is

(a) $\left(\frac{3x+5}{x+1}\right) \cdot e^{x^2}$ (b) $\left(\frac{6x+5}{x+1}\right) \cdot e^{x^2}$
 (c) $\left(\frac{6x+5}{(x+1)^2}\right) \cdot e^{x^2}$ (d) $\left(\frac{5-6x}{x+1}\right) \cdot e^{x^2}$

17. Solution of differential equation

$$\frac{dy}{dx} = y + \int_0^1 y dx, y(0) = 1 \text{ is:}$$

(a) $y = \frac{1}{3-e}(2e^x + e + 1)$ (b) $y = \frac{1}{3+e}(2e^x - e + 1)$
 (c) $y = \frac{1}{3-e}(2e^x - e + 1)$ (d) none



18. The equation of the curve satisfying the equation

$$x(1 - x \ln y) \frac{dy}{dx} + y = 0 \text{ and passing through } \left(1, \frac{1}{e}\right) \text{ is}$$

- (a) $x(ey + \ln y + 1) = 1$ (b) $(ey + \ln x + 1) = 1$
(c) $y(ex + \ln y + 1) = 0$ (d) None of these

19. Solution of the differential equation

$$(e^{x^2} + e^{y^2})y \frac{dy}{dx} + e^{x^2}(xy^2 - x) = 0, \text{ is}$$

- (a) $e^{x^2}(y^2 - 1) + e^{y^2} = c$ (b) $e^{y^2}(x^2 - 1) + e^{x^2} = c$
(c) $e^{y^2}(y^2 - 1) + e^{x^2} = c$ (d) $e^{x^2}(y - 1) + e^{y^2} = c$

20. Solution of the differential equation

$$y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0 \text{ is given by}$$

- (a) $2 \log |x| - \log |y| - \frac{1}{xy} = c$
(b) $2 \log |y| - \log |x| - \frac{1}{xy} = c$
(c) $2 \log |x| + \log |y| + \frac{1}{xy} = c$
(d) $2 \log |y| + \log |x| + \frac{1}{xy} = c$

21. Solution of differential equation

$$(x^2 + y^2 + a^2)y \frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0 \text{ is}$$

- (a) $(x^2 + y^2)^2 + 2a^2(x^2 - y^2) = c$
(b) $(x^2 + y^2)^2 + 2a^2(y^2 - x^2) = c$
(c) $(x^2 - y^2)^2 + 2a^2(x^2 + y^2) + c = 0$
(d) None

22. Solution of the differential equation

$$3x^2y^2 + \cos(xy) - xy \sin(xy) + \frac{dy}{dx}(2x^3y - x^2 \sin(xy)) = 0 \text{ is}$$

- (a) $(x^2y^2 + \cos xy) = c$ (b) $x(x^2y^2 + \cos xy) = c$
(c) $y(x^2y^2 + \cos xy) = c$ (d) None

23. The solution of the differential equation

$$x dx + y dy + \frac{xdy - ydx}{x^2 + y^2} = 0 \text{ is}$$

- (a) $y = x \tan\left(\frac{x^2 + y^2 + c}{2}\right)$
(b) $x = y \tan\left(\frac{x^2 + y^2 + c}{2}\right)$
(c) $y = x \tan\left(\frac{c - x^2 - y^2}{2}\right)$
(d) none of these

24. A curve passing through (2, 3) and satisfying the differential

$$\text{equation } \int_0^x t y(t) dt = x^2 y(x), (x > 0) \text{ is}$$

- (a) $x^2 + y^2 = 13$ (b) $y^2 = \frac{9}{2}x$
(c) $\frac{x^2}{8} + \frac{y^2}{18} = 1$ (d) $xy = 6$

25. If $\int_a^x t y(t) dt = x^2 + y(x)$ then y as a function of x is

- (a) $y = 2 - (2 + a^2)e^{\frac{x^2 - a^2}{2}}$
(b) $y = 1 - (2 + a^2)e^{\frac{x^2 - a^2}{2}}$
(c) $y = 2 - (1 + a^2)e^{\frac{x^2 - a^2}{2}}$
(d) none

26. The function satisfying the equation

$$f(x) = -\int_0^x f(t) \tan t dt + \int_0^x \tan(t - x) dt,$$

$$\text{where } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ is:}$$

- (a) $1 + \cos x$ (b) $1 - \cos x$
(c) $\cos x - 1$ (d) none



27. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is
- (a) $ye^{\frac{x}{y}} = e$ (b) $xe^{\frac{x}{y}} = e$
- (c) $xe^{\frac{y}{x}} = e$ (d) $ye^{\frac{y}{x}} = e$
28. The latus rectum of the conic passing through the origin and having the property that normal at each point (x, y) intersects the x-axis at ((x+1), 0) is :
- (a) 1 (b) 2
- (c) 4 (d) none
29. The curve for which the sum of the lengths of the tangent and subtangent at any of its point is proportional to the product of the co-ordinates of the point of tangency. The proportionality factor is equal to k.
- (a) $y = \ln |c(k^2x^2 - 1)|$ (b) $y = \frac{1}{k} \ln |c(k^2x^2 - 1)|$
- (c) $y = \ln |k(c^2x^2 + 1)|$ (d) $y = \ln |k(c^2x^2 - 1)|$
30. The equation of the curve for which any tangent intersects the y-axis at the point equidistant from the point of tangency and the origin is
- (a) $x^2 - y^2 = cx$ (b) $x^2 + y^2 = cy$
- (c) $x^2 + y^2 = cx$ (d) $x^2 - y^2 = cy$
31. The equation of the curve passing through origin if the middle point of the segment of its normal from any point of the curve to the x-axis lies on the parabola $2y^2 = x$, is
- (a) $y^2 = 2x + 1 - e^{2x}$ (b) $y^2 = 2x - 1 + e^x$
- (c) $y^2 = 2x + 1 + e^x$ (d) None
32. The equation of the curve for which the portion of y-axis cut-off between the origin and the tangent varies as cube of the abscissa of the point of contact, is
- (a) $2y + kx^3 = cx$ (b) $2x + ky^3 = cy$
- (c) $2x^3 + ky = c$ (d) None
33. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is
- (a) 30 min (b) 45 min
- (c) 60 min (d) 80 min
34. An inverted conical tank of 2 m radius and 4 m height is initially full of water, has an outlet at bottom. The outlet is opened at some instant. The rate of flow through the outlet at any time t is $6h^{3/2}$, where h is height of water level above the outlet at time t. Then the time it takes to empty the tank is
- (a) $\frac{2\pi}{11}$ units (b) $\frac{\pi}{9}$ units
- (c) $\frac{2\pi}{8}$ units (d) none of these
35. The orthogonal trajectories of the family of curves $a^{n-1}y = x^n$ are given by (a is the arbitrary constant)
- (a) $x^n + n^2y = \text{constant}$ (b) $ny^2 + x^2 = \text{constant}$
- (c) $n^2x + y^n = \text{constant}$ (d) $n^2x - y^n = \text{constant}$
36. The orthogonal trajectories of the family of circle given by $x^2 + y^2 - 2ay = 0$ (a is parameter), is :
- (a) $x^2 + y^2 - 2kx = 0$ (b) $x^2 + y^2 - 2ky = 0$
- (c) $x^2 + y^2 - 2k_1x - 2k_2y = 0$ (d) none of these
37. The solution of the differential equation $\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y'(0) = 1$ and $y(0) = 0$ is
- (a) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} - 1$
- (b) $-\frac{\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{x}{3}$
- (c) $-\frac{\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{x}{3} + 1$
- (d) None of these



38. The equation of the curve satisfying the differential equation

$$\frac{d^2y}{dx^2}(x^2 + 1) = 2x \frac{dy}{dx} \text{ passing through the point } (0, 1) \text{ and}$$

having slope of tangent at $x = 0$ as 3 is

- (a) $y = x^2 + 3x + 2$ (b) $y^2 = x^2 + 3x + 1$
(c) $y = x^3 + 3x + 1$ (d) None of these

Objective Questions II [One or more than one correct option]

39. The solution of $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ satisfying $y(1) = 1$ is

given by

- (a) a hyperbola
(b) a circle
(c) $y^2 = x(1+x) - 1$
(d) $(x-2)^2 + (y-3)^2 = 5$

40. Solution of the differential equation :

$$(3 \tan x + 4 \cot y - 7) \sin^2 y \, dx - (4 \tan x + 7 \cot y - 5) \cos^2 x \, dy = 0 \text{ is}$$

- (a) $\frac{3}{2} \cot^2 x - 7 \cot x + \frac{7}{2} \tan^2 y - 5 \tan y + 4 \cot x \cdot \tan y = c$
(b) $\frac{3}{2} \tan^2 x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4 \tan x \cdot \cot y = c$
(c) $3 \tan^2 y - 14 \cot x \cdot \tan^2 y + 7 \cot^2 x - 10 \tan y \cot^2 x$
 $+ 8 \cot x \cdot \tan y + 2c \cot^2 x \tan^2 y = 0$
(d) $3 \cot^2 y - 14 \cot x \cdot \cot^2 y + 7 \cot^2 x + 10 \cot y \tan^2 x$
 $+ 8 \tan x \cdot \cot y = 0$

41. Solutions of the differential equation

$$\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0 \text{ are given by}$$

- (a) $y + e^{-x} = C$ (b) $y - e^{-x} = C$
(c) $y + e^x = C$ (d) $y - e^x = C$

42. The solution of $p^2 + (2y \cot x)p = y^2$ where $p = \frac{dy}{dx}$ is

- (a) $y(1 + \cos x) = c$ (b) $y(1 - \cos x) = c$
(c) $x = 2 \sin^{-1} \sqrt{\frac{c}{2y}}$ (d) $x = 2 \sin^{-1}(\sqrt{2y}) + C$

Numerical Value Type Questions

43. If the solution of differential equation

$$x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 12y \text{ is } y = Ax^m + Bx^{-n} \text{ then find the value}$$

of $m + n$, if $m \& n \in \mathbb{N}$.

44. A rumour spreads through a population of 5000 people at a rate proportional to the product of the number of people who have heard it and the number who have not. Suppose that 100 people initiate the rumour and that a total of 500 people know the rumour after 2 days. How many days will it take for half the people to hear the rumour (round off to nearest integer)

$$(\log 9 / \log 49 = 129/229).$$

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

45. Column I	Column II
Diff. equation	Solution curve
(A) $(1+xy)xdy + (1-xy)ydx = 0$	(P) $\frac{x^3}{3} + \frac{y^3}{3}$ $- 2x^2y - 2xy^2 = c$
(B) $(x^2 - 4xy - 2y^2)dx$ $+ (y^2 - 4xy - 2x^2)dy = 0$	(Q) $\frac{-1}{xy} + \log\left(\frac{y}{x}\right) = c$
(C) $e^y \cdot dx + (xe^y - 2y)dy = 0$	(R) $\sqrt{x^2 + y^2} - x/y = c$
(D) $\frac{xdx + ydy}{\sqrt{x^2 + y^2}} = \frac{ydx - xdy}{y^2}$	(S) $xe^y - y^2 = c$

The correct matching is :

- (a) (A-Q; B-P; C-S; D-R)
(b) (A-P; B-S; C-S; D-R)
(c) (A-P; B-S; C-Q; D-R)
(d) (A-Q; B-P; C-R; D-S)



Using the following passage, solve Q.46 and Q.47

Passage

Consider the differential equation $e^x(ydx - dy) = e^{-x}(ydx + dy)$. Let $y = f(x)$ be a particular solution to this differential equation which passes through the point $(0, 2)$.

Let $C \equiv y = \log_{1/4} \left(x - \frac{1}{4} \right) + \frac{1}{2} \log_4(16x^2 - 8x + 1)$, be another curve

46. The range of the function $g(x) = \log_2(f(x))$ is
 (a) $[1, \infty)$ (b) $[2, \infty)$
 (c) $[0, \infty)$ (d) None of these
47. If the area bounded by the curve $y = f(x)$, curve C, ordinate $x = 1/4$ & the ordinate $x = a$ is $4 - \ln 4 + \frac{1}{e^{1/4}} - e^{1/4}$, then value of a is
 (a) $\ln 6$ (b) $\ln 4$
 (c) 4 (d) $\ln 12$

Subjective Type

48. If $(a + bx)e^{y/x} = x$ then prove that

$$x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$$

49. A normal is drawn at a point $P(x, y)$ of a curve meets the x -axis at Q . If PQ is of constant length k , then show that the differential equation of the curve is

$$y \frac{dy}{dx} = \pm \sqrt{k^2 - y^2}$$

50. A curve passing through the point $(1, 1)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of P from the x -axis. Determine the equation of the curve.



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. If $x^2 + y^2 = 1$, then (2000)
- (a) $yy'' - 2(y')^2 + 1 = 0$
- (b) $yy'' + (y')^2 + 1 = 0$
- (c) $yy'' + (y')^2 - 1 = 0$
- (d) $yy'' + 2(y')^2 + 1 = 0$

2. If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$, then $y(1)$ is equal to (2003)

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
- (c) 1 (d) -1

3. If $y = y(x)$ and $\frac{2 + \sin x}{y+1} \left(\frac{dy}{dx} \right) = -\cos x$, $y(0) = 1$, then

$y\left(\frac{\pi}{2}\right)$ equals : (2004)

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
- (c) $-\frac{1}{3}$ (d) 1

4. The solution of primitive equation $(x^2 + y^2)dy = xy dx$, is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e$, then x_0 is (2005)

- (a) $\sqrt{\frac{e^2 + 1}{2}}$ (b) $\sqrt{2(e^2 + 1)}$
- (c) $\sqrt{2(e^2 - 1)}$ (d) $\sqrt{3} e$

5. For the primitive integral equation $ydx + y^2dy = x dy$, $x \in \mathbb{R}$, $y > 0$, $y = y(x)$, $y(1) = 1$, then $y(-3)$ is (2005)
- (a) 3 (b) 5
- (c) 1 (d) 2

6. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $f(x)$ is : (2007)

- (a) $\frac{1}{3x} + \frac{2x^2}{3}$ (b) $-\frac{1}{3x} + \frac{4x^2}{3}$
- (c) $-\frac{1}{x} + \frac{2}{x^2}$ (d) $\frac{1}{x}$

7. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with : (2007)

- (a) variable radii and a fixed centre at $(0, 1)$
- (b) variable radii and a fixed centre at $(0, -1)$
- (c) fixed radius 1 and variable centres along the x-axis
- (d) fixed radius 1 and variable centres along the y-axis

8. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1-x^2}}$

in $(-1, 1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx$

is (2014)

- (a) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (b) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$
- (c) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (d) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

9. If $y = y(x)$ satisfies the differential equation

$$8\sqrt{x} \left(\sqrt{9 + \sqrt{x}} \right) dy = \left(\sqrt{4 + \sqrt{9 + \sqrt{x}}} \right)^{-1} dx, \quad x > 0$$

and $y(0) = \sqrt{7}$, then $y(256) =$ (2017)

- (a) 80 (b) 9
- (c) 16 (d) 3



Objective Questions II [One or more than one correct option]

10. A curve $y = f(x)$ passes through $(1, 1)$ and tangent at $P(x, y)$ cuts the x -axis and y -axis at A and B respectively such that $BP : AP = 3 : 1$, then (2006)
- (a) equation of curve is $xy' - 3y = 0$
- (b) normal at $(1, 1)$ is $x + 3y = 4$
- (c) curve passes through $(2, 1/8)$
- (d) equation of curve is $x^3y - 1 = 0$
11. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statement is (are) true? (2015)
- (a) $y(-4) = 0$
- (b) $y(-2) = 0$
- (c) $y(x)$ has a critical point in the interval $(-1, 0)$
- (d) $y(x)$ has no critical point in the interval $(-1, 0)$
12. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circle is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and $y' \left(\text{here } y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2} \right)$, then which of the following statements is (are) true? (2015)
- (a) $P = y + x$
- (b) $P = y - x$
- (c) $P + Q = 1 - x + y + y' + (y')^2$
- (d) $P - Q = x + y - y' - (y')^2$
13. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point $(1, 3)$. Then the solution curve (2016)
- (a) intersects $y = x + 2$ exactly at one point
- (b) intersects $y = x + 2$ exactly at two points
- (c) intersects $y = (x + 2)^2$
- (d) does **NOT** intersect $y = (x + 3)^2$
14. Let Γ denote a curve $y = f(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to Γ at a point P intersect the y -axis at Y_P . If PY_P has length 1 for each point P on Γ . Then which of the following options is/are correct? (2019)

$$(a) y = -\ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) + \sqrt{1 - x^2}$$

$$(b) xy' + \sqrt{1 - x^2} = 0$$

$$(c) xy' - \sqrt{1 - x^2} = 0$$

$$(d) y = \ln \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) - \sqrt{1 - x^2}$$

Numerical Value Type Questions

15. Let f be a real-valued differentiable function on $\mathbb{R}$ (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to.... (2010)
16. Let $f : [1, \infty) \rightarrow [2, \infty)$ be a differential function such that $f(1) = 2$. If $6 \int_1^x f(t) dt = 3x f(x) - x^3 - 5$, Then $f(2) =$ (2011)

ASSERTION REASON

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
- (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
- (C) If ASSERTION is true, REASON is false.
- (D) If ASSERTION is false, REASON is true.

17. Let a solution $y = y(x)$ of the differential equation $x\sqrt{x^2 - 1} dy - y\sqrt{y^2 - 1} dx = 0$ Satisfy $y(2) = \frac{2}{\sqrt{3}}$.

(2008)

Assertion : $y(x) = \sec \left(\sec^{-1} x - \frac{\pi}{6} \right)$

Reason : $y(x)$ is given by $\frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1 - \frac{1}{x^2}}$

- (a) A (b) B
(c) C (d) D



Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

18. Column I

Column II

(A) $\int_0^{\pi/2} (\sin x)^{\cos x} (\cos x \cot x - \log(\sin x)^{\sin x}) dx$ (p) 1

(B) Area bounded by $4y^2 = -x$ and $x - 1 = -5y^2$ (q) 0

(C) Cosine of the angle of intersection of curves $y = 3^{x-1} \log x$ and $y = x^x - 1$ is (r) $Ke^{y/2}$

(D) Solution of differential equation (s) $4/3$

$\frac{dy}{dx} = \frac{2}{(x+y)}$ is $x + y + 2 = \dots$ (2006)

The correct matching is :

- (a) (A-P; B-P; C-S; D-R)
- (b) (A-P; B-S; C-R; D-P)
- (c) (A-P; B-S; C-R; D-S)
- (d) (A-P; B-S; C-P; D-R)

Subjective Type

19. A curve passes through (2, 0) and slope at point P(x, y) is $\frac{(x+1)^2 + (y-3)}{(x+1)}$. Find equation of curve and area between curve and x-axis in 4th quadrant. (2004)

20. Determine the equation of the curve passing through the origin, in the form $y = f(x)$, which satisfies the differential equation $\frac{dy}{dx} = \sin(10x + 6y)$. (2006)



DIFFERENTIAL EQUATIONS



Please share your valuable feedback by scanning the QR code.



Teacher's Note:



Go down deep into anything and you will find mathematics, if you integrate anything you will find mathematics. If you differentiate anything you will find mathematics.

*"Samajh ayega to mazaa ayega,
Mazaa ayega to samajh ayega".*



Pulkit Sir loves teaching Mathematics, he has been a pioneer in producing Top Ranks in JEE and Olympiads. He has personally taught and mentored AIR 1,6,7 (JEE ADVANCED), and thousands of more students who have successfully cleared these competitive exams in the last few years.



Answer Key



CHAPTER -1 | INDEFINITE INTEGRATION

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------|---------|----------|---------|---------|
| 1. (c) | 2. (b) | 3. (b) | 4. (3) | 5. (a) |
| 6. (d) | 7. (a) | 8. (c) | 9. (d) | 10. (a) |
| 11. (b) | 12. (2) | 13. (d) | 14. (d) | 15. (b) |
| 16. (b) | 17. (c) | 18. (3) | 19. (b) | 20. (a) |
| 21. (6) | 22. (a) | 23. (8) | 24. (a) | 25. (b) |
| 26. (a) | 27. (9) | 28. (8) | 29. (b) | 30. (b) |
| 31. (a) | 32. (d) | 33. (a) | 34. (2) | 35. (b) |
| 36. (36) | 37. (b) | 38. (a) | 39. (b) | 40. (b) |
| 41. (1) | 42. (c) | 43. (c) | 44. (c) | 45. (c) |
| 46. (b) | 47. (a) | 48. (8) | 49. (4) | 50. (b) |
| 51. (a) | 52. (a) | 53. (c) | 54. (a) | 55. (a) |
| 56. (b) | 57. (1) | 58. (d) | 59. (b) | 60. (b) |
| 61. (b) | 62. (c) | 63. (10) | 64. (b) | 65. (c) |
| 66. (d) | 67. (c) | 68. (b) | 69. (a) | 70. (c) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (d) | 4. (a) | 5. (b) |
| 6. (b) | 7. (b) | 8. (b) | 9. (b) | 10. (c) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (d) |
| 16. (a) | 17. (a) | 18. (c) | 19. (c) | 20. (b) |
| 21. (c) | 22. (d) | 23. (a) | 24. (b) | 25. (a) |
| 26. (d) | 27. (c) | 28. (b) | 29. (a) | 30. (a) |
| 31. (a) | 32. (a) | 33. (d) | 34. (a) | 35. (c) |
| 36. (b) | | | | |

CHAPTER - 1 | INDEFINITE INTEGRATION

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (b) 2. (d) 3. (b) 4. (b) 5. (a)
 6. (c) 7. (a) 8. (b) 9. (b) 10. (c)
 11. (b) 12. (a) 13. (d) 14. (a) 15. (a)
 16. (d) 17. (c) 18. (b) 19. (d) 20. (c)
 21. (c) 22. (c) 23. (a) 24. (b) 25. (c)
 26. (b) 27. (b) 28. (c) 29. (c) 30. (a)
 31. (c) 32. (c) 33. (d) 34. (b) 35. (a)
 36. (c) 37. (c) 38. (b) 39. (b) 40. (b)
 41. (a) 42. (b) 43. (a) 44. (b,d)
 45. (a,b,c,d) 46. (c,d) 47. (a,b) 48. (2240)
 49. (002) 50. (1998) 51. (1982) 52. (I) 53. (d)
 54. (c) 55. (c) 56. (b) 57. (d) 58. (a)
 59. (c) 60. (d) 61. (b) 62. (d)

$$63. \ln \left| \frac{xe^x}{1+xe^x} \right| + \frac{1}{1+xe^x} + c$$

$$64. -\frac{1}{2} \log|x+1| + \frac{1}{4} \log(x^2+1) + \frac{3}{2} \tan^{-1} x + \frac{x}{x^2+1} + c$$

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (c) 2. (d) 3. (a) 4. (c) 5. (c)
 6. (a,c) 7. (d)

$$8. (x+1) \tan^{-1} \left(\frac{2x+2}{3} \right) - \frac{3}{4} \log(4x^2+8x+13) + c$$

$$9. \frac{1}{6(m+1)} (2x^{3m} + 3x^{2m} + 6x^m)^{\frac{m+1}{m}} + c$$

Answer Key

CHAPTER -2 | DEFINITE INTEGRATION, AREA UNDER CURVES



EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------|---------|---------|---------|----------|
| 1. (b) | 2. (c) | 3. (c) | 4. (c) | 5. (d) |
| 6. (b) | 7. (b) | 8. (b) | 9. (2) | 10. (c) |
| 11. (2) | 12. (b) | 13. (1) | 14. (b) | 15. (b) |
| 16. (2) | 17. (c) | 18. (a) | 19. (c) | 20. (4) |
| 21. (d) | 22. (d) | 23. (4) | 24. (c) | 25. (3) |
| 26. (3) | 27. (c) | 28. (c) | 29. (c) | 30. (2) |
| 31. (c) | 32. (a) | 33. (c) | 34. (a) | 35. (50) |
| 36. (a) | 37. (b) | 38. (b) | 39. (d) | 40. (29) |
| 41. (19) | 42. (b) | 43. (b) | 44. (d) | 45. (a) |
| 46. (b) | 47. (b) | 48. (2) | 49. (c) | 50. (a) |
| 51. (c) | 52. (b) | 53. (b) | 54. (c) | 55. (a) |
| 56. (c) | 57. (d) | 58. (d) | 59. (a) | 60. (c) |
| 61. (a) | 62. (c) | 63. (d) | 64. (a) | 65. (b) |
| 66. (d) | 67. (b) | 68. (b) | 69. (2) | 70. (2) |
| 71. (a) | 72. (b) | 73. (a) | 74. (c) | 75. (a) |
| 76. (b) | 77. (b) | 78. (c) | 79. (d) | 80. (a) |
| 81. (a) | 82. (c) | 83. (4) | 84. (3) | 85. (b) |
| 86. (6) | 87. (1) | 88. (6) | 89. (3) | 90. (2) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--------------|---------|--------------|---------|------------|
| 1. (b) | 2. (a) | 3. (d) | 4. (c) | 5. (a) |
| 6. (a) | 7. (c) | 8. (b) | 9. (b) | 10. (c) |
| 11. (b) | 12. (a) | 13. (a) | 14. (d) | 15. (d) |
| 16. (a) | 17. (b) | 18. (c) | 19. (c) | 20. (b) |
| 21. (c) | 22. (c) | 23. (d) | 24. (c) | 25. (d) |
| 26. (a) | 27. (d) | 28. (b) | 29. (d) | 30. (d) |
| 31. (d) | 32. (b) | 33. (a) | 34. (d) | 35. (c) |
| 36. (b) | 37. (d) | 38. 18.00 | 39. (a) | 40. (c) |
| 41. (b) | 42. (b) | 43. (d) | 44. (d) | 45. (b) |
| 46. (a) | 47. (c) | 48. (b) | 49. (a) | 50. (b) |
| 51. (b) | 52. (c) | 53. (d) | 54. (c) | 55. (d) |
| 56. (d) | 57. (b) | 58. (001.50) | | 59. (b) |
| 60. (001.00) | | 61. (a) | 62. (d) | 63. (d) |
| 64. (d) | 65. (c) | 66. (a) | 67. (b) | 68. 021.00 |
| 69. (a) | 70. (b) | 71. (b) | 72. (b) | 73. (a) |
| 74. (d) | 75. (c) | 76. (c) | 77. (a) | 78. (b) |
| 79. (b) | 80. (a) | 81. (c) | 82. (b) | 83. (d) |
| 84. (d) | 85. (a) | | | |

CHAPTER - 2 | DEFINITE INTEGRATION, AREA UNDER CURVES

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---|-------------|------------|-------------|-----------|
| 1. (d) | 2. (a) | 3. (d) | 4. (c) | 5. (d) |
| 6. (d) | 7. (c) | 8. (c) | 9. (d) | 10. (d) |
| 11. (c) | 12. (a) | 13. (c) | 14. (d) | 15. (a) |
| 16. (c) | 17. (d) | 18. (d) | 19. (d) | 20. (d) |
| 21. (b) | 22. (c) | 23. (d) | 24. (c) | 25. (a) |
| 26. (d) | 27. (b) | 28. (c) | 29. (a) | 30. (c) |
| 31. (b) | 32. (a) | 33. (d) | 34. (c) | 35. (c) |
| 36. (a) | 37. (d) | 38. (b) | 39. (b) | 40. (d) |
| 41. (c) | 42. (b) | 43. (b) | 44. (a) | 45. (a) |
| 46. (b) | 47. (b) | 48. (a) | 49. (a) | 50. (c) |
| 51. (d) | 52. (b) | 53. (d) | 54. (d) | 55. (b) |
| 56. (c) | 57. (b) | 58. (b) | 59. (c) | 60. (a,c) |
| 61. (a,b,d) | 62. (a,d) | 63. (b, c) | 64. (a,c,d) | |
| 65. (a,b) | 66. $n = 3$ | 67. (a) | 68. (c) | 69. (a) |
| 70. (a) | 71. (d) | 72. (d) | 73. (c) | 74. (b) |
| 75. (a) | 76. (a) | 77. (a) | | |
| 78. $\frac{1}{2} \log 6 - \frac{1}{10}$ | 79. π^2 | | | |

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--|---|-----------|--------------|-----------|
| 1. (b) | 2. (b) | 3. (c) | 4. (c) | 5. (c) |
| 6. (a) | 7. (c) | 8. (a) | 9. (a) | 10. (d) |
| 11. (b) | 12. (a) | 13. (c) | 14. (a) | 15. (c) |
| 16. (b) | 17. (d) | 18. (b) | 19. (a) | 20. (d) |
| 21. (a) | 22. (c) | 23. (c) | 24. (a) | 25. (a,d) |
| 26. (a,b,c) | 27. (a,b,d) | 28. (b,d) | 29. (a,c,d) | |
| 30. (a,c) | 31. (b,c) | 32. (a,c) | 33. (b,d) | 34. (a,b) |
| 35. (a,b,d) | 36. 5051 | 37. (2) | 38. (0) | |
| 39. (3) | 40. (9) | 41. (7) | 42. (1) | 43. (4) |
| 44. (2) | 45. 4.00 | 46. 0.50 | 47. (004.00) | |
| 48. (a) | 49. (d) | 50. (c) | 51. (b) | 52. (a) |
| 53. (d) | 54. (a, b, c) | 55. (c,d) | | |
| 56. $\frac{1}{2}(\ln x)^2$ | 57. $\frac{4\pi}{\sqrt{3}} \tan^{-1}\left(\frac{1}{2}\right)$ | | | |
| 58. $\frac{24}{5} \left(\operatorname{ecos}\left(\frac{1}{2}\right) + \frac{e}{2} \sin\left(\frac{1}{2}\right) - 1 \right)$ | | | | |

Answer Key

CHAPTER -3 | DIFFERENTIAL EQUATIONS



EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|----------|---------|----------|---------|
| 1. (a) | 2. (d) | 3. (2) | 4. (c) | 5. (1) |
| 6. (d) | 7. (c) | 8. (4) | 9. (3) | 10. (a) |
| 11. (c) | 12. (a) | 13. (b) | 14. (c) | 15. (b) |
| 16. (a) | 17. (b) | 18. (c) | 19. (2) | 20. (d) |
| 21. (a) | 22. (c) | 23. (b) | 24. (a) | 25. (b) |
| 26. (a) | 27. (a) | 28. (b) | 29. (b) | 30. (d) |
| 31. (a) | 32. (5) | 33. (d) | 34. (a) | 35. (d) |
| 36. (b) | 37. (c) | 38. (a) | 39. (b) | 40. (c) |
| 41. (b) | 42. (c) | 43. (b) | 44. (a) | 45. (b) |
| 46. (c) | 47. (b) | 48. (b) | 49. (c) | 50. (b) |
| 51. (a) | 52. (b) | 53. (c) | 54. (d) | 55. (b) |
| 56. (a) | 57. (35) | 58. (b) | 59. (20) | 60. (b) |
| 61. (b) | 62. (2) | 63. (b) | | |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------|---------|----------|---------|---------|
| 1. (a) | 2. (b) | 3. (b) | 4. (c) | 5. (c) |
| 6. (a) | 7. (b) | 8. (d) | 9. (d) | 10. (d) |
| 11. (d) | 12. (a) | 13. (c) | 14. (c) | 15. (a) |
| 16. (d) | 17. (b) | 18. (b) | 19. (c) | 20. (a) |
| 21. (a) | 22. (a) | 23. (b) | 24. (c) | 25. (b) |
| 26. (c) | 27. (b) | 28. (b) | 29. (b) | 30. (c) |
| 31. (d) | 32. (b) | 33. (b) | 34. (b) | 35. (d) |
| 36. (b) | 37. (c) | 38. (b) | 39. (d) | 40. (c) |
| 41. (c) | 42. (b) | 43. 3.00 | 44. (a) | |

CHAPTER -3 | DIFFERENTIAL EQUATIONS

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|---------------------------------|-------------|---------|-----------|-----------|
| 1. (b) | 2. (b) | 3. (d) | 4. (c) | 5. (b) |
| 6. (a) | 7. (d) | 8. (b) | 9. (a) | 10. (a) |
| 11. (d) | 12. (a) | 13. (a) | 14. (d) | 15. (a) |
| 16. (b) | 17. (c) | 18. (a) | 19. (a) | 20. (a) |
| 21. (b) | 22. (b) | 23. (c) | 24. (d) | 25. (a) |
| 26. (c) | 27. (a) | 28. (b) | 29. (b) | 30. (c) |
| 31. (a) | 32. (a) | 33. (c) | 34. (d) | 35. (b) |
| 36. (a) | 37. (d) | 38. (c) | 39. (a,c) | 40. (b,c) |
| 41. (a,d) | 42. (a,b,c) | 43. (7) | 44. (5) | |
| 45. (a) | 46. (a) | 47. (b) | | |
| 50. $x^2 + y^2 = 2x$ or $x = 1$ | | | | |

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--|-----------|-----------|-----------|-----------|
| 1. (b) | 2. (a) | 3. (a) | 4. (d) | 5. (a) |
| 6. (a) | 7. (c) | 8. (b) | 9. (d) | 10. (c,d) |
| 11. (a,c) | 12. (b,c) | 13. (a,d) | 14. (b,d) | 15. (9) |
| 16. (6) | 17. (c) | 18. (d) | | |
| 19. $\frac{4}{3}$ sq. unit | | | | |
| 20. $y = \frac{1}{3} \left[\tan^{-1} \left(\frac{5 \tan 4x}{4 - 3 \tan 4x} \right) - 5x \right]$ | | | | |

Note

